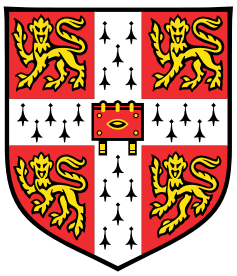
# Data-Centric AI for Reliable and Trustworthy Machine Learning



**Nabeel Seedat**

University of Cambridge

This thesis is submitted for the degree of
*Doctor of Philosophy*

Lucy Cavendish College June 2025

# Declaration

This thesis is the result of my own work and includes nothing which is the outcome of work done in collaboration or content generated by artificial intelligence, except as declared in the preface and specified in the text. It is not substantially the same as any work that has already been submitted, or is being concurrently submitted, for any degree, diploma or other qualification at the University of Cambridge or any other University or similar institution except as declared in the preface and specified in the text. It does not exceed the prescribed word limit for the relevant Degree Committee.

Nabeel Seedat
June 2025

# Abstract

Data-centric AI represents an emerging paradigm shift in machine learning (ML) research, underscoring the vital role of data rather than the de facto model-centric focus. This thesis adopts a holistic, data-centric perspective and advocates for the essential role of data in developing reliable and trustworthy ML systems, particularly in high-stakes settings. We begin by identifying five key data-centric challenges across the ML pipeline—from dataset collection to model deployment—namely: data characterization, test-time incongruity detection, synthetic data augmentation and curation, context-aware model evaluation, and data-enhanced uncertainty quantification. For each, we introduce novel problem formulations, propose principled data-centric ML methods and frameworks, and empirically demonstrate their effectiveness and impact in achieving our goal of reliable and trustworthy ML. First, we formalize data characterization, propose an algorithmic framework that stratifies datasets through the lens of uncertainty estimation and learning dynamics, and demonstrate its utility in tasks such as dataset auditing, data sculpting, feature acquisition, and more, applied to tabular, image and text data. Second, we formalize the notion of test-time incongruity within in-distribution data and propose a model-agnostic algorithm that constructs feature-wise intervals to flag uncertain or inconsistent instances, showing its advantages over model-dependent approaches. Third, we address the challenge of ML in ultra-low data regimes, proposing a synthetic data augmentation framework that synergizes large language model data generation with a principled curation mechanism to produce reliable and realistic synthetic data, thereby unlocking the data logjam in ultra-low data regimes. Fourth, we enhance the evaluation and testing of ML models by formalizing the context-aware testing paradigm and validating our LLM-based instantiation, which generates, operationalizes, and falsifies failure hypotheses to more effectively surface model deficiencies. Fifth, we improve uncertainty estimation (specifically conformal prediction) by leveraging the inherent signal in the data through a self-supervised conformal prediction framework that uses auxiliary representations learned from both unlabeled and labeled data, and validate its improvements to interval efficiency and adaptivity. Together, this thesis fundamentally reimagines ML development, offering an end-to-end, principled vision for data-centric AI supported by concrete methods and frameworks that advance the reliability and trustworthiness of ML systems.

I would like to dedicate this thesis to my loved ones.

# Acknowledgements

First and foremost, all praise and thanks are due to Allah. May peace and blessings be upon the Prophet Muhammad (peace be upon him), whose example reminds us that knowledge is a trust and that its highest purpose is to benefit others. It is through Allah's countless blessings that I was granted the extraordinary opportunity to pursue this PhD at Cambridge — a privilege I could never have imagined possible. I am deeply conscious that whatever I have achieved, including this thesis, is not through my efforts alone, but through His grace, guidance, and the capabilities with which He has blessed me. May this work, in some small way, be a means of seeking His pleasure, and may the knowledge gained through it be used to serve His creation in meaningful ways.

I am profoundly grateful to my supervisor, Mihaela van der Schaar, for believing in my potential and for her guidance, support, and encouragement throughout my PhD. Mihaela gave me the freedom to explore my own research interests and to pursue Data-centric AI before it became a fashionable topic. Her passion for bold, out-of-the-box research that tackles real-world problems pushed me beyond my comfort zone and helped me grow tremendously as a researcher. I am especially glad that this journey culminated in our many joint tutorials on Data-centric AI at some of the largest machine learning conferences.

I am grateful to Fergus Imrie for our close collaboration on so many projects — his mentorship (and friendship) have been invaluable.

I've been fortunate to work with so many fantastic and talented colleagues from whom I've learned a great deal. In alphabetical order, I would like to thank: Nicolas Astorga, Ioana Bica, Jeroen Berrevoets, Boris van Breugel, Jonathan Crabbe, Lasse Hansen, Elizabeth Heremans, Sam Holt, Nicolas Huynh, Alan Jeffares, Jiashuo Liu, Tennison Liu, Max Ruiz Luyten, Thomas Pouplin, Zhaozhi Qian, Paulius Rauba, Evgeny Saveliev, Hao Sun, Mikkel Werling.

To the wider van der Schaar Lab — I will always cherish the countless insightful research discussions, engaging group meetings, and lab lunches. Special thanks to Alicia Curth,

Krzysztof Kacprzyk, Yuchao Qin, Alihan Hüyük, Alex Chan, Qiyao Wei, Julianna Piskorz, Kasia Kobalczyk, Antonin Berthon, Claudio Fanconi, Harry Amad, Rob Davis, Andrew Rashbass, and all others I have had the pleasure of interacting with.

I sincerely thank the Cystic Fibrosis Trust for its generous financial support of my PhD studies.

I would also like to thank my thesis examiners, Prof. Anders Hansen and Prof. Joaquin Vanschoren, for their time and valuable comments on this thesis.

To Faheem, Nureen, Rayyan, Marwan, and Sinan — I am so glad we were together in the UK. Having family nearby made special days away from home brighter and more meaningful.

To my grandparents: though they are no longer with us, the value they placed on education, their prayers, and their unwavering belief in my potential helped me believe that I could take on something as big as this. I wish they could have seen the outcome, especially as Nana once predicted that I would one day become a scientist. May Allah grant them the highest ranks of Jannah.

To my amazing wife, Hameeda, whose love and sacrifice made this journey possible. Thank you for your incredible faith in me, and for supporting my goal to pursue this PhD even when it meant embarking on this adventure so soon after we were married. I am deeply grateful for your patience through all the long hours, and for shouldering so much of the responsibility for our home (including the wonderful food!), which allowed me to focus on my research. Your companionship, love, and motivation have helped me navigate all the ups and downs of both research and life over the past four years. This shared adventure simply would not have been the same, or even possible, without you. May Allah grant us many more adventures together in the future.

Last but not least, I am eternally grateful to my beloved parents, Mohammed Ameen and Yasmin Seedat. Words cannot adequately express my gratitude. It is your love, prayers, sacrifices, both seen and unseen, and all the opportunities you tirelessly worked to provide throughout my life, that made this thesis possible. Your belief in me, even when I doubted myself, has been a constant source of strength. While I can never fully repay you, I hope this work brings you profound joy and pride, as it is a testament to your unwavering belief in me and to the sacrifices you made to nurture that belief into reality.

# Table of contents

**List of figures** xix

**List of tables** xxvii

**1 Introduction** 1
- 1.1 The Case for Data-Centric AI via DC-Check. . . . 2
- 1.2 Goals of this thesis. . . . 3
- 1.3 Problems considered . . . . 4
  - 1.3.1 Characterizing ML Data Quality . . . . 4
  - 1.3.2 Identifying Incongruous Regions in In-Distribution Data . . . . 5
  - 1.3.3 Principled Synthetic Data Augmentation . . . . 5
  - 1.3.4 Improving granular model evaluation . . . . 6
  - 1.3.5 Leveraging data for better uncertainty quantification . . . . 6
- 1.4 Contributions . . . . 6
  - 1.4.1 Data Characterization . . . . 7
  - 1.4.2 Test Data Incongruity Identification . . . . 7
  - 1.4.3 Synthetic Data Augmentation and Curation . . . . 7
  - 1.4.4 Context-Aware Model Evaluation and Testing . . . . 8
  - 1.4.5 Self-Supervised Uncertainty Quantification . . . . 8
- 1.5 Publications . . . . 9

**2 Data Characterization** 13
- 2.1 Introduction . . . . 13
- 2.2 Related work . . . . 15
- 2.3 Formulation . . . . 17
  - 2.3.1 Preliminaries . . . . 17
  - 2.3.2 Uncertainty decomposition during training . . . . 18
  - 2.3.3 Stratification based on data uncertainty . . . . 19

2.3.4 Using Data-IQ with a variety of models, beyond Neural Networks (P4) 21
2.4 Experiments . . . 22
2.4.1 (**P1**) Robust data characterization . . . 22
2.4.2 (**P2**) Principled data collection . . . 25
2.4.3 (**P3**) Reliable Model Deployment . . . 28
2.4.4 Data-IQ beyond tabular settings. . . . 30
2.5 Discussion . . . 34

**3 Test Data Incongruity Identification** **37**
3.1 Introduction . . . 37
3.2 Related work . . . 39
3.3 Data-SUITE . . . 40
3.3.1 Preliminaries . . . 41
3.3.2 Feature PIs . . . 42
3.3.3 Identifying Inconsistent and Uncertain Instances . . . 46
3.4 Experiments . . . 47
3.4.1 Validating coverage & comparing properties . . . 47
3.4.2 Synthetic data stratification w/ downstream task . . . 48
3.4.3 Real dataset stratification w/ downstream task . . . 51
3.4.4 Data-SUITE Ablation . . . 53
3.4.5 Use-Case: Data-SUITE in the hands of users . . . 55
3.5 Discussion . . . 58

**4 Synthetic Data Augmentation and Curation** **59**
4.1 Introduction . . . 59
4.2 CLLM: Synergy of LLM Synthetic Data Generation and Data Curation . . . 62
4.2.1 Data generation with LLMs based on a small $\mathscr{D}_{\text{train}}$ . . . 62
4.2.2 Data curation with learning dynamics . . . 66
4.3 Experiments . . . 70
4.3.1 Overall performance: downstream utility . . . 70
4.3.2 Addressing potential biases from the LLM . . . 73
4.3.3 Ablation: Decoupling prior knowledge and data model . . . 76
4.3.4 Hardness: a proxy signal to flag poor quality synthetic datasets . . . 77
4.4 Discussion . . . 78

**5 Context-Aware Model Evaluation and Testing** **81**
5.1 Introduction . . . 81

5.2 Related work . . . 83
5.3 A context-aware testing framework for ML . . . 84
5.3.1 A multiple hypothesis testing view of ML evaluation . . . 84
5.3.2 The failures of data-only testing . . . 85
5.3.3 Formulating context-aware testing . . . 86
5.4 SMART Testing . . . 87
5.5 Experiments . . . 90
5.5.1 Robustness to False Positives . . . 90
5.5.2 Targeting model failures . . . 92
5.5.3 Robustness to False Negatives . . . 93
5.5.4 Requirements-constrained testing . . . 94
5.5.5 Adaptive testing for a deployment environment . . . 95
5.5.6 Assessing and mitigating potential LLM challenges and biases. . . . 97
5.5.7 Additional experiments . . . 98
5.6 Discussion . . . 98

**6 Self-Supervised Uncertainty Quantification 101**
6.1 Introduction . . . 101
6.2 Related Work . . . 103
6.3 Background: Conformal Prediction . . . 105
6.3.1 Inductive conformal prediction (ICP) . . . 105
6.3.2 Conformal residual fitting (CRF) . . . 106
6.4 Method: Self-Supervised Conformal Prediction . . . 106
6.4.1 Data splits and assumptions . . . 108
6.4.2 Predictive and self-supervised phase (S1) . . . 109
6.4.3 Conformal normalizer phase (S2) . . . 110
6.4.4 Conformalization phase (S3) . . . 110
6.4.5 Other locally adaptive methods . . . 111
6.4.6 Remark on SSCP and coverage guarantees . . . 111
6.5 Experiments . . . 111
6.5.1 Synthetic demonstration . . . 112
6.5.2 Real data experiments . . . 115
6.5.3 Insights . . . 117
6.6 Discussion . . . 119

**7 Conclusion** **121**
7.1 Broader Perspectives . . . 121
7.2 Societal Implications . . . 124
7.3 Future Agenda . . . 127
7.3.1 Concluding remarks. . . . 129

**References** **131**

**Appendix A Appendix for Chapter 2** **151**
A.1 Additional Details . . . 151
A.1.1 Extended Related Work . . . 151
A.1.2 Data-IQ adapted to any model . . . 153
A.1.3 Why does Data-IQ produce a bell-shape? . . . 154
A.1.4 Evaluating Data-IQ and Data Maps over stages . . . 154
A.1.5 Data Maps: difference in our setting . . . 156
A.1.6 Data-IQ thresholds . . . 159
A.1.7 Data-IQ subgroup characterization . . . 160
A.2 Experimental Details . . . 162
A.2.1 Code details . . . 162
A.2.2 Benchmarks, Datasets and Implementation details . . . 162
A.2.3 Datasets . . . 164
A.2.4 Additional experiment details . . . 167
A.3 Additional Experiments . . . 170
A.3.1 Understanding the representation space . . . 170
A.3.2 Data-IQ: Neural Networks vs Other Model Classes . . . 171
A.3.3 Collecting more data - it might make things worse . . . 174
A.3.4 Identifying subsets in an outcome-driven manner: assessing quality 175
A.3.5 Data Insights: additional results . . . 175
A.3.6 Principled feature acquisition (P2): additional results . . . 177
A.3.7 Effect of averaging over the uncertainty metric . . . 182
A.3.8 Effect of additional sources of randomness . . . 182
A.3.9 Effect of hyper-parameters on the recovery of estimated groups . . . 183
A.3.10 Data sculpting: Absolute number of samples . . . 184
A.3.11 Correlation of weights through training . . . 185

**Appendix B Appendix for Chapter 3** **187**
B.1 Additional Details . . . 187

B.1.1 Extended Related Work . . . 187
B.1.2 Data-SUITE Details . . . 188
B.2 Experimental Details . . . 192
B.2.1 Benchmarks & Implementations . . . 192
B.2.2 Code details . . . 192
B.2.3 Synthetic Experiment Details . . . 196
B.2.4 Real-Data Experiment Details . . . 197
B.3 Additional Experiments . . . 200
B.3.1 Regions identified by Data-SUITE as NOT OOD . . . 200
B.3.2 EDA & digestable prototype insights . . . 202
B.3.3 Inconsistent instances $\lambda$ sensitivity . . . 204
B.3.4 Computation time comparison . . . 204
B.3.5 Comparison to domain adaptation . . . 206

**Appendix C Appendix for Chapter 4** **209**
C.1 Additional Details & Extended Related Work . . . 209
C.2 Experimental Details . . . 213
C.2.1 Code details . . . 213
C.2.2 Datasets . . . 213
C.2.3 Data generation. . . . 214
C.2.4 Data curation . . . 215
C.2.5 Downstream task . . . 216
C.2.6 Prompt example . . . 217
C.3 Additional Experiments . . . 219
C.3.1 Curation mechanism also improves open-source LLM generated data 219
C.3.2 Ablation for contextual information on Compas . . . 221
C.3.3 Comparison to random noise baseline . . . 221
C.3.4 Detailed results for Chapter 4.3.1 . . . 222
C.3.5 Full results for performance evaluation . . . 226
C.3.6 Comparing CLLM vs TabPFN . . . 226
C.3.7 CLLM with context in low-resource languages . . . 228
C.3.8 Comparison of curation mechanism vs Data-IQ . . . 228
C.3.9 CLLM in specialized domains. . . . 230
C.3.10 Behavior beyond the low-data regime. . . . 231
C.3.11 Choice of thresholds . . . 232

**Appendix D Appendix for Chapter 5** **235**
D.1 Extended related work . . . 235
D.1.1 Enhanced overview of relevant literature . . . 235
D.1.2 Components of the ML Testing pipeline . . . 238
D.1.3 Comparison of the features of slice discovery methods . . . 239
D.2 SMART Details . . . 240
D.2.1 Contextual understanding . . . 240
D.2.2 Operationalizing variables . . . 241
D.2.3 Feasibility checks . . . 243
D.2.4 Data adjustment queries . . . 244
D.2.5 Requirements, automation, and outputs . . . 245
D.2.6 Moving outside of IID data . . . 245
D.2.7 SMART and multiple testing . . . 246
D.3 Experimental Details . . . 247
D.3.1 Code details . . . 247
D.3.2 Datasets . . . 247
D.3.3 Baselines . . . 247
D.3.4 LLM prompts . . . 248
D.3.5 Specific Experimental details . . . 253
D.4 Additional experiments . . . 257
D.4.1 Effects of LLMs . . . 257
D.4.2 Effects of different tabular machine learning models . . . 260
D.4.3 On the inductive biases of ML model testing . . . 261
D.4.4 Cost of SMART . . . 263
D.4.5 SMART with open-weight models . . . 264
D.4.6 Understanding the importance of feature names . . . 265
D.4.7 Example of a model report . . . 266

**Appendix E Appendix for Chapter 6** **271**
E.1 Comment on SSCP Coverage . . . 271
E.2 Experimental details . . . 272
E.2.1 Predictive and conformal normalization models. . . . 272
E.2.2 Self-supervised learning . . . 272
E.2.3 Code details . . . 273
E.3 Additional Experiments . . . 274
E.3.1 Ablation: unlabeled data experiment . . . 274
E.3.2 Sanity check: alternative sources of signal . . . 274

E.3.3 Conformalized Quantile Regression . . . . . 275
E.3.4 Evaluating alternative self-supervised tasks & the source of gain . . 277

# List of figures


1.1 Overview of the thesis illustrating the different data-centric challenges addressed in this thesis across the entire ML pipeline/lifecycle. Works not covered in the thesis, but addressing additional challenges illustrated in the figures are outlined in 1.5. (Figure is based on DC-Check by Seedat et al. [216]) . . . 3

2.1 Data-IQ systematically characterizes data into subgroups using (i) any ML model, trained in stages (epochs/iterations) and can be checkpointed. At training time, Data-IQ (ii) leverages the model's checkpoints to analyze the training behavior of individual examples and (iii) characterizes each example based on its aleatoric (data) uncertainty & prediction confidence. (iv) At deployment, the subgroups inform model usage by embedding the training set in a representation space and characterizing the new data points in this space. . . . 15

2.2 Example outlining the differences between $v_{\text{al}}$ (Data-IQ) and $v_{\text{ep}}$ (Data Maps) & showing the type of uncertainty matters. . . . 20

2.3 Robustness to variation across models based on Spearman correlation, where Data-IQ has the highest correlation (i.e. consistency) across all datasets. . . . 23

2.4 Data-IQ's robustness to variation across different models (colors). . . . 23

2.5 Assessment on characterization when used with different backbone models shows that Data-IQ is more consistent. . . . 24

2.6 Comparing subgroups identified by Data-IQ (descriptions above). Colors represent the GMM clusters. . . . 25

2.7 Quantifying the value of feature acquisition based on change in ambiguity. Only Data-IQ captures this. . . . 26

2.8 Performance improvement via data sculpting based on subgroups . . . 28

2.9 Data-IQ subgroups can unmask unreliable prediction deferral by uncertainty methods . . . 30

2.10 Subgroups highlighted by Data-IQ on the CelebA dataset . . . . . 34

3.1 Illustration highlighting two problems Data-SUITE addresses . . . . . 38
3.2 Outline of our Data-SUITE framework . . . . . 41
3.3 Data-SUITE blueprint: generator, representer and conformal predictor. . . . 43
3.4 Conformal Predictor in **Data-SUITE**. . . . . 44
3.5 Comparison of methods based on coverage, deficit and excess under various configurations $(D_a, D_b, D_c)$ . . . . . 49
3.6 Example on CUTRACT of how Data-SUITE stratification can be used to improve downstream performance, contrasted with baseline $\mathscr{D}_{\text{test}}$ (blue) or random selection (black). . . . . 53
3.7 Insights of prototypes identified by Data-SUITE. Tables describe the average prototypes for *certain* and *uncertain* instances. . . . . 57

4.1 CLLM uses a small dataset $\mathscr{D}_{\text{train}}$ and a frozen black-box LLM to generate a larger synthetic set $\mathscr{D}_{\text{syn}}$. The curator computes the learning dynamics of samples in $\mathscr{D}_{\text{syn}}$, assessing samples based on their aleatoric uncertainty and predictive confidence, then curates $\mathscr{D}_{\text{syn}}$ with the goal that a downstream model trained on the curated $\mathscr{D}_{\text{curated}}$ will have improved performance. . . . 61
4.2 GPT-4 is able to extrapolate to regions of the oracle (true manifold) even where there is no training data covering them, as can be seen by the overlap with the turquoise dots, with the effect more pronounced when $\mathscr{D}_{\text{train}}$ is small 64
4.3 Subgroups with fewest samples in $\mathscr{D}_{\text{train}}$ benefit the most from data augmentation, on average. . . . . 65
4.4 Contextual information in the prompt is important for extrapolation. . . . . 66
4.5 $\hat{\eta}$ aligns more with $\mathscr{D}_{\text{curated}}$ than $\mathscr{D}_{\text{discarded}}$ for each generative model: the curation step keeps high quality samples tailored to the downstream task. . 69
4.6 The data generated by the LLM captures the distinct features in "atypical" regions of the Oracle manifold, as in-context samples are added to the prompt. This shows that it is flexible enough to adapt its prior knowledge to the nuances of the data. The group encircled in blue represents patients who are $>$ 88 years old, representing around 3.5% of the Oracle. This illustrates that the added in-context samples can successfully guide the LLM to generate these rare samples. . . . . 77

4.7 The proportion of discarded samples is a proxy for test performance. This negative linear relationship where each point is a synthetic dataset generated by GPT-4 (e.g. Adult, Covid, Compas) allows us to flag datasets that will lead to unreliable downstream performance. . . . . 78

5.1 Overview of the SMART Testing Framework showing the four steps. All steps are automatically executed by an LLM. . . . . 87
5.2 SMART uses an LLM to integrate context $\mathscr{C}$, and data context $\mathscr{C}_D$. Relevant and likely failure hypotheses are then generated by LLM (i.e. sampling mechanism). The hypotheses are then operationalized in $\mathscr{D}$ and evaluated. In contrast, data-only methods are not guided by context and requirements, searching more exhaustively in $\mathscr{D}$ for divergent slices. . . . . 88
5.3 A self-falsification module within the SMART framework. A hypothesis generator generates plausible hypotheses and justifications for when the model might fail. This is operationalized with $g_i$ and tested against the empirical data. . . . . 89
5.4 Contextual-awareness in SMART reduces FPs, i.e. reducing the proportion of irrelevant features in slices. SMART is not sensitive to the number of irrelevant features, unlike data-only methods. $\downarrow$ is better . . . . . 91
5.5 Proportion of irrelevant features (y) for each slice discovery method, based on the sample size. Lower is better. . . . . 91
5.6 Number of significant groups discovered (out of a total of 10) based on the training dataset size. SMART can operate under any sample size; the self-falsification mechanism requires a larger sample size to falsify hypotheses. $\uparrow$ is better. . . . . 96

6.1 **Left.** Standard inductive conformal prediction results in constant width prediction intervals. **Center.** Conformal residual fitting produces adaptive intervals but can be inefficient in regions. **Right.** The errors of a self-supervised task are included above the plot with red indicating larger self-supervised errors. SSCP leverages these errors to improve the efficiency of conformal residual fitting. . . . . 103

6.2 SSCP produces adaptive conformal prediction intervals leveraging self-supervised learning to improve the conformal normalizer ($\sigma$) and consists of 3 phases. (S1) where (i) train the predictive regressor $f = g \circ e$, thereafter (ii) the encoder representation ($e$) is transferred to use for the self-supervised task, where we train the self-supervised predictor $h$. (S2) the conformal normalizer phase is then carried out, with the conformal normalizer, augmented with the self-supervised input, trained to predict residuals (representing difficulty of prediction). Finally, (S3) the conformalization phase, computes normalized scores ($\gamma$) on $\mathscr{D}_{\text{cal}}$, with the critical non-conformity score ($\varepsilon$) chosen as the $(1-\alpha)$-smallest. At test time, the adaptive prediction intervals $[l(\mathbf{x}), r(\mathbf{x})]$ are produced by adapting the critical non-conformity score $\varepsilon$ using the conformal normalizer, i.e $\varepsilon\sigma(\hat{\mathbf{x}})$. The "shaded blocks" in each phase are trainable. . . . . 108

6.3 200 observations drawn from our synthetic distribution with the distributions ground truth 90% prediction intervals highlighted with a dashed line. . . . 113

6.4 **Conformal prediction intervals.** Standard CRF (pink) fails on this synthetic task due to a complex relationship between the inputs and the latent dimension. By including the errors of a self-supervised task as an input, SSCP (green) is able to learn much more efficient prediction intervals. . . . . 114

6.5 Relative prediction interval widths of SSCP compared to CRF for varying proportions of labeled data (↓ better). Avg. interval width across datasets is lower with the inclusion of unlabeled data across all $p$. . . . . 117

6.6 Prediction interval width for most challenging samples on Star. SSCP reduces interval width . . . . . 117

6.7 Comparing the prediction intervals in different regions. SSCP provides benefits in the sparser regions. . . . . 118

6.8 Assessing the 90% confidence interval gains — which implies width reductions across datasets, averaged across runs. We find that SSCP has a net reduction in intervals across all datasets, highlighting the robustness of the approach. . . . . 119

A.1 Pseudo-ensemble for GBDTs for usage with Data-IQ, where each sub-model is a checkpoint . . . . . 153

A.2 $\mathscr{P}(x, \theta)$ peaked around $\bar{\mathscr{P}}(x)$ . . . . . 154

A.3 The bell-shapes derived theoretically and practically match. . . . . 155

A.4 Evaluation of Data-IQ and Data Maps over various epochs (stages), even letting the model overfit. Data-IQ is more stable, whereas Data Maps changes shape after overfitting, which is undesirable. . . . . 155
A.5 Data Maps: Epistemic Uncertainty (variability) evolution over epochs per subgroup . . . . . 157
A.6 Data-IQ: Aleatoric Uncertainty evolution over epochs per subgroup . . . . 158
A.7 Comparison of Data Maps when the model is just right and overfits, illustrating how the shape changes . . . . . 158
A.8 SUPPORT Quantifying the value of feature acquisition based on change in ambiguity. Only Data-IQ captures this relationship. . . . . 160
A.9 $\mathscr{P}(x,\theta)$ peaked around $\bar{\mathscr{P}}(x)$ . . . . . 161
A.10 Representation space of training examples broken down by sub-group. We then zoom into the Ambiguous subgroup and illustrate the heterogeneous outcomes, where examples with similar features can have different outcomes. 170
A.11 NN vs XGBoost comparison on the Covid-19 dataset, showing Data-IQ is more consistent . . . . . 172
A.12 NN vs XGBoost comparison on the Fetal dataset, showing Data-IQ is more consistent . . . . . 172
A.13 NN vs XGBoost comparison on the Prostate dataset, showing Data-IQ is more consistent . . . . . 173
A.14 NN vs XGBoost comparison on the Support dataset, showing Data-IQ is more consistent . . . . . 173
A.15 The *Ambiguous* subgroup proportion in fact increases as the dataset size increases, i.e. more examples are added. This is due to the increased probability of feature collisions. . . . . 174
A.16 Fetal subgroups identified by Data-IQ (descriptions above). Colors represent the GMM clusters . . . . . 176
A.17 Support subgroups identified by Data-IQ (descriptions above). Colors represent the GMM clusters . . . . . 176
A.18 Covid subgroups identified by Data-IQ (descriptions above). Colors represent the GMM clusters . . . . . 176
A.19 COVID Quantifying the value of feature acquisition based on change in ambiguity. Only Data-IQ captures this relationship. . . . . 177
A.20 FETAL Quantifying the value of feature acquisition based on change in ambiguity. Only Data-IQ captures this relationship. . . . . 178

A.21 Prostate Quantifying the value of feature acquisition based on change in ambiguity. Only Data-IQ captures this relationship. . . . . 179
A.22 Support Quantifying the value of feature acquisition based on change in ambiguity. Only Data-IQ captures this relationship. . . . . 180
A.23 Data-IQ vs Data Maps compared on text and image modalities . . . . . 181
A.24 Assessment of whether we can distinguish groups with different training dynamics when we average over the metric . . . . . 183
A.25 Data sculpting - absolute number of *Ambiguous* samples removed vs test accuracy . . . . . 185

B.1 D-Vine structure . . . . . 192
B.2 OOD-Uncertain Instances Match Rate or overlap . . . . . 201
B.3 Mean probability of predicted OOD, i.e. confidence . . . . . 201
B.4 OOD-Inconsistent Instances Match Rate or overlap . . . . . 201
B.5 PSA Deep-Dive, which highlights how the difference is not evident at the population level or for certain instances. Only the uncertain instances highlight the heterogeneity. However, it does highlight common support based on the whiskers of the box plot. . . . . 202
B.6 Insights of prototypes identified by Data-SUITE. Tables describe the average prototypes for *certain* and *uncertain* instances. . . . . 203
B.7 Earth Movers Distance Deep-Dive into PSA, which highlights how the difference is not evident at the population level or for certain instances. Only the uncertain instances highlight the heterogeneity. . . . . 205
B.8 $\lambda$ sweep to flag inconsistent instances . . . . . 206

C.1 Learning dynamics computed for 20 samples . . . . . 215
C.2 The random noise baseline does not match the performance of CLLM . . . 221
C.3 Behavior of baselines at high $n$, i.e. beyond the low data regime. . . . . 231
C.4 Assessment of the effects of different thresholds in CLLM . . . . . 232

D.1 Contrasting paradigms of ML model testing along (i) Testing relevance, which accounts for requirements and/or context, and (ii) Degree of automation and adaptability when carrying out testing. We desire a new paradigm of ML testing to address both. . . . . 237
D.2 Components of ML Testing — (i) Test search, (ii) Operationalize, (iii) Evaluation report, with example approaches for each component. . . . . 238

D.3 A strong machine learning testing framework should incorporate textual understanding during testing, should meet requirements, should be automated, and should provide contextual reports with an emphasis on model failures. . 240
D.4 Identifying the importance of feature names as a source of information for context-aware testing. We compare two versions of SMART on CUTRACT and SEER datasets. The first version, "base", is the normal SMART method. The second version, "corrupt", has feature names changed to uninformative ones. The goal is to identify slices with the highest performance discrepancies between the average & subgroup prediction, calculated as the difference in their respective accuracies (y-axis). We see that showing feature names enables us to find subgroups with a higher performance discrepancy from baseline performance than without feature names. ↑ is better. . . . . 266

E.1 Repeating the partially labeled experiments from Chapter 6.5.2 with CQR. Evaluating independent vs. shared representations and the inclusion of additional unlabeled data for various proportions of labeled data. Lower proportions of labeled data achieve the largest gains over standard CQR. . . 277
E.2 Comparing the prediction intervals in different regions. All variants of SSCP provide benefits in the sparser regions. The intervals are similar for SSCP (VIME) and SSCP (AE) in the denser regions, with SSCP (VIME) generally performing better. This is especially the case in the sparser regions with narrower intervals. . . . . 279
E.3 Comparing the metrics - ordered from the largest to smallest. We note that the benefit of SSCP for both VIME and AE is largely in reducing the excess width of the intervals on those samples for which it is greatest (i.e. the most inefficient intervals). . . . . 280
E.4 Comparison of the distributions of the metrics. We can clearly see that the addition of the self-supervision helps to reduce the long tails of the distributions corresponding to the most inefficient intervals. . . . . 280

# List of tables


2.1 Comparison of accuracy performance rank and (dataset quality). Synthetic dataset w/ better quality (↑ easy)produces the best real data test performance. 28
2.2 Comparison of different model improvement/robustness techniques . . . . 29

3.1 MSE based on instance stratification for different methods. Data-SUITE outperforms other methods, whilst data-centric methods in general outperform model-centric methods. . . . . 51
3.2 Comparison of real-world datasets . . . . . 52
3.3 MPI metric across datasets for different methods . . . . . 53
3.4 Coverage of constituent components of Data-SUITE . . . . . 54
3.5 Downstream MSE for ablations of constituent components of Data-SUITE . 54

4.1 Deep dive into the top 5 demographic subgroups in the Covid dataset with the largest gains, across 10 seeds, for $|\mathscr{D}_{\text{train}}| = 20$. GPT-4 improves performance on the smallest groups. . . . . 65
4.2 Including contextual information in the prompt improves precision (P), recall (R), and utility (U) in low-sample settings (exemplified on the Covid-19 dataset). . . . . 66
4.3 AUC (%) averaged over 4 downstream models on $\mathscr{D}_{\text{test}}$. Curation improves performance for all methods across all sample sizes $n$, as indicated by ↑. CLLM w/ GPT-4 (Cur.) provides the strongest performance for both private/proprietary datasets and public datasets . . . . . 71
4.4 Average rank of approaches across the different datasets and seeds. CLLM w/ GPT-4 ranks first across all $n$, and curation improves all the generative models. . . . . 72
4.5 Curation can help address bias. . . . . 75
4.6 Downstream accuracy when varying the number of in-context samples in the prompt to generate the augmented datasets. . . . . 76

5.1 Example hypotheses on model failure, justifications, and operationalizations generated by the SMART framework on a healthcare dataset. The p-values show whether the model's performance significantly differs from average performance with $|\Delta Acc|$ measuring the effect size, i.e. the absolute difference between the model's accuracy on slice $\mathscr{S}$ and its overall accuracy on $\mathscr{D}$ . . . . . 88
5.2 Identifying slices with the highest performance discrepancies. We show differences in accuracies ($|\Delta Acc|$) between the top identified divergent slice and average performance across four classifiers (over 5 runs). ↑ is better. . . 92
5.3 False Negative Rate (FNR) for different methods at various settings. ↓ is better. 93
5.4 Requirement satisfaction showing how many times the top 10 generated slices satisfied the requirements (Req) and how many slices had statistically significantly different performance from average (Sig ) on a testing dataset. The maximum is 10. ↑ is better. . . . . 95
5.5 Number of slices identified (out of a maximum of 10) that had significantly divergent performance from average (higher is better). $S_\alpha$ counts the number of significantly divergent groups at $\alpha = 0.05$; $S_\alpha/n$ applies the Bonferroni correction. ↑ is better. . . . . 96
5.6 Proportion of times the corrupted minority subgroup is correctly identified as the top underperforming subgroup. . . . . 97

6.1 Assessing the impact of the SSL task on improving CRF, even *without* additional data. We find in all cases, averaged over 5 runs, that the additional SSL information can help to improve width, deficit, and excess (↓ better). . 116

A.1 Comparison of related work, highlights that *ONLY* Data-IQ naturally address all desiderata. (♡): satisfied naturally by construction, (♢): needs our framework . . . . . 151
A.2 Summary of features for the SEER Dataset [46] . . . . . 164
A.3 Summary of features for the CUTRACT Dataset [174] . . . . . 165
A.4 Summary of features for the Fetal Dataset [13] . . . . . 165
A.5 Summary of features for the Covid-19 Dataset [15] . . . . . 166
A.6 Summary of features for the Support dataset [114] . . . . . 169
A.7 Comparison of robustness/consistency across different models on the basis of the Spearman rank correlation . . . . . 173
A.8 Clustering results . . . . . 175

A.9 Comparison of robustness/consistency across different models on the basis of the Spearman rank correlation . . . 183
A.10 Assessment of stability in group assignment across different parameterizations. The score reflects the average overlap of id characterization. . . . 184

B.1 Comparison of related work . . . 187
B.2 Different configurations of the synthetic data . . . 197
B.3 Summary of features for the SEER Dataset [46] . . . 197
B.4 Summary of features for the CUTRACT Dataset [174] . . . 198
B.5 Summary of features for the ADULT Dataset [11] . . . 199
B.6 Summary of features for the ELECTRICITY Dataset [80] . . . 199
B.7 Comparison of computation time across methods to the nearest minute . . . 206
B.8 Comparison of average performance improvement for Data-SUITE vs MMD (Domain Adaptation) . . . 207

C.1 Summary of the datasets used. * Denotes private/proprietary datasets. . . . 213
C.2 AUC averaged over 4 downstream models on $\mathscr{D}_{\text{test}}$ for GPT-4, GPT-3.5, Mistral-7b, LLAMA-13b, LLAMA-70b and Mixtral-8x7b (Curated and Uncurated) . . . 220
C.3 Including contextual information in the prompt improves precision (P), recall (R), and utility (U) in low-sample settings (results shown for Compas) . . . 221
C.4 AUC for the RF model on $\mathscr{D}_{\text{test}}$ where curation improves performance for all methods across all sample sizes $n$, as indicated by ↑. . . . 222
C.5 AUC for the XGB model on $\mathscr{D}_{\text{test}}$ where curation improves performance for all methods across all sample sizes $n$, as indicated by ↑. . . . 223
C.6 AUC for the DT model on $\mathscr{D}_{\text{test}}$ where curation improve performance for all methods across all sample sizes $n$, as indicated by ↑. . . . 224
C.7 AUC for the LR model on $\mathscr{D}_{\text{test}}$ where curation improves performance for all methods across all sample sizes $n$, as indicated by ↑. . . . 225
C.8 AUC averaged over four downstream models on $\mathscr{D}_{\text{test}}$ where curation improves performance for all methods across all sample sizes $n$, as indicated by ↑. CLLM w/ GPT-4 (Curated) dataset provides the strongest performance for both private/proprietary datasets and public datasets . . . 226
C.9 Comparison of CLLM vs TabPFN. We report the AUC averaged over 4 downstream models on $\mathscr{D}_{\text{test}}$. . . . 227
C.10 Test AUC in settings where the names of features have been translated to Swahili and Hausa, with and without curation . . . 228

C.11 Comparing the CLLM curation mechanism to Data-IQ curation [210]. We report the AUC averaged over 4 downstream models on $\mathscr{D}_{\text{test}}$. . . . . 230
C.12 CLLM performance on the Higgs dataset . . . . . 231

D.1 Comparison of ML testing paradigms in terms of how tests are defined, the type of search space (i.e. relevance based on context and requirements), sensitivity to multiple testing and automation to enable scalability. . . . . 235
D.2 Comparison of SMART testing with software testing works . . . . . 238
D.3 Comparison of slice discovery methods . . . . . 239
D.4 Summary of the datasets used. . . . . 247
D.5 Number of slices identified (out of a maximum of 10) that had significantly divergent performance from average (higher is better). $S_\alpha$ counts the number of significantly divergent groups at $\alpha = 0.05$; $S_{\alpha/n}$ applies the Bonferroni correction. . . . . 257
D.6 The differences in accuracies between the top slice identified for each method on a testing dataset. The p-value is computed to determine the significance of the difference in accuracy. For the accuracy, higher values imply a greater ability to detect divergent slices (hence, higher is better). For the p-value, the lower value is better. Averages +- standard deviations are shown across 5 runs with random seeds and data splits. . . . . 258
D.7 Identifying slices with the highest performance discrepancies. We demonstrate differences in accuracy ($|\Delta Acc|$) between the top divergent slice identified and average performance across two state-of-the-art deep learning classifiers (over five runs) on the SEER dataset. ↑ is better. 0.00 implies the evaluation method does not support the model. . . . . 258
D.8 Comparison of Hypotheses and Justifications generated by GPT-4 and GPT-3.5 for a breast cancer prediction task. The hypothesis targets likely model failures. . . . . 259
D.9 The differences in accuracies between the top slice identified for each method on a testing dataset. The p-value computes the p-value associated with the difference in the accuracy. For the accuracy, higher values imply a greater ability to detect divergent slices (hence, higher is better). For the p-value, the lower is better. Averages +- standard deviations are shown across 5 runs with random seeds and data splits. . . . . 260
D.10 The differences in accuracies between the top slice identified for each method on a testing dataset when the dataset's continuous features are discretized. . 261

D.11 Number of discovered slices on a synthetic dataset with no prior relationships in three data generating process scenarios. slices, capped at most 20. Results are averaged over 50 runs and ± standard deviations are shown. . . . . 263
D.12 Number of conditions per discovered slice (false positives) in three data generating process scenarios. Results are averaged over 50 runs and ± standard deviations are shown. Lower is better. . . . . 263
D.13 Cost of SMART (USD) for different GPT LLMs and different numbers of hypotheses generated. The cost is estimated based on token counts. Note: GPT-4o models are even cheaper. . . . . 264
D.14 Comparison hypotheses by GPT-4 and overlap w/ open-weight models . . . 264
D.15 Example hypotheses and justifications when dataset column names are hidden. The hypotheses and justifications for features that do not encode meaningful information lose practical meaning. Context-aware testing benefits from rich feature names that can be used to guide meaningful hypothesis generation. . . . . 265

E.1 Avg. width of intervals with and without unlabeled data (↓ is better) . . . . 274
E.2 Sanity check comparing alternative self-supervised signals to SSCP (VIME). We report mean average interval width over 5 runs. (↓ better) . . . . . 275
E.3 Repeating the fully labeled experiments from Chapter 6.5.2 with CQR. We report the mean average interval width (↓ better) over 5 runs. These results indicate that using CQR augmented with a self-supervised loss (SSCP-CQR) often improves vanilla CQR. . . . . 276
E.4 Assessing the impact of the SSL task on improving CRF for two different self-supervised tasks (i.e., VIME and AE). We find in general that the self-supervision helps with VIME being more performant than AE as the self-supervised task. (↓ better) . . . . . 278

# Chapter 1

# Introduction

Rapid progress in machine learning (ML) has fueled remarkable advances across diverse domains, including healthcare, finance, manufacturing and e-commerce. However, a critical gap persists between algorithmic advances and reliable real-world deployment. High-profile ML failures across various domains, from gender and racial biases in image processing algorithms [36] to object tracking systems latching onto spurious correlations [247] and healthcare diagnostic tools failing on demographically distinct populations [18], reveal a troubling pattern: exceptional performance on benchmark datasets frequently fails to translate into reliable and trustworthy ML systems.

These failures are not random occurrences but symptoms of a systematic issue rooted in the predominantly model-centric approach to ML development. As highlighted in [42], approximately 85% of industrial AI/ML systems are projected to "deliver erroneous outcomes due to bias in data or algorithms," while only 53% successfully bridge the gap between prototypes and production.

The reason for this is that ML research and development typically focuses on advances in model architectures, optimization techniques, and algorithmic refinement, while the fundamental role of data in determining ML system reliability remains vastly underappreciated. This is despite the fact that many ML failures are not due to the choice of model architecture or hyperparameters, but rather from data-centric issues within the ML pipeline—spanning data collection, preprocessing, training, evaluation, and deployment.

This insight has prompted the emergence of *Data-centric AI* [156] — a paradigm shift and area of ML research that places the role of data, rather than *only* models, at the center of ML development. Specifically, recognizing that the "data work" often dismissed as merely operational [203, 160] is, in fact, fundamental to developing truly reliable and trustworthy ML systems in dynamic, real-world settings.

## 1.1 The Case for Data-Centric AI via DC-Check.

Recognizing the limitations of the prevailing model-centric paradigm, this thesis advocates for and advances the role of Data-centric AI in the development of reliable and trustworthy ML. As mentioned previously, while advances in model architectures and optimization techniques have driven impressive progress, these gains often fails to translate into robust real-world performance. At the core of this disconnect is a tendency to treat data as a static, one-time asset rather than as a dynamic, evolving component of the ML pipeline.

To address this gap, we introduced DC-Check [216], a framework designed to formalize the data-centric perspective toward ML and to think systematically about the different capabilities necessary across the ML development process.

In doing so, our framework defines data-centric AI formally as:

> Data-centric AI research encompasses research into methods, tools and frameworks to systematically characterize, evaluate, and monitor the underlying data used to both train and evaluate ML models.

To bring this definition to life and help understand what ML methods and algorithms are needed to realize the vision of data-centric AI, we introduce a structured framework for how to think critically and consistently about data considerations throughout the ML pipeline and what ML tools could help address these considerations for reliable and trustworthy ML.

Specifically, we propose DC-Check [216], a checklist-style methodology that helps practitioners and researchers proactively identify and mitigate data-related risks across the four key stages of the ML lifecycle: Data, Training, Testing, and Deployment (Figure 1.1). Each stage addresses distinct but interconnected challenges:

- Data: How datasets are selected, curated, and documented to ensure appropriate coverage and representation.
- Training: How data informs how we train models. From data-informed model selection to group robust and data noise-aware training mechanisms.
- Testing: How models are evaluated using data going beyond average testing, such as performance across diverse subgroups, edge cases, and stress-testing scenarios.
- Deployment: How systems behave at test and deployment time with regards to changing data that requires monitoring, adaptation, and retraining due to data changes, as well as enabling trustworthy predictions given the dynamic nature of data.

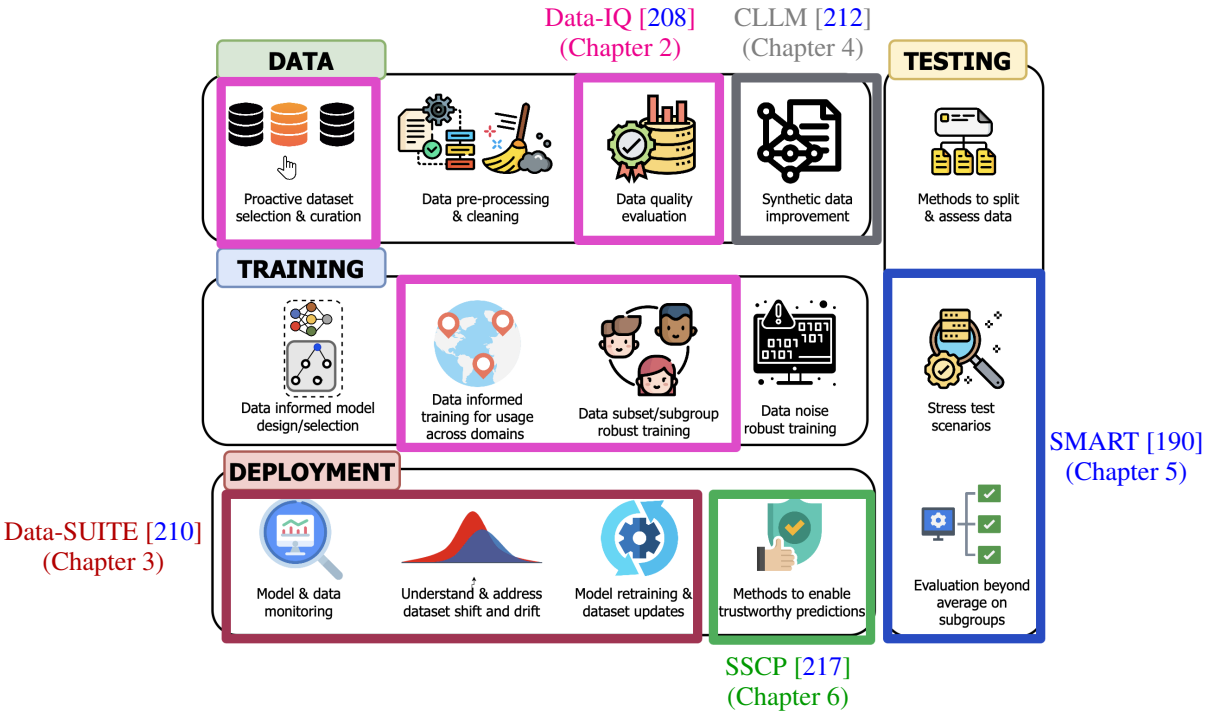



Fig. 1.1 Overview of the thesis illustrating the different data-centric challenges addressed in this thesis across the entire ML pipeline/lifecycle. Works not covered in the thesis, but addressing additional challenges illustrated in the figures are outlined in 1.5. (Figure is based on DC-Check by Seedat et al. [216])

Through the DC-Check framework [216], we propose that data-centric AI must be understood from an end-to-end pipeline perspective, rather than a narrow focus limited only to early stages of dataset preparation alone. In particular, we note that data-related challenges recur and evolve throughout the lifecycle of ML systems, and hence, addressing them systematically, from collection through to deployment, is essential for building robust, trustworthy AI. By structuring this approach and highlighting its key dimensions, DC-Check provides a conceptual and practical foundation for operationalizing the different dimensions of data-centric ML research. Consequently, DC-Check and the challenges raised therein serve as a guiding framework for the ML methods and frameworks algorithms proposed in this thesis.

## 1.2 Goals of this thesis.

Building on the foundation established by DC-Check [216], the goal of this thesis is to go beyond high-level guidance and address fundamental challenges needed to advance Data-centric AI through the development of novel methodologies and tools. Although DC-Check [216] provided a conceptual blueprint for data-centric thinking, highlighting four critical

stages throughout the ML pipeline: Data, Training, Testing, and Deployment—it focused primarily on what practitioners and researchers should consider when aiming to build reliable and trustworthy systems.

However, a pressing need remains: to move from problem identification to instantiation. i.e. to formalize how these data-centric principles can be systematically operationalized in practice. The remainder of this thesis takes up that challenge. Consequently, this thesis proposes novel data-centric ML frameworks and methodologies toward realizing the goal of reliable and trustworthy ML.

## 1.3 Problems considered

This section introduces the core problems addressed in this thesis, each corresponding to specific challenges in data-centric AI for reliable and trustworthy ML. These core problems were carefully selected to address critical gaps identified in our DC-Check framework [216], spanning different aspects of the ML pipeline from initial data characterization to deployment-time assessment. Moreover, each represents a fundamental challenge where conventional model-centric approaches have proven insufficient.

### 1.3.1 Characterizing ML Data Quality

Machine learning models rely on data to learn patterns, yet datasets often contain diverse instances with varying degrees of informativeness, learnability, uncertainty, and noise. This heterogeneity can manifest itself as samples such as mislabeled samples or ambiguous instances, which, when included in a training dataset, can lead to degraded model performance and reliability. Specifically, while we might have high model performance on average, this can mask systematic underperformance on specific subgroups of data, a direct consequence of these data challenges. Consequently, a central challenge in data-centric AI is developing principled data characterization techniques, allowing practitioners to identify and subsequently address problematic training data. A key difficulty lies in distinguishing between data uncertainty arising from intrinsic randomness (aleatoric uncertainty) and uncertainty due to limited knowledge (epistemic uncertainty). Traditional data preprocessing pipelines fail to account for these nuanced variations, often treating all data instances equally. However, real-world settings, particularly in healthcare, finance, and scientific research, demand a more structured approach to identifying and handling uncertain or uninformative data points. The focus of Chapter 2 is hence on systematic data characterization: How can we systematically identify and characterize examples based on their inherent data qualities? Specifically, how

might we leverage a model's training/learning dynamics to characterize data (and as a result, its learnability). Thereafter, how would this influence the sculpting of datasets (i.e. data curation), inform principled feature acquisition, enable dataset comparison, and ultimately guide reliable model deployment?

### 1.3.2 Identifying Incongruous Regions in In-Distribution Data

Reliable machine learning hinges not only on data quality, but also on the assumption of consistency of data distributions between training and test/deployment time. Although significant research has focused on identifying out-of-distribution data, an equally important yet understudied problem concerns heterogeneous regions within in-distribution data. Even when data technically remains within the training distribution, feature space heterogeneity can manifest in varying levels of incongruence arising from different sub-populations, biases, or temporal changes. Such data incongruities can then affect the subsequent reliability of model predictions. The focus of Chapter 3 is on identifying in-distribution inconsistencies: How can we develop a structured approach to detect and characterize in-distribution inconsistencies and incongruities to enable practitioners to both improve model deployment reliability and gain insights into their data for future data collection?

### 1.3.3 Principled Synthetic Data Augmentation

Machine learning's transformative potential is often constrained by a pervasive roadblock: insufficient data. Real-world domains often face data scarcity, whether due to expensive annotation processes such as in healthcare, limited digital infrastructure in developing regions, or underrepresentation of minority groups within larger datasets. Data augmentation serves as one approach to tackle this challenge. However, in low-data regimes, conventional tabular data generative models widely used for data augmentation struggle as the limited training samples provide a limited data manifold from which to learn. The result is that the synthetic augmented data lack sufficient diversity to cover the true underlying data manifold. The focus of Chapter 4 is hence to address the challenge of ML in low-data regimes, developing novel and principled approaches to synthetic data augmentation in low-data regimes. In doing so, the goal is to address the data logjam, such that we can unlock the use of ML in areas currently underserved.

### 1.3.4 Improving granular model evaluation

Despite solid average performance, ML models often prove unreliable in various real-world scenarios. For example, underperforming on different subgroups or struggling with data that deviates from the training distribution. The predominant model evaluation or testing paradigm relies exclusively on held-out data, evaluating models either on average or by stratifying the test data into different subgroups. This data-only approach operates under the restrictive assumption that empirical data is the sole input for testing, disregarding valuable contextual information that could guide the testing process. This leads to critical problems: high false positive rates from identifying spurious underperforming subgroups, high false negative rates from failing to identify true underperforming subgroups, and testing of subgroups that lack practical meaning. The focus of Chapter 5 is improving a cornerstone of reliable ML: better and more robust model evaluation. In particular, we focus on addressing the aforementioned challenges with current ML model evaluation approaches to ensure that models meet real-world reliability standards. The goal is to bring ML closer to the evaluation standards of other mature industries, such as the automotive or aviation industry.

### 1.3.5 Leveraging data for better uncertainty quantification

Uncertainty quantification is crucial for trustworthy ML, particularly in high-stakes domains like healthcare and finance. In particular, robust uncertainty estimates can provide insight into when model predictions should be trusted or deferred for human oversight. Traditional methods for uncertainty quantification, such as conformal prediction, offer distribution-free guarantees on prediction confidence. However, they are typically inefficient in regions of data scarcity. The focus of Chapter 6 is to address a critical limitation of standard uncertainty quantification techniques: their inability to leverage additional data (labeled or unlabeled) as auxiliary information derived from the data itself that can enhance prediction interval efficiency. For example, in settings with data scarcity, leveraging unlabeled samples can provide additional structure to improve uncertainty quantification.

## 1.4 Contributions

This section outlines the primary contributions of this thesis within each chapter.

### 1.4.1 Data Characterization

In this chapter, we develop a systematic framework that characterizes examples into subgroups with respect to their outcomes. First, we introduce Data-IQ, a framework that studies the learning dynamics of individual training samples through model training. Thereafter, Data-IQ analyzes and stratifies these training trajectories through a principled lens of uncertainty quantification-based metrics into easy, ambiguous and hard to learn instances. Second, we then validate these Data-IQ groups to understand their utility across different use cases, including data curation/selection, data understanding, dataset comparison, feature acquisition and model generalization. Third, we highlight the flexibility of Data-IQ to different ML model classes, as well as, different problem domains (tabular, image and text).

### 1.4.2 Test Data Incongruity Identification

In this chapter, we propose a paradigm shift from model-centric uncertainty to data-centric uncertainty to detect and understand in-distribution incongruous examples. First, we introduce Data-SUITE, a data-centric framework to identify these regions, independent of a task-specific model. Specifically, we evaluate the congruence of test instances with respect to the training set. Second, we show the capability of this framework to answer two practically useful questions: (1) which test instances will be reliably predicted by a model trained with the training instances? and (2) can we identify incongruous regions of the feature space so that data owners understand the data's limitations or guide future data collection? Third, we empirically validate Data-SUITE's performance and coverage guarantees, providing practitioners with confidence in the reliability of the identified regions. Fourth, we demonstrate across multiple real-world datasets with different types of incongruence that Data-SUITE best identifies incongruous data regions where downstream models may be unreliable, translating to improved model performance compared to alternative approaches.

### 1.4.3 Synthetic Data Augmentation and Curation

In this chapter, we develop a novel approach for data augmentation in low-data regimes, based on large language model (LLM) synthetic data generation and principled data curation. First, we introduce Curated LLM (CLLM), which synergistically combines the prior knowledge of LLMs for data generation with a principled curation mechanism to improve data augmentation in low-data regimes. Second, we empirically demonstrate on multiple real-world datasets that CLLM enables superior downstream performance compared to six widely used tabular data generative models and data augmentation techniques, with the largest gains observed for

underrepresented subgroups. Third, we provide comprehensive insights into both components of CLLM, namely the LLM generation and curation mechanism. This sheds new light on why the approach is beneficial, including how it addresses potential biases in the generated data.

### 1.4.4 Context-Aware Model Evaluation and Testing

In this chapter, we challenge the predominant data-only paradigm of ML testing and introduce Context-Aware Testing (CAT) as a principled alternative. First, we identify critical gaps in data-only ML testing approaches from the perspective of hypothesis testing around high false positive rates and high false negative rates. Second, we formalize the Context-Aware Testing paradigm, providing a principled framework to incorporate contextual information to guide the generation of targeted tests, thus addressing the fundamental question of "what should we test for?" Third, we propose SMART Testing, the first instantiated context-aware testing system that employs LLMs to hypothesize likely and relevant model failures and empirically refute them with data using a novel self-falsification mechanism. Fourth, we empirically demonstrate across diverse settings that SMART automatically identifies more relevant and impactful failures than alternative approaches, challenges the de facto data-only paradigm, and generates comprehensive model reports that provide stakeholders with actionable insights into identified failure modes.

### 1.4.5 Self-Supervised Uncertainty Quantification

In this chapter, we extend conformal prediction with self-supervised learning to improve predictive uncertainty quantification in settings with limited labeled data. First, we propose Self-Supervised Conformal Prediction (SSCP), a framework that leverages self-supervision as additional auxiliary information derived from the data itself. This auxiliary information is then used to improve the efficiency and adaptivity of the conformal prediction interval. Second, we empirically demonstrate that self-supervised losses improve residual estimates, thereby enhancing prediction interval adaptiveness, especially for more challenging examples and in sparser regions of the data space where traditional approaches struggle. Third, we show that labeled data can be repurposed for the self-supervised task to improve the efficiency of conformal prediction intervals, offering benefits even without additional data. Fourth, we demonstrate the value of unlabeled data to improve the efficiency of conformal prediction intervals, thereby offering a novel approach to uncertainty quantification that maximizes the utility of all available data.

## 1.5 Publications

Parts of this thesis have been published in peer-reviewed conferences, workshops and journals. Below we list the primary publications covered in this thesis and related publications not covered in this thesis. (* indicates equal contribution).

For Chapter 1, the motivations and figures are based on the following publication:

- Navigating Data-centric Artificial Intelligence with DC-Check: Advances, challenges, and opportunities [216].
  IEEE Transactions on Artificial Intelligence, 2024.
  **N.Seedat**, F.Imrie, M.van der Schaar

For Chapter 2 ("Data Characterization"), the directly corresponding publication is:

- Data-IQ: Characterizing subgroups with heterogeneous outcomes in tabular data [208].
  Conference on Neural Information Processing Systems (NeurIPS), 2022.
  **N.Seedat**, J.Crabbe, I.Bica, M.van der Schaar

Further publications that are related to data characterization in different problem settings, but are not directly included in the thesis, are as follows:

- TRIAGE: Characterizing and auditing training data for improved regression [209].
  Conference on Neural Information Processing Systems (NeurIPS), 2023.
  **N.Seedat**, J.Crabbe, Z.Qian, M.van der Schaar

- Dissecting Sample Hardness: A Fine-Grained Analysis of Hardness Characterization Methods for Data-Centric AI [215].
  International Conference on Learning Representations (ICLR), 2024.
  **N.Seedat**, F.Imrie, M.van der Schaar

- Reimagining synthetic tabular data generation through data-centric AI: A comprehensive benchmark.
  Conference on Neural Information Processing Systems (NeurIPS), 2023 [78].
  L.Hansen*, **N.Seedat***, M.van der Schaar, A. Petrovic

- You can't handle the (dirty) truth: Data-centric insights improve pseudo-labeling [211].
  Journal of Data-centric Machine Learning Research (DMLR), 2024.
  International Conference on Learning Representations (ICLR) - Workshop on Data-centric Machine Learning, 2024.
  **N.Seedat***, N.Huynh*, F.Imrie, M.van der Schaar

- When is Off-Policy Evaluation (Reward Modeling) Useful in Contextual Bandits? A Data-Centric Perspective [234].
  Journal of Data-centric Machine Learning Research (DMLR), 2024.
  International Conference on Learning Representations (ICLR) - Workshop on Data-centric Machine Learning, 2024.
  H.Sun*, A.Chan*, **N.Seedat**, A.Huyuk, M.van der Schaar.

- Going Beyond Static: Understanding Shifts with Time-Series Attribution [140].
  International Conference on Learning Representations (ICLR), 2025.
  J.Liu, **N.Seedat**, M.van der Schaar.

For Chapter 3 ("Test Data Incongruity Identification"), the directly corresponding publication is:

- Data-SUITE: Data-centric identification of in-distribution incongruous examples [210].
  International Conference on Machine Learning (ICML), 2022.
  **N.Seedat**, J.Crabbe, M.van der Schaar

Further publications that are related to the task of data inconsistency and incongruity identification, but are not directly included in the thesis, are as follows:

- DAGnosis: Localized Identification of Data Inconsistencies using Structures [93].
  International Conference on Artificial Intelligence and Statistics (AISTATS), 2024.
  N.Huynh*, J. Berrevoets*, **N.Seedat**, J.Crabbe, Z.Qian, M. van der Schaar

For Chapter 4 ("Synthetic Data Augmentation and Curation"), the directly corresponding publication is:

- Curated LLM: Synergy of LLMs and Data Curation for tabular augmentation in low-data regimes [212].
  International Conference on Machine Learning (ICML), 2024.
  **N.Seedat***, N.Huynh*, B.van Breuegl, M.van der Schaar

For Chapter 5 ("Context-Aware Model Evaluation and Testing"), the directly corresponding publication is:

- Context-Aware Testing: A New Paradigm for Model Testing with Large Language Models [190].
  Conference on Neural Information Processing Systems (NeurIPS), 2024.
  **N.Seedat***, P.Rauba*, M.Luyten, M.van der Schaar

Further publications that are related to model evaluations, but are not directly included in the thesis, are as follows:

- Can you rely on your model evaluation? Improving model evaluation with synthetic test data [242].
  Conference on Neural Information Processing Systems (NeurIPS), 2023.
  B.van Breugel*, **N.Seedat***, F.Imrie, M.van der Schaar

For Chapter 6 ("Self-Supervised Uncertainty Quantification"), the directly corresponding publication is:

- Improving adaptive conformal prediction using self-supervised learning [217].
  International Conference on Artificial Intelligence and Statistics (AISTATS), 2023.
  **N.Seedat***, A.Jeffares*, F.Imrie, M. van der Schaar

Further publications that are related to uncertainty quantification, but are not directly included in the thesis, are as follows:

- Relaxed Quantile Regression: Prediction Intervals for Asymmetric Noise [183].
  International Conference on Machine Learning (ICML), 2024.
  T.Pouplin*, A.Jeffares* **N.Seedat**, M. van der Schaar

- What is Flagged in Uncertainty Quantification? Latent Density Models for Uncertainty Categorization.
  Conference on Neural Information Processing Systems (NeurIPS), 2023 [235].
  H Sun, B van Breugel, J Crabbe, **N.Seedat**, M. van der Schaar

- U-PASS: An Uncertainty Guided Deep Learning Pipeline for Automated Sleep Staging [84].
  Computers in Biology and Medicine, 2024.
  E.Heremans, **N.Seedat**, B. Buyse, D.Testelmans, M. van der Schaar, M. De Vos.

- Automated remote sleep monitoring needs uncertainty quantification [85].
  Journal of Sleep Research, 2024.
  E.Heremans, L. Van den Bulcke, **N.Seedat**, A.Devulder, P.Borzée, B.Buyse, D.Testelmans, M. Van Den Bossche, M.van der Schaar, M. De Vos

Other research publications related to Data-centric AI, but not included in this thesis, include the following works:

- Generalization:a key challenge for responsible AI in patient-facing clinical applications [67].
  Nature Digital Medicine, 2024.
  L.Goetz*, **N.Seedat***, R.Vandersluis, M. van der Schaar
- Bootstrapping Self-Improvement of Language Model Programs for Zero-Shot Schema Matching [220].
  International Conference on Machine Learning (ICML), 2025.
  Conference on Neural Information Processing Systems (NeurIPS), Gen AI for Health and TRL Workshops, 2024.
  **N.Seedat**, M. van der Schaar
- Active Learning with LLMs for Partially Observed and Cost-Aware Scenarios [10].
  Conference on Neural Information Processing Systems (NeurIPS), 2024.
  N.Astorga, T.Liu, **N.Seedat**, M. van der Schaar
- Self-Healing Machine Learning: A framework for autonomous adaptation in real-world environments [189].
  Conference on Neural Information Processing Systems (NeurIPS), 2024.
  P.Rauba, **N.Seedat**, K. Kacprzyk, M. van der Schaar
- Large Language Models to enhance Bayesian Optimization [142].
  International Conference on Learning Representations (ICLR), 2024.
  T.Liu*, N.Astorga*, **N.Seedat**, M. van der Schaar
- Differentiable and transportable structure learning [21].
  International Conference on Machine Learning (ICML), 2023.
  J.Berrevoets, **N.Seedat**, F.Imrie, M. van der Schaar
- Continuous-Time Modeling of Counterfactual Outcomes Using Neural Controlled Differential Equations [213].
  International Conference on Machine Learning (ICML), 2022 .
  **N.Seedat***, F.Imrie*, A.Bellot, Z.Qian, M. van der Schaar **N.Seedat**, M. van der Schaar
- Unlocking Historical Clinical Trial Data with ALIGN: A Compositional Large Language Model System for Medical Coding [219].
  **N.Seedat**, C.Tozzi, AH.Ardiaca, M. van der Schaar, J.Weatherall, A.Taylor
- Towards Human-Guided, Data-Centric LLM Co-Pilots [206].
  E.Saveliev*, J.Liu*, **N Seedat***, A.Boyd, M van der Schaar.

# Chapter 2

# Data Characterization

## 2.1 Introduction

Most machine learning (ML) models are optimized using empirical risk minimization (ERM) during training, with the goal of maximizing average performance [138]. However, in real-world settings, while models may perform well on average, they also might underperform on specific subgroups of data [191, 81, 158]. Most of the current ML literature has focused on this problem in the context of computer vision, where underperforming subgroups are typically associated with data examples that have spurious correlations [138, 230] or mislabeling [177].

In this chapter, we focus on tabular data, the most ubiquitous data format in medicine and finance, where data is based on relational databases [23, 263]. Specific to the tabular setting, we formalize an understudied source of underperformance, namely *heterogeneity of outcomes*. This phenomenon is vital in healthcare, where patients with *similar features* can have *different outcomes* [92, 59, 262]. For example, [160] showed that prognostic models for risk prediction that perform well on average then underperform on specific cancer types due to heterogeneity of risk (outcome). Previous works have audited subgroups belonging to sensitive attributes (e.g. demographics, race or gender), as it is well-known that ML models generally underperform on these subgroups [98, 32]. However, this approach is limiting, as it needs the sensitive attributes to be pre-specified, as well as not capturing the case where complex feature interactions may lead to underperformance.

To address this, we take a different approach to automatically stratify data into subgroups, usable with *any ML model trained in stages (epochs/iterations)*; e.g., applicable to neural networks, gradient boosting etc. Specifically, in this chapter, we study the behavior of individual examples during training, called *training or learning dynamics*. This allows us to formalize that examples can lie on the spectrum from easy to hard to learn. To anchor

ideas, let us consider the task of patient mortality prediction. Based on their features, sicker patients more often have a mortality event. Thus, they are easy to learn for any model and will be predicted *correctly* with *high confidence* (*Easy*). However, a subgroup might have a heterogeneous outcome: survival despite their poor prognosis. This heterogeneity could result from randomness, making it practically impossible for a model to learn. These examples will be predicted *incorrectly* yet with *high confidence* (or equivalently have low confidence for the correct class) (*Hard*). In tabular data, there are also examples with inherent ambiguity, where the predicted probability for the correct class remains low. They appear in cases where the current features are insufficient to distinguish the example correctly, regardless of the model used [86, 145] (*Ambiguous*).

Identifying these subgroups is practically valuable, as improving the accuracy and robustness of ML models often depends on the characteristics and quality of the data [95, 76, 193, 203]. As mentioned in [203], the "data work" is often undervalued as merely operational, yet failing to account for it can have immense practical harm [203, 160]. Consequently, our goal is to build a systematic framework with the following desired properties (**P1-P4**), motivated by the considerations of practitioners at various stages of the ML pipeline. In satisfying **P1-P4**, we seek to address the "dire need for an ML-aware data quality that is not only principled, but also practical for a larger collection (...) of ML models" [193]:

**(P1) Robust data characterization**: The characterization of data examples should be robust, such that it is consistent across similar performing models that have different architectures/parametrizations.
**(P2) Principled data collection**: The characterization should be informative and actionable, providing practitioners with insights that enable both quantitative feature collection and dataset selection.
**(P3) Reliable model deployment**: The characterization should enable reliable model usage, both by unmasking unreliable subgroups or using the subgroups to tailor the data for better performance.
**(P4) Plug & play**: The characterization should be applicable to a variety of ML models widely used on tabular data, including neural networks, gradient boosting (and variants), etc.

To fulfill *P1-P4*, we propose **Data-IQ**, a systematic framework that characterizes examples based on the inherent qualities (IQ) of the data at both training and deployment time. As outlined in Figure 2.1, Data-IQ leverages confidence and in the *"data-centric AI"* spirit focused on the data: aleatoric uncertainty (i.e. uncertainty inherent to the data). This permits Data-IQ to provide ML-aware data quality that is principled and practical for a variety of ML models. In doing so, we make the following contributions:

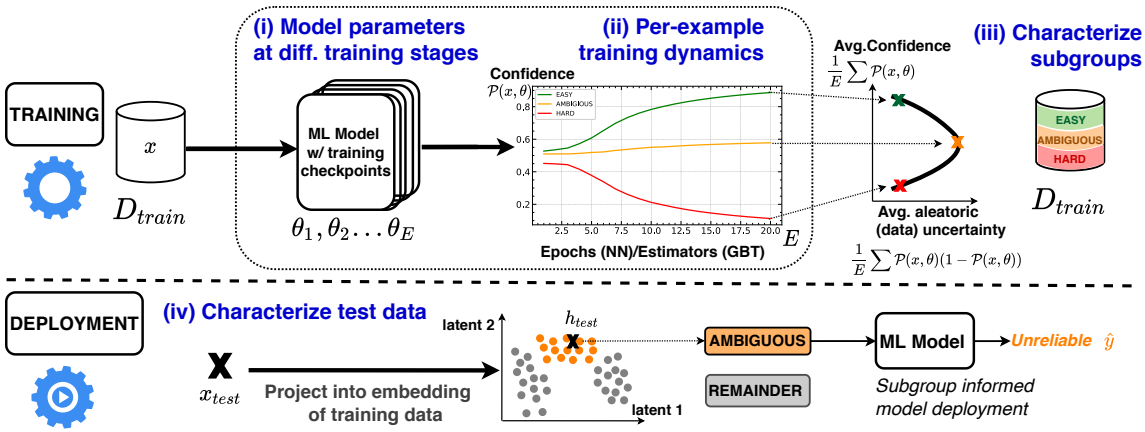



Fig. 2.1 Data-IQ systematically characterizes data into subgroups using (i) any ML model, trained in stages (epochs/iterations) and can be checkpointed. At training time, Data-IQ (ii) leverages the model's checkpoints to analyze the training behavior of individual examples and (iii) characterizes each example based on its aleatoric (data) uncertainty & prediction confidence. (iv) At deployment, the subgroups inform model usage by embedding the training set in a representation space and characterizing the new data points in this space.

**Contributions:** ① Data-IQ models the aleatoric (data) uncertainty, which permits subgroup identification that is most robust to variation across different yet similar performing models/parameterizations, compared to other baselines, i.e. *P1*. ② Data-IQ aids with principled data collection *P2* in two ways: Firstly, it permits quantifying the value of an acquired feature by measuring how the feature reduces the aleatoric uncertainty of the example. This information enables a more principled approach to feature acquisition. Secondly, it permits the comparison of datasets based on the proportion of ambiguous examples. We demonstrate that the proportions link to how well a model trained with the dataset generalizes. ③ Experimentally, the subgroups identified by Data-IQ can inform reliable model deployment, i.e. P3. We highlight cases where assessment on average might mask unreliable performance, including data sculpting, model robustness, and uncertainty estimation methods. ④ Data-IQ by construction is "plug-and-play" i.e. *P4* with *any* ML model that can be checkpointed, granting practitioners the flexibility to apply Data-IQ to their model of choice.

## 2.2 Related work

The work in this chapter primarily engages with the literature on data characterization and contributes to the nascent area of data-centric AI [156, 180]. An extended discussion of related work is found in Appendix A.1.1.

**Data characterization.** The literature to characterize data samples has used a variety of different metrics. However, their goals have typically been different, such as identifying spurious correlations or mislabeling, compared to Data-IQ, whose goal is to characterize subgroups with respect to the outcome predictions. Furthermore, none of these methods completely addresses all the desired properties (P1- P4). The closest to our work on data quality is Data Maps [237]. A key contrast to Data-IQ is that Data Maps uses confidence and prediction variability as metrics to stratify and flag instances. In Chapter 2.3, we show that this prediction variability corresponds to the model uncertainty (i.e. epistemic uncertainty). Alternatively, Data-IQ takes a different and more principled approach, capturing the inherent data uncertainty (known as *aleatoric uncertainty*) [107]. To further contrast, epistemic uncertainty is reducible by collecting more data; however, in comparison, aleatoric uncertainty is irreducible even with more samples. This is due to the fact that the aleatoric uncertainty captures properties inherent to the data [44, 107, 60]; hence, only additional informative features can reduce the aleatoric uncertainty [60]. Later in Figure 2.4, we show on real data that capturing the aleatoric uncertainty allows Data-IQ to be more robust to variation across different models, compared to Data Maps (*P1*). This allows practitioners to characterize their data in such a way that ensures that insights into the data are more consistent. We further show theoretically in Chapter 2.3.3 why the characterization by Data-IQ indeed provides a more principled definition for *Ambiguous* examples compared to Data Maps.

In addition to Data Maps, other related data characterization methods address specific computer vision problems: identifying mislabeled images using the area under the margin (AUM) [177], gradient norm to identify "important examples" to aid pruning during training [173], or underperformance due to spurious image correlations [138]. The tabular data setting considered in this chapter requires new methods due to the specific problem of heterogeneous outcomes for examples with similar features (i.e. "feature collision"). The ambiguity in the tabular, "feature collision" sense is different or non-existent in modalities such as images, as all the information is contained in an image (information cannot be added in the same way as tabular data).

**Data-Centric AI.** The assessment of data quality is a critical but often overlooked problem in ML [203]. While the focus in ML is typically on optimizing models, the task of ensuring high-quality data (or even improving one's data) can be equally valuable for improving performance [203, 95]. Even when it is considered, the process of assessing datasets is often adhoc or artisinal [203, 156, 38, 210]. The recent growth of data-centric AI research aims to build systematic tools for "data collection, labeling, and quality monitoring

processes for datasets to be used in machine learning" [210, 180]. Data-IQ contributes to this nascent body of work, specifically around ML-aware data quality monitoring [193].

## 2.3 Formulation

We now provide a detailed formulation of Data-IQ and motivate our proposed example stratification that uses aleatoric uncertainty and confidence. We then describe how Data-IQ stratifies examples into subgroups at both training and testing time. Finally, we show that Data-IQ's formulation allows for usage with *any* ML model trained in stages, e.g., neural networks, GBDTs, etc, unlike other approaches.

### 2.3.1 Preliminaries

We consider the typical supervised learning setting, where the aim is to assign an input $x \in \mathscr{X} \subseteq \mathbb{R}^{d_X}$ to a class $y \in \mathscr{Y} \subseteq \mathbb{N}$. We have a dataset $\mathscr{D}$ with $N \in \mathbb{N}^*$ examples, i.e., $\mathscr{D} = \{(x_i, y_i) \mid i \in [N]\}$ drawn IID from an unknown distribution. Our goal is then to learn a model $f_\theta : \mathscr{X} \to \Delta^{|\mathscr{Y}|-1}$, parameterized by $\theta \in \Theta$, where $\Delta^{|\mathscr{Y}|-1}$ denotes the $(|\mathscr{Y}|-1)$-dimensional probability simplex. Typically, the parameters $\theta$ are learned to minimize empirical risk, by minimizing the average training loss, i.e., $\text{ERM}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \ell(x_i, y_i; \theta)$. When referring specifically to the training split of the dataset, we denote it by $\mathscr{D}_{\text{train}} \subseteq \mathscr{D}$.

This brings us to the essence of the problem: "not all examples are created equally". e.g. patients with similar features might have heterogeneous outcomes, reflected in their labels $y$ being different. These correspond to subgroups within $\mathscr{D}_{\text{train}}$ on which a predictive model might systematically underperform. We formalize this concept of latent/unobserved heterogeneous by assigning to each example $x^n$ a latent *subgroup* label $g^n \in \mathscr{G}$, where $\mathscr{G} = \{\textit{Easy}, \textit{Ambiguous}, \textit{Hard}\}$. This label is determined by the training dynamics of a model on $\mathscr{D}_{\text{train}}$. Before giving a precise description of how those group labels are assigned, it is useful to detail the context. Several works have established that the training/learning dynamics of a model contains valuable signals about the quality of the data itself [7, 6, 134]. For instance, it takes more epochs/iterations for a model to assign the correct label to noisier/more difficult to learn training examples. With Data-IQ, we build on those observations and assign a label $g^n$ to each example $x^n$ by studying its training dynamic, which is then used to estimate the aleatoric uncertainty and predictive confidence of each example. We next detail how this is done and how this contrasts with existing approaches.

### 2.3.2 Uncertainty decomposition during training

Recall that practitioners desire flexibility in the choice of the model. Hence, we focus on *any* ML model that is trained in stages and can be checkpointed over the course of model training (parameterized by $\theta$) and on a given example *from the training set* $(x,y) \in \mathscr{D}_{\text{train}}$. Assume that the model $f_\theta$ corresponds to a conditional categorical distribution, assigning a probability to each class given the input $x$: $f_\theta(x) = P(Y \mid X = x, \vartheta = \theta)$. During iterative training, the model parameters $\theta$ vary, where over $E \in \mathbb{N}^*$ epochs/iterations, these parameters take $E$ different values at each checkpoint, i.e. $\theta_1, \theta_2, \ldots, \theta_E$. Since our analysis relies on the model's training dynamics, we want to take those different parameters into account. For the sake of notation, we introduce a random variable $\vartheta$ that has an empirical distribution over this set of parameters captured through the training process $\vartheta \sim \text{Unif}\{\theta_1, \ldots, \theta_E\}$, i.e. $\vartheta$ takes each checkpoint value $\theta_e$ with probability $1/E$. The variability of the model's parameters at training time is then reflected by the variance $\mathbb{V}_\vartheta[\cdot]$.

The uncertainty we model is based on the random variable $Y \mid X = x$ that represents the possible labels given the input $x$. Since the ground-truth label $y$ is available for training data examples, we would like to distinguish between 2 cases: ① the predicted label corresponds to the ground-truth label $Y = y$ and ② the predicted label is different from the ground-truth label $Y \neq y$.

To this end, we introduce a binary random variable $\tilde{Y}$ that is set to one when the predicted label equals the ground-truth label ($\tilde{Y} = 1$ if $Y = y$) and that is zero otherwise ($\tilde{Y} = 0$ if $Y \neq y$). As discussed earlier, we are interested in the uncertainty of the predictive random variable $\tilde{Y} \mid X = x$. This uncertainty is modeled by the variance $v(x,y) = \mathbb{V}_{\tilde{Y}|X,Y}\left[\tilde{Y} \mid X = x, Y = y\right]$. We will now show that this quantity can be evaluated with the model predictions.

We start by noting that the definition of $\tilde{Y}$ implies that $\tilde{Y} \mid X = x, \vartheta = \theta$ is a Bernoulli random variable with parameter[1] $P(x,\theta) = P(Y = y \mid X = x, \vartheta = \theta) = [f_\theta(x)]_y$. From this observation, we can decompose $v(x)$ with the law of total variance and make each term explicit:

$$\begin{aligned} v(x,y) &= \underbrace{\mathbb{V}_\vartheta\left(\mathbb{E}[\tilde{Y} \mid X = x, \vartheta]\right)}_{\text{Bernoulli Mean: } P(x,\vartheta)} + \underbrace{\mathbb{E}_\vartheta\left[\mathbb{V}(\tilde{Y} \mid X = x, \vartheta)\right]}_{\text{Bernoulli Var: } P(x,\vartheta)(1-P(x,\vartheta))} \\ &= \underbrace{\mathbb{V}_\vartheta\left(P(x,\vartheta)\right)}_{\text{Epistemic uncertainty: } v_{\text{ep}}(x,y)} + \underbrace{\mathbb{E}_\vartheta\left[P(x,\vartheta)(1-P(x,\vartheta))\right]}_{\text{Aleatoric uncertainty: } v_{\text{al}}(x,y)}. \end{aligned} \tag{2.1}$$

In Equation 2.1, we have split the overall uncertainty into two components: *epistemic* and *aleatoric uncertainty*. This type of decomposition is similar to those in the context of

[1] In this case, $[f_\theta(x)]_y$ denotes the component $y$ of the probability vector $f_\theta(x)$.

Bayesian neural networks [124, 218]. To understand the distinction between uncertainties, it is useful to closely examine the variances in the *first* equality of Equation 2.1. For *epistemic uncertainty* $v_{\text{ep}}$, variance is evaluated on the *model parameters* $\vartheta$. Hence, epistemic uncertainty originates from the fact that the model's predictions oscillate when we change its parameters. For the *aleatoric uncertainty* $v_{\text{al}}$, the variance is evaluated on the *predicted label* $\tilde{Y} \mid X, \vartheta$. Hence, the variability originates from the inability to predict the correct label with high confidence. While existing work uses epistemic uncertainty to stratify examples, we argue that aleatoric uncertainty is a better and principled choice to capture the inherent data uncertainty.

### 2.3.3 Stratification based on data uncertainty

We now explain how the above notion of uncertainty permits Data-IQ to assign a group label $g \in \mathscr{G}$ to each training example $x$. First, we use the empirical distribution $\vartheta \sim P_{\text{emp}}(\{\theta_e \mid e \in [E]\})$ to explicitly write the two types of uncertainties in Equation 2.1,where $\bar{P}(x) = 1/E \sum_{e=1}^{E} P(x, \theta_e)$:

$$v_{\text{ep}}(x,y) = \frac{1}{E}\sum_{e=1}^{E}\big[P(x,y,\theta_e) - \bar{P}(x,y)\big]^2 \quad v_{\text{al}}(x,y) = \frac{1}{E}\sum_{e=1}^{E} P(x,y,\theta_e)\big(1 - P(x,\theta_e)\big), \quad (2.2)$$

**Stratification at training time.** Before giving a precise definition of the group labels, let us give an intuitive definition for each group. ① *Easy*: examples that have low data uncertainty that the model can correctly predict with high confidence, ② *Ambiguous*: examples that have high data uncertainty, hence the model is unable to predict with confidence and ③ *Hard*: examples that have low data uncertainty that the model is unable to predict (i.e. predicted incorrectly yet with high confidence or equivalently have low confidence for the correct class). We note that we need the model's prediction for the ground-truth class to delineate *Easy* and *Hard* examples. In practice, we use the model's average confidence for the ground-truth class $\bar{P}(x)$ defined previously for this purpose. We make use of this concept to detail how labels are assigned to training examples $(x,y) \in \mathscr{D}_{\text{train}}$:

$$g((x,y),\mathscr{D}_{\text{train}}) = \begin{cases} \textit{Easy} & \text{if } \bar{P}(x,y) \geq C_{\text{up}} \wedge v_{\text{al}}(x,y) < Q_{50}[v_{\text{al}}(\mathscr{D}_{\text{train}})] \\ \textit{Hard} & \text{if } \bar{P}(x,y) \leq C_{\text{low}} \wedge v_{\text{al}}(x,y) < Q_{50}[v_{\text{al}}(\mathscr{D}_{\text{train}})] \\ \textit{Ambiguous} & \text{otherwise} \end{cases} \quad (2.3)$$

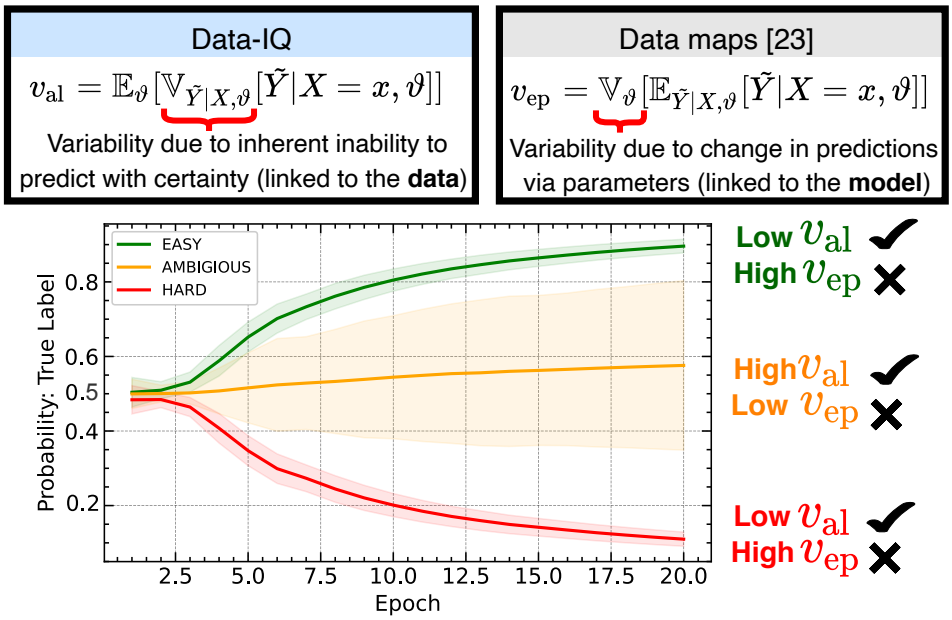



Fig. 2.2 Example outlining the differences between $v_{al}$ (Data-IQ) and $v_{ep}$ (Data Maps) & showing the type of uncertainty matters.

where $0 < C_{low} < C_{up} < 1$ are lower and upper confidence thresholds, respectively, and $Q_n$ the n-th percentile. We provide a practical method to set $C_{up}$ and $C_{low}$, applicable to any dataset in Appendix A.1.

In contrast to Data-IQ, which uses Aleatoric uncertainty $v_{al}(x)$, Data Maps [237] identifies ambiguous training examples $(x,y) \in \mathscr{D}_{train}$ as those with high epistemic uncertainty $v_{ep}(x)$. We consider a typical scenario to see how this characterization might cause problems, illustrated in Figure 2.2. Consider an example $x$ in which the model cannot classify confidently during the entire training $P(x,\theta_e) = 0.5\ \forall e \in [E]$. In this case, the epistemic uncertainty $v_{ep}(x,y)$ vanishes, as the prediction is consistently unconfident (i.e. low variability of the model predictions). This implies that Data Maps would consider this example as non-ambiguous, despite the ambiguous model prediction for this example. This problem can be traced back to the definition of epistemic uncertainty, which measures the sensitivity of a model prediction with respect to the model's parameters.

A more principled definition for ambiguous examples should capture examples for which the model cannot predict the appropriate label with high confidence (i.e. data uncertainty). This is precisely what the aleatoric uncertainty $v_{al}(x)$ captures (Data-IQ). Furthermore, it is easy to verify that the previous example $P(x,\theta_e) = 0.5\ \forall e \in [E]$ maximizes the aleatoric uncertainty (see Figure 2.2). Since high aleatoric uncertainty captures ambiguous examples for various values of the model's parameters, we believe that it better reflects the inherent

quality of the data. In that sense, we expect this quantity to be more stable and robust to variation for different ML model parameters/architecture changes (*P1*). We experimentally validate the consistency in Chapter 2.4.

**Stratification at inference time.** Prior data charaterization methods are only applicable at training time (i.e. to audit the training dataset). To address this limitation and improve the practical utility, we also propose an approach to stratify examples into subgroups at deployment time using Data-IQ. However, if we try to apply the above training data stratification mechanism for incoming data at deployment time, we face a problem: $\bar{P}(x,y)$ requires the ground-truth class $y$ — which is unavailable at deployment time.

For this reason, we follow an alternative approach based on representation learning that does not require access to ground-truth labels. The idea is the following: we construct a low-dimensional UMAP embedding [148] $h : \mathscr{X} \rightarrow \mathscr{H}$ of the training set's examples $x_{\text{train}} \in \mathscr{D}_{\text{train}}$. In doing this, we note two things (see Appendix A.3): ① *Ambiguous* examples have distinctive features and are clustered in embedding space. Thus, it is possible to distinguish the *Ambiguous* examples using the embedding. ② It is not possible to reliably distinguish *Easy* examples from *Hard* examples based on their embeddings. The reason is that *Hard* examples are a minority with outcome randomness that have similar features as the *Easy* examples. Combining these observations, we note that it is possible to identify *Ambiguous* test examples. This label is assigned by computing the related embedding $h(x_{\text{test}})$ and comparing this embedding with the nearest neighbor embedding from the training set, i.e. $d[h(x_{\text{test}}), h(x_{\text{train}})] \forall x_{\text{train}} \in \mathscr{D}_{\text{train}}$, where $d$ is Euclidean distance in embedding space. For models like neural networks with an implicit representation space, the same analysis can be done using the model's representation space.

### 2.3.4 Using Data-IQ with a variety of models, beyond Neural Networks (P4)

The baseline methods discussed are primarily applicable only to neural networks. However, practically in tabular settings (e.g. healthcare/finance etc), practitioners often use other highly performant iterative learning algorithms such as Gradient Boost Decision Trees (GBDTs), XGBoosts or variants [226, 23]. Data-IQ's formulation by construction is naturally adaptable to *any* ML model trained in stages that can be checkpointed. This satisfies *P4*, which allows practitioners the flexibility to use Data-IQ with their application-specific model of choice. Appendix A.1 provides guidelines, space and time considerations, and discusses the specifics of how Data-IQ is easily adapted, for example, to GBDTs.

## 2.4 Experiments

We now present a detailed empirical evaluation demonstrating that Data-IQ satisfies (**P1**) Robust data characterization, (**P2**) Principled data collection and (**P3**) Reliable Model Deployment, introduced in Chapter 2.1. Recall that (**P4**) Plug and play is satisfied by the formulation of Data-IQ. Additionally, we conduct experiments highlighting the flexibility of Data-IQ beyond tabular data when used with text and image data.

**Datasets.** We primarily conduct experiments on four real-world medical datasets, with diverse characteristics (different sizes, binary/multiclass, varying degrees of task difficulty etc) and highlight real-world applicability with heterogeneous patient outcomes: (1) Covid-19 dataset of Brazilian patients [15], (2) Prostate cancer datasets from both the US [46] and UK [174], (3) Support dataset of seriously ill hospitalized adults [114], (4) Fetal state dataset of cardiotocography [13]. We describe the datasets in greater detail in Appendix A.2, along with further experimental details.

### 2.4.1 (P1) Robust data characterization

**Robustness to variation.** As per *P1*, we desire that Data-IQ identifies subgroups in a manner robust to variation across different models. This would allow a practitioner to obtain consistent insights about their data even when using different model architectures/parameterizations. When comparing the different alternative data characterization methods, we note that each method has its own specific metric used to characterize examples (see Appendix A.2). To assess robustness to variation, we compare the consistency of the different characterization metrics when evaluated on models with different architectures/parameterizations. All models are trained to convergence, with early stopping on a validation set.

Quantitatively, we compute the Spearman rank correlation between all model combinations, see Figure 2.3. We observe that Data-IQ is the most consistent and robust to variation across different models, having the highest score on all datasets, satisfying *P1*. Furthermore, the baseline methods themselves are also not consistent in performance ordering across datasets, which is undesirable. Ultimately, the robustness means that practitioners can feel confident in the consistency of data insights derived using Data-IQ.

To specifically compare Data-IQ and Data Maps [237], we examine 3 distinct models that achieve similar performance on the Covid-19 [15] tabular dataset, and we produce a characterization of the training dataset using each model in Figure 2.4. We note that Data Maps groups can be recovered from 2.3 by replacing the aleatoric uncertainty $v_{\text{al}}$ from Data-IQ with its epistemic counterpart $v_{\text{ep}}$. The y-axis is the same for both methods and corresponds to $\bar{P}(x)$. The x-axis corresponds to $v_{\text{al}}(x)$ for Data-IQ and to $v_{\text{ep}}(x)$ for Data

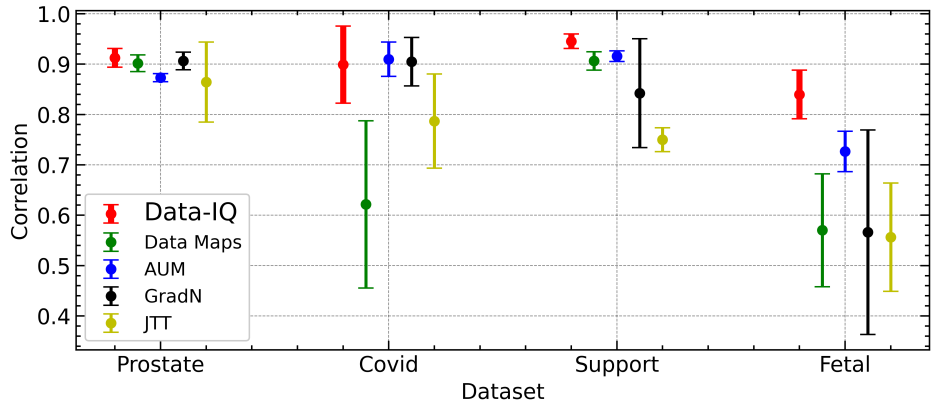
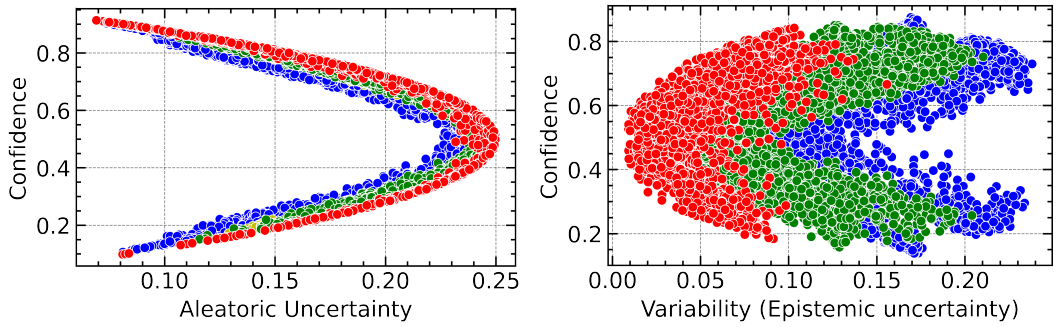



Fig. 2.3 Robustness to variation across models based on Spearman correlation, where Data-IQ has the highest correlation (i.e. consistency) across all datasets.

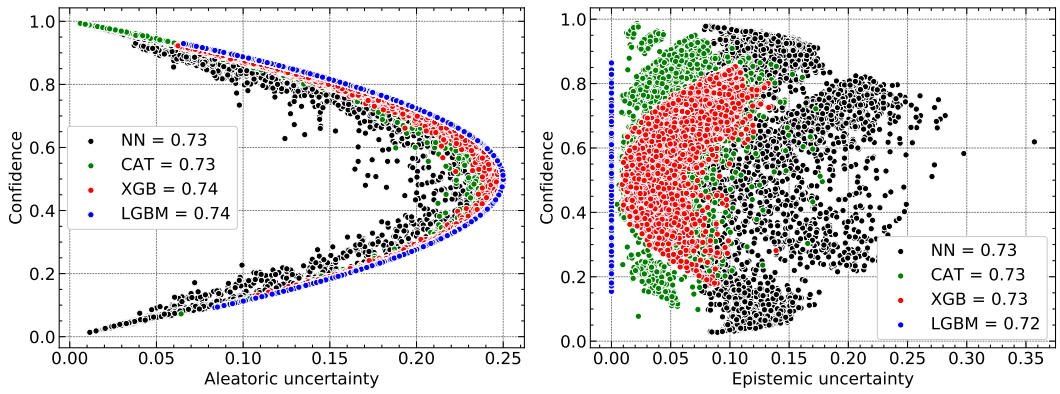



(a) Data-IQ (b) Data Maps [237]

Fig. 2.4 Data-IQ's robustness to variation across different models (colors).

Maps. Each model is assigned a color in Figure 2.4. We note three things ① Data-IQ's characterization of the data is significantly more stable across models. ② Linked to the points in Chapter 2.3, Data Map's high and low confidence examples, in fact, have high epistemic uncertainty $v_{\text{ep}}$, which can lead to incorrect conclusions when attempting to use Data Maps to characterize data. ③ Data-IQ always distributes the data around a bell shape, which standardizes its interpretation. We provide a theoretical analysis to explain this bell-shaped observation in Appendix A.1.

**Data-IQ: Neural Networks vs Other model classes.** Data-IQ can be used with *any* ML model trained in stages linked to *P4*: Plug and Play. Methods such as XGBoost, LightGBM, and CatBoost are widely used by practitioners on tabular data, often more so than neural networks [23]. Ideally, based on *P1*, we desire that the characterization of examples is consistent for similar performing models, irrespective of whether the model is a neural network or an XGBoost model.




(a) Data-IQ (b) Data Maps [237]

Fig. 2.5 Assessment on characterization when used with different backbone models shows that Data-IQ is more consistent.

To assess the robustness of both Data-IQ and Data Maps, we train a neural network, XGBoost, LightGBM and CatBoost models to achieve the same performance and then perform the characterization for all models. We can clearly see in Figure 2.5 (Support) that Data-IQ has a similar characterization across all four models. In contrast, for Data Maps, the characterizations are significantly different for the different model classes. The implication of this result is that by Data-IQ capturing the uncertainty inherent to the data (aleatoric uncertainty), leading to a more consistent and stable characterization of the data itself. This highlights that Data-IQ characterizes the data in a manner that is not as sensitive to the choice of model when compared to Data Maps. For additional analysis, see Appendix A.3.

**Data insights from subgroups.** Given the distinct differences between the subgroups, we seek to understand what factors make these subgroups different and how they can provide insight into the dataset. Such insights are especially useful in clinical settings. Our results for the prostate cancer dataset are illustrated in Figure 2.6 (with other datasets in Appendix A.3). To visualize the different groups of patients within each subgroup, we cluster the samples within each subgroup (*Easy*, *Ambiguous* and *Hard*) using a Gaussian Mixture Model (GMM) similar to [230], selecting the optimal clusters based on the Silhouette score. We assess cluster quality vs alternatives in Appendix A.3.

In general, across different datasets, the subgroups are characterized as follows: (1) *Easy*: Severe patients with a death outcome and less severe patients with a survival outcome. (2) *Ambiguous*: Patients with similar features but different outcomes. This could suggest that the features we have at hand are insufficient to separate the differences in outcomes. (3) *Hard*: Severe patients with a survival outcome and less severe patients with a death outcome. i.e. opposite outcomes as expected due to randomness in the outcomes.

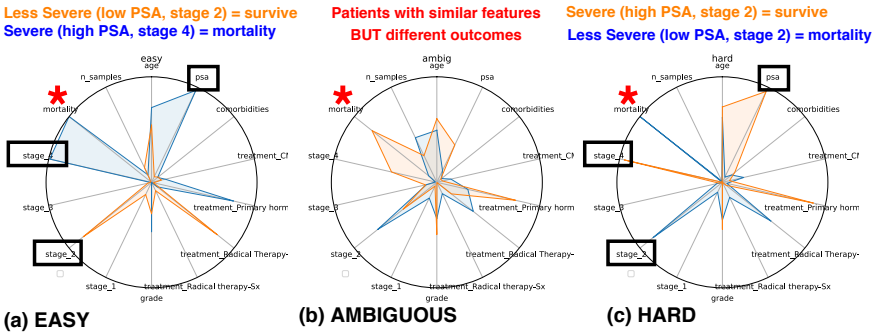



Fig. 2.6 Comparing subgroups identified by Data-IQ (descriptions above). Colors represent the GMM clusters.

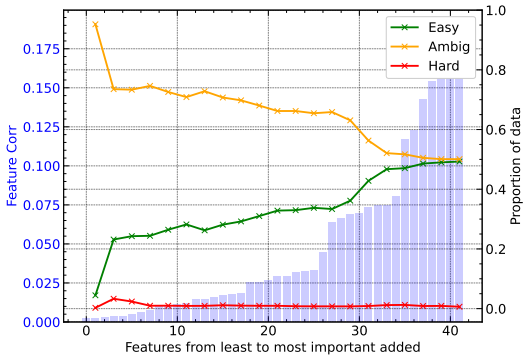
### 2.4.2 (P2) Principled data collection

**Principled feature acquisition.** As per Chapter 2.4.1, the *Ambiguous* subgroup has examples with similar features, yet different outcomes. Recall that this case of ambiguity in the tabular setting is very different from ambiguity in other modalities, such as images. The ambiguity is due to insufficient features to adequately separate the examples. We link this to the concept that the *Ambiguous* subgroup has a high aleatoric uncertainty that is irreducible, even if we collect more data examples. Rather, aleatoric uncertainty can only be reduced by acquiring more informative features [60].

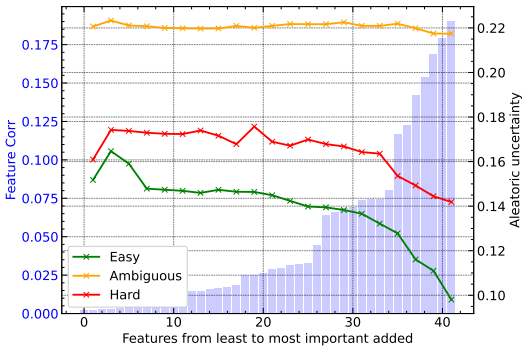
We leverage this idea and show that Data-IQ's example characterization provides a principled approach to assessing the benefit of acquiring a specific feature. This idea of feature acquisition is different from feature selection, where all features are present, and then we select the most "important feature". Additionally, this is different from active learning, which quantifies the value of acquiring examples, not features.

With the above in mind, an informative and/or valuable feature should decrease example ambiguity (i.e. aleatoric uncertainty). Hence, a decrease in the proportion of *Ambiguous* examples can serve as a proxy for the feature's potential value to the dataset. Understanding the value of features is useful in settings such as healthcare, where feature acquisition comes at a cost. To showcase the potential, we construct a semi-synthetic experiment, where we rank and sort the features based on correlation with the target.

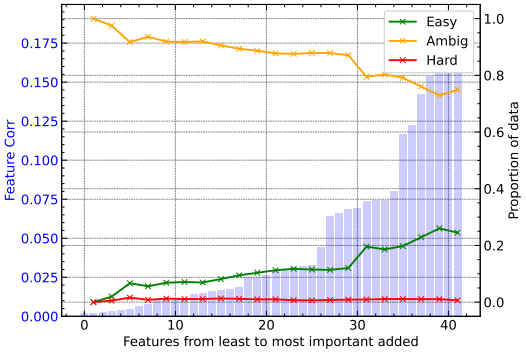
We then train different models, where we sequentially "acquire" features of increasing value (based on correlation). Figure 2.7 shows results for the Support dataset. For Data-IQ, Figure 2.7 (a) shows that as we acquire "valuable" features, the proportion of the *Ambiguous* subgroup drops, whilst the *Easy* subgroup increases, with significant changes for the important features. This shows that Data-IQ's subgroup characterization can be used to quantify a feature's value based on its ability to decrease ambiguity. In contrast, Data Maps,



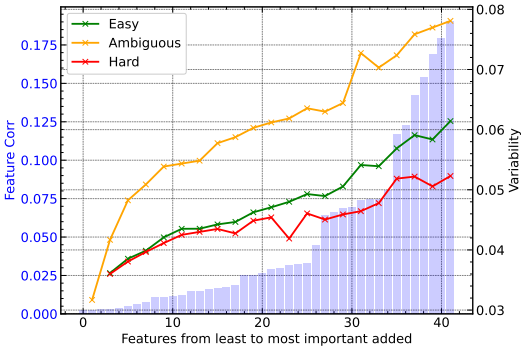

(a) Data-IQ subgroup ambiguity proportion is reduced as more informative features are acquired.




(b) Data-IQ aleatoric uncertainty remains stable for Ambiguous, reduces for others as features are acquired.




(c) Data Maps subgroup proportions largely unaffected as features acquired.




(d) Data Maps variability increases across subgroups as features acquired.

Fig. 2.7 Quantifying the value of feature acquisition based on change in ambiguity. Only Data-IQ captures this.

Figure 2.7 (c), exhibits a minimal response to feature acquisition, suggesting that it may not be sensitive enough to capture the feature's value.

Furthermore, for Data-IQ, we see that the examples that remain as *Ambiguous*, after features are collected, maintain a consistent aleatoric uncertainty. This is desired as it demonstrates for those examples which remain *Ambiguous*, that indeed the features collected are not informative enough to reduce their inherent (aleatoric) uncertainty, i.e. those remaining still need better features (see Figure 2.7 (b)). While for Data Maps, the added features, in fact, increase the variability for *all* subgroups, making it harder to stratify (see Figure 2.7 (d)). This links to the fact that Data Maps subgroups can't capture the value of the acquired features. We further show experimentally in Appendix A.3 that for *Ambiguous* examples, it is not simply a case of increasing the size of the dataset (i.e. more examples). In fact, this can increase the proportion of *Ambiguous* examples due to the increased probability of feature collisions as the dataset size increases. Ultimately, this motivates the usefulness of principled feature acquisition (which Data-IQ can guide) as a way to decrease dataset ambiguity.

**Principled dataset comparison.** Extending beyond feature-level, an understudied scenario involves systematically selecting between datasets in two cases: (1) purchasing data from data markets [120, 187, 188] and (2) organizations where the data is siloed, with lengthy access processes [49, 68]. In both cases, synthetic versions of the real dataset have begun to be used [49]. For now, we ignore privacy concerns and focus on data fidelity and quality, which compares the real and synthetic datasets using statistical measures. However, as per [2], the conclusions can vary between different metrics. In practice, competing "synthetic" datasets can be generated by different ML synthetic data generators or by different synthetic data providers. Thus, while they model the same underlying distribution, depending on the process used to generate the synthetic data, one version might be superior to the other. We now ask: Can Data-IQ help us to systematically select between synthetic datasets? We consider the setting where the real data is *not accessible*. Hence, we cannot use existing reference-based evaluation metrics, yet we still wish to compare the synthetic datasets (e.g., comparing between different providers' datasets).

We simulate this scenario by generating synthetic data using 2 different synthetic approaches, representing 2 synthetic data providers/vendors: (V1) CTGAN [259] and (V2) Gaussian Copula [136]. We then characterize the dataset subgroup proportions using Data-IQ. We hypothesize that datasets with greater proportions of *Easy* examples generalize better (i.e. more learnable samples). As is common, we validate fidelity by training with synthetic data and testing with real data [50, 176], where the best fidelity data produces the best model performance on real data (test set). Table 2.1 shows that datasets with the highest quality,

Table 2.1 Comparison of accuracy performance rank and (dataset quality). Synthetic dataset w/ better quality (↑ easy)produces the best real data test performance.

| Dataset | (V1) CTGAN | (V2) Gaussian Copula |
|---|---|---|
| Prostate | **Rank 1 (63% Easy)** | Rank 2 (30% Easy) |
| Covid | **Rank 1 (70% Easy)** | Rank 2 (63% Easy) |
| Support | **Rank 1 (59% Easy)** | Rank 2 (38% Easy) |
| Fetal | Rank 2 (40% Easy) | **Rank 1 (51% Easy)** |

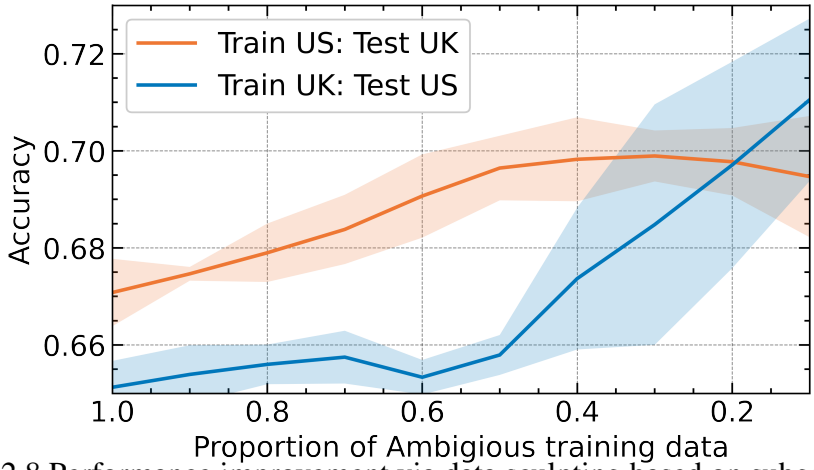



Fig. 2.8 Performance improvement via data sculpting based on subgroups

as measured by Data-IQ, indeed produce the best performance on real test data (i.e., Rank 1). In addition, it shows that the same "vendor" does not always produce the best dataset, highlighting the value of comparative assessment. Ultimately, these two aspects demonstrate that when the real data is not available, Data-IQ is a useful tool in the hands of practitioners wishing to assess data quality, especially when selecting between different datasets.

### 2.4.3 (P3) Reliable Model Deployment

**Less is more: data sculpting based on subgroups.** What role do the data subgroups, specifically *Ambiguous* examples, play in ensuring model generalization? Using multi-country prostate cancer data, we train a baseline model on US data (SEER) and assess generalization when deployed on patients in the UK (CUTRACT) and vice versa. In Figure 2.8, we see that test time generalization performance monotonically increases as we decrease the proportion of *Ambiguous* training data (see Appendix A.3.10 for absolute numbers). Ultimately, this illustrates the value of sculpting the training dataset by removing ambiguous examples as a way to improve the reliability of a deployed model.

Table 2.2 Comparison of different model improvement/robustness techniques

| Dataset | Group | Baseline | Group-DRO (**Data-IQ**) | Group-DRO (George) | JTT |
|---|---|---|---|---|---|
| Prostate | Overall | 0.837 | **0.840** ↑ | 0.565 ↓ | 0.723 ↓ |
| | Ambiguous | 0.740 | **0.741** ↑ | 0.598 ↓ | 0.538 ↓ |
| | The Rest | **0.935** | **0.935** | 0.535 ↓ | 0.896 ↓ |
| Covid | Overall | 0.729 | **0.732** ↑ | 0.687 ↓ | 0.455 ↓ |
| | Ambiguous | 0.629 | **0.633** ↑ | 0.609 ↓ | 0.477 ↓ |
| | The Rest | **0.832** | **0.832** | 0.766 ↓ | 0.438 ↓ |
| Support | Overall | **0.734** | **0.734** | 0.660 ↓ | 0.611 ↓ |
| | Ambiguous | 0.621 | **0.622** ↑ | 0.576 ↓ | 0.521 ↓ |
| | The Rest | **0.858** | **0.858** | 0.754 ↓ | 0.711 ↓ |
| Fetal | Overall | 0.768 | **0.833** ↑ | 0.829 ↑ | 0.829 ↑ |
| | Ambiguous | 0.575 | **0.701** ↑ | 0.695 ↑ | 0.698 ↑ |
| | The Rest | 0.970 | **0.974** ↑ | 0.970 | 0.970 |

**Group-DRO: Not a silver bullet when used in tabular settings.** Once subgroups of underperformance are identified, it is generally assumed that methods such as Group Distributionally Robust Optimization (DRO) [201] can be applied to improve model performance and robustness. We compare Group-DRO with groups identified by Data-IQ and as baselines: George [230] and Just-Train-Twice (JTT) [138]. As per the previous experiments, the largest underperforming group (in proportion) is the *Ambiguous* examples. Similar to the literature, we evaluate the performance change from a baseline model after Group-DRO is applied.

The results in Table 2.2 show that Group-DRO using Data-IQ's groups both improves overall performance and improves performance on the *Ambiguous* group. Contrastingly, for the other baselines, the performance actually degrades.

Nevertheless, while using Data-IQ can boost performance, it is evident that simply applying Group-DRO is not a silver bullet to equalize subgroup performance, given the sometimes small improvement. The rationale is that the tabular data setting is different from the spurious correlation setting in computer vision, where Group-DRO typically shines. Ultimately, we believe, based on the feature acquisition results, that in tabular settings, practitioners would be better served by acquiring more informative features to improve performance and reduce ambiguity.

**Subgroup-informed usage of uncertainty estimation.** Uncertainty estimation methods are essential in safety-critical areas such as healthcare [207], yet are typically assessed on average. We ask the question: Since subgroups have different performance properties, are uncertainty estimates equally reliable for each subgroup? As is common in the literature [129, 258], if an uncertainty estimate is reliable and informative of predictive performance,

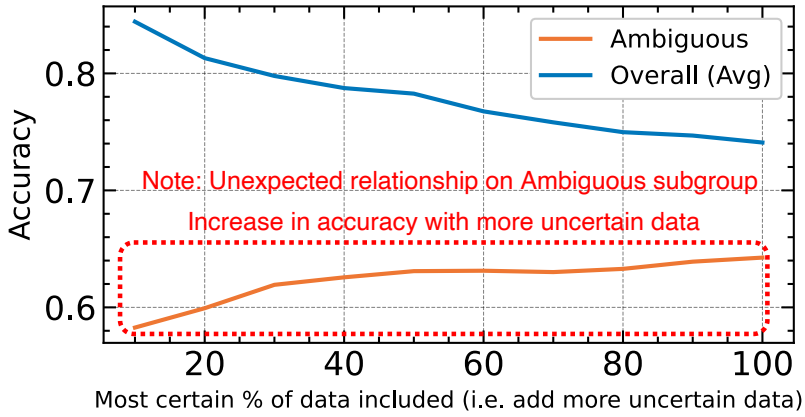



Fig. 2.9 Data-IQ subgroups can unmask unreliable prediction deferral by uncertainty methods

it can be used to defer "uncertain examples". This is done by rank sorting examples based on uncertainty and thresholding proportions of tolerated uncertainty [129, 258, 121, 153]. Ideally, as the threshold proportion of examples increases (i.e. more uncertain samples are included), we should see a monotonic decrease in accuracy. We assess this by training a Bayesian Neural Network (BNN)[63] to obtain uncertainty estimates. We then compute the performance across different threshold proportions $\tau \in \{0.1, 0.2, \ldots, 1\}$ for the *Ambiguous* subgroup specifically, and as is commonly done, across the entire dataset (i.e. *average*).

Figure 2.9 shows a *specific* example, wherein *average* examples exhibit the monotonic decreasing relationship, as expected. However, the *Ambiguous* examples categorized by Data-IQ, contrary to expectations, show an increase in accuracy as more uncertain examples are included. This suggests that uncertainty estimates in this case are not as informative of predictive performance for the *Ambiguous* examples. This result illustrates the potential for practitioners to use Data-IQ at deployment time to understand which examples require auditing before deferring, as the "average" monotonic decreasing behavior, based on the uncertainty estimates, may not always hold. Ultimately, the result further highlights how subgroup characterization via Data-IQ could assist practitioners in unmasking unreliable performance not evident on average.

### 2.4.4 Data-IQ beyond tabular settings.

We have primarily assessed the utility of Data-IQ in the tabular setting. That said, we now qualitatively evaluate the utility of Data-IQ on text data (*NLP*) and images (*computer vision*), respectively.

**Data-IQ on text data (NLP)**

We first illustrate the potential utility of Data-IQ on text data for NLP. Specifically, we examine a sentiment classification task (IMBb 50k). We train a bidirectional Recurrent Neural Network (RNN) on the IMBDb 50K dataset, where the task is to predict the sentiment based of the reviews. We then use Data-IQ to analyze the training dynamics and categorize the training data into subgroups - *Easy*, *Ambiguous* and *Hard*

We find that the *Easy*, *Ambiguous* and *Hard* groups represent different phenomena in the text data. We summarize these below.

**EASY: Average review length - 163 words**

- Easy to predict reviews. e.g. very negative sentiment with a negative label.

**NEGATIVE (0)**: This movie is pathetic in every way possible. Bad acting, horrible script (was there one?), terrible editing, lousy cinematography, cheap humor. Just plain horrible. I had seen 'The Wishmaster' a couple weeks before this movie and I thought it was a dead-ringer for worst movie of the year. Then, I saw 'The Pest' and suddenly 'The Wishmaster' didn't seem so bad at all. Bad Bad Bad. Excruciatingly bad.

**POSITIVE (1)** When I saw this movie for the first time I was both surprised and a little shocked by the blatant vibrance of the story. It is a very artistic drama with incredible special effects, spectacular acting, not to mention a very excellent job in the makeup department. Jennifer Lopez has pulled herself out of past roles that dug into her career with this movie, portraying a very sensitive child psychologist who works with a team of engineers to enter the minds of comatose patients to treat them. Vincent D'onofrio played amazingly well. His portrayal of a sadist serial killer was perfect to a T. The sheer emotion conveyed by his performance is astounding. Vince Vaughn isn't my favorite, but still performed exceptionally well. The symbolism and artistry was intriguing and titillating, sometimes surprising, and other times shocking. Overall, I say this is a wonderful movie, with excellent acting and beautiful artwork

**NEGATIVE (0)**: I've tried to watch this so-called comedy, but it's very hard to bear. This is a bad, narrow-minded, cliché-ridden movie. Definitively not funny, but very much boring and annoying, indeed. Bad script, bad acting. It's a complete waste of time - and there remains nothing more to say, I'm afraid 1 out of 10 points

**AMBIGUOUS: Average review length - 283 words**

- Linguistic ambiguity. A mix of positive and negative comments with a review. Other Ambiguous reviews even have text that simply describes the story without any positive/negative sentiment.

- Ambiguous reviews are much longer (in words).

**NEGATIVE (0)** I saw this movie on the film festival of Rotterdam (jan '06) and followed the discussion between director and public afterwards. Many people reacted shocked and protesting. He will get a lot of negative critics. But: the world is cruel like this, and it's not funny. People don't like it. That itself doesn't mean that the movie is bad. I can see that difference. Don't shoot the messenger that shows us the world outside our 'hubble'! Nevertheless I think this a bad movie. Film-technically it's a good one. Nice shots and script, most good fitting music, great actors. The director pretends to make a psychological movie, - the psychology however is of poor quality. Describing such a powerful violence itself is not the art. The art would be a powerful description of the psychological process behind that violence. How does a shy boy come to such a cruelty? The director pretends to describe that, - but is not good in that. The director used several times the word the 'selfishness' of people, mentioning for instance the teacher. Only: this teacher wasn't selfish,- just someone in several roles, caring for his pupils, ánd worried about his script. I think it's a simplification to call him selfish. The atmosphere in the village is creepy, and the mother made awful mistakes ('you terribly let me down') but it doesn't become believable for me, that there is caused súch a lot of pain, that the shyest boy comes to such terrible things. In fact, reality is far more complex than the way, this film describes and it needs far better descriptions. The interesting thing would be: how does it work? Describe that process for me please, so that we understand.With the written phrase on the end, the director said to point to an alternative way of life. It was the other extreme, and confirmed for me that director and scriptwriter are bad psychologists, promoting black/white-thinking. The connection between violence in films and in society has been proved. Use such a violence gives the responsibility to use it right. There are enough black/white-thinkers in the world, causing lots of war and misery. I hope, this movie won't be successful.

**NEGATIVE (0)**
When the film started I got the feeling this was going to be something special. The acting and camera work were undoubtedly good. I also liked the characters and could have grown to empathise with them. The film had a good atmosphere and there was a hint of fantasy. However, as the film went on, the plot never appeared to takeoff and just rolled on scene by scene. I was unable to understand the connection between the stories. All I could see was the characters occasionally bumping into each other and references to ships in bottles. Without that connection, I was just left with a few unremarkable short stories.

**POSITIVE (1)** Artistically speaking, this is a beautiful movie–the cinematography, music and costumes are gorgeous. In fact, this movie is prettier than those directed by Akira Kurasawa himself. In this case, he only wrote the movie as it was made several years after his death.So, as far as the writing goes, the dialog was well-written and the story, at times, was interesting. However, the story was also rather depressing yet uninvolving in some ways–after all, it's the story of a group of women who work in a brothel. It's interesting that although prostitution has been seen as a much more acceptable business in Japan, the women STILL long for a better life. This reminds me a lot of the movie Streets Of Shame, though Streets Of Shame's characters are a lot less likable and more one-dimensional.So, overall it gets a 7–mostly due to everything BUT the writing. It's too bad that the weakest link in this movie is the story by the great Kurasawa.

**HARD: Average review length - 124 words**

- Hard-to-predict reviews. e.g. very positive sentiment with a negative label. This might reflect mislabeling.

**NEGATIVE (0)** this one of the best celebrity's reality shows a ever saw. we can see the concerts we can see the life of Britney, i love the five episodes. i was always being surprised by Britney and the subjects of the show i think that some people don't watch the show at all we can how a great person she his. she his really funny really gentle and she loves her fans and we can see how she loves her work. i just don't give a 10 because of k-fed he his a real jerk he doesn't seem to like Britney at all. I they make a second season of this great show because it shows at some people how Britney really is. Go Britney your the best and you will never leave our hearts.

**POSITIVE (1)** Okay, first of all I got this movie as a Christmas present so it was FREE! FIRST - This movie was meant to be in stereoscopic 3D. It is for the most part, but whenever the main character is in her car the movie falls flat to 2D! What!!?!?! It's not that hard to film in a car!!! SECOND - The story isn't very good. There are a lot of things wrong with it. THIRD - Why are they showing all of the deaths in the beginning of the film! It made the movie suck whenever some was going to get killed!!! Watch it for a good laugh , but don't waste your time buying it. Just download it or something for cheap

**NEGATIVE (0)** this is a great movie. I love the series on tv and so I loved the movie. One of the best things in the movie is that Helga finally admits her deepest darkest secret to Arnold!!! that was great. i loved it it was pretty funny too. It's a great movie! Doy!

**Data-IQ on images (computer vision)**

We now illustrate the potential utility of Data-IQ on image data for computer vision tasks. Specifically, we train a convolutional neural network (CNN) on the CelebA dataset, where the task is to predict the celebrity's gender based on the image of the celebrity. We then use Data-IQ to analyze the training dynamics and categorize the data into subgroups - *Easy*, *Ambiguous* and *Hard*. We find that the *Easy*, *Ambiguous* and *Hard* groups represent different phenomena in the image data. We summarize the findings in Figure 2.10

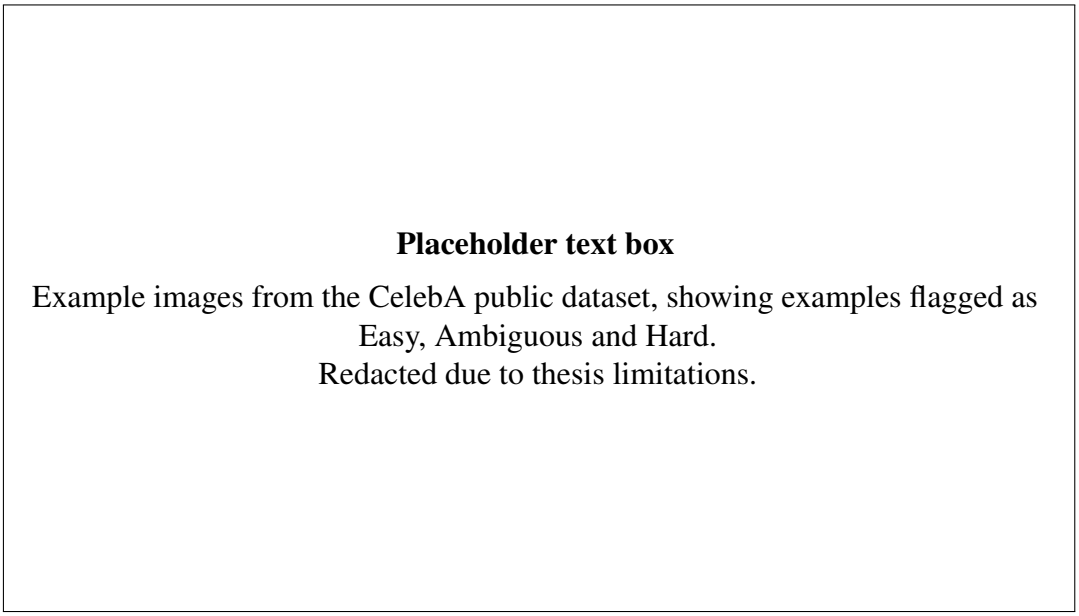



Fig. 2.10 Subgroups highlighted by Data-IQ on the CelebA dataset

## 2.5 Discussion

In this chapter, we introduce Data-IQ, a systematic framework that can be used with *any* ML model with checkpoints to characterize examples into subgroups with respect to the outcome. Through several experiments, we demonstrate that the usage of aleatoric uncertainty, which captures properties more inherent to the data, is indeed more principled, being more robust to variation across models and/or parameterizations. Data-IQ's consistency is unmatched by any compared baseline. Data-IQ should not automate and replace the intuition of a data scientist. Rather, as we have demonstrated, Data-IQ should serve as a systematic "data-centric ML" tool that assists and empowers data scientists with the "data" work at training time, whilst also guiding reliable model usage at deployment time.

**Limitations and future opportunities.** ① While Data-IQ characterizes examples, the current formulation does not allow us to understand which attributes are responsible for the characterization per example. An interesting extension is around dataset explainability, allowing practitioners to better probe their data. ② In high-stakes settings such as healthcare, to mitigate possible adverse effects (e.g. difficulty of *Easy* vs *Hard*), Data-IQ should be used with a "human-in-the-loop", allowing experts to complement and validate findings with domain knowledge. In particular, this is an important step before sculpting/pruning a dataset based on characterization by Data-IQ. ③ A related and important risk is that Hard or Ambiguous samples may not be random, but instead systematically concentrated in minority or underrepresented subgroups. In medical data, for example, rare disease presentations, atypical demographic profiles, or genetically distinct subpopulations may exhibit genuinely heterogeneous outcomes that are correctly labelled but locally unlearnable given the current dataset size — causing Data-IQ to classify them as Hard. Pruning these samples based on Data-IQ characterization alone could therefore inadvertently reduce the representation of already marginalised groups, compounding existing disparities rather than correcting them. To guard against this, we strongly recommend that practitioners cross-reference Data-IQ subgroup assignments with any available sensitive attribute information before making curation decisions. Where Hard examples cluster within a protected group, the appropriate response may be to collect more data from that group rather than to remove the existing examples. This is consistent with Data-IQ's broader use case as a tool for principled feature acquisition and targeted data collection, rather than purely for pruning.

# Chapter 3

# Test Data Incongruity Identification

## 3.1 Introduction

Machine Learning (ML) models are well-known to rely on the quality of training data [170]. Hence, when ML models are deployed in the real world, the reliability of predictions depends on the test/deployment data's congruence with respect to the training data. Significant literature has focused on identifying deployment time data instances that lie outside the training data's distribution (i.e. out of distribution). This includes label shifts [192, 90] or input feature shifts where these instances fall out of the support of the training set's distribution [266]. However, a much less studied yet equally important problem is identifying heterogeneous regions of in-distribution (ID) data.

Data in the wild can be ID yet have heterogeneous regions in feature space. This manifests in varying levels of *incongruence*, in cases of different sub-populations, data biases or temporal changes [132, 65, 159]. We illustrate each of these types of incongruence with real-world data (Table 3.2).

To address this issue, in this chapter, we present a *data-centric* framework to characterize at deployment time, such *incongruous* regions of *ID data*. We define two groups, namely, (i) *inconsistent* and (ii) *uncertain*, with respect to the training distribution. We contextualize the difference based on prediction intervals (PI) (See Chapter 3.3.3). When feature values lie outside of a PI, we term it inconsistent; alternatively, we characterize the level of feature uncertainty based on the PI's width.

At this point, one might ask, if the data is ID, why should we worry? Not accounting for these incongruous ID regions of the feature space can be problematic when deploying models in high-stakes settings such as healthcare, where spurious model predictions can be deadly [205, 246]. That said, even in settings where poor predictions are not risky, consistent exploratory data analysis (EDA) and retroactive auditing of such data is time-consuming for

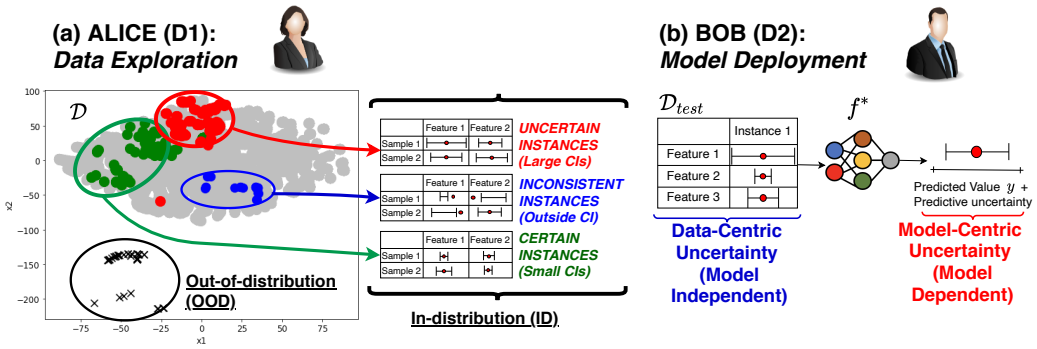



Fig. 3.1 Illustration highlighting two problems Data-SUITE addresses

data scientists [179, 105]. Hence, systematically identifying these incongruous regions has immense practical value.

Consequently, we build a framework to empower data scientists to address the aforementioned challenges related to insightful exploratory data analysis (EDA) and reliable model deployment, anchored by the following desiderata:

**(D1) Insightful Data Exploration**: Alice has a new dataset $\mathscr{D}$ and wants to explore and gain insights into it with respect to a training set without necessarily training a model. It would be useful if, *independent of a predictive model*, she could both identify the incongruous regions of the feature space (e.g., sub-population bias or under-representation), as well as obtain easily digestible prototype examples of each region. This could provide guidance on where to collect more data and, if this is not possible, to understand the data's limitations.

**(D2) Reliable Model Deployment**: Bob has a trained model $f^*$ and now deploys it to another site/location. For new data $\mathscr{D}_{\text{test}}$, it would be useful if he could identify incongruous regions, for which he should NOT trust $f^*$ to make reliable and trustworthy predictions.

**(D3) Practitioner confidence**: Both Alice and Bob want to feel confident when using any tool. Guarantees of coverage of predictive intervals (e.g. PIs) could assist in this regard.

These examples, shown in Figure 3.1, highlight the need to understand incongruence in data. As we shall discuss in the related work, there has been significant work on uncertainty estimation, with a focus on the uncertainty of a model's predictions (*model-centric*). Estimating predictive uncertainty can address Bob's use case (*D2*). However, since it requires a predictive model, it is not naturally suited to Alice's insights use-case (*D1*). Further, most predictive uncertainty methods do not provide guarantees on coverage (*D3*).

Therefore, in satisfying all the desiderata, we take a different approach and advocate for modeling the uncertainty in the data features [1]. This is different from model predictive uncertainty, as we construct PIs (at feature level) without reference to any downstream model. A benefit of the flexibility is that we can flag instances and draw insights that are not model-specific (i.e. *model independent*). We focus on tabular data, a common format in medicine, finance, manufacturing etc, where data is based on relational databases [23, 263]. That said, compared to image data, tabular data has an added challenge since specific features may be uncertain while others are not; hence, characterizing an instance as a whole is non-trivial.

In addressing this, we introduce **Data S**earching for **U**ncertain and **I**nconsistent **T**est **E**xamples (**Data-SUITE**), a *data-centric* framework to identify incongruous regions of data using PI's and make the following contributions:

**Contributions:** ① Data-SUITE is a paradigm shift from model-centric uncertainty and, to the best of our knowledge, the first to characterize ID regions in a systematic *data-centric, model-independent* manner. Not only is this more flexible, but it also enables us to gain insights that are not model-specific. ② Data-SUITE's pipeline-based approach to constructing feature-wise PIs enables specific properties that permit us to flag uncertain and inconsistent instances, making it possible to identify incongruous data regions. ③ Data-SUITE's performance and properties, such as coverage guarantees, are validated to satisfy *D3*. ④ Further motivating the paradigm shift, we empirically highlight the performance benefit of a data-centric approach compared to a model-centric approach. ⑤ As a portrayal of reliable model deployment (*D2*), we show on real-world datasets with different types of incongruence that Data-SUITE best identifies incongruous data regions, translating to the best performance improvement. ⑥ Finally, we illustrate with multiple use-cases how Data-SUITE can be used as a model-independent tool to facilitate insightful data exploration, hence satisfying *D1*.

## 3.2 Related work

The work in this chapter primarily engages with the literature on uncertainty quantification and contributes to the nascent area of data-centric ML. We also highlight key differences of this work from the literature on noisy labels.

**Uncertainty quantification.** There are numerous Bayesian and non-Bayesian methods proposed for uncertainty quantification, including Gaussian processes [257], Quantile Regres-

[1] Here, "feature" uncertainty refers to the degree of incongruity with the training distribution, rather than the uncertainty of the measured value (e.g. measurement noise)

sion [116], Bayesian Neural Networks [63, 71], Deep Ensembles [125], MC Dropout [57] and Conformal Prediction [251, 14]. These methods typically model predictive uncertainty, i.e., measuring the certainty in a specific model's prediction. The predominant focus on predictive uncertainty is different from the notion of uncertainty in our setting, which is feature-level or data uncertainty. We specifically highlight that we quantify data uncertainty independent of a task-specific model. Additionally, the aforementioned methods often do not assess the coverage or provide guarantees of the uncertainty interval [254] (i.e., how often the interval contains the true value). The concept of coverage will be outlined further in Chapters 3.3 and 3.4.

**Data-Centric ML.** Ensuring high data quality is a critical but often overlooked problem in ML, where the focus is on optimizing models [203, 95]. Even when it is considered, the process of assessing datasets is ad hoc or artisanal [203, 156]. However, there has been recent discussion around data-centric ML, which involves tools applied to the underlying data used to train and evaluate models, independent of the task-specific, predictive models. Our work contributes to this nascent body of work - presenting Data-SUITE, which, to the best of our knowledge, is the first systematic data-centric framework to model uncertainty in datasets. Specifically, we model the uncertainty in the feature (data) values themselves (*data-centric*), which contrasts with modeling the uncertainty in predictions (*model-centric*).

**Noisy labels.** Learning with noisy data is a widely studied problem, and we refer the reader to [4, 231] for an in-depth review. Specifically, in the machine learning context, there is a focus on label noise. We argue that work on noisy labels is not directly related, as the typical goal is to learn a model that is robust to the label noise, which is different from our goal of modeling the uncertainty in the features. Additionally, the success of the literature on noisy labels is tightly coupled to the task-specific predictive model, which is different from our model-independent setting.

## 3.3 Data-SUITE

We now give a detailed formulation of Data-SUITE. We present a block diagram of our framework, Data-SUITE, in Figure 3.2.

We start with a problem formulation and outline the motivation for working with feature prediction intervals (PIs). Then, we describe how these PIs are built by leveraging copula modeling, representation learning, and conformal prediction. Finally, we demonstrate how these PIs permit us to flag uncertain and inconsistent instances.

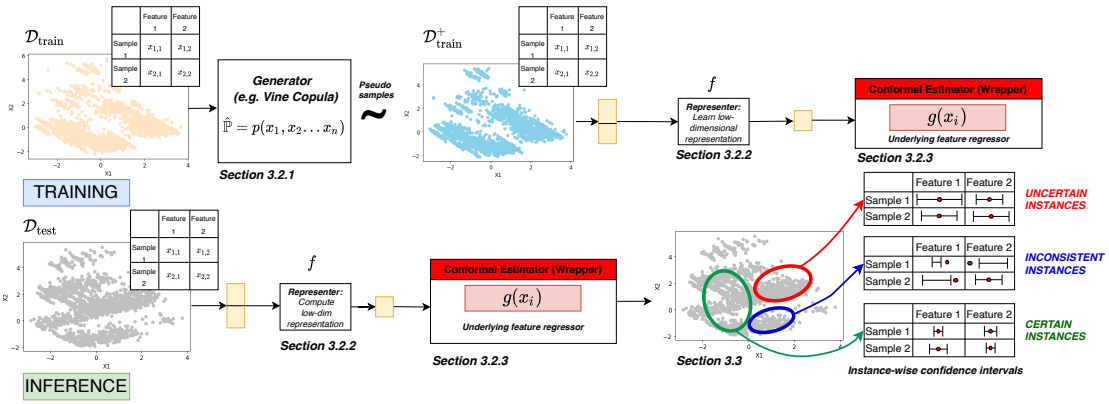



Fig. 3.2 Outline of our Data-SUITE framework

### 3.3.1 Preliminaries

We consider a feature space $\mathscr{X} = \prod_{i=1}^{d_X} [a_i, b_i] \subset \mathbb{R}^{d_X}$, where $[a_i, b_i]$ is the range for feature $i$. Note that we make the range of each feature explicit; this will be necessary in defining the formalism. We assume that we have a set of $M \in \mathbb{N}^*$ *training instances* $\mathscr{D}_{\text{train}} = \left\{ x^{(m)} \mid m \in [M] \right\}$ sampled from an unknown distribution $\mathscr{P}$, where $[M]$ denotes the positive integers between 1 and $M$. These instances typically correspond to training data for a model on a downstream task such as classification.

We assume that we are given new test instances $\mathscr{D}_{\text{test}}$. Our goal is to flag the subset of instances from $\mathscr{D}_{\text{test}}$ that are quantitatively different from instances of $\mathscr{D}_{\text{train}}$ without necessarily being OOD. To do this, we use $\mathscr{D}_{\text{train}}$ to build PIs $[l_i(x), r_i(x)] \subseteq [a_i, b_i]$ for each feature $i \in [d_X]$ of each test instance $x \in \mathscr{D}_{\text{test}}$. As we show in Chapter 3.3.3, these PIs permit us to systematically flag test instances whose features are uncertain or inconsistent with respect to $\mathscr{D}_{\text{train}}$. For now, let us motivate the usage of feature PIs: (1) With a model of uncertainty and inconsistency at the feature level, it is possible to identify regions of the feature space $\mathscr{X}$ where bias and/or low coverage occurs with the training data $\mathscr{D}_{\text{train}}$. (2) Since PIs are built with $\mathscr{D}_{\text{train}}$ and without reference to any predictive downstream model, the flagged instances in $\mathscr{D}_{\text{test}}$ are likely to be problematic for any downstream model trained on top of $\mathscr{D}_{\text{train}}$. Hence, we are able to draw conclusions that are not model-specific. These two points are illustrated in our experiments from Chapter 3.4. Let us now detail how the PIs are built.

### 3.3.2 Feature PIs

We now build PIs $[l_i(x), r_i(x)] \subseteq [a_i, b_i]$ for each feature $i \in [d_X]$ of each test instance $x \in \mathscr{D}_{\text{test}}$. It goes without saying that the PIs should satisfy some properties, i.e.

*(P1) Coverage:* We would like to guarantee that for a fresh random draw $X \sim \mathscr{P}$ independent of the calibration set, the feature $X_i$ lies within the interval such that $\mathbb{P}(X_i \in [l_i(X), r_i(X)]) \geq 1 - \alpha$, where the significance level $\alpha \in (0,1)$ can be chosen. This is then considered across all features to characterize the instance (see Chapter 3.3.3).
*(P2) Instance-wise:* The PI should be *adaptive at an instance level*. i.e, we do not wish $r_i(x) - l_i(x)$ to be constant w.r.t $x \in \mathscr{X}$. In this way, the PIs allow the ordering of various test instances $x \in \mathscr{D}_{\text{test}}$ according to their uncertainty. This property is particularly desirable in healthcare settings where we wish to quantify the variable uncertainty for individual patients, rather than for a population as a whole.
*(P3) Feature-wise:* We build PIs $[l_i(x), r_i(x)]$ for each feature $i \in [d_X]$ as opposed to an overall confidence region $R(x) \subset \mathscr{X}$. Although less general than the latter approach, feature-wise PIs are more interpretable, allowing the attribution of inconsistencies and uncertainty to individual features.
*(P4) Downstream coupling:* Instances with smaller PIs are more reliably predicted by a downstream model trained on $\mathscr{D}_{\text{train}}$. More precisely, our PIs should have a negative correlation between PI width and downstream model performance. In this way, PIs allow us to draw conclusions about the incongruence of test instances $x \in \mathscr{D}_{\text{test}}$.

To construct feature PIs that satisfy these properties, we introduce a new framework leveraging copula modeling, representation learning and conformal prediction. The blueprint of our method is presented in Figure 3.3. Concretely, our method relies on 3 building blocks: a *generator* that augments the initial training set $\mathscr{D}_{\text{train}}$; a *representer* that leverages the augmented training set $\mathscr{D}^{+}_{\text{train}}$ to learn a low-dimensional representation $f : \mathscr{X} \to \mathscr{H}$ of the data and a *conformal predictor* that predicts instance-wise feature PIs $[l_i(x), r_i(x)]$ on the basis of each instance's representation $f(x) \in \mathscr{H}$. By construction, this method fulfills properties (P2) and (P3). As we will see in the following, the conformal predictor guarantees (P1). We demonstrate (P4) empirically in Chapter 3.4. Chapter 3.4.4 quantifies the significance of each block via an ablation study. Let us now detail each block.

**Generator.** The purpose of the generator is to augment the initial training set $\mathscr{D}_{\text{train}}$ with instances that are consistent with the initial distribution $\mathscr{P}$. Many data augmentation techniques can be used for this block. Since our focus is on tabular data, we found *copula modeling* to be particularly useful. Copulas leverage Sklar's theorem [229] to estimate

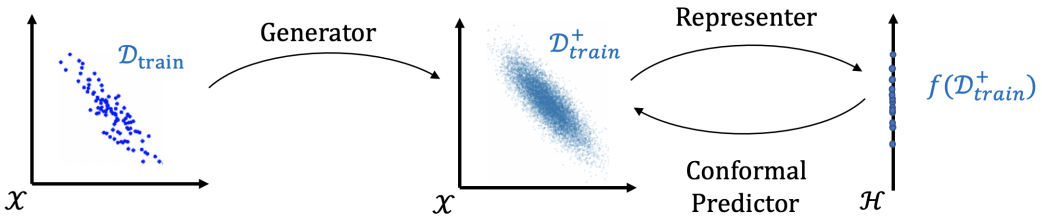



Fig. 3.3 Data-SUITE blueprint: generator, representer and conformal predictor.

multivariate distributions with univariate marginal distributions. In our case, we use vine copulas [17] to build an estimate $\hat{\mathscr{P}}$ for the distribution $\mathscr{P}$ on the basis of $\mathscr{D}_{\text{train}}$. We then build an augmented training set $\mathscr{D}^{+}_{\text{train}}$ by sampling from the copula density $\hat{\mathscr{P}}$. Interestingly, our method does not need to access $\mathscr{D}_{\text{train}}$ once the copula density $\hat{\mathscr{P}}$ is available. It is perfectly possible to use only instances from $\hat{\mathscr{P}}$ to build the augmented dataset $\mathscr{D}^{+}_{\text{train}}$. This could be useful for data sharing if access to the training set $\mathscr{D}_{\text{train}}$ is restricted to the user. Further details and motivations on copulas are found in Appendix B.1.2. Note that a copula may not be ideal for very high-dimensional (large $d_X$) data in domains such as computer vision or genomics. In those cases, copula modeling can be replaced by domain-specific augmentation techniques.

**Representer.** A trivial way to verify the coverage guarantee (P1) would be to use the true values of the features to build the PIs: $[l_i(x), r_i(x)] = [x_i - \delta, x_i + \delta]$ for some $\delta \in \mathbb{R}^{+}$. The problem with this approach is two-fold: (1) it does not leverage the distribution $\mathscr{P}$ underlying the training set $\mathscr{D}_{\text{train}}$, and (2) it results in an uninformative reconstruction with PIs that do not capture the specificity of each instance, hence contradicting (P2). To provide a more satisfactory solution, we propose to represent the augmented training data $\mathscr{D}^{+}_{\text{train}}$ with a representation function $f : \mathscr{X} \to \mathscr{H}$ that maps the data into a lower-dimensional latent representation space $\mathscr{H} \subseteq \mathbb{R}^{d_H}$, $d_H < d_X$. The purpose of this representer is to capture the structure of the low-dimensional manifold underlying $\mathscr{D}^{+}_{\text{train}}$. At test time, the conformal predictor (detailed next) uses the lower representations $f(x) \in \mathscr{H}$ to estimate a reconstruction interval for each feature $x_i$. This permits a satisfactory solution to the two aforementioned problems: (1) the PIs are reconstructed in terms of latent factors that are useful to describe the training set $\mathscr{D}_{\text{train}}$ and (2) the predicted PIs vary according to the representation $f(x) \in \mathscr{H}$ of each test instance $x \in \mathscr{D}_{\text{test}}$. In essence, our approach is analogous to autoencoders. As we will explain soon, the crucial difference is the decoding step: our method outputs PIs for the reconstructed input. In this work, we use Principal Component Analysis (PCA),

the workhorse for tabular data, to learn the representer $f$. Note that more general encoder architectures can be used in relevant settings, such as computer vision.

**Conformal Predictor.** We now turn to the core of the problem: estimating feature-wise PIs. As previously mentioned, the PIs $[l_i(x), r_i(x)], i \in [d_X]$ will be *computed on the basis of the latent representation* $f(x)$ for each $x \in \mathscr{D}_{\text{test}}$. The idea is simple: for each feature $i \in [d_X]$, we train a regressor $g_i : \mathscr{H} \to [a_i, b_i]$ to reconstruct an estimate of the initial features $x_i$ from the latent representation $f(x)$ of the associated training instance: $(g_i \circ f)(x) \approx x_i$. We stress that the regressor $g_i$ does not observe the true observed $x_i$ but only of the latent representation $f(x)$, as illustrated in Figure 3.4. Of course, the feature regressor by itself provides point-wise estimates for the features. In order to turn these into PIs, we use conformal prediction as a wrapper around the feature regressor [251].

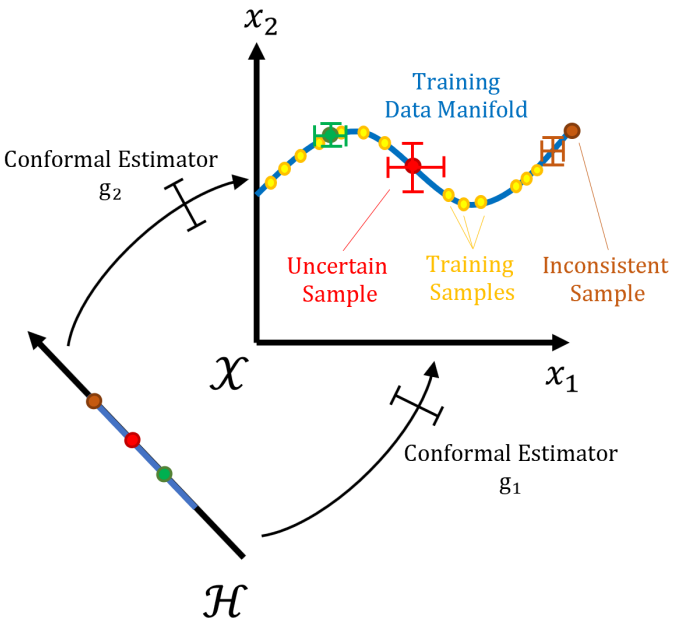



Fig. 3.4 Conformal Predictor in **Data-SUITE**.

Specifically, we formalize our problem in the framework of Inductive Conformal Prediction (for motivations, see Appendix B.1.2). Hence, under the formulation, we start by splitting the augmented training set into a proper training set and a calibration set: $\mathscr{D}^+_{\text{train}} = \mathscr{D}^+_{\text{train2}} \sqcup \mathscr{D}^+_{\text{cal}}$ where $\mathscr{D}^+_{\text{train2}}$ is the *proper training set*, for which we use the latent representation of the proper training set, to train the feature regressor $g_i, i \in [d_X]$ for the reconstruction task. Then, the latent representation of the calibration set is used to compute the non-conformity score ($\mu$), which estimates how different a new instance looks from other instances. In practice, we use the absolute error non-conformity score $\mu_i(x) = |x_i - (g_i \circ f)(x)|$.

We obtain an empirical distribution of non-conformity scores $\{\mu_i(x) \mid x \in \mathscr{D}^+_{\text{cal}}\}$ over the calibration instances for each feature $i \in [d_X]$. This is used to obtain the critical non-conformity score $\varepsilon_{\mu,i}$, which corresponds to the $\lceil(|\mathscr{D}^+_{\text{cal}}|+1)(1-\alpha)\rceil$-th smallest residual from the set $\{\mu_i(x) \mid x \in \mathscr{D}^+_{\text{cal}}\}$ [249]. We then apply the method to any unseen incoming data to obtain PIs for the data point i.e. $[l_i(x), r_i(x)] = [(g_i \circ f)(x) - \varepsilon_{\mu,i}, (g_i \circ f)(x) + \varepsilon_{\mu,i}]$.

However, in this form, the PIs are constant for all instances, where the width of the interval is determined by the residuals of the most difficult instances (i.e. the largest residuals).

We adapt our conformal prediction framework to obtain the desired adaptive intervals (*P2*) using a normalized non-conformity function ($\gamma$), see Equation 3.1 [25, 103]. The numerator is computed as before based on $\mu$, however, the denominator normalizes per instance. We learn the normalizer per feature $i \in [d_X]$.

To do so, we compute the log residuals per feature for all instances in the respective proper training set $\mathscr{D}_{\text{train2}}$. We produce tuples per feature: $\{(f(x), \ln|x_i - (g_i \circ f)(x)|) \mid x \in \mathscr{D}^+_{\text{train2}}\}$.

These are used to train a *different* model, $\sigma_i : \mathscr{H} \to \mathbb{R}^+$ (e.g., MLP), to predict the log residuals. We can then apply $\sigma_i$ to *test instances* to capture the difficulty in predicting said instance. Note that we apply an exponential to the predicted log residual for the test instance, converting to the true scale and ensuring positive estimates.

$$\gamma_i(x) \equiv \frac{|x_i - (g_i \circ f)(x)|}{\sigma_i(f(x))}, \tag{3.1}$$

We can then obtain the critical non-conformity score $\varepsilon_{\gamma,i}$ applied to the empirical distribution of normalized non-conformity scores $\{\gamma_i(x) \mid x \in \mathscr{D}^+_{\text{cal}}\}$, in the same way as before based on residuals. The instance-specific adaptive intervals are then obtained as per Equation 3.2, where $g_i$ is the underlying feature regressor and $\sigma_i$ is the instance-wise normalizing function.

$$[l_i(x), r_i(x)] = [(g_i \circ f)(x) - \varepsilon_{\gamma,i}\sigma_i(f(x)), (g_i \circ f)(x) + \varepsilon_{\gamma,i}\sigma_i(f(x))] \tag{3.2}$$

**Remarks on theoretical guarantees.** Under the exchangeability assumption detailed in the Appendix B.1.2, the validity of coverage guarantees (P1) is fulfilled with our definition. In our implementation, we use $\alpha = .05$.

### 3.3.3 Identifying Inconsistent and Uncertain Instances

Now that we have PIs $[l_i(x), r_i(x)] \subset [a_i, b_i]$ for each feature $x_i, i \in [d_X]$ of the instance $x$, we can evaluate whether instances from a dataset fall within the predicted range. If it falls outside the predicted range, we characterize the inconsistency (see Definition 3.3.1)

**Definition 3.3.1** (Inconsistency)**.** Let $x \in \mathscr{D}_{\text{test}}$ be a test instance for which we construct a $(1-\alpha)$-PI, $[l_i(x), r_i(x)]$, for each feature $x_i, i \in [d_X]$ for some predetermined $\alpha \in (0,1)$. For each $x_i, i \in [d_X]$, the *feature inconsistency* is a binary indicator if $x_i$ falls out of the PI.

$$\nu_i(x) \equiv 1(x_i \notin [l_i(x), r_i(x)]) \tag{3.3}$$

The *instance inconsistency* $\nu(x)$ is obtained by averaging over the feature inconsistencies $\nu_i(x)$.

$$\nu(x) \equiv \frac{1}{d_X} \sum_{i=1}^{d_X} \nu_i(x)$$

The instance $x$ is *inconsistent* if the fraction of inconsistent features is above a predetermined threshold[2] $\lambda \in [0,1]$: $\nu(x) > \lambda$.

There can also be degrees of uncertainty in the feature value for features that fall within the PI, which can reflect the instance as a whole. Indeed, if the PI $[l_i(x), r_i(x)]$ is large, the feature $x_i$ is likely to fall within its range. Nonetheless, we should keep in mind that large PIs correspond to a large uncertainty for the related feature. This will also typically happen when the instance $x \in \mathscr{D}_{\text{test}}$ differs from the training set $\mathscr{D}_{\text{train}}$ used to build the PI. We now introduce a quantitative measure that expresses the degree of uncertainty of the instance with respect to $\mathscr{D}_{\text{train}}$ (see Definition 3.3.2).

**Definition 3.3.2** (Uncertainty)**.** Let $x \in \mathscr{D}_{\text{test}}$ be a test instance for which we construct a $(1-\alpha)$ PI $[l_i(x), r_i(x)]$ for each feature $x_i, i \in [d_X]$ for some predetermined $\alpha \in (0,1)$. For each $x_i, i \in [d_X]$, we define the *feature uncertainty* $\Delta_i(x)$ as the feature PI width normalized by the feature range:

$$\Delta_i(x) \equiv \frac{r_i(x) - l_i(x)}{b_i - a_i} \tag{3.4}$$

[2] In our implementation, we use $\lambda = 0.5$.

The *instance uncertainty* $\Delta(x)$ is obtained by averaging over all feature uncertainties:

$$\Delta(x) \equiv \frac{1}{d_X} \sum_{i=1}^{d_X} \Delta_i(x) \quad \in (0, 1].$$

***Remark.*** Instance uncertainties are strictly larger than zero since feature uncertainties are computed over all features. Hence, this characterization offers a natural split between *certain* and *uncertain* instances if we sort the instances based on uncertainty magnitude.

## 3.4 Experiments

We now present a detailed empirical evaluation demonstrating that Data-SUITE satisfies (**D1**) Insightful Data Exploration,(**D2**) Reliable Model Deployment and (**D3**) Practitioner confidence, introduced in Chapter 3.1. We tackle these in reverse order, as practitioner confidence is a prerequisite for the adoption of **D1** and **D2**.

Recall that the notion of uncertainty in Data-SUITE is different from predictive uncertainty (model-centric). We empirically compare these two paradigms using existing methods for predictive uncertainty. That said, a natural additional question is whether model-centric uncertainty estimation methods can simply be applied to this setting and provide uncertainty estimates for feature values.

We benchmark the following widely used Bayesian and non-Bayesian methods (under BOTH the model-centric & data-centric paradigms): Bayesian Neural Networks (BNN) [63], Deep Ensembles (ENS) [125], Gaussian Processes (GP) [257], Monte-Carlo Dropout (MCD) [57] and Quantile Regression (QR) [116]. We also ablate and test Data-SUITE's constituent components independently: conditional sampling from copula (COP), Conformal Prediction on raw data (CONF) [251, 14]. For implementation details, see Appendix B.2.1.

### 3.4.1 Validating coverage & comparing properties

We first validate the PIs to ensure that the coverage guarantees are satisfied, such that users can have confidence that the true value lies within the predicted PIs (**D3**). We assess the PIs based on the following metrics defined in [155] - (1) Coverage: how often the PI contains the true value, (2) Deficit: the extent of PI shortfall (i.e., the severity of the errors) and (3) Excess: the extent of PI excess width to capture the true value.

$$\text{Coverage} = \mathbb{E}\left[\mathbf{1}\{X_i \in [l_i(X), r_i(X)]\}\right], \tag{3.5}$$

$$\text{Deficit} = \mathbb{E}\left[\mathbf{1}\{X_i \notin [l_i(X), r_i(X)]\} \cdot \min\{|X_i - l_i(X)|,\ |X_i - r_i(X)|\}\right], \tag{3.6}$$

$$\text{Excess} = \mathbb{E}\left[\mathbf{1}\{X_i \in [l_i(X), r_i(X)]\} \cdot \min\{X_i - l_i(X),\ r_i(X) - X_i\}\right]. \tag{3.7}$$

**Synthetic data.** We assess the properties of different methods using synthetic data - as the ground truth values are available, even when encoding incongruence. The synthetic data with features, $\mathbf{X} = [X_1, X_2, X_3]$, is drawn IID from a multivariate Gaussian distribution, parameterized by mean vector $\mu$ and a positive definite covariance matrix $\Sigma$ (details in Appendix B.2.3). We sample $M = 1000$ points for both $\mathscr{D}_{\text{train}}^{synth}$ and $\mathscr{D}_{\text{test}}^{synth}$ and encode incongruence in $\mathscr{D}_{\text{test}}^{synth}$ using a multivariate additive model $\hat{\mathbf{X}} = \mathbf{X} + \mathbf{Z}$, where $\mathbf{Z} \in \mathbb{R}^{M \times d_X}$, is the perturbation matrix (with $d_X = 3$).

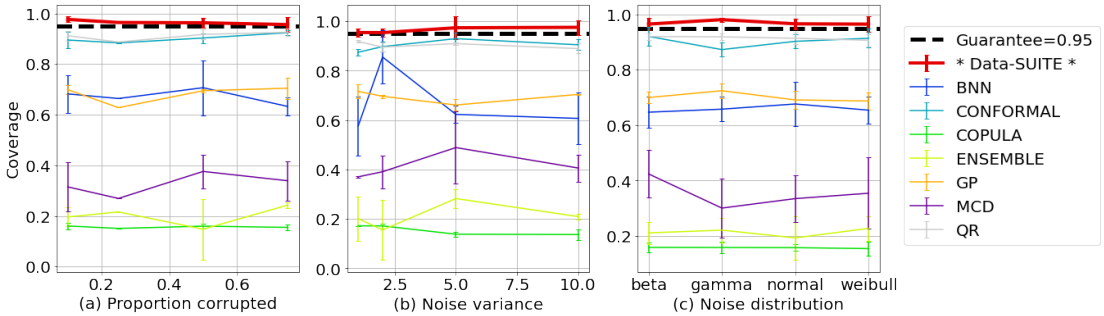
We conduct experiments with different configurations: (1) $D_a$: Multivariate Gaussian with variance 2 and varying proportion of perturbed instances. (2) $D_b$: Multivariate Gaussian with varying variance and fixed proportion of perturbed instances (50%) and (3) $D_c$: Varying distribution $\in \{Beta, Gamma, Normal, Weibull\}$.

Figure 3.5 outlines the mean coverage, deficit and excess averaged over five runs for $\mathscr{D}_{\text{test}}^{synth}$ under the different configurations ($D_a, D_b, D_c$). There is a clear variability amongst the different methods, suggesting that specific methods are more suitable. Data-SUITE outperforms the other methods based on coverage and deficit across all configurations. We note that the methods with poor coverage are typically "incorrectly" confident, i.e. small intervals with low coverage and high deficit.

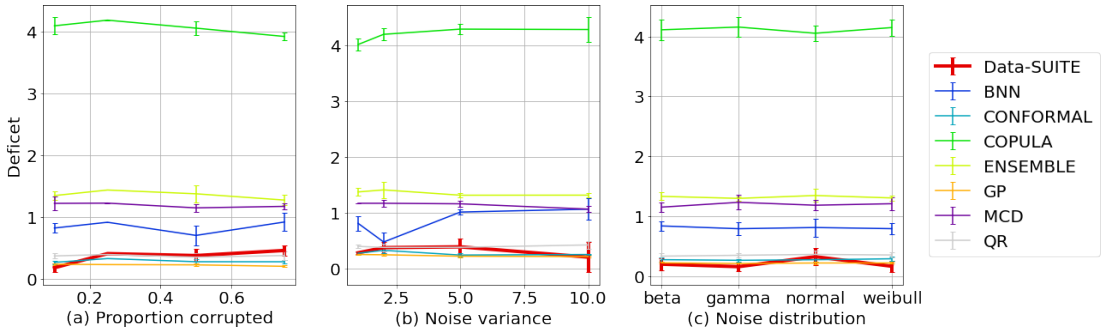
Figure 3.5 also demonstrates a meaningful relationship in that coverage and deficit are inversely related (high coverage is associated with a low deficit), as with deficit and excess. Although high coverage and low deficit ideally occur with low excess, we observe that high levels of coverage occur in conjunction with high levels of excess. Critically, however, in satisfying **D3**, Data-SUITE maintains the 95% coverage guarantees across all configurations, unlike other methods.

### 3.4.2 Synthetic data stratification w/ downstream task

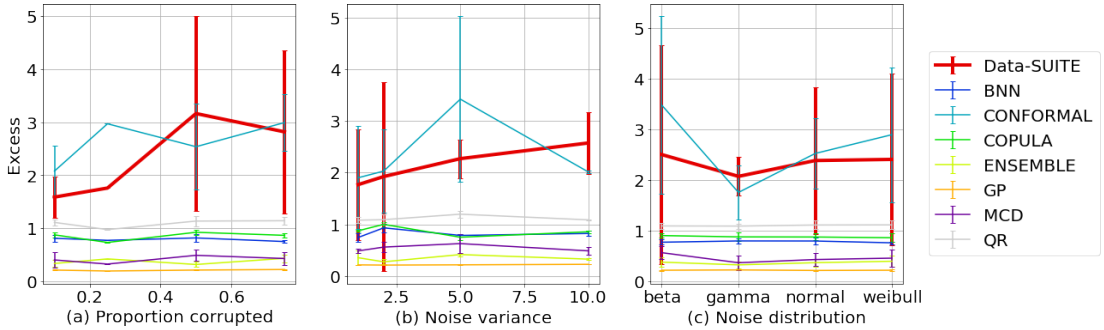
While it is essential to validate the properties of a method, the most useful goal is whether the intervals can be used to identify instances that a downstream predictive model will reliably predict. With Data-SUITE, we stress that this is done in a model-independent manner (i.e. no knowledge of the downstream model).




(a) i. Coverage & 95% guarantee (1-$\alpha$)




(b) ii.Deficit




(c) iii.Excess

Fig. 3.5 Comparison of methods based on coverage, deficit and excess under various configurations ($D_a, D_b, D_c$)

We train a downstream regression model using $\mathscr{D}_{\text{train}}^{synth}$, where features $X_1, X_2$ are used to predict $X_3$. We first compute a baseline mean squared error (MSE) on a held-out validation set of $\mathscr{D}_{\text{train}}^{synth}$ and the complete test set $\mathscr{D}_{\text{test}}^{synth}$ ($\hat{\mathbf{X}} = \mathbf{X} + \mathbf{Z}$). Thereafter, we construct predictive intervals for $\mathscr{D}_{\text{test}}^{synth}$ using all benchmark methods (either uncertainty intervals or PIs). The intervals are then used to sort instances based on width.

In addition, we answer the question of whether a *data-centric* or *model-centric* approach yields the best performance. For the data-centric paradigm, we construct intervals for the features $X_1, X_2$. As a result, instances are then categorized in a model-independent manner based on data-level PIs. In contrast, the model-centric paradigm is tightly coupled with a task-specific model, categorizing instances using predictive uncertainty based on prediction $X_3$.

We then compute MSE for the 100 most certain instances as ranked by each method (smallest widths). For Data-SUITE, we also compute the MSE for those instances identified as *inconsistent* (outside PIs). The best method is one in which the *certain* sorted instances produce MSE values closest to the clean train MSE (baseline), i.e., it has the lowest MSE.

Table 3.1 shows the MSE for configurations $D_a$ and $D_b$. As one example of satisfying **D2**, Data-SUITE has the best performance and identifies the top 100 *certain* instances (at deployment time) that yield the best downstream model performance, with the lowest MSE across all configurations.

In addition, as expected, the *inconsistent* instances are unreliably predicted. The poor performance for ablations of Data-SUITE components suggests the necessity of the interconnected framework (more in Chapter 3.4.4)

Additionally, we see for the same base methods (e.g. BNN, MCD etc), that the data-centric paradigm outperforms the model-centric paradigm in identifying the "best" instances to give the lowest MSE. This result highlights the performance advantage of a flexible, model-independent data-centric paradigm compared to the model-centric paradigm.

Table 3.1 MSE based on instance stratification for different methods. Data-SUITE outperforms other methods, whilst data-centric methods in general outperform model-centric methods.

| | | Proportion ($D_a$) | | | Variance ($D_b$) | |
|---|---|---|---|---|---|---|
| | PERTURBATION | .1 | .25 | .5 | 1 | 2 |
| | *Train Data (BASELINE)* | *.067* | *.059* | *.068* | *.065* | *.068* |
| | Test Data | .222 | .513 | .889 | .275 | .889 |
| **Data-centric** | **Data-SUITE (All, Uncertainty)** | **.069** | **.122** | **.197** | **.104** | **.197** |
| | Data-SUITE (All, Inconsistent) | .595 | 1.608 | 2.322 | .791 | 2.322 |
| | Data-SUITE (CONF) | .125 | .396 | .846 | .293 | .846 |
| | Data-SUITE (COP) | .220 | .277 | .451 | .236 | .451 |
| | BNN | .192 | .216 | .704 | .173 | .704 |
| | ENS | .125 | .311 | .565 | .204 | .565 |
| | GP | .112 | .153 | .296 | .158 | .296 |
| | MCD | .173 | .391 | .692 | .201 | .692 |
| | QR | .116 | .228 | .635 | .193 | .635 |
| **Model-centric** | | | | | | |
| | BNN (Predictive) | .208 | .220 | .692 | .195 | .692 |
| | ENS (Predictive) | .143 | .226 | .625 | .257 | .625 |
| | GP (Predictive) | .147 | .472 | .584 | .237 | .584 |
| | MCD (Predictive) | .206 | .255 | .684 | .213 | .684 |
| | QR (Predictive) | .187 | .477 | .671 | .223 | .671 |

### 3.4.3 Real dataset stratification w/ downstream task

We now demonstrate how Data-SUITE can be used practically with real data to stratify instances for improved downstream performance (satisfying **D2**). Specifically, our aim is to use Data-SUITE to assist with more reliable and performant model deployment across a variety of scenarios.

To this end, we select three real-world datasets with different types of incongruence, as presented in Table 3.2. For details, see Appendix B.2.4.

**Evaluation.** We stratify $\mathscr{D}_{\text{test}}$ into *certain* and *uncertain* instances based on the interval width predicted by each method. e.g. the most uncertain has the largest width.

***Are the identified instances OOD?*** At this point, one might be tempted to assume that the identified instances are simply OOD. We show in reality that this is unlikely the case. We apply existing algorithms to detect OOD and outliers: Mahalanobis distance [127], SUOD [268], COPOD [135] and Isolation Forest [139]. For each of the detection methods, we compute the overlap between the predicted OOD/Outlier instances and the uncertain and inconsistent instances as identified by Data-SUITE.

We found minimal overlap between methods ranging between 1-18%. Additionally, the OOD detection methods were often unconfident in their predictions, with average confi-

dence scores ranging between 5-50%. Both results suggest that the identified uncertain and inconsistent instances are unlikely OOD. For more, see Appendix B.3.1.

Table 3.2 Comparison of real-world datasets

| Dataset | Incongruence Type | Downstream Task | Stratification |
|---|---|---|---|
| **Seer (US) & Cutract (UK)** | Geographic UK-US (Cross-site medical) | Predict mortality from prostate cancer | $\mathscr{D}_{\text{train}}^{Seer}$: Seer (US), $\mathscr{D}_{\text{test}}^{Cut}$: Cutract (UK) |
| **Adult** | Demographic Bias (Gender & Income) | Predict income over \$50K | $\mathscr{D}_{\text{train}}^{Adult}$, $\mathscr{D}_{\text{test}}^{Adult}$ balanced split |
| **Electricity** | Temporal (Consumption patterns) | Predict electricity price rise/fall | $\mathscr{D}_{\text{train}}^{Elec}$ :1996 , $\mathscr{D}_{\text{test}}^{Elec}$ : 1997-1998 |

Each stratification method is used to identify different instances for each group, hence we aim to quantify which method identifies instances that provide the most improvement to downstream performance. We do this by computing the accuracy of the *certain* and *uncertain* stratifications on a downstream random forest trained on $\mathscr{D}_{\text{train}}$. Ideally, correct instance stratification results in greater accuracy for *certain* compared to *uncertain* instances.

As an overall comparative metric, we compute the *Mean Performance Improvement - MPI* (Equation 3.8). MPI is the difference in accuracy (*Acc*) between *certain* and *uncertain* instances, as identified by a specific method, averaged over a set of threshold proportions $\mathscr{T}$. The best-performing method would clearly identify the most appropriate *certain* and *uncertain* instances, which would translate to the largest MPI.

$$MPI = \frac{1}{|\mathscr{T}|} \sum_{\tau \in \mathscr{T}} \Big( Acc(Cert_\tau) - Acc(Uncert_\tau) \Big) \tag{3.8}$$

where $\mathscr{T} = \{0.05\text{k} \mid k \in [20]\}$, $Cert_\tau$ = Set of proportion $\tau$ most *certain* instances, $Uncert_\tau$ = Set of proportion $\tau$ most *uncertain* instances.

Figure 3.6 illustrates an example of Data-SUITE applied to the CUTRACT dataset. The metric *MPI* (Equation 3.8) is the mean difference between certain (green) and uncertain (red) curves. The results demonstrate performance improvement when evaluating with the stratified *certain* and *uncertain* instances (compared to performance evaluated on the baseline $\mathscr{D}_{\text{test}}$ or random sampling of instances). The result further demonstrates that the identified *inconsistent* instances have worse performance compared to *uncertain* instances.

Table 3.3 shows the *MPI* scores across methods. In satisfying **D2**, of improving deployed model performance, Data-SUITE consistently outperforms other methods, providing the greatest performance improvement, with the lowest variability across datasets. The result

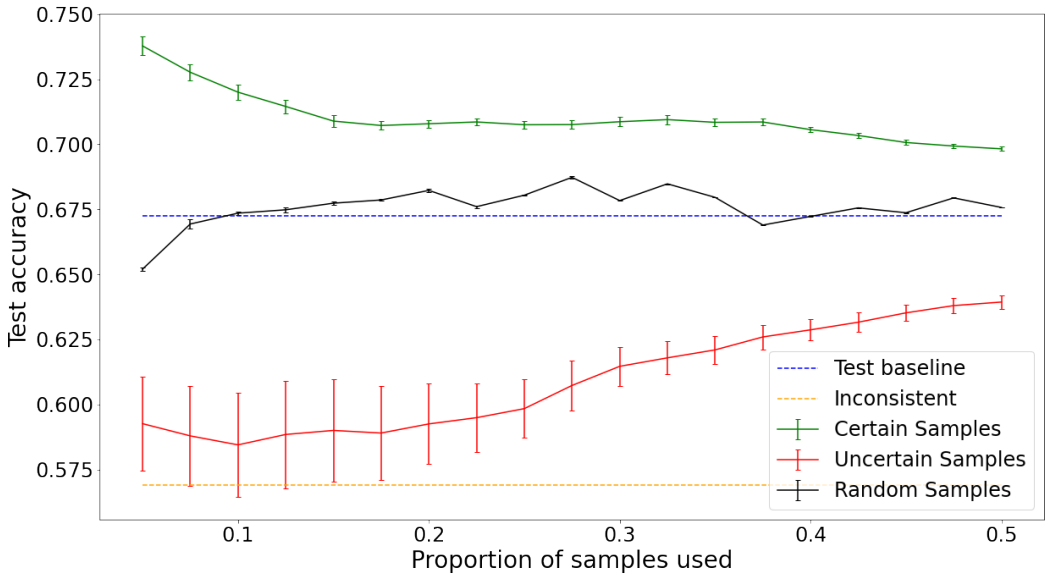



Fig. 3.6 Example on CUTRACT of how Data-SUITE stratification can be used to improve downstream performance, contrasted with baseline $\mathscr{D}_{\text{test}}$ (blue) or random selection (black).

suggests that Data-SUITE identifies the most appropriate *certain* and *uncertain* instances, accounting for the performance improvement. Overall, the quality of stratification by Data-SUITE has not been matched by any benchmark uncertainty estimation method.

Table 3.3 MPI metric across datasets for different methods

| | SEER-CUTRACT | Adult | Electricity |
|---|---|---|---|
| **Data-SUITE** | **0.11 ± 0.015** | **0.64 ± 0.03** | **0.26 ± 0.03** |
| BNN | 0.08 ± 0.02 | -0.15 ± 0.01 | -0.005 ± 0.01 |
| CONFORMAL | 0.05 ± 0.01 | -0.07 ± 0.07 | 0.12 ± 0.03 |
| ENSEMBLE | 0.01 ± 0.02 | -0.03 ± 0.02 | -0.02 ± 0.02 |
| GP | 0.05 ± 0.04 | 0.56 ± 0.02 | 0.04 ± 0.04 |
| MCD | 0.01 ± 0.01 | -0.16 ± 0.01 | 0.15 ± 0.03 |
| QR | -0.10 ± 0.03 | 0.12 ± 0.06 | 0.15 ± 0.06 |

### 3.4.4 Data-SUITE Ablation

Data-SUITE adopts a pipeline-based approach. Hence, to better understand the effect of each component, we perform an ablation study of different constituent components. The constituent components are then compared to the complete pipeline, which we denote as Data-SUITE (ALL). The two components that we explicitly test are:

- Data-SUITE (CONF): Test the conformal predictor without the representer. The conformal prediction process and instance stratification are as defined previously.
- Data-SUITE (COP): Test the copula by itself. For this ablation, we fit the copula on $\mathscr{D}_{\text{train}}$. Then, to compute intervals per feature, we condition on the remaining features and sample 100 estimates from the copula. For example, for feature $x_1$, we condition on $x_2 ... x_n$. The uncertainty estimate is then the variance of these samples, which is then used to stratify the samples as before.

Firstly, we compare the coverage for each of the constituent components as per Table 3.4 for different configurations of the synthetic experiment. We see that Data-SUITE (All) is the only method to maintain coverage guarantees across all configurations. That said, we see a significant divergence between Data-SUITE (CONF) and Data-SUITE (COP), which suggests that the conformal estimator is the most important component. The performance gap to Data-SUITE (All) is then likely the representer.

Table 3.4 Coverage of constituent components of Data-SUITE

| | Proportion ($D_a$) | | | Variance ($D_b$) | |
|---|---|---|---|---|---|
| PERTURBATION | .1 | .25 | .5 | 1 | 2 |
| **Data-SUITE (All)** | **.97** | **.96** | **.96** | **.95** | **.96** |
| Data-SUITE (CONF) | .89 | .88 | .90 | .87 | .90 |
| Data-SUITE (COP) | .16 | .15 | .16 | .17 | .16 |

Secondly, we compare the downstream MSE for each of the constituent components as per Table 3.5 for different configurations of the synthetic experiment. Recall that a lower MSE (i.e. closer to Train Data (BASELINE)) is desired. Data-SUITE (All) outperforms the constituent components and is less sensitive to perturbations. Interestingly, despite the higher coverage for Data-SUITE (CONF) vs Data-SUITE (COP), when evaluating for a downstream task, Data-SUITE (COP), in fact, produces results that are less sensitive to perturbations.

Table 3.5 Downstream MSE for ablations of constituent components of Data-SUITE

| | Proportion ($D_a$) | | | Variance ($D_b$) | |
|---|---|---|---|---|---|
| PERTURBATION | .1 | .25 | .5 | 1 | 2 |
| *Train Data (BASELINE)* | *.067* | *.059* | *.068* | *.065* | *.068* |
| Test Data | .222 | .513 | .889 | .275 | .889 |
| **Data-SUITE (All)** | **.069** | **.122** | **.197** | **.104** | **.197** |
| Data-SUITE (CONF) | .125 | .396 | .846 | .293 | .846 |
| Data-SUITE (COP) | .220 | .277 | .451 | .236 | .451 |

Both results show reduced performance for the constituent components of Data-SUITE, with each component having different individual sensitivities. This highlights the necessity of the inter-connected Data-SUITE (All) framework.

### 3.4.5 Use-Case: Data-SUITE in the hands of users

We now demonstrate how users can practically leverage Data-SUITE to better understand their data. We do so by profiling the incongruous regions identified by Data-SUITE and highlighting the insights that can be garnered, independent of a model. This satisfies **D1**, where the quantitative profiling provides valuable insights that could assist data owners in characterizing where to collect more data and, if this is not possible, in understanding the limitations of the data.

For visual purposes, we embed the identified *certain* and *uncertain* instances into 2-D space as shown in Figure 3.7. We clearly see that the *certain* and *uncertain* instances are distinct regions and that they lie ID as evidenced by the embedding projection. This reinforces the quantitative findings of the previous experiment (i.e., not OOD).

We further highlight the centroid, "average prototypes" of the *certain* and *uncertain* regions. These are digestible examples of the region, which can easily be understood by stakeholders. For the SEER-CUTRACT analysis, in addition to prototypes for $\mathscr{D}_{\text{test}}^{Cut}$ regions, we can also find the nearest neighbor SEER (USA) prototypes for each instance. Comparing the average and nearest neighbor prototypes helps us tease out the incongruence between the two geographic sites.

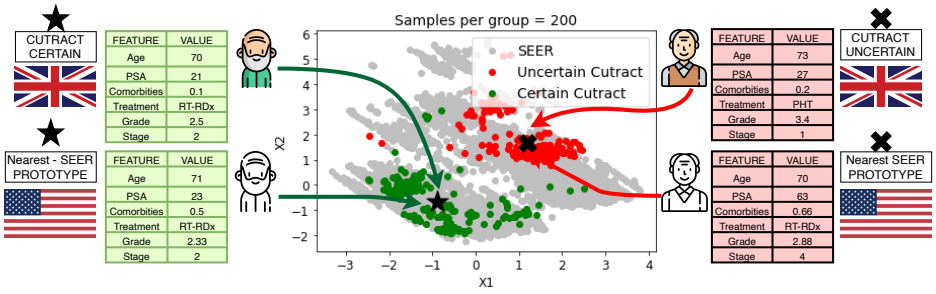
Overall, Figure 3.7, in their respective captions, highlights the most valuable insights, quantitatively garnered on the basis of Data-SUITE across all three datasets. Below are deep dives to illustrate the utility.

***SEER-CUTRACT.*** Data-SUITE identifies distinct *certain* (Green) and *uncertain* (Red) regions of ID data, as shown in Figure 3.7 (i). Beyond average prototypes for $\mathscr{D}_{\text{test}}$ regions i.e., CUTRACT (UK), we also find the nearest neighbor SEER (USA) prototypes for each instance in the identified regions. Comparing these prototypes helps us to tease out the differences between the two geographic sites. We note three specific insights that can assist end-users. (1) *certain* instances represent less severe patients than the *uncertain* instances (see PSA values). (2) *certain* instances: CUTRACT (UK) and SEER (USA) prototypes are similar in all feature values. (3) *uncertain* instances: CUTRACT (UK) and SEER (USA) prototypes have differences in certain feature values (PSA, more co-morbidities, different treatment, staging score).

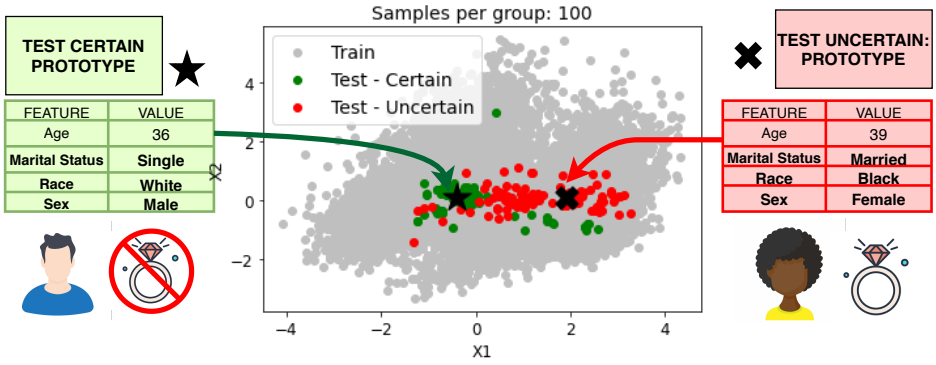
***Adult.*** The dataset has a known bias between gender and income, which Data-SUITE successfully identifies. However, an added dimension between the most *certain* and *uncertain* instances, as shown in Figure 3.7 (ii) is that marital status and race are also relevant biases. We capture this finding with prototypes, (i) *certain* prototype: younger, single, white, males (ii) *uncertain* prototype: older, married, black, females. These prototypes can inform end-users, such as data scientists and stakeholders, of the dangers in naively building models with this dataset without first considering these issues.

***Electricity.*** The dataset has concept drift over time, which Data-SUITE identifies as the *uncertain* instances, which have increased the transfer of energy between NSW and Victoria, increased NSW energy price, and decreased demand relative to the *certain* instances. These prototypes can inform users of this change in consumption habits, despite the data still lying in-distribution. Additionally, as per Figure 3.7 (iii), the *certain* instances capture the time-frame closer to the training data (Start 1997-Mid 1997), whilst the *uncertain* instances are later in time (Mid 1997-Early 1998), hence more likely affected by the concept drift.

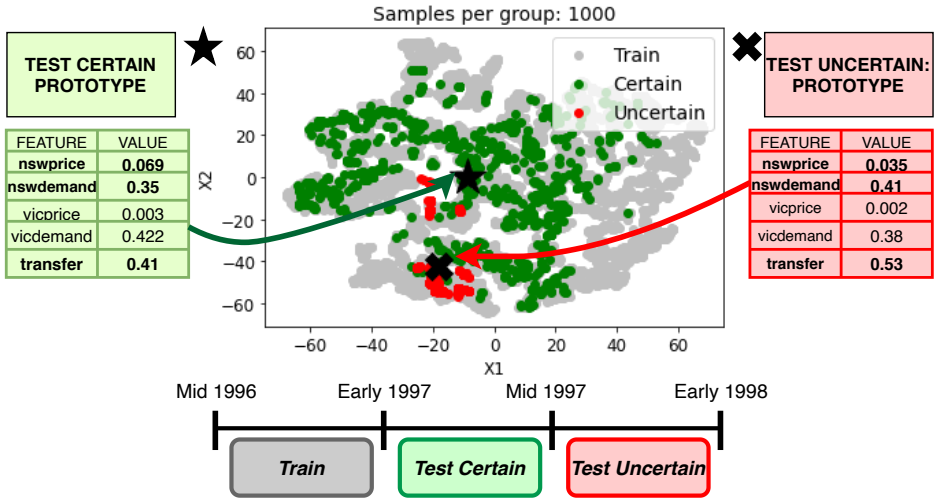
Finally, we conduct a more detailed analysis of the regions in Appendix B.3.2 to outline further potential practitioner usage.




(a) i. SEER-CUTRACT: CUTRACT *certain* instances are similar to their SEER nearest prototypes, whilst CUTRACT *uncertain* instances are different from their nearest SEER prototypes (e.g. PSA).




(b) ii. Adult: The *certain* and *uncertain* instances represent two different demographics, aligning with the known dataset biases toward females. The *uncertain* instances specifically highlight a sub-group of black females who are married.




(c) iii. Electricity: The *certain* instances are similar to the training set in features and time. The *uncertain* instances identified represent a later time period, wherein concept drift has likely occurred.

Fig. 3.7 Insights of prototypes identified by Data-SUITE. Tables describe the average prototypes for *certain* and *uncertain* instances.

## 3.5 Discussion

Automation should not replace the expertise and judgment of a data scientist in understanding the data, nor will it replace the ingenuity required to build better models. In this spirit, we developed Data-SUITE and illustrated its capability across multiple datasets to empower data scientists to perform more insightful data exploration and enable more reliable model deployment. We address these use cases for the understudied problem of in-distribution heterogeneity and propose a flexible data-centric solution, independent of a task-specific model. Data-SUITE permits the stratification of test data into inconsistent and uncertain instances with respect to training data. This stratification has been shown to be in line with downstream performance and to provide valuable insights for profiling incongruent test instances in a rigorous and quantitative way. The quality of this stratification by Data-SUITE is not matched by any benchmark uncertainty estimation method (data-centric or not). The promising result opens up future avenues to advance the agenda, taking it a step further, both explaining and correcting the identified instances, as well as guiding future data collection.

# Chapter 4

# Synthetic Data Augmentation and Curation

## 4.1 Introduction

**No data, No Machine Learning.** Machine learning (ML) has transformed numerous industries, but its wider adoption is hindered by a pervasive roadblock: insufficient data. Specifically, the use of ML algorithms presumes the availability and access to large datasets for training, be it labeled or unlabeled. Unfortunately, real-world domains are often data scarce: (i) in healthcare and finance, collecting annotations can be expensive or practically impossible; (ii) in developing and low-to-middle income countries (LMICs), digital infrastructure (such as electronic healthcare records (EHRs)) can be limited or nonexistent [1, 9, 163, 152, 3, 40] and (iii) within large datasets, there can be (ethnic) minorities that are underrepresented. This lack of data has serious consequences, namely sidelining these settings to the peripheries of ML advancements and preventing the development of accurate models. This begs the question, how can we build a reliable ML model in this *low-data regime* with so few samples? Solving this problem is a major opportunity that would unlock the potential of ML across society, domains, and regions.

**Aim.** To address this important yet undervalued low-data problem, we aim to augment the *small labeled dataset* ($n < 100$) with synthetic samples. We focus on tabular data, as defining augmentations is non-trivial and can easily result in nonsensical or invalid samples. Moreover, tabular domains like healthcare are often where data scarcity is acute.

**Related work.** Data augmentation is widely used to address the low-data problem, and different approaches to address data scarcity have been proposed in tabular data contexts. Prior methods are either based on generative models [64, 22, 252, 146, 238] such as GANs

[259], VAEs [259], Normalizing Flows [169], Score-based models [119, 109], or alternatively are traditional interpolative methods such as SMOTE [31, 252, 146]. However, in low-data regimes ($n < 100$), a challenge is that the limited training data may not describe the full data distribution well, despite it being an i.i.d. draw. Consequently, this harms the capability of conventional augmentation methods, as the resulting augmented synthetic data may not be sufficiently diverse and accurate, restricting the generalizability of predictive models trained on such data.

Recent work has shown the potential of fine-tuning Large Language Models (LLMs) for tabular data generation [24]. While LLMs offer some degree of prior knowledge, there are two challenges to using LLMs in our setting. First, it is computationally expensive to fine-tune LLMs while needing specialized hardware —luxuries often not available in LMICs, thereby limiting applicability in such settings. Second, fine-tuning often assumes a large number of samples. In our low-data setting, it could lead to overfitting and low-quality generated samples and, hence, poor downstream models—as we show for [24] in Chapter 4.3. Tangential to data augmentation, prior work has tackled data scarcity in the tabular setting via the lens of transfer learning or few-shot learning by using a knowledge graph [147, 199] (which might not be available) or a pre-trained model [133, 100, 82]. However, unlike data augmentation, these approaches are not flexible, as they tie the data customer to use a certain downstream predictor. We provide an extended discussion on this point in Appendix C.1.

**Curated LLM.** To address the shortcomings of the aforementioned augmentation approaches, we propose `Curated LLM` (CLLM). First, CLLM leverages the in-context learning capabilities of LLMs for synthetic data generation, thereby reducing the computational burden compared to fine-tuning. We also posit for the low-data regime; the diverse pretraining corpus of LLMs carries valuable prior knowledge, which may offer more diversity in the data generation compared to other conventional tabular generators. *Of course, LLMs are not perfect*. Consequently, balancing the utility of LLMs against the risk of noisy, irrelevant data is important to ensure reliable downstream performance. Hence, this necessitates a systematic assessment of the generated data. In fact, this issue is vital for *any* generative model.

This motivates the second key aspect of CLLM, i.e. a post-generation data curation mechanism. This addresses the *overlooked* aspect that not all of the synthetic samples are useful to downstream model performance, with some samples even harmful. We anchor our approach with ideas from learning theory that show the behavior of individual data samples during training, called learning dynamics, which provides a salient signal about the value of samples to a learner [7, 6, 134]. To provide intuition, samples with variable predictions might be considered ambiguous, or other samples might never be learned correctly and could

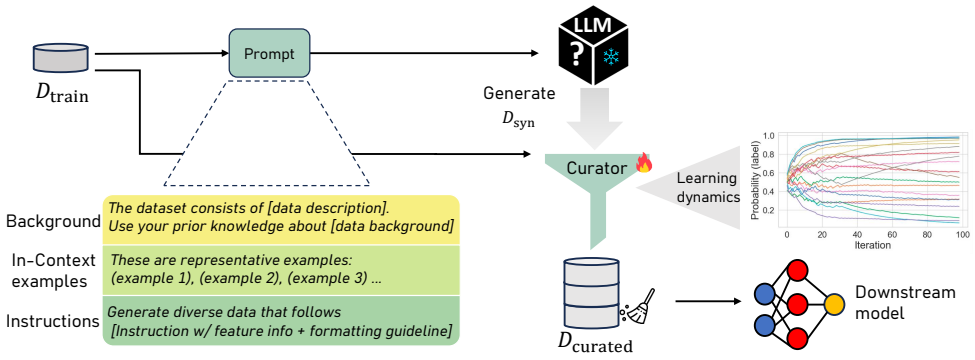



Fig. 4.1 CLLM uses a small dataset $\mathscr{D}_{\text{train}}$ and a frozen black-box LLM to generate a larger synthetic set $\mathscr{D}_{\text{syn}}$. The curator computes the learning dynamics of samples in $\mathscr{D}_{\text{syn}}$, assessing samples based on their aleatoric uncertainty and predictive confidence, then curates $\mathscr{D}_{\text{syn}}$ with the goal that a downstream model trained on the curated $\mathscr{D}_{\text{curated}}$ will have improved performance.

harm a model. In CLLM, we study the learning dynamics of the synthetic data samples with respect to a model trained on the small real dataset. We then analyze these dynamics by computing two key metrics: confidence and aleatoric (data) uncertainty. These metrics form the basis for curating the synthetic samples. We then aim to enable a highly performant downstream model when trained on the curated dataset.

**Contributions:** CLLM is a novel data augmentation approach allying the strengths of LLMs with a robust data curation mechanism to improve data augmentation in the *low-data regime* ($n < 100$), bringing several contributions: ① **Improved performance:** We empirically demonstrate on 7 real-world datasets that CLLM enables superior downstream performance compared to 6 widely used tabular data generative models and data augmentation techniques. ② **Value of curation:** We show that the *overlooked* aspect of synthetic data curation improves downstream performance across the generative models. This highlights the flexibility and broad utility of our curation mechanism for data augmentation. ③ **Insights:** We dissect the two aspects of CLLM(LLM and data curation) along a variety of dimensions, providing insights and understanding into why the approach is beneficial. We show that the largest gains are for underrepresented subgroups and in low-data settings. These contributions, which address the data logjam, pave the way toward wider usage of ML across society, domains, and regions.

## 4.2 CLLM: Synergy of LLM Synthetic Data Generation and Data Curation

**Problem formulation.** Let $(X,Y)$ be a pair of random variables taking values in a feature space $\mathscr{X}$ and a label space $\mathscr{Y} = \{1,\ldots,k\}$. We assume access to only a small labeled dataset $\mathscr{D}_{\text{train}} = \{(x_i,y_i)\}_{i=1}^{n}$ (where $n < 100$, denoting the low-data setting), representing an i.i.d. sample drawn from the true data-generating distribution $p_R(x,y)$. We also assume access to a pre-trained LLM to generate data samples. Let $p_\Phi(x,y)$ denote the distribution over $(X,Y)$ induced by the LLM generation procedure, where $\Phi$ represents the conditioning variables that we control (e.g., the input prompts and in-context examples). Our goal is to generate a synthetic dataset $\mathscr{D}_{\text{syn}}$ to augment the small $\mathscr{D}_{\text{train}}$ and subsequently use the augmented dataset to train a classifier $f : \mathscr{X} \to \Delta^{k-1}$. Successful augmentation will provide a better classifier $f$ than if we had trained $f$ on the small $\mathscr{D}_{\text{train}}$. We measure downstream performance on a separate held-out dataset of real test data, $\mathscr{D}_{\text{test}}$.

**Our Approach.** To address this challenge, we introduce CLLM, an approach for synthetic data augmentation in low-data regimes. As shown in Figure 4.1, CLLM leverages LLMs to **generate** a synthetic dataset $\mathscr{D}_{\text{syn}}$ using a small dataset $\mathscr{D}_{\text{train}}$ (Chapter 4.2.1). It exploits the LLMs' prior knowledge via in-context learning (ICL) and contextual information. CLLM then **curates** $\mathscr{D}_{\text{syn}}$ by analyzing the learning dynamics of samples in $\mathscr{D}_{\text{syn}}$ based on predictive confidence and aleatoric (data) uncertainty. These metrics are obtained by training a supervised model on $\mathscr{D}_{\text{train}}$. We leverage them to define a curated dataset $\mathscr{D}_{\text{curated}}$, which is used to train a downstream classifier (Chapter 4.2.2).

In each sub-section, we describe and motivate the design of the different aspects of CLLM (LLM and curation mechanism). Furthermore, we provide insights and understanding into their role in improving data utility, which we later quantify on multiple real-world datasets in Chapter 4.3.

### 4.2.1 Data generation with LLMs based on a small $\mathscr{D}_{\text{train}}$

As outlined in Chapter 4.1, in the low-data regime, conventional tabular generative models (e.g. CTGAN, TVAE) are constrained by the limited $\mathscr{D}_{\text{train}}$ and may not generate sufficiently diverse and/or accurate synthetic data. To address this, we propose to leverage LLMs, building on their large-scale pretraining. We first outline the appealing properties of LLMs for tabular data generation when we have very few samples and then describe design choices to exploit these.

- **Prior knowledge.** LLMs are pre-trained with a vast corpus of information [35, 227]. When prompted to generate samples with limited real data, LLMs can leverage this encoded prior information about similar problems and feature-label relationships to enhance both the accuracy and diversity of generation.
- **Contextual understanding.** LLMs can process background and contextual information about the problem via natural language [261]. For example, a high-level description of the task, features and their meanings can be conveniently described through natural language. Such information is unavailable to conventional generators that only utilize numerical examples.
- **Few-shot capabilities.** LLMs have demonstrated proficiency in generalizing to different tasks with just a few examples [27, 255, 151]. In the context of generation, we envision the idea of in-context generation using limited real examples.

To benefit from these capabilities, we craft the LLM prompt with three different parts (see Figure 4.1): (1) *Background*: text description of the dataset and task (e.g. predict Covid mortality). Additionally, we include a description of what each feature means, explicitly prompting the LLM to use prior knowledge about these features. (2) *Examples*: we serialize the tabular samples in $\mathscr{D}_{\text{train}}$ as example demonstrations and provide both the features and the label in text format. (3) *Instructions*: To generate a synthetic dataset $\mathscr{D}_{\text{syn}}$, we instruct the LLM to leverage the contextual information and the provided examples as an i.i.d. draw from the distribution. We instruct the LLM to identify structural and feature-label relationships in the data and generate diverse data following the structure and format of the provided examples. We provide more details on the prompts in Appendix C.2.

**Motivation for a frozen LLM.** Using a frozen black-box LLM (e.g. GPT-4 or GPT-3.5) is computationally cheaper and requires less specialized hardware (i.e. GPUs) compared to fine-tuning an LLM. This relates to the settings described in Chapter 4.1, such as LMICs, where we may not have the computational resources to fine-tune an LLM.

Even in settings where fine-tuning is possible, we show empirically in Chapter 4.3 that LLM fine-tuning (e.g. GReaT baseline) is suboptimal in low-data settings ($n < 100$) compared to providing in-context examples coupled with curation.

**Dissecting the LLM's generative features.** We now investigate various dimensions to understand and illustrate empirically the appealing features of LLMs as data generators in the low-data regime and how our design choices unlock them. We use the Brazil *Covid-19* dataset [16] as a running example and focus on GPT-4 as the LLM.

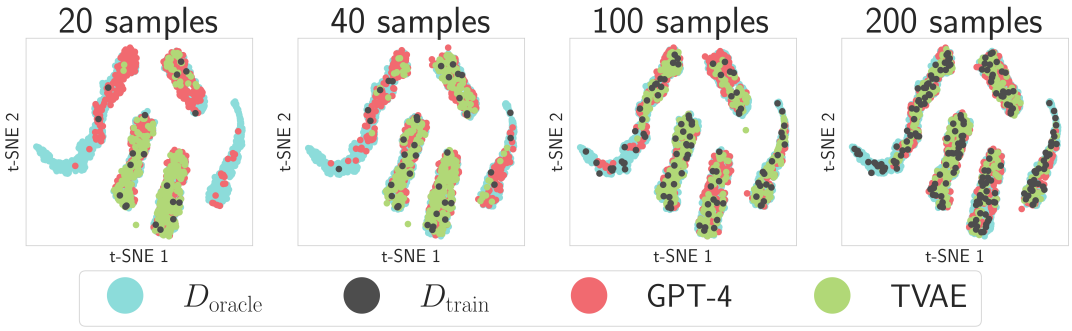



Fig. 4.2 GPT-4 is able to extrapolate to regions of the oracle (true manifold) even where there is no training data covering them, as can be seen by the overlap with the turquoise dots, with the effect more pronounced when $\mathscr{D}_{\text{train}}$ is small

---

► **Extrapolation to unseen regions of the manifold.** We compare the samples generated by GPT-4 to TVAE, a widely used tabular data generator. We consider $\mathscr{D}_{\text{oracle}}$, a held-out dataset from the same distribution as $\mathscr{D}_{\text{train}}$, such that $|\mathscr{D}_{\text{oracle}}| \gg |\mathscr{D}_{\text{train}}|$, thereby providing an approximation for the true manifold. The t-SNE plots in Figure 4.2 shows, when $\mathscr{D}_{\text{train}}$ is very small ($n = 20$ samples), hence its samples do not cover all regions of $\mathscr{D}_{\text{oracle}}$. For example, $\mathscr{D}_{\text{train}}$ does not contain samples from specific demographic subgroups (e.g. people aged 40 or below). As expected, TVAE only generates samples constrained by the limited $\mathscr{D}_{\text{train}}$. In contrast, GPT-4 is able to extrapolate and generate samples even in unseen regions of $\mathscr{D}_{\text{train}}$, thereby better covering $\mathscr{D}_{\text{oracle}}$. This stems from its *contextual understanding* of the features, unlocking the use of its *prior knowledge*. It leads to better coverage in the low-data regime, consequently aiding in superior downstream performance, as shown in Table 4.3. As $n$ increases ($\geq 100$), $\mathscr{D}_{\text{train}}$ provides better coverage, which naturally benefits both GPT-4 and TVAE. Overall, this result shows how prior knowledge encoded in LLMs addresses shortcomings of conventional generative approaches (e.g. TVAE) in the low-data regime.

► **GPT-4 benefits underrepresented groups the most.** Having illustrated the extrapolation capabilities of GPT-4, we now ask: *where does augmentation benefit downstream performance the most?* We evaluate performance gains for different demographic subgroups, such as age groups and ethnic groups (Amarela - East Asian, Prada - Mixed). Figure 4.3 shows the performance gain obtained by training a classifier on data generated by GPT-4 compared to training on the small $\mathscr{D}_{\text{train}}$. The greatest gains, on average, are for subgroups for which we have *no data* in $\mathscr{D}_{\text{train}}$, yet GPT-4 can extrapolate and generate samples for these subgroups. This further validates the rationale of extrapolation via prior knowledge as a key source of gain for GPT-4.

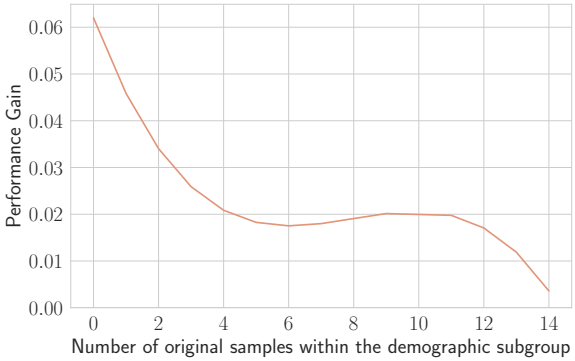



Fig. 4.3 Subgroups with fewest samples in $\mathscr{D}_{\text{train}}$ benefit the most from data augmentation, on average.

Table 4.1 shows fine-grained results (across 10 different seeds) for the 5 subgroups that benefit the most from data augmentation, which are small-sized demographic subgroups. This finding has real-world implications for equity, showing we can improve performance for underrepresented subgroups even when we lack data or collecting data is difficult/costly.

Table 4.1 Deep dive into the top 5 demographic subgroups in the Covid dataset with the largest gains, across 10 seeds, for $|\mathscr{D}_{\text{train}}| = 20$. GPT-4 improves performance on the smallest groups.

| Subgroup | $n_{\text{samples}}$ in $\mathscr{D}_{\text{train}}$ (min - max) | Avg. Acc. Gain v. $\mathscr{D}_{\text{train}}$ GPT-4 | TVAE |
|---|---|---|---|
| Age_40 | 0-6 | **6.38** +- 2.09 | -3.37 +- 2.86 |
| Liver | 0-1 | **3.85** +- 3.37 | -13.1 +- 3.38 |
| Renal | 0-3 | **4.52** +- 2.01 | -18.0 +- 3.22 |
| Amarela | 0-1 | **8.71** +- 1.40 | -2.03 +- 2.88 |
| Parda | 3-11 | **5.07** +- 1.50 | -6.57 +- 1.61 |

▶ **Importance of contextual information in the prompt.** A natural question is : *how important is the prompt to elicit the prior knowledge of the LLM?* We explore two variants: (1) *Prompt w/ context*: provides contextual information including background about the dataset, feature names and descriptions (our approach) and (2) *Prompt w/ no context*: only provides the numerical in-context examples (ablation). Figure 4.4 qualitatively shows that not including contextual knowledge in the prompt gives lower coverage of $\mathscr{D}_{\text{oracle}}$ with less extrapolation beyond $\mathscr{D}_{\text{train}}$. We quantify this in Table 4.2 using Precision (Quality) and Recall (Diversity) metrics [202], as well as Utility (Downstream performance). GPT-4 *with contextual information* has superior precision and recall in the low data setting. Furthermore, we show that *the lack* of contextual information in the prompt significantly harms the precision (quality) of the data even compared to TVAE. This highlights that LLMs need

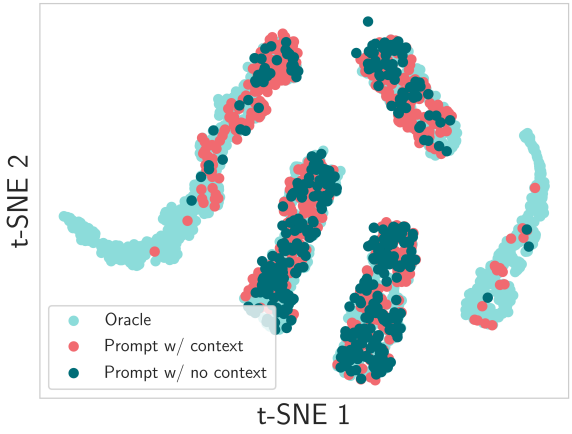



Fig. 4.4 Contextual information in the prompt is important for extrapolation.

guidance, as we are only able to get the extrapolation and performance benefits by including contextual information, further motivating our design choices in the prompt. We conduct the same experiment with the Compas dataset in Appendix C.3.2.

Table 4.2 Including contextual information in the prompt improves precision (P), recall (R), and utility (U) in low-sample settings (exemplified on the Covid-19 dataset).

| $n_{\text{samples}}$ in $\mathscr{D}_{\text{train}}$ | GPT-4 w/ context | | | GPT-4 no context | | | TVAE | | |
|---|---|---|---|---|---|---|---|---|---|
| | P | R | U | P | R | U | P | R | U |
| 20 | $\mathbf{0.41}_{(\mathbf{0.04})}$ | $\mathbf{0.87}_{(\mathbf{0.03})}$ | $\mathbf{0.74}_{(\mathbf{0.01})}$ | $0.13_{(0.0)}$ | $0.82_{(0.01)}$ | $0.66_{(0.01)}$ | $0.33_{(0.07)}$ | $0.50_{(0.03)}$ | $0.59_{(0.02)}$ |
| 40 | $\mathbf{0.40}_{(\mathbf{0.01})}$ | $\mathbf{0.91}_{(\mathbf{0.01})}$ | $\mathbf{0.76}_{(\mathbf{0.0})}$ | $0.11_{(0.0)}$ | $0.89_{(0.0)}$ | $0.69_{(0.0)}$ | $0.27_{(0.01)}$ | $0.68_{(0.01)}$ | $0.62_{(0.03)}$ |
| 100 | $\mathbf{0.42}_{(\mathbf{0.01})}$ | $0.86_{(0.02)}$ | $\mathbf{0.75}_{(\mathbf{0.01})}$ | $0.11_{(0.01)}$ | $\mathbf{0.90}_{(\mathbf{0.01})}$ | $0.74_{(0.01)}$ | $0.39_{(0.02)}$ | $0.67_{(0.03)}$ | $0.64_{(0.06)}$ |
| 200 | $0.44_{(0.02)}$ | $0.85_{(0.02)}$ | $\mathbf{0.75}_{(\mathbf{0.0})}$ | $0.08_{(0.01)}$ | $\mathbf{0.90}_{(\mathbf{0.0})}$ | $0.60_{(0.01)}$ | $\mathbf{0.47}_{(\mathbf{0.0})}$ | $0.73_{(0.01)}$ | $0.65_{(0.02)}$ |

### 4.2.2 Data curation with learning dynamics

When prompted with $\Phi$ (which contains the in-context samples of $\mathscr{D}_{\text{train}}$), the LLM generates samples from a distribution $p_\Phi(X,Y)$ that approximates $p_R(X,Y)$, implicitly exploiting the LLM's large-scale pretraining and few-shot capabilities. LLMs are, of course, not perfect and could generate noisy samples, hence, this distribution may be inaccurate [1].

To make this distribution more relevant to the downstream task, we propose a data curation mechanism. Specifically, we focus on the noisy feature-label relationship $p_\Phi(Y|X)$,

[1] We could finetune the model on the scarce $\mathscr{D}_{\text{train}}$ we have, but it is likely to still lead to overfitting due to the extreme data scarcity and LLM parameter size.

for which we expect $p_\Phi(Y|X) \neq p_R(Y|X)$ given the small size of $\mathscr{D}_{\text{train}}$. This motivates us to curate $\mathscr{D}_{\text{syn}}$ and discard likely mislabeled samples.

We anchor our approach with ideas from learning theory that show that the behavior of individual samples during model training (called *learning dynamics*) contains signals about the nature of the samples themselves [7, 6, 134].

Some samples are easily and confidently predicted over different model checkpoints, whereas other samples might be challenging to learn (e.g. due to mislabeling) and hence might be incorrectly predicted for the given label.

Consequently, we operationalize *learning dynamics* as the basis of our proposed curation mechanism. Specifically, we analyze samples in $\mathscr{D}_{\text{syn}}$ by studying their learning dynamics computed with a classifier trained on $\mathscr{D}_{\text{train}}$. We then categorize and filter samples in $\mathscr{D}_{\text{syn}}$, and produce a curated dataset $\mathscr{D}_{\text{curated}} \subseteq \mathscr{D}_{\text{syn}}$.

**Learning dynamics.** We now formalize how we compute learning dynamics for individual samples. Assume that a classifier $f$ is trained in an iterative scheme (e.g. neural networks or XGBoost trained over iterations) on $\mathscr{D}_{\text{train}}$, which makes it possible to analyze the learning dynamics of samples in $\mathscr{D}_{\text{syn}}$ over these iterations. The classifier $f$ should be at least as flexible as the model that the practitioner intends to use for the downstream task. $f$ is trained from scratch on $\mathscr{D}_{\text{train}}$ and goes through $e \in [E]$ different checkpoints leading to the set $\mathscr{F} = \{f_1, f_2, \ldots, f_E\}$, such that $f_e$ is the classifier at the $e$-th checkpoint. Let $[f_e(x)]_y$ denote the predicted probability for class $y$ and sample $x$.

Our goal is to assess the learning dynamics of samples in $\mathscr{D}_{\text{syn}}$ over these $E$ training checkpoints while we train $f$ on $\mathscr{D}_{\text{train}}$. For this, we define $H$, a random variable following a uniform distribution $\mathscr{U}_{\mathscr{F}}$ over the set of checkpoints $\mathscr{F}$. Specifically, given $H = h$ and a sample $(x, y)$, we define the correctness in the prediction of $H$ as a binary random variable $\hat{Y}_{\mathscr{F}}(x, y)$ with the following conditional: $\mathbb{P}(\hat{Y}_{\mathscr{F}}(x, y) = 1 | H = h) = [h(x)]_y$ and $\mathbb{P}(\hat{Y}_{\mathscr{F}}(x, y) = 0 | H = h) = 1 - \mathbb{P}(\hat{Y}_{\mathscr{F}}(x, y) = 1 | H = h)$.

**Curation metrics.** Equipped with a probabilistic interpretation of the predictions of a model, we now define two characterization metrics that we use for curation: (i) average confidence and (ii) aleatoric (data) uncertainty, inspired by [124, 210].

**Definition 4.2.1** (Average confidence)**.** For any set of checkpoints $\mathscr{F} = \{f_1, ..., f_E\}$, the average confidence for a sample $(x, y)$ is defined as the following marginal:

$$\begin{aligned} \bar{P}_{\mathscr{F}}(x, y) &:= \mathbb{P}(\hat{Y}_{\mathscr{F}}(x, y) = 1) \\ &= \mathbb{E}_{H \sim \mathscr{U}_{\mathscr{F}}}[\mathbb{P}(\hat{Y}_{\mathscr{F}}(x, y) = 1 | H)] \\ &= \frac{1}{E} \sum_{e=1}^{E} [f_e(x)]_y \end{aligned}$$

**Definition 4.2.2** (Aleatoric uncertainty)**.** For any set of checkpoints $\mathscr{F} = \{f_1, ..., f_E\}$, the aleatoric uncertainty for a sample $(x, y)$ is defined as:

$$\begin{aligned} v_{al,\mathscr{F}}(x,y) &:= \mathbb{E}_{H \sim \mathscr{U}_{\mathscr{F}}}[Var(\hat{Y}_{\mathscr{F}}(x,y)|H)] \\ &= \frac{1}{E}\sum_{e=1}^{E}[f_e(x)]_y(1-[f_e(x)]_y) \end{aligned}$$

Intuitively, for binary classification ($k = 2$), the aleatoric uncertainty for a sample $x$ is maximized when $[f_e(x)]_y = \frac{1}{2}$ for all checkpoints $f_e$, akin to random guessing. Recall that aleatoric uncertainty captures the inherent data uncertainty; hence, it is a principled way to capture issues such as mislabeling. This contrasts with epistemic uncertainty, which is model-dependent and can be reduced simply by increasing model parameterization [91].

Having defined sample-wise confidence and aleatoric uncertainty, we categorize samples in $\mathscr{D}_{\text{syn}}$ as *Selected* or *Discarded*: for a sample $(x, y)$, a set of training checkpoints $\mathscr{F}$, and two thresholds $\tau_{\text{conf}}$ and $\tau_{\text{al}}$, we define the category $c(x, y, \mathscr{F})$ as *Discarded* if $\bar{P}_{\mathscr{F}}(x, y) < \tau_{\text{conf}}$ and $v_{al,\mathscr{F}}(x, y) < \tau_{\text{al}}$, and *Selected* otherwise.

Hence, a *Discarded* sample is one for which we have very low confidence in predicting its associated label, whereas we also have low inherent data uncertainty. Finally, given a function $f$ associated with the set of checkpoints $\mathscr{F}$, we define the curated set $\mathscr{D}_{\text{curated}} = \{(x, y)|(x, y) \in \mathscr{D}_{\text{syn}}, c(x, y, \mathscr{F}) = \textit{Selected}\}$. We also define $\mathscr{D}_{\text{discarded}} = \mathscr{D}_{\text{syn}} \setminus \mathscr{D}_{\text{curated}}$.

To summarize, the objective of the curation step is that augmenting with the curated synthetic data leads to a better classifier $f_{\mathscr{D}_{\text{curated}}}$ (trained on $\mathscr{D}_{\text{train}} \cup \mathscr{D}_{\text{curated}}$) for the downstream task, compared to the baseline classifier $f_{\mathscr{D}_{\text{syn}}}$ (trained on $\mathscr{D}_{\text{train}} \cup \mathscr{D}_{\text{syn}}$). Thus, our objective is $M(f_{\mathscr{D}_{\text{curated}}}) > M(f_{\mathscr{D}_{\text{syn}}})$, where $M$ is a performance measure (for example, accuracy). In Chapter 4.3, we empirically show how performance on this curated dataset is superior both for LLM-generated data as well as other classes of generative models.

**Dissecting the role of curation.** We now empirically demonstrate the role of curation in correcting the noisy feature-label relationship present in $\mathscr{D}_{\text{syn}}$, highlighting two insights: (i) curation discards samples which are atypical in their label with respect to their neighbors in $\mathscr{D}_{\text{syn}}$ (ii) discarded samples can be considered "mislabeled", and we quantify their atypicality using a large held-out dataset $\mathscr{D}_{\text{oracle}}$.

► **Discarded samples conflict on the label with their neighbors in $\mathscr{D}_{\text{syn}}$.** We audit every synthetic sample $(x, y)$ generated by GPT-4 (across 7 datasets) and compute the proportion of its $k$ nearest neighbors in $\mathscr{D}_{\text{syn}}$ that share the same label $y$. The agreement with the neighbors assesses the typicality of a sample's $y$ given $x$, where naturally lower agreement

is linked to mislabeling, which we aim to detect via curation. Taking $k = 10$, we obtain an average agreement of $a_{\text{curated}} = \mathbf{0.74}$ for $\mathscr{D}_{\text{curated}}$, compared to $a_{\text{discarded}} = \mathbf{0.58}$ for $\mathscr{D}_{\text{discarded}}$. This shows that the samples removed by our curation mechanism are those which, despite having similar features $x$, do not agree with the labels of their surrounding neighbors. This corroborates ideas in [8] of how proximity violations are useful to guide remedial action to improve models. Not removing these mislabeled samples injects noise into the downstream classifier, thus reducing performance.

▶ **Assessing discarded samples with $\mathscr{D}_{\text{oracle}}$.** Ideally, the samples we select should better align with the true feature-label distribution. Since we don't have access to this distribution explicitly, we compute a proxy for $\eta(x) = \arg\max_y \mathbb{P}(Y = y|X = x)$, which we call $\hat{\eta}$. It is obtained by training a classifier on a held-out dataset $\mathscr{D}_{\text{oracle}}$—the same size as $\mathscr{D}_{\text{test}}$ and an order of magnitude larger than $\mathscr{D}_{\text{train}}$. For each method, we then report the accuracy of $\hat{\eta}$ on both the curated $\mathscr{D}_{\text{curated}}$ and discarded $\mathscr{D}_{\text{discarded}}$ datasets (see Figure 4.5).

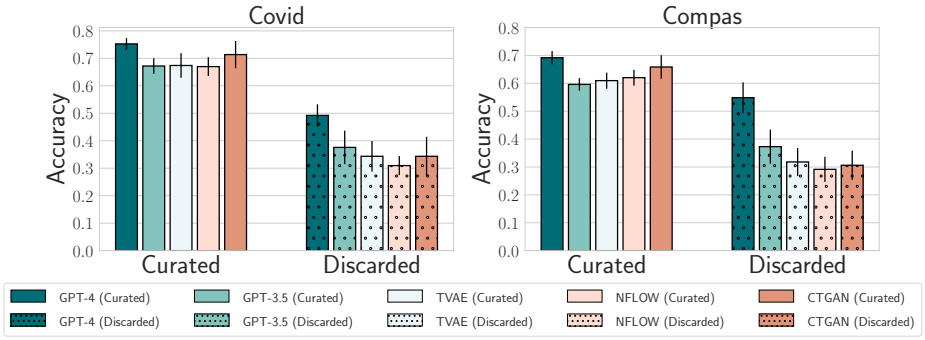



Fig. 4.5 $\hat{\eta}$ aligns more with $\mathscr{D}_{\text{curated}}$ than $\mathscr{D}_{\text{discarded}}$ for each generative model: the curation step keeps high quality samples tailored to the downstream task.

---

We highlight two key observations. First, the curated datasets for all the generative models exhibit a higher agreement with the proxy $\hat{\eta}$ than the discarded datasets. This aligns with the desideratum of only keeping samples that exhibit the correct feature-label relationships. This provides a rationale for why curation helps improve discriminative performance, as samples in $\mathscr{D}_{\text{curated}}$ are much more likely to have the correct feature-label relationship.

Second, GPT-4 has a higher agreement with $\hat{\eta}$ on $D_{\text{discarded}}$ compared to other generators. This illustrates that the LLM's prior knowledge enables it to better capture the distribution $\mathbb{P}(Y|X = x)$. Note that generative baselines (e.g. TVAE) model the joint $\mathbb{P}(X,Y)$, *without any context*, of which is the set of features and which is the label. In contrast, we can define in the LLM prompt which column is the target $Y$, allowing the LLM to better capture the feature-label relationships. This complements findings from Figure 4.2, showing GPT-4 extrapolates to unseen regions of the feature manifold, captured by the support of $\mathbb{P}(X)$.

## 4.3 Experiments

We now perform an end-to-end quantitative evaluation of CLLM across multiple real-world datasets in terms of **downstream utility**, demonstrating the value of allying the generative capabilities of LLMs with our curation mechanism.

Chapter 4.3.1 compares the downstream performance of models when trained on uncurated vs curated data for a variety of augmentation approaches. Having evaluated CLLM on a range of datasets, we also demonstrate and examine the following aspects. Chapter 4.3.2 addresses the issue of potential biases from the LLM and how our curation prevents the biases from propagating into the augmented data. Chapter 4.3.3 ablates the two components of CLLM, namely the prior knowledge and data model, to understand their contributions to performance. Chapter 4.3.4 illustrates how our characterization of samples during the curation step can help to flag synthesized datasets (e.g, via the LLM), which, if used for training, will result in poor downstream performance via a *hardness proxy*.

**Experimental setup.** We compare CLLM (with GPT-4 [161] and GPT-3.5 [27]) against a variety of baselines for tabular synthetic data generation and augmentation: CTGAN [259], TVAE [259], Normalizing Flows [169], TabDDPM [119], SMOTE [31] and GReaT [24], which fine-tunes an LLM. We evaluate performance on 7 real-world datasets with different feature counts and representative of the diverse domains where CLLM can have an impact. For each dataset, we vary the number of samples available in $\mathscr{D}_{\text{train}}$. Given stochasticity, we repeat each experiment for 10 seeds.

While we do not know the exact makeup of the pretraining data for LLMs like GPT-4, there is the possibility that open-source data might be included. This poses the risk of memorization as the primary source of performance gain. To disentangle the role of memorization, we select 4 real-world medical datasets (Maggic [178], Covid [16], SEER [47], CUTRACT [174]) that require an authorization process to access, hence are unlikely to form part of the LLMs training corpus. We use common open-source datasets (Adult and Drug from the UCI repository [11] and Compas [5]) that are highly reflective of data-scarce domains. Further experimental details can be found in Appendix C.2.

### 4.3.1 Overall performance: downstream utility

We assess overall performance based on *Utility* of the augmented data, which we evaluate in terms of AUC on the real $\mathscr{D}_{\text{test}}$, when using four different types of downstream models (see Appendix C.2). This setup mirrors the widely adopted Train-on-Synthetic-Test-on-Real

Table 4.3 AUC (%) averaged over 4 downstream models on $\mathscr{D}_{\text{test}}$. Curation improves performance for all methods across all sample sizes $n$, as indicated by ↑. CLLM w/ GPT-4 (Cur.) provides the strongest performance for both private/proprietary datasets and public datasets

| | Real data | | CLLM (OURS) | | | | Baselines | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | GPT-4 | | GPT-3.5 | | CTGAN | | TabDDPM | | GReaT | | NFLOW | | SMOTE | | TVAE | |
| Dataset | $\mathscr{D}_{\text{oracle}}$ | $\mathscr{D}_{\text{train}}$ | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. |
| covid (n=20) | 74.41 | 68.50 | 73.78 | **73.87** ↑ | 69.85 | 71.41 ↑ | 59.00 | 63.67 ↑ | 66.84 | 66.85 ↑ | 57.38 | 66.46 ↑ | 62.87 | 68.56 ↑ | 66.95 | 66.82 | 61.69 | 66.11 ↑ |
| cutract (n=20) | 72.23 | 70.12 | 71.15 | **72.50** ↑ | 69.97 | 71.54 ↑ | 64.01 | 67.98 ↑ | 66.05 | 66.59 ↑ | 52.38 | 67.02 ↑ | 64.44 | 70.42 ↑ | 68.41 | 69.24 ↑ | 68.94 | 70.22 ↑ |
| maggic (n=20) | 67.41 | 57.13 | 60.70 | **61.48** ↑ | 57.54 | 58.69 ↑ | 52.75 | 54.51 ↑ | 54.59 | 55.39 ↑ | 50.29 | 55.64 ↑ | 54.72 | 57.38 ↑ | 55.84 | 56.15 ↑ | 54.08 | 56.19 ↑ |
| seer (n=20) | 87.92 | 80.67 | 84.53 | **84.82** ↑ | 83.34 | 83.71 ↑ | 74.34 | 78.73 ↑ | 80.59 | 80.60 ↑ | 47.57 | 74.43 ↑ | 76.06 | 79.98 ↑ | 79.23 | 80.02 ↑ | 74.53 | 78.73 ↑ |
| compas (n=20) | 67.51 | 63.11 | **68.01** | 67.91 | 62.07 | 64.43 ↑ | 55.67 | 62.56 ↑ | 57.67 | 60.87 ↑ | 53.33 | 63.59 ↑ | 59.49 | 64.62 ↑ | 61.06 | 61.59 ↑ | 58.30 | 62.58 ↑ |
| adult (n=20) | 84.17 | 77.45 | 50.39 | 71.48 ↑ | 49.23 | 72.37 ↑ | 72.23 | 76.86 ↑ | 74.35 | 75.04 ↑ | 67.00 | 77.25 ↑ | 67.46 | 76.48 ↑ | 73.75 | 73.67 | 73.20 | 76.90 ↑ |
| drug (n=20) | 77.81 | 70.84 | 75.08 | **75.29** ↑ | 71.68 | 72.14 ↑ | 68.31 | 72.65 ↑ | 68.12 | 69.68 ↑ | 58.78 | 68.89 ↑ | 62.13 | 67.75 ↑ | 70.16 | 70.16 | 66.60 | 69.18 ↑ |
| covid (n=40) | 74.41 | 70.77 | 73.40 | **73.95** ↑ | 70.42 | 71.93 ↑ | 63.63 | 68.46 ↑ | 70.50 | 70.44 | 56.50 | 68.68 ↑ | 66.41 | 70.48 ↑ | 68.66 | 68.44 | 61.03 | 67.35 ↑ |
| cutract (n=40) | 72.23 | 69.18 | 69.87 | **71.72** ↑ | 68.47 | 69.56 ↑ | 63.01 | 67.87 ↑ | 65.63 | 67.27 ↑ | 54.39 | 68.44 ↑ | 61.40 | 67.98 ↑ | 67.86 | 67.95 ↑ | 59.79 | 66.62 ↑ |
| maggic (n=40) | 67.41 | 58.26 | 59.29 | **60.77** ↑ | 57.50 | 59.15 ↑ | 55.00 | 56.78 ↑ | 55.24 | 56.94 ↑ | 48.81 | 56.64 ↑ | 54.68 | 58.58 ↑ | 57.40 | 57.44 ↑ | 55.04 | 57.33 ↑ |
| seer (n=40) | 87.92 | 82.93 | 84.29 | **84.93** ↑ | 83.46 | 84.44 ↑ | 80.05 | 83.67 ↑ | 82.59 | 81.37 | 54.93 | 81.11 ↑ | 79.88 | 84.36 ↑ | 80.79 | 82.21 ↑ | 78.69 | 83.62 ↑ |
| compas (n=40) | 67.51 | 62.34 | 67.57 | **67.85** ↑ | 61.34 | 62.84 ↑ | 56.29 | 61.02 ↑ | 58.85 | 60.11 ↑ | 58.88 | 64.37 ↑ | 58.61 | 63.54 ↑ | 60.83 | 60.95 ↑ | 55.94 | 61.04 ↑ |
| adult (n=40) | 84.17 | 79.44 | 48.31 | 73.82 ↑ | 49.21 | 74.27 ↑ | 71.82 | 79.11 ↑ | 71.51 | 77.99 ↑ | 66.77 | 78.81 ↑ | 71.13 | 79.71 ↑ | 77.90 | 78.84 ↑ | 72.58 | **80.02** ↑ |
| drug (n=40) | 77.81 | 71.86 | 74.30 | **75.79** ↑ | 71.33 | 72.76 ↑ | 69.46 | 72.74 ↑ | 71.08 | 73.07 ↑ | 64.89 | 73.64 ↑ | 62.51 | 70.97 ↑ | 69.23 | 69.78 ↑ | 65.22 | 70.30 ↑ |
| covid (n=100) | 74.41 | 71.57 | 73.77 | **74.71** ↑ | 70.71 | 72.76 ↑ | 69.05 | 72.13 ↑ | 71.60 | 73.22 ↑ | 63.52 | 72.04 ↑ | 64.25 | 72.64 ↑ | 70.08 | 70.78 ↑ | 69.05 | 71.96 ↑ |
| cutract (n=100) | 72.23 | 70.96 | 70.20 | **72.51** ↑ | 69.97 | 71.94 ↑ | 67.94 | 72.42 ↑ | 70.53 | 71.98 ↑ | 55.72 | 69.14 ↑ | 67.59 | 72.42 ↑ | 68.79 | 69.68 ↑ | 66.89 | 71.52 ↑ |
| maggic (n=100) | 67.41 | 59.65 | 58.98 | **61.32** ↑ | 55.71 | 58.90 ↑ | 57.20 | 59.34 ↑ | 57.26 | 58.28 ↑ | 49.54 | 57.91 ↑ | 56.36 | 60.11 ↑ | 58.89 | 58.99 ↑ | 56.17 | 58.86 ↑ |
| seer (n=100) | 87.92 | 83.95 | 84.45 | **85.37** ↑ | 83.92 | 85.08 ↑ | 81.60 | 85.14 ↑ | 83.04 | 84.83 ↑ | 70.32 | 83.83 ↑ | 81.16 | 85.03 ↑ | 81.82 | 82.49 ↑ | 78.88 | 84.50 ↑ |
| compas (n=100) | 67.51 | 62.56 | 68.02 | **68.19** ↑ | 60.10 | 62.47 ↑ | 60.01 | 63.73 ↑ | 58.32 | 61.34 ↑ | 59.97 | 64.19 ↑ | 60.02 | 64.04 ↑ | 61.44 | 61.73 ↑ | 59.97 | 62.82 ↑ |
| adult (n=100) | 84.17 | 81.24 | 46.09 | 74.57 ↑ | 47.56 | 73.97 ↑ | 74.29 | 80.45 ↑ | 75.93 | 78.22 ↑ | 77.09 | **81.66** ↑ | 70.70 | 81.04 ↑ | 80.56 | 81.10 ↑ | 74.04 | 80.23 ↑ |
| drug (n=100) | 77.81 | 73.58 | 76.24 | **76.74** ↑ | 69.46 | 71.05 ↑ | 68.19 | 73.28 ↑ | 72.43 | 73.79 ↑ | 67.26 | 75.28 ↑ | 62.67 | 73.12 ↑ | 70.90 | 71.53 ↑ | 68.22 | 73.59 ↑ |
| covid (n=200) | 74.41 | 72.33 | 73.40 | **74.62** ↑ | 70.70 | 73.12 ↑ | 71.07 | 73.89 ↑ | 72.47 | 74.44 ↑ | 65.55 | 73.07 ↑ | 65.04 | 72.90 ↑ | 71.68 | 71.87 ↑ | 67.89 | 72.38 ↑ |
| cutract (n=200) | 72.23 | 71.75 | 71.39 | 73.01 ↑ | 70.28 | 72.39 ↑ | 69.28 | 72.41 ↑ | 71.83 | **74.03** ↑ | 66.66 | 72.49 ↑ | 68.77 | 73.16 ↑ | 70.23 | 70.80 ↑ | 66.61 | 71.87 ↑ |
| maggic (n=200) | 67.41 | 61.39 | 58.92 | **61.41** ↑ | 57.33 | 60.16 ↑ | 58.48 | 61.33 ↑ | 56.26 | 57.20 ↑ | 50.74 | 59.60 ↑ | 55.95 | 60.75 ↑ | 60.73 | 60.78 ↑ | 57.18 | 60.23 ↑ |
| seer (n=200) | 87.92 | 84.63 | 84.39 | 85.56 ↑ | 83.48 | 84.80 ↑ | 82.04 | 85.34 ↑ | 84.39 | **86.57** ↑ | 82.15 | 86.03 ↑ | 77.73 | 85.19 ↑ | 83.38 | 84.15 ↑ | 79.71 | 85.26 ↑ |
| compas (n=200) | 67.51 | 63.27 | 67.02 | **68.15** ↑ | 60.48 | 63.39 ↑ | 60.58 | 64.32 ↑ | 60.60 | 63.52 ↑ | 61.11 | 65.08 ↑ | 56.58 | 63.60 ↑ | 61.99 | 62.80 ↑ | 60.15 | 63.99 ↑ |
| adult (n=200) | 84.17 | 82.12 | 40.96 | 75.84 ↑ | 49.89 | 76.11 ↑ | 78.18 | 82.32 ↑ | 81.66 | 83.17 ↑ | 80.06 | **83.32** ↑ | 74.31 | 82.64 ↑ | 82.26 | 82.39 ↑ | 75.21 | 82.02 ↑ |
| drug (n=200) | 77.81 | 76.10 | 75.58 | 76.06 ↑ | 70.66 | 72.81 ↑ | 71.31 | 75.98 ↑ | 69.61 | 71.79 ↑ | 72.35 | **77.41** ↑ | 65.25 | 75.26 ↑ | 74.38 | 74.78 ↑ | 68.39 | 74.33 ↑ |

(TSTR) [50]. Additionally, we compare the performance to training on the small $\mathscr{D}_{\text{train}}$, as well as training on the large held-out $\mathscr{D}_{\text{oracle}}$, the latter serving as an upper bound.

**GPT-4 + Curation has the best overall performance.** Table 4.3 shows the performance of the proposed CLLM (GPT-4 and GPT-3.5) vs baselines — both *with* and *without* our curation mechanism. We find that the GPT-4 + Curation variant of CLLM outperforms baselines in almost all settings (20/28). Interestingly, its performance is close to or even exceeds the performance of $\mathscr{D}_{\text{oracle}}$. Table 4.4 further shows that GPT-4 + Curation ranks first on average vs all generative methods.

**Sample size sensitivity.** We now investigate the performance gains of CLLM as we vary the number of samples $n$ in $\mathscr{D}_{\text{train}}$, in Table 4.3 and Table 4.4. Performance improvements and high ranking across datasets for CLLM (GPT-4+Curation) are especially noticeable in the low-data regime (i.e. $n < 100$). In this regime, the limited size of $\mathscr{D}_{\text{train}}$ severely constrains the other baseline methods. In contrast, as illustrated in Chapter 4.2.1, CLLM can

Table 4.4 Average rank of approaches across the different datasets and seeds. CLLM w/ GPT-4 ranks first across all $n$, and curation improves all the generative models.

| Method | n=20 | n=40 | n=100 | n=200 |
|---|---|---|---|---|
| CLLM w/ GPT-4 | **2.71 ± 1.44** | **2.14 ± 1.06** | **2.29 ± 1.19** | **3.29 ± 1.38** |
| GPT-4 | 3.86 ± 1.73 | 4.29 ± 1.83 | 6.00 ± 1.77 | 7.57 ± 1.65 |
| CLLM w/ GPT-3.5 | 4.14 ± 0.94 | 4.14 ± 0.71 | 6.86 ± 1.24 | 7.57 ± 0.70 |
| NFLOW (curated) | 6.00 ± 1.21 | 4.71 ± 0.80 | 4.00 ± 0.57 | 4.71 ± 0.63 |
| GPT-3.5 | 6.71 ± 1.52 | 7.29 ± 1.26 | 11.57 ± 0.94 | 12.57 ± 0.57 |
| TVAE (curated) | 7.14 ± 1.17 | 7.86 ± 1.30 | 6.43 ± 0.40 | 6.71 ± 0.52 |
| SMOTE (curated) | 7.71 ± 0.33 | 8.14 ± 0.91 | 7.71 ± 1.19 | 7.43 ± 1.07 |
| SMOTE | 7.86 ± 0.55 | 9.57 ± 0.80 | 9.57 ± 1.09 | 9.00 ± 1.03 |
| TabDDPM (curated) | 8.29 ± 0.98 | 8.00 ± 0.93 | 6.00 ± 0.95 | 5.14 ± 1.68 |
| CTGAN (curated) | 8.29 ± 1.42 | 7.14 ± 0.91 | 4.14 ± 0.62 | 3.71 ± 0.39 |
| GReaT (curated) | 8.57 ± 1.50 | 6.57 ± 1.21 | 6.29 ± 1.38 | 3.57 ± 0.92 |
| TabDDPM | 10.14 ± 1.19 | 9.86 ± 1.15 | 10.00 ± 1.03 | 10.29 ± 1.02 |
| TVAE | 12.14 ± 0.89 | 14.00 ± 0.70 | 13.71 ± 0.39 | 14.43 ± 0.40 |
| NFLOW | 12.86 ± 0.47 | 14.14 ± 0.37 | 14.00 ± 0.45 | 15.29 ± 0.33 |
| CTGAN | 13.86 ± 0.68 | 13.14 ± 0.47 | 12.86 ± 0.37 | 12.00 ± 0.53 |
| GReaT | 15.71 ± 0.26 | 15.00 ± 0.53 | 14.57 ± 1.03 | 12.71 ± 0.96 |

leverage GPT-4's prior knowledge to extrapolate beyond the small $\mathscr{D}_{\text{train}}$, thereby improving downstream performance. As expected, the performance gap between CLLM and other methods decreases as the size of $\mathscr{D}_{\text{train}}$ grows (e.g. $n = 200$), where sufficient training data helps other generators achieve good performance. We further decouple the prior knowledge of the LLM and the number of in-context samples in Appendix 4.3.3, showing the importance of the in-context samples to guide the LLM's generation.

**Curation generally helps all generative models.** Our curation mechanism consistently benefits all generative models for the different $n$. It ensures that only high-quality samples are retained, which is crucial for good data augmentation and downstream performance and has been overlooked in previous works. This explains why the combination of the best generative model and curation (CLLM) gives the best results and highest rankings in the low-data regime (e.g. $n = 20$). In addition to GPT-4 and GPT-3.5, we show the versatility of our proposed curation mechanism to provide benefit with other open-source LLM backbones, including Mistral-7b [96], LLAMA-2 [241] and Mixtral [97] (see Appendix C.3.1).

**Performance benefits maintained for private and public datasets.** One may hypothesize that the strong LLM (e.g. GPT-4) performance is explained by datasets being part of the LLMs' training corpus, hence possibly being memorized. We show in Table 4.3 that it is unlikely, as we retain strong performance for both open-source datasets, as well as

private medical datasets which require authorization processes for access and are unlikely to be part of the LLM pretraining dataset.

*Remark on ICL vs fine-tuning.* Our results in Table 4.3 and Table 4.4 indicate that ICL is better than fine-tuning (GReaT baseline) in the low-data regime. This highlights the difficulty of fine-tuning in this regime, where it is easy to overfit to $\mathscr{D}_{\text{train}}$. As we increase the number of samples, this baseline, coupled with curation, improves to the level of CLLM.

### 4.3.2 Addressing potential biases from the LLM

There are many challenges that might arise from the use of LLMs. An example is potential biases from the LLM, which might stem from their training data and might affect our synthetic data generated by the LLM.

Let us first describe some considerations that highlight how a practitioner using CLLM can address this issue.

1. **Choice of LLM:** Different LLMs may exhibit different levels of bias due to their respective pretraining and alignment with human feedback. With CLLM, the practitioner has a lot of flexibility in the choice of the LLM. This is an important aspect when it comes to bias. The practitioner may want to minimize the risk of having bias in the augmented data by selecting one particular LLM for their task based on prior knowledge about different LLMs and their respective potential bias [58].

2. **Fairness analysis on the augmented data:** Practitioners can directly assess potential bias in the data generated by CLLM: (i) by doing a standard fairness analysis and computing different fairness metrics [48, 79, 149] on the generated data, as can be done more generally when building a model with any given dataset. (ii) If some bias is detected, the practitioner can use any off-the-shelf debiasing method [29, 54], or use a training objective to make the downstream model fair [20, 265]. Note that this approach is possible because CLLM is a data augmentation method, which gives a lot of flexibility for the practitioner to adjust the data according to their needs.

3. **CLLM can reduce representation bias:** While these aforementioned considerations involve choices external to CLLM, we also want to emphasize that our CLLM approach has the potential to already address bias indirectly. As shown in Figure 4.3, underrepresented groups in the population benefit the most from CLLM. This suggests that we could use CLLM to remove representation bias, by leveraging CLLM to augment $\mathscr{D}_{\text{train}}$ with synthetic data from underrepresented groups. Table 4.1 and Table 4.2 show that

data generated with CLLM better aligns with the ground-truth distribution compared to using a widely used conventional generative approach (TVAE).

4. **Curation aligns the feature/label relationships:** The curation step of CLLM can already help indirectly address issues of bias in feature-label relationships if training data $\mathscr{D}_{\text{train}}$ is unbiased, even if it is not its primary purpose. The curation mechanism discards samples in $\mathscr{D}_{\text{syn}}$ that do not obey the same feature/label relationship as in $\mathscr{D}_{\text{train}}$. The results shown in Figure 4.5 demonstrate that the curation step aligns the feature/label relationship between the curated set and the ground-truth distribution. Thus, assuming that $\mathscr{D}_{\text{train}}$ is not biased and has the correct $Y|X$, discriminatory correlations stemming from the LLM bias could be detected in the form of feature/label relationships which differ from those in the gold standard $\mathscr{D}_{\text{train}}$.

We now provide additional empirical evidence on how curation can help with bias in a synthetic setup.

**Synthetic setup**: We consider a synthetic setup where $X$ and $Y$ are random variables such that $X = [X_1, X_2]^\top \in \mathbb{R}^2$ denotes the features, and $Y \in \{0,1\}$ is the label. Furthermore, we let $X_1$ (the first coordinate of $X$) be a sensitive attribute. We then create a biased distribution such that a downstream predictor $\hat{Y}$ trained on this biased distribution will violate the fairness criterion of equalized odds [79].

Formally, we consider a mixture of Gaussians to define $X$, i.e. $X = ZA + (1-Z)B$ where $Z \sim Ber(1/2)$, $A \sim \mathscr{N}([-1.5, 0], I_2)$ and $B \sim \mathscr{N}([1.5, 0], I_2)$. Furthermore, we let $Y = Z$.

In order to introduce some bias in the data, we consider the variable $Y'$ such that $Y' = 1 - Y$ if $|X_1| > 2.5$, else $Y' = Y$.

Let $\hat{Y}$ denote a downstream predictor, supposedly trained on samples drawn from the distribution of $(X, Y')$ (instead of the ground-truth distribution of $(X, Y)$). Intuitively, for $y \in \{0,1\}$, because of the definition of $Y'$, we can expect $\mathbb{P}(\hat{Y} = y \mid |X_1| > 2.5, Y = y)$ to be much smaller than $\mathbb{P}(\hat{Y} = y \mid |X_1| \leq 2.5, Y = y)$, hence strongly violating equality of odds. To quantify that, we will compute the absolute equality of odds differences defined as

$$\Delta_{Y=1} = |\mathbb{P}(\hat{Y} = 1 \mid |X_1| \leq 2.5, Y = 1) - \mathbb{P}(\hat{Y} = 1 \mid |X_1| > 2.5, Y = 1)|$$

and

$$\Delta_{Y=0} = |\mathbb{P}(\hat{Y} = 0 \mid |X_1| \leq 2.5, Y = 0) - \mathbb{P}(\hat{Y} = 0 \mid |X_1| > 2.5, Y = 0)|$$

We investigate whether our curation mechanism can help address the bias. We generate a training dataset $\mathscr{D}_{\text{train}}$ of size $n = 20$, by sampling independent samples from the distribution of $(X, Y)$. We generate an augmented dataset $\mathscr{D}_{\text{syn}}$ of size 1000, by sampling from the biased

distribution of $(X, Y')$. We then curate $\mathscr{D}_{\text{syn}}$ using the curation mechanism of CLLM, and obtain $\mathscr{D}_{\text{curated}} \subset \mathscr{D}_{\text{syn}}$. Finally, we train three XGBoost models on each of the three datasets $\mathscr{D}_{\text{train}}$, $\mathscr{D}_{\text{syn}}$, and $\mathscr{D}_{\text{curated}}$.

**Results.** We evaluate the performance of these downstream models on a held-out $\mathscr{D}_{\text{test}}$.

We then report the average test accuracy, along with $\Delta_{Y=0}$ and $\Delta_{Y=1}$ for 10 different seeds in Table 4.5, which demonstrates that the curation mechanism helps ensure that the bias present in the augmented dataset $\mathscr{D}_{\text{syn}}$ does not propagate to the downstream model, as can be seen with the low values of $\Delta_{Y=0}$ and $\Delta_{Y=1}$ for the models trained on $\mathscr{D}_{\text{curated}}$.

Table 4.5 Curation can help address bias.

| **Dataset used for downstream training** | **Test accuracy (%)** (↑) | $\Delta_{Y=1}$ (↓) | $\Delta_{Y=0}$ (↓) |
|---|---|---|---|
| $\mathscr{D}_{\text{syn}}$ (Biased) | 76.7 | 0.89 | 0.90 |
| $\mathscr{D}_{\text{train}}$ (Unbiased) | 90.0 | 0.17 | **0.07** |
| $\mathscr{D}_{\text{curated}}$ | **91.3** | **0.13** | 0.08 |

**Beyond bias: factual incorrectness, reasoning failures, and model drift.** While the aforementioned discussion focuses on representational and statistical biases, a broader risk of black-box LLMs is that they may generate *factually incorrect* records (e.g., medically implausible combinations of covariates), exhibit *reasoning failures* that break domain constraints, or change behavior over time due to vendor updates (*model drift*). These risks are particularly salient for tabular generation because errors can appear "locally plausible" at the level of individual features, yet is globally inconsistent when considering clinical constraints or causal relationships.

CLLM partially mitigates these risks by design: (i) the prompt conditions generation on the dataset schema, feature semantics, and in-context demonstrations, and (ii) the curation stage empirically filters samples that do not preserve the observed $Y \mid X$ relationships in $D_{\text{train}}$, which can remove a subset of LLM-induced artifacts. However, curation is not a formal guarantee of factual correctness or fairness, especially when the training set is itself small or biased.

In practice, we recommend additional safeguards when deploying CLLM in high-stakes domains: (a) *schema and constraint validation* (hard rules; range checks; cross-field consistency constraints), (b) *distributional sanity checks* comparing marginals and pairwise dependencies between $D_{\text{train}}$ and $D_{\text{syn}}$, (c) *model/version pinning* with audit logs to ensure reproducibility under LLM updates, and (d) *human-in-the-loop review* for flagged slices or sensitive subpopulations.

### 4.3.3 Ablation: Decoupling prior knowledge and data model

The performance of CLLM can be attributed to two components: the background knowledge of the LLM and its capacity to build a strong data model. In this subsection, we provide insights to understand the effect of the LLM's background knowledge (e.g. prior).

Table 4.6 Downstream accuracy when varying the number of in-context samples in the prompt to generate the augmented datasets.

| In-context samples | Downstream accuracy |
|---|---|
| $n = 1$ (Prior) | $70.20 \pm 1.60$ |
| $n = 20$ | $73.87 \pm 0.50$ |
| $n = 40$ | $73.95 \pm 0.67$ |
| $n = 100$ | $74.71 \pm 0.34$ |
| $\mathscr{D}_{\text{oracle}}$ | $74.6 \pm 0.15$ |

We consider the Covid dataset (private medical dataset, to avoid memorization issues) and generate synthetic data with GPT-4. We ablate the prompt used in our work (detailed in Appendix C.2.6) and solely provide one in-context example in the prompt in order to give the LLM the minimal amount of information about the desired structure of the dataset. This forces the LLM to rely on its own prior (background knowledge), and removes the effect of in-context examples, which could be used to build a data model. We report the results for the prior and CLLM in Table 4.6.

From these results, we conclude that the LLM prior permits to obtain good downstream performance but is outperformed by $\mathscr{D}_{\text{oracle}}$ by a margin of 4.4%. Hence, we cannot solely rely on the prior. Furthermore, downstream performance increases as the number of in-context samples increases. This shows it is important to include in-context samples if we wish to obtain downstream performance close to $\mathscr{D}_{\text{oracle}}$, as the LLM can build a good data model. This implies that while the LLM uses background knowledge of similar datasets, it still needs in-context samples to refine its prior by creating a good data model.

We then quantify and visualize the strength of the prior by studying how much the LLMs' output distribution adapts to the in-context samples provided. We evaluate data generated by the prior of the LLM ($n = 1$), and for $n = 20, 40, 100$ on the Covid dataset. In particular, we observe in Figure 4.6 that there is a region in the oracle data that is not captured by the LLM's prior output (the left part of the leftmost blob, circled in blue in Figure 4.6). However, as the number of in-context real examples increases in the prompt of the LLM, we observe that this steers the LLM to generate data covering this region.

This region is associated with the subgroup of people older than 87 years old, and having many severe co-morbidities (e.g. Diabetes, Cardiovascular diseases) and many respiratory symptoms. This subgroup in the Oracle dataset represents less than 3.5% of the data and

is completely ignored by the GPT-4 prior. In particular, the prior defaults to more typical patients in the range 70-80 years old.

On the contrary, as $n$ increases, the LLM is guided by the in-context samples and generates samples from this subgroup, which are "rarer" or different from the general population.

This shows that the LLM captures the distinct features of this particular region and is not overwhelmed by the prior. Instead, the data in the form of in-context samples adapts it and aligns the augmented dataset with the ground-truth distribution.

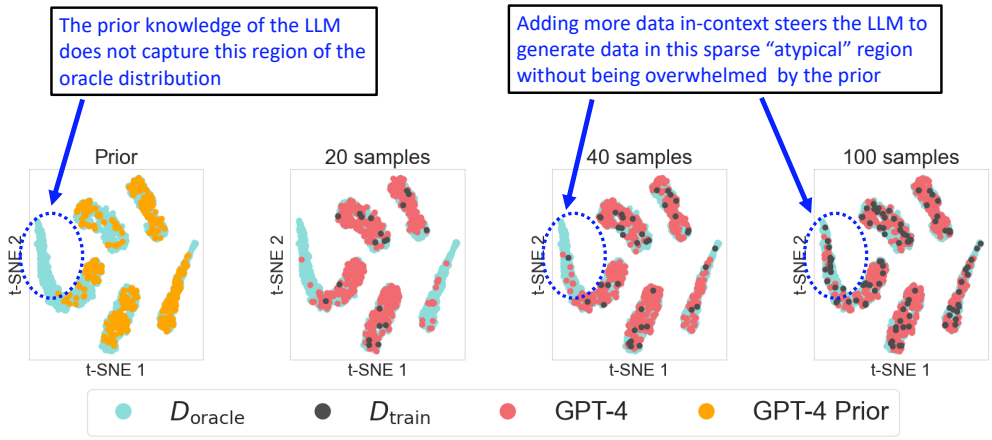



Fig. 4.6 The data generated by the LLM captures the distinct features in "atypical" regions of the Oracle manifold, as in-context samples are added to the prompt. This shows that it is flexible enough to adapt its prior knowledge to the nuances of the data. The group encircled in blue represents patients who are $>$ 88 years old, representing around 3.5% of the Oracle. This illustrates that the added in-context samples can successfully guide the LLM to generate these rare samples.

### 4.3.4 Hardness: a proxy signal to flag poor quality synthetic datasets

Having a systematic way to assess datasets generated by LLMs is essential because their black-box nature provides little control over their generation quality. This contrasts with conventional generators for which training loss is an exploitable signal. Hence, we ask: could we have a signal to identify a potential problematic dataset generated by an LLM without an exhaustive manual review? For example, GPT-4 produced low-quality synthetic data for the Adult dataset (across the different sample sizes), resulting in poor downstream performance. Although curation improves it, downstream performance is still suboptimal.

Addressing this question is important since datasets are rarely created by the ML model builder in real-world ML workflows but rather by specialist data teams or data owners

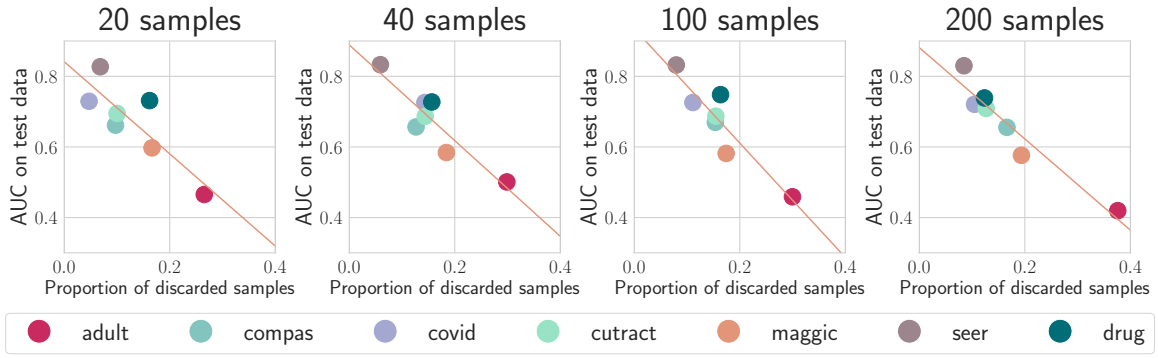



Fig. 4.7 The proportion of discarded samples is a proxy for test performance. This negative linear relationship where each point is a synthetic dataset generated by GPT-4 (e.g. Adult, Covid, Compas) allows us to flag datasets that will lead to unreliable downstream performance.

[61, 204, 68]. Thus, having a signal to preemptively flag a potentially suboptimal generated dataset spares investment in both storing the subpar data and/or training a model likely to underperform on real data.

To address this, we posit that $\mathscr{D}_{\text{syn}}$ should intuitively be considered imperfect if curation discards many of its samples, since the number of discarded samples measures the quality of samples with respect to the small but gold-standard $\mathscr{D}_{\text{train}}$. Hence, we investigate the relationship between test performance (AUC) and the proportion of samples discarded by curation. Figure 4.7, where each point is a synthetic dataset generated by GPT-4 (e.g. Adult, Compas), shows a strong negative linear relationship between these two quantities. This holds for different $n$ with slopes fairly stable around $-1.4$. This relationship corroborates the poor quality of the dataset generated by GPT-4 on the Adult dataset, providing a useful proxy that $\mathscr{D}_{\text{syn}}$ is unlikely to lead to good downstream performance.

## 4.4 Discussion

In this chapter, we introduce CLLM, an approach for synthetic data augmentation in the low-data setting. CLLM exploits the prior knowledge of LLMs along with our curation mechanism for improved downstream performance.

As empirically shown, CLLM outperforms traditional generative models—most noticeably on underrepresented subgroups, for which data augmentation is of utmost importance. CLLM is grounded in the ICL capability of LLMs. Further improvements may be achieved through different tuning and prompting of the LLM, as shown in different domains [150, 141].

Improving LLM tuning and prompting is beyond the scope of our work, but we regard this as a promising avenue for future work.

While the key contribution of this work is the curation of LLM outputs, overall downstream performance gains are still fundamentally tied to the LLM backbone. Specifically, a practical consideration is that using less parameterized LLMs leads to poorer uncurated data. That said, our curation mechanism naturally addresses this aspect and improves downstream performance (see Appendix C.3.1).

It is important to note, however, that the curation mechanism itself carries an analogous risk. Because curation discards synthetic samples classified as low-confidence and low-uncertainty — i.e., samples the model trained on $D_{\text{train}}$ predicts incorrectly yet with high certainty — there is a possibility that synthetic samples representing rare but valid minority presentations are disproportionately discarded. This may occur if $D_{\text{train}}$, being small, provides insufficient coverage of a minority subgroup's feature space for the base model to recognise the generated samples as plausible. In such cases, curation may reinforce rather than remedy the scarcity of minority data. Practitioners should therefore monitor the demographic composition of $D_{\text{curated}}$ relative to $D_{\text{syn}}$, particularly in healthcare applications where subgroup-level performance disparities are consequential. Future work could explore group-aware curation criteria that apply differential thresholds for rare subgroups, ensuring that diversity is preserved alongside quality.

Finally, while CLLM addresses the data logjam issue, increasing access to ML across regions, domains, and societies is also about more than just technology. We believe broader engagement and discussion with various stakeholders is crucial to responsibly expand ML access, thereby realizing the benefits of ML in an equitable way.

# Chapter 5

# Context-Aware Model Evaluation and Testing

## 5.1 Introduction

The ability to rigorously test and validate machine learning (ML) models is crucial for their reliable deployment in real-world applications. Despite solid aggregate performance, ML models are unreliable in a variety of real-world scenarios, such as on different subgroups [158, 236, 66, 28, 242, 216], or encountering data deviating from its training distribution [175, 118, 172, 67], often resulting in significant financial or societal consequences due to model unreliability. These failures point to deficiencies in the way we test such ML models. To understand this in greater detail, let us address two questions: how are we currently testing ML models and can we do better?

**How are we testing?** The predominant *de facto* paradigm of testing ML models relies on using only held-out data to evaluate the model either on average or by assessing the performance of models on different subgroups within the dataset. Such *data-only methods* optimize a given objective function to find subgroups (or slices of data) within the dataset where a trained model underperforms relative to aggregate performance. We refer to this relative underperformance as a *model failure*. However, data-only methods operate under the restrictive assumption that the available empirical data is the sole input for testing ML model performance. In practice, this is almost always violated. It is common to have *a priori* knowledge of where models are likely to fail, given the problem context.

The restrictive assumption of data-only methods comes at a significant cost. Specifically, data-only methods, which have been the go-to approach for ML testing, are required to iterate and test the performance across a large number of possible subgroups. Each subgroup

test is equivalent to evaluating a separate hypothesis about the model's performance on that specific slice of data. This raises a subtle but important problem—from the perspective of hypothesis testing, such methods *implicitly test multiple hypotheses* (Chapter 5.3.1). The more subgroups we test, the more hypotheses we implicitly evaluate. Consequently, this results in critical problems associated with multiple testing: (i) ***high false positive*** and (ii) ***high false negative rates*** (Chapter 5.3.2). A third complementary challenge is that each data slice tested is (iii) ***not necessarily practically meaningful***. The practical implications of these drawbacks are quite concerning—they hinder our ability to accurately identify where the model performs poorly, thereby undermining the reliability of our model evaluations.

**Can we do better?** To address these three challenges, we propose to *loosen the restrictive assumption of reliance only on data* and propose a new paradigm of testing called Context-Aware Testing (CAT) to offer an alternative to the dominant data-only paradigm (Chapter 5.3.3). Specifically, CAT provides a principled approach to incorporate external knowledge (or *context*) into the ML testing process. With the view that each evaluated data slice corresponds to an implicit hypothesis test, we propose that ML evaluation on observational datasets can be achieved via a *context-guided sampling mechanism* (Definition 5.3.1). This sampling procedure uses context as an inductive bias to prioritize specific data slices to test for those with a higher chance of surfacing meaningful model failures. Therefore, CAT fundamentally helps to answer the question of "what should we test for?".

To anchor ideas, let's consider an example of building an ML model to predict prostate cancer [47]. Data-only methods employ a search procedure over the dataset to find divergence across a large number of possible feature combinations which may lead to (i) ***high false positive rates*** by identifying spurious underperforming subgroups (e.g. based on eye color or patient ID); (ii) ***high false negative rates*** by failing to identify true underperforming subgroups due to the large number of combinations tested and applied testing corrections; and (iii) ***testing subgroups which are not practically meaningful*** (e.g. interaction between eye color and height). In contrast, a CAT-based approach would define and target task-relevant subgroups, limiting the number of tests conducted with better false positive control and greater statistical power. As we empirically show in Chapter 5.5, obtaining many false positives and false negatives is overwhelmingly common in current testing practices.

In bringing CAT to reality, we develop the framework called **SMART** [1] **Testing**, which performs automated ML model evaluation by actively identifying potential failure cases (Chapter 5.4). SMART uses large language models (LLMs) to generate contextually relevant failure hypotheses to test. We further introduce a *self-falsification mechanism* to automatically validate the generated failure hypotheses using the available data, allowing efficient pruning

[1] systematic, modular, automated, requirements-responsive, transferable

of spurious hypotheses. Finally, SMART generates comprehensive model reports that provide insights into the identified failure modes, their impact, and potential root causes, enabling stakeholders to make informed decisions.

**Contributions.** ① We identify critical gaps in the predominant data-only ML testing paradigm, illustrating they miss important dimensions. ② We formalize the *Context-Aware Testing* paradigm, providing a principled framework to incorporate context in addition to data into the testing process, which is then used to guide the generation of targeted tests. ③ We propose the first context-aware testing system, *SMART* Testing, which employs LLMs to hypothesize likely and relevant model failures and empirically refutes them with data using a novel *self-falsification mechanism*. ④ We demonstrate the value of context for effective testing, challenging the de facto data-only paradigm by showing how SMART identifies impactful model failures while avoiding false positives across diverse settings when compared to data-only testing. Additionally, SMART identifies failures in important societal groups and generates comprehensive model reports.

## 5.2 Related work

To highlight the need for SMART Testing, we contrast it with other alternative ML testing paradigms, specifically, data-only testing methods that address the same testing problem as SMART. We provide an overview in Table D.1 and an extended discussion in Appendix D.1.

***Data-only testing methods*** [39, 200, 162, 171]: Address the question: "what should we test?". Data-only methods search the data to find "slices" where the model's predictions underperform compared to average performance, deeming those slices as model failures. Although automated, data-only approaches operate *only* on raw data without accounting for the problem context. Consequently, they must search across a large space of potential failures, usually covering all subsets of features and their distinct values. While an exhaustive search may seem beneficial, as we show in Chapter 5.3, the reality is that performing many tests on a finite dataset risks discovering slices where model failure is irrelevant or due to random variability, i.e. the multiple testing problem. SMART Testing addresses this challenge by prioritizing relevant and likely model failures through contextual awareness.

***Orthogonal testing dimensions*:** While SMART addresses what to test, several other dimensions of model testing exist that are orthogonal to our approach (detailed in Appendix D.1). (i) Behavioral Testing [195, 198, 242] evaluates model behaviors (i.e. responses to data) by operationalizing tests along pre-defined dimensions (often defined by humans) — rather than discovering the test dimensions. SMART fundamentally differs by addressing

the core issue of "what to test". (ii) Software functional testing [37, 225], aims to primarily test functional correctness (e.g. input-output functionality such as monotonicity), rather than testing for failures. In addition, test cases are either pre-specified or specified with an approximation of the underlying model to probe for functional correctness.

## 5.3 A context-aware testing framework for ML

The prevailing paradigm for testing ML models relies on *data-only* methods, which exclusively use data to surface model failures. We first explore the limitations of data-only methods by analyzing ML testing through the lens of a multiple-hypothesis testing problem. Specifically, we highlight why data-only methods are uniquely prone to finding *false positive* and *false negative* model failures. To address this, we introduce a new paradigm of testing called **context-aware testing**, which relies on external knowledge, or *context*, as an inductive bias to better identify where models fail.

### 5.3.1 A multiple hypothesis testing view of ML evaluation

**Preliminaries.** We denote the feature space as $\mathscr{X}$, the label space as $\mathscr{Y}$, and $\mathbb{P}$ as the joint probability distribution over $\mathscr{X} \times \mathscr{Y}$. We wish to test a fixed, trained black-box model $f : \mathscr{X} \to \mathscr{Y}$, using a finite dataset $\mathscr{D} \subset \mathscr{X} \times \mathscr{Y}$ usually split into $\mathscr{D}_{\text{train}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_{\text{train}}}$ and $\mathscr{D}_{\text{test}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_{\text{test}}}$. As is common in ML, a validation set $\mathscr{D}_{\text{val}}$ is often split from the test set. We assume the existence of a loss function $\ell : \mathscr{Y} \times \mathscr{Y} \to \mathbb{R}$ that measures the discrepancy between the model's prediction and the true labels point-wise.

The primary goal of ML testing is to identify *meaningful failure modes*—subgroups (data slices) of the data distribution where the model's performance is significantly worse than its average behavior. Formally, let $\mathscr{S} \subseteq \mathscr{X} \times \mathscr{Y}$ denote a *data slice* and let $\mathbb{P}_{\mathscr{S}} := \mathbb{P}(\cdot \mid (X,Y) \in \mathscr{S})$ be the conditional distribution induced by $\mathscr{S}$. We aim to identify slices where the slice-specific expected loss $\mu_{\mathscr{S}} := \mathbb{E}_{(X,Y)\sim\mathbb{P}_{\mathscr{S}}}[\ell(f(X),Y)]$ significantly exceeds the population-level expected loss $\mu_{\mathbb{P}} := \mathbb{E}_{(X,Y)\sim\mathbb{P}}[\ell(f(X),Y)]$[2].

**Testing ML models is a multiple-hypothesis testing problem.** We are interested in identifying failure modes that generalize beyond the training dataset. We interpret the empirical dataset $\mathscr{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ as a sample from a population distribution $\mathbb{P}$ over $\mathscr{X} \times \mathscr{Y}$, and our goal is to make an *inferential* claim about slice performance with respect to $\mathbb{P}$. Suppose we have a candidate data slice $\widehat{\mathscr{S}} \subseteq \mathscr{D}$, i.e. the observed restriction of a population-

[2] While technically any subset of the dataset can be considered a data slice, we are practically interested in *meaningful* failures, such as model failures on vulnerable socio-economic groups.

level slice $\mathscr{S} \subseteq \mathscr{X} \times \mathscr{Y}$ to the sample $\mathscr{D}$. We compute the empirical slice loss

$$\widehat{\mu}_{\widehat{\mathscr{S}}} := \frac{1}{|\widehat{\mathscr{S}}|} \sum_{(\mathbf{x}_i, y_i) \in \widehat{\mathscr{S}}} \ell\big(f(\mathbf{x}_i), y_i\big)$$

and compare it to the empirical loss over the full dataset

$$\widehat{\mu}_{\mathscr{D}} := \frac{1}{|\mathscr{D}|} \sum_{(\mathbf{x}_i, y_i) \in \mathscr{D}} \ell\big(f(\mathbf{x}_i), y_i\big).$$

To make an inferential claim about the model's performance on the corresponding population slice, we formulate the hypothesis test

$$H_0 : \mu_{\mathscr{S}} = \mu_{\mathbb{P}} \qquad \text{vs.} \qquad H_1 : \mu_{\mathscr{S}} \neq \mu_{\mathbb{P}},$$

where $\mu_{\mathscr{S}}$ and $\mu_{\mathbb{P}}$ are the true slice-specific and population-level losses defined above (and the empirical quantities $\widehat{\mu}_{\widehat{\mathscr{S}}}$, $\widehat{\mu}_{\mathscr{D}}$ serve as their plug-in estimators). In practice, this can be done using an appropriate frequentist test, rejecting $H_0$ when $p < \alpha$ for a pre-specified significance level $\alpha \in (0, 1)$.

However, in realistic testing scenarios, we evaluate the model's performance not just on a single slice but on a large collection of candidates $\{\mathscr{S}_j\}_{j=1}^m$. This amounts to conducting many simultaneous hypothesis tests. Accounting for multiple testing is important. A naive testing procedure that does not adjust for multiplicity could surface a large number of spurious failure modes simply by chance (Type I error). Conversely, controlling the false discovery rate [19] might involve adjusting the per-test significance threshold to $\alpha' \ll \alpha$, potentially sacrificing power to detect true failures (Type II error).

The multiple hypothesis testing viewpoint reveals a key challenge in ML model evaluation: To reliably surface meaningful failures, we require a principled procedure for generating a relatively small number of promising hypotheses (candidate slices) to test.

### 5.3.2 The failures of data-only testing

Existing ML testing methodologies are *data-only* in that they only use the available empirical data to test for model failures and vary in their optimization objective [162, 39, 200, 131, 171]. However, data-only methods operate under the restrictive assumption that the available empirical data is the sole input for testing ML model performance. In practice, this is almost never the case. It is common to have an *a priori* understanding of where models are

likely to fail given the data distribution, model class, training algorithm, and deployment context. This restrictive assumption results in three challenges: ▶ **(i) High false positive rate**: data-only methods search over a large space of data slices and each evaluation amounts to an implicit hypothesis test (Chapter 5.3.1). Therefore, the probability of observing a false failure increases with every test performed. ▶ **(ii) High false negative rate**: The naive testing procedure can be made robust by correcting for the number of tests performed, which reduces the statistical power to detect true failures. ▶ **(iii) Lack of meaningful failures**: Data-only methods are fundamentally limited by the fact that not all statistically significant slices are practically meaningful. We empirically validate these claims in Chapter 5.5.

💡 The core limitation of data-only testing is the lack of a principled failure mode *discovery* procedure that can incorporate prior knowledge to guide the search for meaningful errors.

### 5.3.3 Formulating context-aware testing

To address the limitations of data-only testing, we introduce **Context-Aware Testing** (CAT), a principled framework for identifying meaningful model failures using context. This could be the context implicitly encoded in the dataset (i.e. via meaningful feature names) or available external input (i.e. external contextual knowledge, such as a string of input information from a human expert). Let $\mathscr{C}$ denote the space of all possible contextual information. We decompose context into two components: $\mathscr{C} = \mathscr{C}_e \times \mathscr{C}_D$, where $\mathscr{C}_e$ represents external contextual input (e.g., domain knowledge provided by a user, which may be empty, denoted $\emptyset \in \mathscr{C}_e$) and $\mathscr{C}_D$ represents dataset-derived context. A specific context instance is thus $c = (c_e, c_D) \in \mathscr{C}$. Our core insight is that we can use $\mathscr{C}$ as an inductive bias to help guide the selection of which slices to test for.

**Definition 5.3.1** (Context-Aware Testing)**.** Let $\mathscr{X}$, $\mathscr{Y}$, $\mathscr{P}$, $f$, $\mathscr{D} = \{(x_i, y_i)\}_{i=1}^N \in (\mathscr{X} \times \mathscr{Y})^N$, and $\ell$ be defined as in the standard supervised learning setup. Let $\mathscr{C}$ be a space of contexts.

Context-aware testing is then defined by two procedures:

1. A context-guided slice sampling mechanism $\pi : \mathscr{C} \times (\mathscr{X} \times \mathscr{Y})^N \times \mathbb{N} \to 2^{2^{\mathscr{X} \times \mathscr{Y}}}$ such that $\pi(c, \mathscr{D}, m) = \{\mathscr{S}_1, \ldots, \mathscr{S}_m\}$, where $\mathscr{S}_j \subseteq \mathscr{X} \times \mathscr{Y}$. Here, $c$ is used as an inductive bias for function $\pi$ to prioritize slices likely to contain meaningful failures, and $m \in \mathbb{N}$ is the number of slices to evaluate.

2. A multiple hypothesis testing procedure: $\forall \mathscr{S}_i \in \pi(c, \mathscr{D}, m)$, test $H_0 : \mu_{\mathscr{S}_i} = \mu_{\mathbb{P}}$ vs. $H_1 : \mu_{\mathscr{S}_i} \neq \mu_{\mathbb{P}}$, where $\mu_{\mathscr{S}_i} := \mathbb{E}_{(X,Y)\sim\mathbb{P}_{\mathscr{S}_i}}[\ell(f(X), Y)]$ and $\mu_{\mathbb{P}} := \mathbb{E}_{(X,Y)\sim\mathbb{P}}[\ell(f(X), Y)]$.

A meaningful failure is characterized by statistical significance and practical relevance.

The targeted sampling mechanism $\pi$ uses the context $\mathscr{C}$ as an inductive bias to prioritize the testing of slices that are (i) *more relevant* to the deployment context and (ii) *more likely* to exhibit significant performance gaps. This principled slice selection offers several key advantages over data-only methods: ▶ *(i) Improved false positive control*: by limiting the number of tests conducted to $m$, CAT controls the risk of spurious discoveries that arise when naively testing all possible slices. ▶ *(ii) Improved true positive rate*: The targeted selection of slices likely to contain failures maintains test power by avoiding the need to aggressively correct for multiple testing. ▶ *(iii) Meaningful failures*: context-guided sampling identifies failures that are both statistically significant and practically relevant.

Context-aware testing overcomes the limitations of data-only methods by employing a principled, context-guided slice sampling mechanism $\pi$ to prioritize the discovery of meaningful model failures.

The core technical challenge in practically realizing CAT is the development of an effective context-conditional sampling algorithm $\pi$. Next, we propose a concrete instantiation of $\pi$ using an LLM to generate plausible failure modes and guide the testing of ML models.

## 5.4 SMART Testing

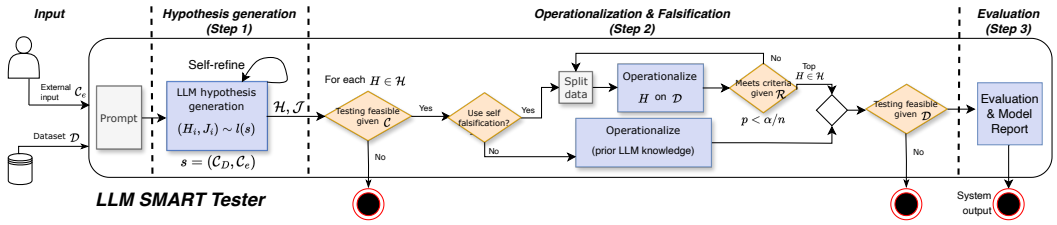



Fig. 5.1 Overview of the SMART Testing Framework showing the four steps. All steps are automatically executed by an LLM.

We now instantiate the CAT framework outlined in Chapter 5.3.3 with a method called **SMART** testing (**s**ystematic, **m**odular, **a**utomated, **r**equirements-responsive, and **t**ransferable). SMART generates relevant and likely hypotheses about potential model failures and empirically evaluates these hypotheses on available data. SMART follows a four-step procedure: (1) Hypothesis generation; (2) Operationalization; (3) Self-falsification; (4) Reporting. Table 5.1 provides an early illustration of what it practically means to generate hypotheses and test them with SMART. The procedure below details how the four steps are implemented.

Table 5.1 Example hypotheses on model failure, justifications, and operationalizations generated by the SMART framework on a healthcare dataset. The p-values show whether the model's performance significantly differs from average performance with $|\Delta Acc|$ measuring the effect size, i.e. the absolute difference between the model's accuracy on slice $\mathscr{S}$ and its overall accuracy on $\mathscr{D}$

| Hypothesis | Justification | Operationalization | p-value | $\lvert\Delta Acc\rvert$ | Evidence |
|---|---|---|---|---|---|
| $f$ underperforms on elderly patients. | Elderly patients may have more complex health situations due to age-related comorbidities, which could make predictions less accurate. | age >= 72 | 0.000 | 0.194 | Supported |
| $f$ underperforms on patients with multiple comorbidities. | The presence of multiple comorbidities could complicate the prediction model due to interactions between different health conditions. | comorbidities >= 2 | 0.900 | 0.0270 | Not supported |
| $f$ underperforms on patients undergoing conservative management. | Conservative management might be chosen for patients with more complex or less predictable cases, which could lead to worse predictive performance. | treatment_conservative _management == 1 | 0.000 | 0.200 | Supported |

**Step 1: Hypothesis Generation.** Recall from Chapter 5.3.3, we wish to define a sampling mechanism $\pi$ to sample slices $\mathscr{S}$ which are both relevant and have a high relative likelihood of failure — where $\pi$ should be both contextually-aware and able to integrate requirements to guide sampling.

We posit that LLMs have the potential to satisfy these properties due to the following capabilities: ▶ *Contextual understanding*: LLMs have been pre-trained with a vast corpus of information and hence have extensive prior knowledge around different contexts and settings [35, 227, 212, 10]. ▶ *Integrate requirements*: LLMs are adept at integrating requirements or additional information about the problem via natural language [261, 212]. ▶ *Hypothesis proposers*: In proposing likely failure modes, LLMs have also been shown to be "phenomenonal hypothesis proposers" [185].

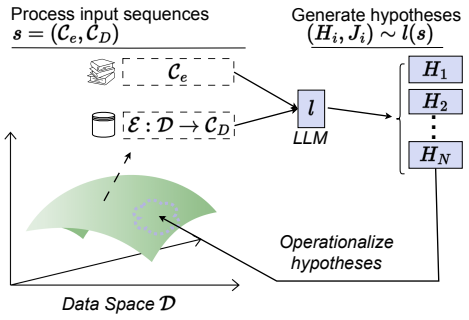



Fig. 5.2 SMART uses an LLM to integrate context $\mathscr{C}$, and data context $\mathscr{C}_D$. Relevant and likely failure hypotheses are then generated by LLM (i.e. sampling mechanism). The hypotheses are then operationalized in $\mathscr{D}$ and evaluated. In contrast, data-only methods are not guided by context and requirements, searching more exhaustively in $\mathscr{D}$ for divergent slices.

An LLM is defined as a probabilistic mapping $l : \Sigma^* \to P(\Sigma)$, where $\Sigma$ denotes the vocabulary space, and $P(\Sigma)$ represents the probability distributions over $\Sigma$. The model processes input sequences $s \in \Sigma^*$, each a concatenation of tokens representing external (contextual) input $c_e$ (which can be null) and dataset contextualization $c_D$, formalized as $s = (c_e, c_D)$. We extract the contextualized description $c_D$ from the dataset $\mathscr{D}$ using an extractor function $\mathscr{E} : \mathscr{D} \to \mathscr{C}_D$, such that $c_D = \mathscr{E}(\mathscr{D})$, which captures essential dataset characteristics (e.g. feature relationships and high-level dataset information). Additionally, we highlight that the LLM will implicitly extract context based on the context encoded in the dataset (e.g., via meaningful feature names). Based on input

$s$, the LLM predicts a distribution over $\Sigma$ from which hypotheses of model failure and corresponding justifications are sampled.

As depicted in Figure 5.2, given $\mathscr{C}_e$ and $\mathscr{C}_D$, we sample the $k$ most likely hypotheses of failures, $\mathscr{H} = \{H_1, H_2, \ldots, H_k\}$, and corresponding justifications $\mathscr{J} = \{J_1, J_2, \ldots, J_k\}$, to provide explainability. This process is formalized using the LLM's mapping $l$ as follows:

$$(H_i, J_i) \sim l(s), \text{where } s = (\mathscr{C}_e, \mathscr{C}_D),\ \forall i \in \{1, 2, \ldots, k\}.$$

**Step 2: Operationalization.** The process of operationalizing each hypothesis $H_i \in \mathscr{H}$ involves translating its natural language expression into a form that can directly operate on the training dataset $\mathscr{D}_{train}$ (an example is provided in Table 5.1). To achieve this, we define an interpreter function $I : \mathscr{H} \to (\mathscr{X} \to \{0,1\})$ that maps each natural language hypothesis $H_i$ to a corresponding binary-valued indicator function $g_i : \mathscr{X} \to \{0,1\}$ on the feature space $\mathscr{X}$, where $g_i(\mathbf{x}) = 1$ if $\mathbf{x}$ satisfies the criteria and $g_i(\mathbf{x}) = 0$ otherwise. Each function $g_i$ induces a data slice $\mathscr{S}_i \subseteq \mathscr{D}_{\text{train}}$ consisting of data points that satisfy the criteria of hypothesis $H_i$, such that $\mathscr{S}_i = \{(x,y) \in \mathscr{D}_{\text{train}} : g_i(x) = 1\}$. Therefore, each hypothesis, after operationalization, corresponds to a specific slice that is being tested. Steps 1 and 2 serve to practically instantiate $\pi$ from Chapter 5.3.3.

**Step 3: Self-falsification** We introduce a novel self-falsification mechanism to empirically evaluate (or *refute*) the generated hypotheses[3]. Specifically, for each *feasible* hypothesis $H_i \in \mathscr{H}$— i.e., one whose operationalization $g_i$ yields a non-empty slice $\mathscr{S}_i = \{(\mathbf{x}, y) \in \mathscr{D}_{\text{train}} : g_i(\mathbf{x}) = 1\} \neq \emptyset$ with sufficient sample size for statistical testing—we attempt to falsify the hypothesis with observed empirical data [4]. This involves evaluating the model $f$ over the data slice $\mathscr{S}_i$ operationalized from $H_i$. We then assess whether the slice performance on $f$ has a significant deviation from the model's overall performance. For instance, in Table 5.1, this is done by computing $|\Delta\text{Acc}|$ and the p-value. The significance of this deviation is determined through frequentist statistical testing, i.e., when $p < \alpha$ for any $\alpha$, which might also

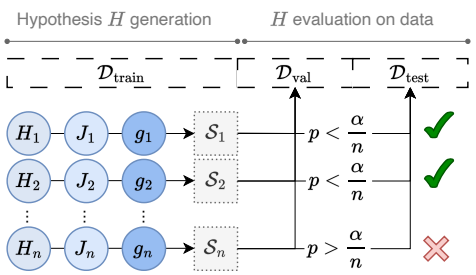



Fig. 5.3 A self-falsification module within the SMART framework. A hypothesis generator generates plausible hypotheses and justifications for when the model might fail. This is operationalized with $g_i$ and tested against the empirical data.

[3] self-falsification is rooted in scientific philosophy and the seminal ideas of Popper [182] who proposed the principle of falsification—the ability to refute hypotheses with empirical observations—as driving science.

[4] We assume data for self-falsification is available. If not, discovered slices could guide new data collection.

be adjusted for multiple hypothesis testing. This step effectively "reshuffles/reranks" the hypotheses based on their likelihood on the observed data. For example, when benchmarking, we select hypotheses based on statistical significance. We rank all hypotheses $H_i \in \mathscr{H}$ by their p-values $p_i$ in ascending order and select the top $k$ hypotheses satisfying $p_i < \alpha : \mathscr{H}^* = \{H_i \in \mathscr{H} : p_i < \alpha \text{ and } H_i \text{ is among the } k \text{ smallest p-values}\}$.

***Remark*:** As shown in Figure 5.1, we can exclude the self-falsification from SMART in cases of small sample sizes. We denote this ablation of SMART as $SMART_{NSF}$.

**Step 4. Model Performance Evaluation and Reporting.** Finally, SMART automatically generates a report of the overall performance of the model under varying conditions generated by the LLM, including a summary report, a complete summary of the tests carried out, intermediate and final results, and potential failure modes of the ML model.

## 5.5 Experiments

We now quantitatively evaluate SMART Testing and demonstrate the viability of this new ML testing paradigm (i.e. CAT), in contrast to data-only testing. We structure the experiments to highlight the different desired capabilities of ML testing.

**Baselines.** We compare SMART with a variety of data-only testing baselines namely: Autostrat [162], Slicefinder [39], Sliceline [200], Pysubgroup [131] and Divexplorer [171]. We also evaluate $SMART_{NSF}$ (i.e. no self-falsification) as an ablation. We exemplify the LLM in SMART by GPT-4 [161] and our ML model to audit is logistic regression. We investigate other LLMs in the Appendix D.4.1 and other tabular models in Appendix D.4.2, observing similar results. We provide additional experimental details in Appendix D.3.

### 5.5.1 Robustness to False Positives

**Goal.** We aim to underscore the role of contextual awareness in preventing false positives in model testing. In particular, we consider the scenario when dealing with tabular data that may contain many irrelevant or uninformative features [106], persisting even post-feature selection [245, 74]. We contrast SMART, which explicitly accounts for context, to data-only approaches, which are context-unaware and can only operate on numerical data.

**Setup.** We fit a predictive model to the training dataset, varying the number of irrelevant, synthetically generated features contained in the dataset. The irrelevant features are drawn from different distributions. We then quantify the proportion of conditions in the identified slices that *falsely* include the irrelevant synthetic features. We evaluate using five real-world tabular datasets spanning diverse domains, namely finance, healthcare, criminal justice, and

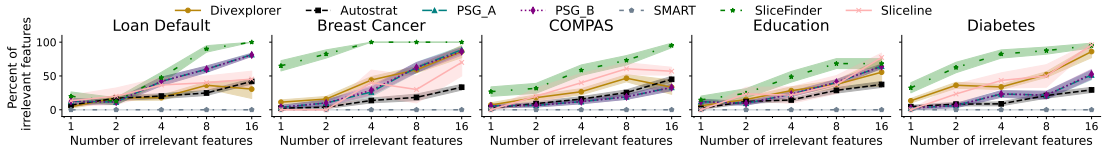



Fig. 5.4 Contextual-awareness in SMART reduces FPs, i.e. reducing the proportion of irrelevant features in slices. SMART is not sensitive to the number of irrelevant features, unlike data-only methods. ↓ is better

education: loan, breast cancer, diabetes, COMPAS recidivism [5], and OULAD education [123]. These datasets have varying characteristics, from sample size to number of features, and are representative of different contexts pertinent to tabular ML to demonstrate the effectiveness of SMART across various real-world contexts

**Analysis.** Figure 5.4 shows the proportions of irrelevant features included in slices for increasing numbers of irrelevant features. Data-only methods are unaware of context and are shown to spuriously include high proportions of irrelevant features in their slices; i.e. *false positives (FPs)*. Additionally, these false discoveries increase for data-only methods as the number of irrelevant features increases. Consequently, the sensitivity to FPs of data-only methods risks that slices identified are neither empirically relevant nor meaningful. In contrast, SMART, by virtue of contextual awareness when generating hypotheses, is *not* sensitive to extraneous and contextually irrelevant features. Importantly, SMART also maintains its robustness to FPs even with an increasing number of irrelevant features.

**Sample size sensitivity.** We also assess sensitivity to the number of data samples, varying the sample size in the training dataset and observing how that affects the number of slices discovered for each method. We show that SMART remains robust to variations irrespective of sample size. In contrast, data-only methods have variable performance with false discoveries sensitive to the sample size. The result is shown in Figure 5.5.

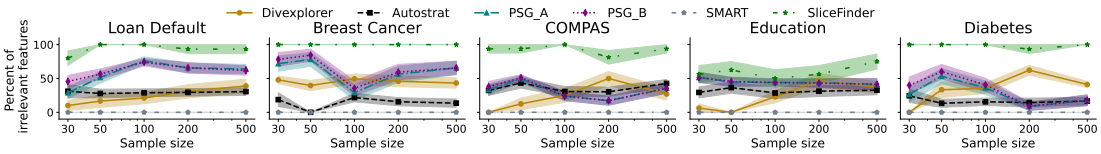



Fig. 5.5 Proportion of irrelevant features (y) for each slice discovery method, based on the sample size. Lower is better.

**Avoiding non-existent failure slices.** It is important that model testing does not flag failures when there are none. In Appendix D.4.3, we demonstrate that SMART's contextual awareness means that it is resistant to spurious failure slices that have no underlying relationships. In contrast, data-only approaches implicitly assume the existence of problematic

slices, which we empirically show makes them prone to spuriously flagging non-existent failure slices.

**Takeaway.** SMART's contextual awareness ensures testing relevance, thereby reducing false positives across different scenarios, in contrast to data-only methods.

### 5.5.2 Targeting model failures

**Goal.** We assess whether the identified failure slices persist when evaluated on new, unseen data across different tabular models. This evaluates the generalization of the identified model failures across multiple different ML models.

Table 5.2 Identifying slices with the highest performance discrepancies. We show differences in accuracies ($|\Delta Acc|$) between the top identified divergent slice and average performance across four classifiers (over 5 runs). ↑ is better.

| | **LogisticRegression** | **SVM** | **XGBClassifier** | **MLPClassifier** |
|---|---|---|---|---|
| Autostrat | $0.24 \pm 0.02$ | $0.24 \pm 0.02$ | $0.09 \pm 0.09$ | $0.24 \pm 0.02$ |
| *PSG_B* | $0.23 \pm 0.01$ | $0.23 \pm 0.01$ | $0.11 \pm 0.07$ | $0.23 \pm 0.01$ |
| *PSG_A* | $0.23 \pm 0.01$ | $0.23 \pm 0.01$ | $0.11 \pm 0.07$ | $0.23 \pm 0.01$ |
| Divexplorer | $0.05 \pm 0.11$ | $0.09 \pm 0.13$ | $0.14 \pm 0.15$ | $0.02 \pm 0.05$ |
| Slicefinder | $0.01 \pm 0.00$ | $0.01 \pm 0.00$ | $0.00 \pm 0.00$ | $0.01 \pm 0.01$ |
| Sliceline | $0.26 \pm 0.06$ | $0.26 \pm 0.06$ | $0.18 \pm 0.09$ | $0.26 \pm 0.06$ |
| $\text{SMART}_{\text{NSF}}$ | $0.17 \pm 0.01$ | $0.17 \pm 0.01$ | $0.09 \pm 0.05$ | $0.17 \pm 0.01$ |
| **SMART** | $\mathbf{0.37 \pm 0.03}$ | $\mathbf{0.37 \pm 0.03}$ | $\mathbf{0.26 \pm 0.06}$ | $\mathbf{0.37 \pm 0.03}$ |

**Setup.** We use the prostate cancer dataset [174] and aim to discover slices indicative of underperformance. Thereafter, we assess the generalizability of the identified failures across four different ML models (logistic regression, SVM, XGBoost, and MLP). Specifically, we compare the top identified slice from each method and compute the absolute difference between the accuracies of that slice and the remainder of the dataset. We posit that a testing framework should be able to identify slices with high-performance discrepancies on unseen data across multiple ML models.

**Analysis.** Table 5.2 shows that SMART slices exhibit the greatest discrepancies in model accuracies on unseen test data across different ML models, indicating that the discovered failure modes are generalizable. We find that SMART surfaces slices with greater performance differences compared to $SMART_{NSF}$ (ablation without self-falsification), highlighting the importance of the introduced self-falsification mechanism. In contrast, data-only methods fail to identify slices where the performance discrepancy is as large as SMART. This limitation can be attributed to the tendency of data-only methods to overfit the training data, leading to high false discovery rates.

**Takeaway.** SMART discovers failure slices where the model substantially underperforms, generalizing to unseen test data across different ML models. In contrast, data-only approaches fail to find slices where the difference in accuracies is as large, highlighting the lack of generalizability and reliability of their findings.

### 5.5.3 Robustness to False Negatives

In our setup, false negatives (FNs) are directly tied to true positives (TPs). i.e. the more true positives we find, the fewer FNs we miss. Across multiple experiments (Figure 5.4, Table 5.4, Table 5.2, Table D.9), we consistently show that SMART identifies TPs at substantially higher rates than data-only. For example, in Table D.9, SMART identifies an average of 9.6 out of 10 subgroups where the ML model significantly underperforms. The fact that data-driven methods discover fewer such subgroups implies that they are missing the ones SMART uncovers.

**Goal.** That said, we conduct an additional experiment to directly assess the false negative rate. We utilize a synthetic setup wherein we can control issues, as FNs are naturally unspecified in real data.

**Setup.** We simulate a dataset to predict recidivism ($Y$) based on five covariates: gender, race, age, income, and education. $\log\left(\frac{P(Y_i|X_i=j)}{P(Y_i\neq j)}\right) = \alpha_j - (\delta_1 X_{\text{age}} + \delta_2 C_{\text{income}} + \delta_3 C_{\text{education}}) + \varepsilon$, where $\varepsilon \sim \mathcal{N}(\mu_0, \sigma_0)$. We then train a predictor function $\hat{f}$ on the data and synthetically introduce underperformance on certain corrupted subgroups. For an individual $i$, if they belong to corrupted subgroup $j$, the prediction $\hat{Y}_i$ is equal to $\hat{f}(X_i)$ with probability $1-p$, and a random prediction sampled from a Bernoulli distribution with probability $p$. If the individual does not belong to subgroup $j$, the prediction is simply $\hat{f}(X_i)$. Finally, we measure how often each testing method identifies that the model $\hat{f}$ underperforms on a subgroup.

Table 5.3 False Negative Rate (FNR) for different methods at various settings. ↓ is better.

| | **FNR (n=1)** | **FNR (n=2)** | **FNR (n=3)** |
|---|---|---|---|
| Autostrat | $0.75 \pm 0.44$ | $0.88 \pm 0.22$ | $0.92 \pm 0.15$ |
| pysubgroup_beam | $0.65 \pm 0.49$ | $0.75 \pm 0.26$ | $0.68 \pm 0.17$ |
| pysubgroup_apriori | $0.65 \pm 0.49$ | $0.75 \pm 0.26$ | $0.68 \pm 0.17$ |
| Divexplorer | $0.05 \pm 0.22$ | $0.40 \pm 0.35$ | $0.72 \pm 0.31$ |
| Sliceline | $0.25 \pm 0.44$ | $0.50 \pm 0.36$ | $0.70 \pm 0.28$ |
| **SMART** | $\mathbf{0.00 \pm 0.00}$ | $\mathbf{0.05 \pm 0.15}$ | $\mathbf{0.38 \pm 0.27}$ |

**Analysis.** Table 5.3 demonstrates results where we synthetically manipulate/corrupt the performance of an ML model on a single subgroup (n=1), two subgroups (n=2), and three subgroups (n=3) out of a total of five. The results are averaged over 20 runs, where the

corrupted groups are randomly selected within each run. We find that SMART is consistently least susceptible to false negatives across all corrupted variables when compared to data-only methods, which struggle especially once more than one variable is corrupted. This serves to corroborate our earlier results.

**Takeaway.** SMART is less prone to FNs compared to data-only methods across all settings.

### 5.5.4 Requirements-constrained testing

**Goal.** Requirements are a crucial yet neglected part of ML model testing, such as verifying performance on societally relevant dimensions or verifying specific aspects to meet compliance requirements. However, *no previous testing framework* has incorporated the notion of satisfying requirements when defining the test slices. This experiment illustrates how SMART integrates requirements provided in natural language, which then influences the hypotheses generated to satisfy testing requirements.

**Setup**. We cover three real-world requirements that end-users might have: one based on demographics and two based on sample size. *Requirement 1:* Each of the top 10 identified unique slices should involve the age of a person. *Requirement 2:* The sample size of the top 10 identified unique slices must have at least 150 observations. *Requirement 3:* The sample size of the top 10 identified unique slices has to be small, within 10 and 150 observations. We exemplify the experiment using a real-world prostate cancer dataset from the UK [174], as healthcare often mandates certain testing requirements (e.g. Collins et al. [41]).

**Analysis**. Table 5.4 shows that SMART, which directly integrates requirements (via natural language), satisfies the requirements a greater number of times compared to data-only methods, which only satisfy requirements by chance (hence the low number of times). Beyond satisfying requirements, the SMART slices also represent model failures that almost always have statistically significant performance differences from average when evaluated on test data. Finally, while $SMART_{NSF}$ (ablation without self-falsification) can satisfy requirements, the number of statistically significant slices is lower than SMART, thus underscoring the value of our self-falsification mechanism. That said, $SMART_{NSF}$ still outperforms data-only baselines.

**Takeaway.** SMART, unlike data-only methods, identifies slices that have significant performance differences whilst also satisfying requirements — an important dimension not even considered by previous testing methods.

Table 5.4 Requirement satisfaction showing how many times the top 10 generated slices satisfied the requirements (Req) and how many slices had statistically significantly different performance from average (Sig ) on a testing dataset. The maximum is 10. ↑ is better.

| | $R_1$: Age | | $R_2$: Min sample size | | $R_3$: Max sample size | |
|---|---|---|---|---|---|---|
| | Req | Sig | Req | Sig | Req | Sig |
| Autostrat | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $1.00 \pm 0.00$ | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ |
| PSG_B | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ |
| PSG_A | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ |
| Divexplorer | $3.20 \pm 2.11$ | $0.95 \pm 1.56$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $2.70 \pm 2.57$ | $1.05 \pm 1.75$ |
| Slicefinder | $4.50 \pm 1.72$ | $1.75 \pm 0.99$ | $4.55 \pm 1.28$ | $3.05 \pm 0.80$ | $3.20 \pm 1.33$ | $1.10 \pm 0.94$ |
| Sliceline | $1.20 \pm 0.40$ | $1.20 \pm 0.40$ | $1.35 \pm 0.48$ | $1.35 \pm 0.48$ | $0.15 \pm 0.48$ | $0.15 \pm 0.48$ |
| **SMART**_NSF | $\mathbf{9.90 \pm 0.30}$ | $5.65 \pm 0.65$ | $6.00 \pm 0.00$ | $6.00 \pm 0.00$ | $4.15 \pm 0.65$ | $2.60 \pm 0.86$ |
| **SMART** | $9.70 \pm 0.56$ | $\mathbf{9.50 \pm 0.59}$ | $\mathbf{9.85 \pm 0.36}$ | $\mathbf{9.80 \pm 0.51}$ | $\mathbf{8.15 \pm 1.11}$ | $\mathbf{6.10 \pm 1.70}$ |

### 5.5.5 Adaptive testing for a deployment environment

**Goal**. Deploying an ML model often entails going beyond IID, such as a different deployment environment. We consider the case of deploying a model to a different country where there is a covariate shift [5] and evaluate the capabilities of testing frameworks to adapt across the different environment and identify model failures.

**Setup**. We use real-world prostate cancer datasets from different countries' cancer registries with known distribution shifts: SEER (US) [47] and CUTRACT (UK) [174]. We train a predictor $f$ on UK data, while our target deployment environment is the US.

**Analysis.** ► ***Identifying model failures.*** Table 5.5 shows that SMART better evaluates models *at deployment time* using the information provided. SMART identifies a much greater number of statistically significant model failures (*almost all possible*), both within the same environment (UK) and when shifting to a different one (US), even after adjusting for multiple comparisons using Bonferroni correction.

► ***Sample size sensitivity.*** We also assess *sensitivity to sample size*, see Figure 5.6. Both variants of SMART are shown to consistently outperform data-only counterparts in identifying a much greater number of significant model failures. *Within domain (UK)*: as expected, we find that for lower sample sizes, $SMART_{NSF}$ (without the self-falsification mechanism) is superior; however, given enough data, we then find that SMART benefits from the self-falsification.

*Deployment environment (US)*: we find that self-falsification similarly requires sufficient samples; which we note is expected behavior. Interestingly, in the deployment setting (US), $SMART_{NSF}$ generally identifies the greatest number of significant failures (*almost*

[5] $p(X)$ changes, while $p(Y|X)$ remains the same

Table 5.5 Number of slices identified (out of a maximum of 10) that had significantly divergent performance from average (higher is better). $S_\alpha$ counts the number of significantly divergent groups at $\alpha = 0.05$; $S_\alpha/n$ applies the Bonferroni correction. ↑ is better.

| | $\mathscr{D}^{\text{UK}}_{\text{train}}$ | | $\mathscr{D}^{\text{UK}}_{\text{test}}$ | | $\mathscr{D}^{\text{US}}_{\text{test}}$ | |
|---|---|---|---|---|---|---|
| | $S_\alpha$ | $S_{\alpha/n}$ | $S_\alpha$ | $S_{\alpha/n}$ | $S_\alpha$ | $S_{\alpha/n}$ |
| Autostrat | $1.00 \pm 0.00$ | $1.00 \pm 0.00$ | $1.00 \pm 0.00$ | $1.00 \pm 0.00$ | $0.95 \pm 0.22$ | $0.95 \pm 0.22$ |
| $\text{PSG}_B$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $1.75 \pm 0.43$ | $1.50 \pm 0.50$ |
| $\text{PSG}_A$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $1.75 \pm 0.43$ | $1.45 \pm 0.50$ |
| Divexplorer | $1.65 \pm 1.88$ | $0.45 \pm 0.92$ | $2.00 \pm 2.28$ | $0.85 \pm 1.53$ | $3.90 \pm 2.57$ | $2.75 \pm 2.21$ |
| Slicefinder | $3.65 \pm 0.96$ | $2.75 \pm 0.62$ | $3.85 \pm 1.11$ | $2.70 \pm 0.46$ | $6.80 \pm 1.03$ | $5.95 \pm 1.02$ |
| Sliceline | $1.00 \pm 0.00$ | $1.00 \pm 0.00$ | $1.35 \pm 0.48$ | $1.35 \pm 0.48$ | $1.35 \pm 0.48$ | $1.35 \pm 0.48$ |
| **SMART**_NSF | $8.30 \pm 0.46$ | $8.00 \pm 0.00$ | $8.45 \pm 0.50$ | $8.00 \pm 0.00$ | $\mathbf{9.20 \pm 0.60}$ | $\mathbf{8.35 \pm 0.48}$ |
| **SMART** | $\mathbf{9.60 \pm 0.49}$ | $\mathbf{9.25 \pm 0.54}$ | $\mathbf{9.45 \pm 0.50}$ | $\mathbf{8.85 \pm 0.57}$ | $8.85 \pm 0.79$ | $\mathbf{8.35 \pm 0.79}$ |

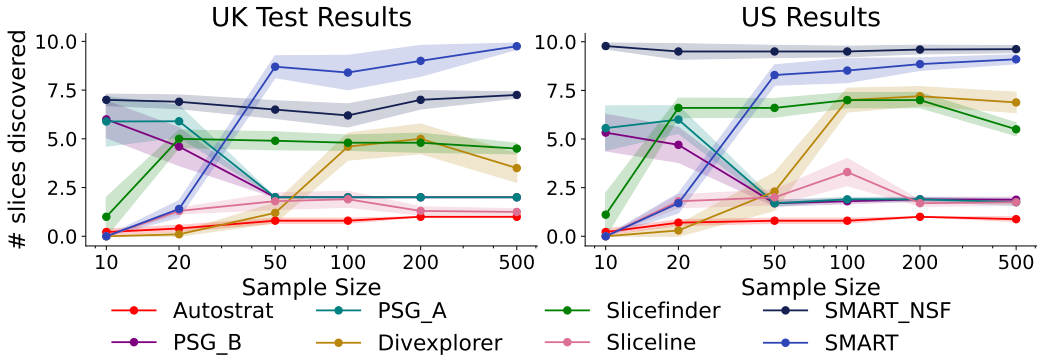



Fig. 5.6 Number of significant groups discovered (out of a total of 10) based on the training dataset size. SMART can operate under any sample size; the self-falsification mechanism requires a larger sample size to falsify hypotheses. ↑ is better.

*all possible*) across different sample sizes. This suggests that under covariate shift, using inductive knowledge (via the LLM) or domain expertise might be more useful to find meaningful model failures.

Overall, the results highlight the flexibility of SMART to handle different scenarios and sample sizes. From a practical perspective, while both SMART variants outperform data-only methods, the implication is that there is nuance in using different SMART variants for different scenarios.

**Takeaway.** SMART identifies more significant divergent failure slices in a deployment setting, outperforming data-only methods across environments and sample sizes.

Table 5.6 Proportion of times the corrupted minority subgroup is correctly identified as the top underperforming subgroup.

| | **Corrupted White** | | **Corrupted black** | |
|---|---|---|---|---|
| $\tau$ | $P_{white}$ | $P_{black}$ | $P_{white}$ | $P_{black}$ |
| 0.01 | $0.78 \pm 0.04$ | $0.15 \pm 0.04$ | $0.16 \pm 0.04$ | $0.78 \pm 0.04$ |
| 0.02 | $0.88 \pm 0.03$ | $0.08 \pm 0.03$ | $0.10 \pm 0.03$ | $0.87 \pm 0.03$ |
| 0.05 | $0.98 \pm 0.01$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.98 \pm 0.01$ |
| 0.10 | $0.98 \pm 0.01$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.99 \pm 0.01$ |
| 0.20 | $0.98 \pm 0.01$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $1.00 \pm 0.00$ |
| 0.30 | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $1.00 \pm 0.00$ |
| 0.50 | $1.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $1.00 \pm 0.00$ |

### 5.5.6 Assessing and mitigating potential LLM challenges and biases.

**Background.** In many settings, we want SMART to be part of a human-in-the-loop model evaluation, particularly to address challenges with LLMs, such as biases or missing dimensions. Let us first discuss how SMART addresses some issues by design and then perform an experimental assessment. ▶ **Using data to mitigate LLM challenges**: We use data in two ways *(i) Data usage in the generation of hypotheses:* Before generating explicit hypotheses of where the model is likely to fail, we provide the LLM with additional information about the data description and model failures of the training dataset. The hypotheses sampled are therefore reflective of the inductive bias of the LLM as well as being conditioned on the data itself; *(ii) Data usage in falsifying hypotheses:* core to SMART is the self-falsification mechanism where we iteratively generate hypotheses and test them on a validation dataset. Data is therefore used to filter out hypotheses that are not supported by the data. Hence, even if "incorrect" hypotheses about group failures are proposed, this step ensures they are discarded. ▶ **SMART provides clear and transparent testing**: This is done in two ways: *(i) SMART's model reports:* that document specific failure cases with natural language justifications (see Appendix D.4.7 for examples). Automatically generated reports can be a useful tool for humans-in-the-loop experts to audit and validate the testing process, such as evaluating whether tests should be added or removed. For example, a domain expert (e.g. a clinician) could review the report to assess whether the identified failure modes are truly relevant and concerning in that specific context. *(ii) Tests include justifications:* The justifications for the tests allow human users to inspect the model's testing procedures, understand the reasons, and audit for biases or missed dimensions.

**Goal.** Going beyond the mitigation strategies by design, we also assess SMART's robustness to prior biases. Specifically, we assess common ethnicity-related biases of LLMs.

**Setup.** We use the same data-generating process as Chapter 5.5.3. We then train a predictor function $\hat{f}$ and simulate a scenario where we intentionally corrupt a model's

predictions for a proportion $\tau$ of a minority subgroup ("white" or "black" ethnicity). $\tau$ ranges from 0 (no corruption) to 1 (completely random predictions for the subgroup). SMART aims to identify the top underperforming subgroup without bias based on historical patterns. We measure the proportion of times the top subgroup contains the "white" ($P_{\text{white}}$) or "black" ($P_{\text{black}}$) minority, averaged over 20 runs and 5 seeds, separately corrupting "white" and "black" ethnicities.

**Analysis.** We show in Table 5.6 that SMART is able to identify where models underperform even in scenarios such as ethnicity bias, where LLMs exhibit prior biases from the training dataset. This links to the above discussion that SMART mitigates such biases by design, both using the training dataset to guide hypothesis generation and using the self-falsification mechanism to empirically evaluate hypotheses (and discard those that aren't reflective of the data).

**Takeaway.** SMART mitigates potential biases in the LLM, both by using the real data to guide hypothesis generation, as well as using the self-falsification mechanism to filter spurious hypotheses.

### 5.5.7 Additional experiments

We conduct additional experiments to validate SMART further. Appendix D.4.1 demonstrates the feasibility with different LLMs, and Appendix D.4.2 assesses SMART on various tabular models. Finally, Appendix D.4.3 shows that in cases with no model failures, data-only methods surface failures by chance, whereas SMART, guided by context, avoids this issue.

## 5.6 Discussion

Responding to recent calls for better model testing [197, 216], we formalize **Context-Aware Testing**; a new testing paradigm, actively seeking out relevant and likely model failures based on contextual awareness — going beyond data alone. We develop **SMART Testing**, using LLMs to hypothesize likely and relevant model failures, providing *improved* and *automated* testing of tabular ML models, compared to data-only methods in various scenarios.

**Model reports.** SMART produces comprehensive and automated model reports documenting failure cases and justifications, thereby providing data scientists, ML engineers and stakeholders increased visibility into model failures. We provide an example SMART report in Appendix D.4.7.

**Further risks from LLM dependence**. Beyond the bias risks associated with LLMs, two additional failure modes of LLMs deserve explicit consideration in the SMART context. First, factual incorrectness: an LLM may generate hypotheses grounded in plausible-sounding but incorrect domain knowledge — for example, asserting a clinical relationship between two variables that is not medically supported. While the self-falsification mechanism will discard hypotheses that are not empirically supported by the data, it cannot detect hypotheses that are both factually wrong and spuriously supported by finite samples. This motivates domain expert review of generated hypotheses and justifications, particularly in high-stakes settings. Second, temporal instability: as LLMs are rapidly updated by their providers, the set of hypotheses generated by SMART may differ across model versions, raising reproducibility concerns. We recommend that practitioners document the specific LLM version used and, where feasible, validate SMART outputs against multiple LLM backbones (as partially assessed in the Appendix. These considerations reinforce the importance of treating SMART as a tool to augment, not replace, human expert judgment in model evaluation.

**Practical considerations**. Given the potential utility of SMART, we highlight the following five practical considerations:

1. *Hypothesis generation.* While CAT offers a principled framework for context-guided testing, LLMs present challenges in hypothesis generation. Although SMART has mechanisms to address these (see Chapter 5.5.6), it cannot guarantee the absence of biases.

2. *Use with small datasets.* SMART may be limited in some cases by insufficient real data to operationalize and test hypotheses, suggesting future work could explore targeted data collection or synthetic data generation to enhance testing.

3. *Extensions to other modalities.* SMART is formalized to test tabular ML models due to the interpretable and structured features in tabular data to guide hypothesis generation. While extensions to other modalities, such as image and text, are beyond the scope of this work, this is a promising future research direction that would require addressing the lack of explicitly interpretable features, possibly via external metadata and developing new ways to operationalize hypotheses on unstructured data. That said, tabular data is ubiquitous in real-world applications [23, 226], with approximately 79% of data scientists working on tabular problems daily, vastly surpassing other modalities [104, 78]. This highlights the immediate impact and relevance of SMART.

4. *Need for interpretable/meaningful feature names.* Feature names play an important role in finding model failures (see Appendix D.4.6). The need for interpretable/meaningful

feature names (e.g. column labels such as sex, age, race etc) as a source of context is similar to human requirements of interpretable feature names to understand what the data refers to. While feature names are typically the de facto in research and industry datasets, on the rare occasions when they are not, this will affect the performance of SMART.

5. *Cost of SMART.* SMART is extremely accessible and cheap to use, approximately $<$0.10 USD for 5 hypotheses and $<$0.5 USD for 100 hypotheses for state-of-the-art models (see Appendix D.4.4).

# Chapter 6

# Self-Supervised Uncertainty Quantification

## 6.1 Introduction

Much of machine learning (ML) research can be summarized as the task of minimizing a model's predictive error on unseen data. In the regression setting, this might correspond to minimizing the mean squared error. However, for real-world applications, particularly in high-stakes domains such as healthcare and finance, we seek various forms of model trustworthiness. One such factor considered in this chapter is estimates of, or even better guarantees, concerning a model's predictive uncertainty at deployment time, i.e. quantifying the uncertainty in a model's prediction.

Conformal prediction [250] has received significant attention in recent years for this reason. This powerful framework provides valid prediction intervals with finite-sample, frequentist guarantees on the marginal coverage of the prediction intervals. More precisely, for a random vector and label $(\mathbf{X}, Y) \in \mathbb{R}^d \times \mathbb{R}$ drawn from a joint distribution $P_{\mathbf{X}Y}$ and a conformal predictor $\hat{C} : \mathbb{R}^d \to 2^{\mathbb{R}}$, then for a user-defined level of coverage $(1-\alpha)$, conformal prediction provides guarantees of the form

$$\mathbb{P}\{Y \in \hat{C}(\mathbf{X})\} \geq 1 - \alpha \tag{6.1}$$

Conformal prediction requires minimal assumptions on the data (i.e. exchangeability) and may be applied to any model upon which we can define a *non-conformity score* quantifying the disagreement between the model output and ground truth target (e.g. mean squared error). It can even be applied post hoc to already trained models, assuming the exchangeability of a

calibration dataset with the test/deployment data of interest. For more details, see Shafer and Vovk [223].

A widely used paradigm is inductive conformal prediction (ICP) [165, 128], which formulates conformal prediction using a dataset to train the predictive model and a calibration dataset on which to compute a critical non-conformity score used to define the width of the prediction interval. However, despite the guarantees of marginal coverage, these intervals are fixed and do not adapt to the difficulty of a sample. In practice, we often desire wider intervals for challenging samples and narrower intervals for easier samples. Locally adaptive conformal prediction [166, 103, 128] aims to address this with a conformal normalization model, which predicts the residual errors of the predictive model (also known as conformalized residual fitting - CRF). The predicted residuals (representing sample prediction difficulty) are then used to adapt the intervals.

In this chapter, we tackle the case of ICP with locally adaptive intervals and seek to improve the quality of the prediction intervals. Although we primarily focus on the CRF approach, the core ideas can be trivially adapted to other methods for local adaptivity, such as Conformalized Quantile Regression (CQR), as we discuss in Chapter 6.4.5 [196]. However, a key benefit of the CRF approach is that the method does not directly modify the underlying base predictive model when adapting the prediction intervals. This enables post hoc conformalization in cases where we cannot augment or retrain such as a fixed/pre-trained regressor (e.g. behind an API) or in the case of a theoretically derived model (e.g. quantifying model error around the Navier-Stokes equations in fluid mechanics).

Specifically, we consider how to improve the conformal normalization function in order to improve the prediction intervals. To do so, we diverge from the typical conformal prediction setting and consider how auxiliary self-supervised tasks can be leveraged to improve conformal predictors. Self-supervised learning has been employed effectively to learn representations to improve performance for downstream tasks; however, we also diverge from this standard setting of model pretraining and representation learning and address an under-explored avenue of using the error from a self-supervised task to help determine more challenging regions in input space such that conformal intervals can adapt appropriately. In essence, we aim to show how data unlabeled/labeled can be repurposed to improve uncertainty quantification- in this case, through the lens of self-supervision.

Consequently, we propose *Self-Supervised Conformal Prediction* (**SSCP**), a framework that leverages information from self-supervised pretext tasks to improve prediction intervals (see Figure 6.1 for a contrast to alternative approaches). We note that the auxiliary self-supervision information does not affect the theoretical guarantees of conformal prediction (see Chapter 6.4.6). To the best of our knowledge, this is the first work to integrate and

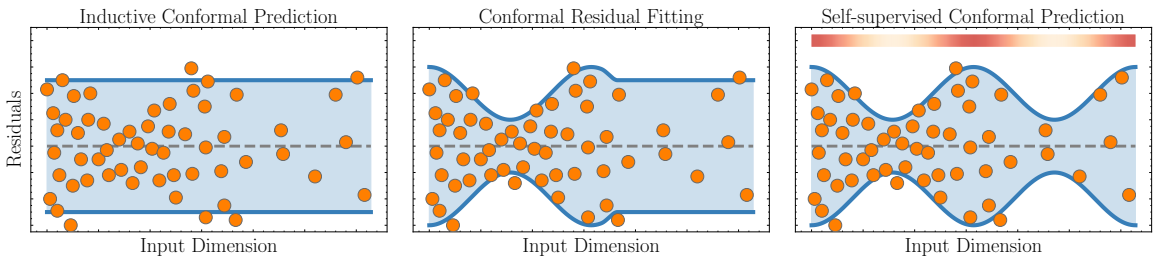



Fig. 6.1 **Left.** Standard inductive conformal prediction results in constant width prediction intervals. **Center.** Conformal residual fitting produces adaptive intervals but can be inefficient in regions. **Right.** The errors of a self-supervised task are included above the plot with red indicating larger self-supervised errors. SSCP leverages these errors to improve the efficiency of conformal residual fitting.

examine the benefit of self-supervised errors to improve conformal prediction, making the following contributions:

**Contributions:** ① **Self-supervision improves the residual estimates, especially in challenging and sparser regions of the data space:** We empirically demonstrate that self-supervised losses improve residual estimates thereby improving the prediction interval adaptiveness, especially for more challenging examples and sparser regions of the data space. ② **Labeled data can be repurposed:** We empirically demonstrate that labeled data can be repurposed for the self-supervised task to improve the quality of conformal prediction intervals. ③ **Unlabeled data improves CP:** We empirically demonstrate the value of unlabeled data to improve the quality of conformal prediction intervals beyond a standard application of self-supervision representation learning.

## 6.2 Related Work

**Adaptive Conformal Prediction.** While standard conformal prediction does provide marginal coverage (Equation 6.1), in practice we often desire conditional coverage $\mathbb{P}\{Y \in \hat{C}(\mathbf{X}) \mid \mathbf{X} = \mathbf{x}\} \geq 1 - \alpha$ for all $\mathbf{x} \in \mathscr{X}$. Although it has been shown that exact conditional coverage cannot be guaranteed in finite samples [56, 248], many works have improved the situation by relaxing the requirement of guarantees on coverage or by maintaining marginal coverage while seeking to improve adaptivity to a given test point.

Locally adaptive conformal prediction [166, 168, 102] uses an independent model to induce adaptiveness. Hence, its mechanism is independent of the base predictive model. This

flexibility means that it can be applied to any predictive model architecture or to an already trained model.

In contrast, typically, other methods are not independent of the predictive model as they require specialized model architectures to obtain adaptive intervals, they require additional assumptions, and/or they sacrifice coverage guarantees. For example, Conformalized Quantile Regression (CQR) [196] has a modified neural network that predicts the two quantiles. Parallel lines of research consider non-parametric density estimation-based approaches [34, 94] and conformal histogram regression [221], which discretely approximates the conditional density.

Finally, conformal prediction splits the dataset into training and calibration datasets. Although this may not be an issue with sufficient labeled data, in many settings (such as healthcare), we only have access to small/limited labeled datasets. Hence, splitting the data might significantly affect performance. To mitigate this issue in such settings, we often have access to large amounts of unlabeled data that goes unused from the perspective of conformal prediction. Hence, to the best of our knowledge, SSCP is the first work to take advantage of such unlabeled data for conformal prediction in the context of self-supervised learning. We note that SSCP does not require additional splits in the labeled data over the underlying ICP method.

**Self-Supervised Learning.** Self-supervised learning has been effectively utilized in many domains, from computer vision [33, 73] to NLP [108, 27] to learn informative representations from raw input features with the goal of improving the performance of downstream models. In the tabular domain, masking and reconstruction methods are often popular pretext tasks, using methods such as autoencoder reconstruction, VIME [263], where the task is to recover a mask vector in addition to the original sample from a corrupted version of the input, and other variants, e.g. [126].

Although self-supervision has had great success, its use beyond model pretraining and representation learning has been largely unexplored. In this chapter, we bridge this gap by studying how self-supervised losses can be used to provide additional information about the difficulty of a sample, making use of this information (from data) to improve the quality of conformal prediction intervals.

## 6.3 Background: Conformal Prediction

In this background discussion, we provide a brief summary of both ICP and CRF, two methods that are fundamental to our proposed framework. For proofs of the theoretical properties referenced, we refer the reader to [248].

We begin with a brief description of the general setting. Consider a supervised learning setting with features $\mathscr{X} \subseteq \mathbb{R}^d$ and scalar labels $\mathscr{Y} \subseteq \mathbb{R}$. We wish to learn a prediction set $\hat{C} : \mathscr{X} \to 2^{\mathbb{R}}$ such that for a desired coverage level $1-\alpha \in (0,1)$, (6.1) holds for a test point $(\mathbf{X}, Y) \in \mathscr{X} \times \mathscr{Y}$. To train such a model, we assume access to labeled data $\mathscr{D}_{\text{labeled}} \subseteq \mathscr{X} \times \mathscr{Y}$. We consider regression predictive tasks; hence, the label $y \in \mathscr{Y}$ is a scalar.

Throughout this work, we consider *inductive* (or *split*) conformal prediction. The standard practice in the literature is to subdivide the labeled data into disjoint training, calibration, and test sets: $\mathscr{D}_{\text{labeled}} = \mathscr{D}_{\text{train}} \sqcup \mathscr{D}_{\text{cal}} \sqcup \mathscr{D}_{\text{test}}$. One might wish to also consider including an (optional) fourth subdivision $\mathscr{D}_{\text{res}}$, however, we defer considering this detail until Chapter 6.4.1. In this setting, the conformal predictor $f$, around whose predictions we desire prediction intervals, is trained on $\mathscr{D}_{\text{train}}$.

### 6.3.1 Inductive conformal prediction (ICP)

Given the trained predictive model $f : \mathscr{X} \to \mathscr{Y}$, we wish to produce conformal predictive intervals as a measure of uncertainty, $[l(\mathbf{x}), r(\mathbf{x})]$ for each test instance $(\mathbf{x}, y) \in \mathscr{D}_{\text{test}}$, where $l(\mathbf{x})$ and $r(\mathbf{x})$ denote the left and right endpoints, respectively. These intervals should satisfy marginal coverage:$\mathbb{P}\{Y \in [l(\mathbf{X}), r(\mathbf{X})]\} \geq 1-\alpha$, equivalently $\mathbb{E}[\mathbf{1}\{Y \in [l(\mathbf{X}), r(\mathbf{X})]\}] \geq 1-\alpha$, where the significance level $\alpha \in (0,1)$ can be chosen[1] i.e. valid intervals with desired marginal coverage. In performing ICP, we only require the mild assumption of exchangeability between $\mathscr{D}_{\text{cal}}$ and $\mathscr{D}_{\text{test}}$ as defined in Def. 6.3.1.

**Definition 6.3.1** (Exchangeability)**.** A dataset of $n$ observations is exchangeable if the data points do not follow any particular order, i.e., all $n$ permutations are equiprobable. Exchangeability is a weaker assumption than i.i.d.; indeed, i.i.d. observations satisfy exchangeability.

To obtain these intervals, we perform a calibration step. Specifically, $\mathscr{D}_{\text{cal}}$ is used to compute a non-conformity score $\mu$, which estimates how different a new instance looks from other instances. In practice, absolute error is a popular non-conformity score (i.e. $\mu(\mathbf{x}, y) = |y - f(\mathbf{x})|$). We then obtain an empirical distribution of non-conformity scores $\{\mu(\mathbf{x}_i, y_i)\}_{(\mathbf{x}_i, y_i) \in \mathscr{D}_{\text{cal}}}$ over the calibration instances.

[1] We select $\alpha = 0.1$ in our experiments.

The critical non-conformity score $\varepsilon$ corresponds to the $\lceil(|\mathscr{D}_{\text{cal}}|+1)(1-\alpha)\rceil$-th smallest score from the set $\{\mu(\mathbf{x}_i,y_i)\}_{(\mathbf{x}_i,y_i)\in\mathscr{D}_{\text{cal}}}$. Finally, we use $\varepsilon$ to construct the intervals $[l(\mathbf{x}),r(\mathbf{x})]=[f(\mathbf{x})-\varepsilon,f(\mathbf{x})+\varepsilon]$.

### 6.3.2 Conformal residual fitting (CRF)

Unfortunately, the widths of the ICP prediction intervals described in the previous section are constant for all instances. While they do provide valid coverage, they do not reflect the difficulty of individual samples and are inefficient as a result (see Figure 6.1). Moreover, from a decision-making perspective, constant width intervals do not aid decision-making as sample uncertainty cannot be sorted.

To remedy this, CRF proposes to produce locally adaptive intervals based on an estimate of the predictive model's residuals. Specifically, to enable locally adaptive intervals (i.e. locally adaptive conformal prediction), a normalized non-conformity function $\gamma$ is used (Equation 6.2). The numerator is computed as before based on $\mu$. However, the function $\sigma$ in the denominator permits us to normalize the non-conformity score per sample (i.e., as an estimate of difficulty), where $\sigma:\mathscr{X}\to\mathbb{R}_{>0}$ is a *conformal normalization model*.

$$\gamma(\mathbf{x},y):=\frac{|y-f(\mathbf{x})|}{\sigma(\mathbf{x})}, \tag{6.2}$$

where $\sigma$ takes only $\mathbf{x}$ in the standard CRF setting; in Section 6.4, we extend its domain to include auxiliary features.

The critical non-conformity score $\varepsilon$ is then computed similarly to before using the $\lceil(|\mathscr{D}_{\text{cal}}|+1)(1-\alpha)\rceil$-th smallest score of the empirical distribution of normalized non-conformity scores $\{\gamma(\mathbf{x},y)\mid(\mathbf{x},y)\in\mathscr{D}_{\text{cal}}\}$.

To train $\sigma$, a sensible approach is to use an additional, typically disjoint dataset $\mathscr{D}_{\text{res}}$ upon which we can evaluate the trained predictive model $f$ to obtain predictions that will be used as targets. This allows us to construct the set of tuples $\{(\mathbf{x}_i,|y_i-f(\mathbf{x}_i)|)\}_{(\mathbf{x}_i,y_i)\in\mathscr{D}_{\text{res}}}$.

This set is used to train the *conformal normalization model* $\sigma$, which is then applied at test time to estimate the predictive model's residuals for each sample $\mathbf{x}$ where $(\mathbf{x},y)\in\mathscr{D}_{\text{test}}$.

## 6.4 Method: Self-Supervised Conformal Prediction

We now describe our proposed method leveraging self-supervision to improve locally adaptive conformal prediction, which we refer to as Self-Supervised Conformal Prediction (SSCP).

Recall that CRF adapts prediction intervals by predicting the residuals of the main predictive model using an auxiliary residual model. By predicting regions in which the predictive model will make greater or smaller errors, the intervals can be appropriately adapted. However, predicting residuals can be challenging; indeed, presumably they would not be residuals in the first place otherwise. Therefore, we propose to improve the performance of the residual model with the additional signal.

Our method is motivated by the following key question: Can self-supervised tasks provide this added signal? With an appropriately selected self-supervised task, we get not only its prediction at test time but also its ground-truth target. Therefore, we have access to the self-supervised error *even* at test time. If this error has a relationship with the error of the main model, then it can provide a useful input feature to the residual model. While correlated errors will often occur naturally due to factors such as noise or complexity in the data distribution, we can encourage this relationship with the errors by careful selection of the self-supervised task and/or shared representation between the main model and the self-supervised model.

Furthermore, the self-supervised model can simply be trained on the labeled training data as an extra step, or even better, it can also leverage any additional unlabeled data that is available. We highlight that this method can be applied orthogonally and *in addition to* any standard self-supervised representation learning of the main model.

Our proposed SSCP method consists of three steps, which we also summarize graphically in Figure 6.2 and algorithmically in Algorithm 1.

- *(S1) Predictive and self-supervised phase*: First, we train a predictive model to solve the regression task, as normal. Then, we share the learned representations from the predictive model and solve the self-supervised pretext task on top of these representations.
- *(S2) Conformal normalizer phase*: Train a conformal normalizer to learn the residuals of the predictive model using the data features as well as the self-supervised errors on the same instances.
- *(S3) Conformalization phase*: Finally, we apply CRF using the prediction model, self-supervised model, and residual model trained in the previous steps.

In the following sections, we describe each step in more detail. First, however, we take a brief detour to describe the data splits and any assumptions on their distributions.

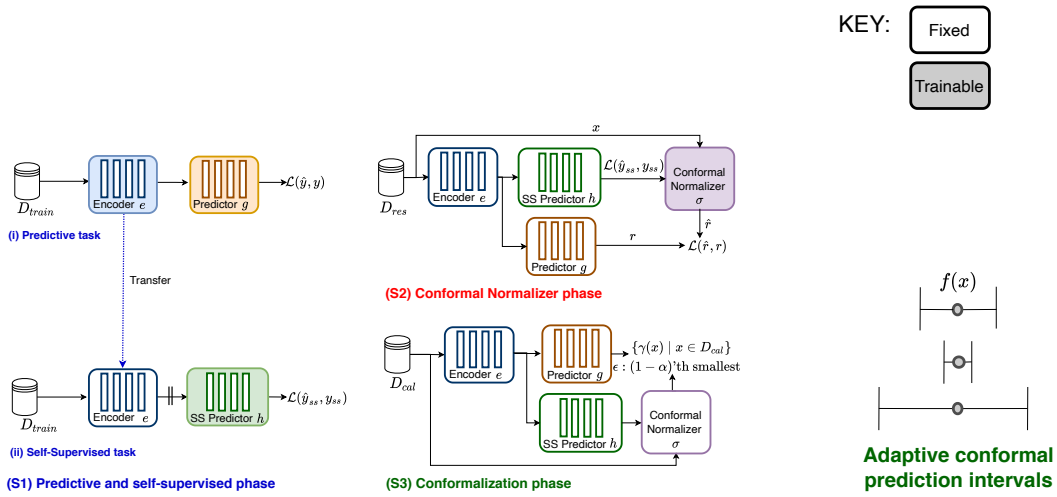



Fig. 6.2 SSCP produces adaptive conformal prediction intervals leveraging self-supervised learning to improve the conformal normalizer ($\sigma$) and consists of 3 phases. (S1) where (i) train the predictive regressor $f = g \circ e$, thereafter (ii) the encoder representation ($e$) is transferred to use for the self-supervised task, where we train the self-supervised predictor $h$. (S2) the conformal normalizer phase is then carried out, with the conformal normalizer, augmented with the self-supervised input, trained to predict residuals (representing difficulty of prediction). Finally, (S3) the conformalization phase, computes normalized scores ($\gamma$) on $\mathscr{D}_{\text{cal}}$, with the critical non-conformity score ($\varepsilon$) chosen as the $(1-\alpha)$-smallest. At test time, the adaptive prediction intervals $[l(\mathbf{x}), r(\mathbf{x})]$ are produced by adapting the critical non-conformity score $\varepsilon$ using the conformal normalizer, i.e $\varepsilon\sigma(\hat{\mathbf{x}})$. The "shaded blocks" in each phase are trainable.

### 6.4.1 Data splits and assumptions

We begin by briefly summarizing the required data splits to perform SSCP. We also highlight that our method requires no further data splits beyond CRF and is therefore *more* data efficient due to its reuse of labeled data and ability to use unlabeled data for the self-supervised tasks. We denote the labeled data as $\mathscr{D}_{\text{labeled}} \subseteq \mathscr{X} \times \mathscr{Y}$ and the self-supervised (unlabeled) data as $\mathscr{D}_{\text{unlabeled}} \subseteq \mathscr{X}$. Let $\pi_{\mathscr{X}} : \mathscr{X} \times \mathscr{Y} \to \mathscr{X}$ denote projection onto the feature coordinate. The predictive model is trained on (a subset of) $\mathscr{D}_{\text{labeled}}$, while the self-supervised model is trained on the feature pool

$$\mathscr{D}^{\text{ssl}}_{\text{train}} := \pi_{\mathscr{X}}(\mathscr{D}_{\text{labeled}}) \cup \mathscr{D}_{\text{unlabeled}} \subseteq \mathscr{X}.$$

In many practical cases, $|\mathscr{D}_{\text{unlabeled}}| \gg |\mathscr{D}_{\text{labeled}}|$. Neither model is required to use its full available pool (e.g., the predictive model may use only $\mathscr{D}_{\text{labeled}}$, while the self-supervised model uses $\mathscr{D}^{\text{ssl}}_{\text{train}}$).

Although not typically discussed in the literature, the residual model should ideally be trained on disjoint data from the predictive model. We refer to this as $\mathscr{D}_{\text{res}}$ and, by using a separate dataset, we ensure that the residual model does not learn on the potentially overfit $\mathscr{D}_{\text{train}}$ residuals.[2] Then, as usual, we have calibration and test data $\mathscr{D}_{\text{cal}}$ and $\mathscr{D}_{\text{test}}$, respectively. The only assumption required on the data to achieve valid coverage is that these two datasets are exchangeable. We do not require assumptions on the remaining data, although, as in any learning problem, model performance is likely to be affected by a divergence between their distributions.

### 6.4.2 Predictive and self-supervised phase (S1)

**Predictive task.** The goal of the predictive stage is to obtain a regressor $f : \mathscr{X} \to \mathscr{Y}$ mapping inputs to targets. Typically, this is an ML model (e.g. a neural network) trained on $\mathscr{D}_{\text{train}}$. However, we place no assumptions on this model, which could be *any* predictive model. An example of such a scenario beyond standard models might appear in a healthcare setting in which our model is the decision process of a doctor around which we would like to obtain valid prediction intervals.

In this work, we focus on the more conventional setting of a data-driven model trained on $\mathscr{D}_{\text{train}}$ to minimize the supervised loss $\mathscr{L}$ (usually mean squared error),

$$\min_f \mathbb{E}_{(\mathbf{x},y)\sim\mathscr{D}_{\text{train}}}\left[\mathscr{L}(y, f(\mathbf{x}))\right].$$

We select this model to be a neural network consisting of an encoder $e$ and a predictor $g$ such that $f = g \circ e$. This provides us with the ability to share an encoding representation between the supervised and self-supervised tasks. In doing so, we enforce a relationship between the two models.

We also highlight that a standard application of self-supervised learning could be applied in learning the predictive model *in addition to* the SSCP procedure we describe in this work. This would consist of applying standard self-supervision prior to the predictive task (S1). Our SSCP framework is orthogonal to such an application.

[2] We note that training both $f$ and $\sigma$ on the same data is possible, but comes with the associated overfitting risk of training $\sigma$ on *training* residuals instead of *testing* residuals. We used a principled experimental design by including this additional split, resulting in $\mathscr{D}_{\text{res}}$, but practitioners may obtain better performance by *double dipping* on $\mathscr{D}_{\text{train}}$.

**Self-supervised task.** The self-supervised task consists of training some model $f_{\text{ss}}$ on $\mathscr{D}_{\text{train}}^{\text{ssl}}$ but this time for some pretext task. We also place no restrictions on the choice of self-supervised tasks. However, we wish that its errors have a useful relationship with those of the predictive model.

In this work, we focus on tabular data where tasks such as input reconstruction-based tasks (such as autoencoders or VIME [263]) are popular. Depending on the task, various domain-specific tasks have been proposed in the self-supervised literature, which may be leveraged in this step. We also use a neural network $f_{\text{ss}} = h \circ e$ where $e$ is the pre-trained encoder of the predictive model and $h$ is a task-specific decoder. We fix the weights of $e$, such that it retains the desired connection with the predictive task. We then optimize $h$ to solve the pretext task, minimizing the self-supervised loss $\mathscr{L}_{ss}$ as part of the following objective:

$$\min_{h} \mathbb{E}_{\mathbf{x} \sim \mathscr{D}_{\text{train}}^{\text{ssl}}} \left[ \mathscr{L}_{\text{ss}}(\mathbf{x}, h(e(\mathbf{x}))) \right]. \tag{6.3}$$

### 6.4.3 Conformal normalizer phase (S2)

The *conformal normalization model* $\sigma$ is directly responsible for the adaptiveness of the prediction intervals. Hence, by improving $\sigma$, we can in turn improve the adaptiveness of the intervals. As discussed previously, this model is fitted to $\mathscr{D}_{\text{res}}$ to avoid learning the $\mathscr{D}_{\text{train}}$ residuals, which may not generalize.

In this work, we propose to incorporate $\ell_{\text{ss}}$ as an additional feature of $\sigma$. If the self-supervised loss for a specific sample has a relationship with the difficulty of the predictive task, then, by adding $\ell_{\text{ss}}(\mathbf{x})$ as a feature, we pass this information to $\sigma$ when predicting the normalizing residuals. To do so, we augment the input space of $\sigma$ as $\hat{\mathbf{x}} = [\mathbf{x}, \ell_{\text{ss}}(\mathbf{x})] \in \mathscr{X} \times \mathbb{R}_{\geq 0}$, with the normalizer's signature now $\sigma : \mathscr{X} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{>0}$. With slight abuse of notation, we continue to denote the augmented normalizer by $\sigma$. We then train $\sigma$ as usual. The decision of which residual normalizer model to use should be selected on a problem-specific basis. We successfully apply a random forest (Chapter 6.5.1) and a neural network (Chapter 6.5.2) in our experiments.

### 6.4.4 Conformalization phase (S3)

Finally, given our trained predictive model $f$, self-supervised model $f_{\text{ss}}$, and residual model $\sigma$, we can apply the same calibration procedure as standard CRF on $\mathscr{D}_{\text{cal}}$ (as described in Chapter 6.3.2) to obtain the $(1-\alpha)$ quantile non-conformity score $\varepsilon$. Then, for a new test example $\mathbf{x}$, we obtain adaptive and valid intervals.

$$[l(\mathbf{x}), r(\mathbf{x})] = [f(\mathbf{x}) - \varepsilon\,\sigma(\hat{\mathbf{x}}),\ f(\mathbf{x}) + \varepsilon\,\sigma(\hat{\mathbf{x}})], \tag{6.4}$$

where $\hat{\mathbf{x}} = [\mathbf{x}, \ell_{\text{ss}}(\mathbf{x})]$. We summarize the complete SSCP framework algorithmically in Algorithm 1.

**Algorithm 1:** Self-supervised Conformal Prediction

**Input:** $\mathscr{D}_{\text{train}}, \mathscr{D}_{\text{unlabeled}}, \mathscr{D}_{\text{res}}, \mathscr{D}_{\text{cal}}, e, g, h, \sigma$
Train $f = g \circ e$ on $\mathscr{D}_{\text{train}}$ (Chapter 6.4.2, S1 (i));
Train $f_{\text{ss}} = h \circ e$ on $\mathscr{D}^{\text{ssl}}_{\text{train}} = \pi_{\mathscr{X}}(\mathscr{D}_{\text{train}}) \cup \mathscr{D}_{\text{unlabeled}}$ (Chapter 6.4.2, S1 (ii));
Compute self-supervised errors $\ell_i = \ell_{\text{ss}}(\mathbf{x}_i)$ for all $(\mathbf{x}_i, y_i) \in \mathscr{D}_{\text{res}}$;
Construct augmented dataset $\mathscr{D}'_{\text{res}} = \{([\mathbf{x}_i, \ell_i], |y_i - f(\mathbf{x}_i)|)\}_{i=1}^{|\mathscr{D}_{\text{res}}|}$;
Train $\sigma$ on $\mathscr{D}'_{\text{res}}$ (Chapter 6.4.3, S2);
Apply standard CRF calibration ;
**Output:** $e$, $g$, $h$, $\sigma$

### 6.4.5 Other locally adaptive methods

The SSCP framework described throughout this chapter has primarily focused on CRF as a method for achieving locally adaptive intervals. However, simple adaptations to the procedure permit alternative methods to be considered. A powerful alternative that has emerged in recent years is CQR [196]. In Appendix E.3.3, we describe the adaptation required to integrate CQR into our SSCP framework and provide an empirical evaluation of CQR as an alternative method.

### 6.4.6 Remark on SSCP and coverage guarantees

Conformal prediction provides theoretical guarantees on the validity of coverage of the prediction intervals based on the exchangeability assumption (see Def. 6.3.1). SSCP inherits these same guarantees. Just as in the case of CRF, SSCP provides valid coverage provided the calibration set and test set satisfy exchangeability (e.g. see [240] for proof of these guarantees). We provide an extended comment on this point in Appendix E.1.

## 6.5 Experiments

We now empirically assess the benefit provided by SSCP compared to CRF. We begin with a synthetic example to highlight *why* adding a self-supervised loss to the conformal

normalization model can improve the prediction intervals. Then, we demonstrate SSCP's performance on a diverse range of real-world datasets, which differ in terms of the number of samples, features, and difficulty.

**Evaluation metrics** We desire that the self-supervised information improves the quality of the conformal prediction intervals on average. We assess quality based on the following commonly used metrics [196, 155]. Note that in all cases, we still empirically obtain intervals with valid coverage. Decreases in these metrics whilst still maintaining coverage implies superior adaptation and improvement to the quality of the prediction intervals:

- Width: The size of the conformal prediction interval, i.e., $\mathbb{E}[r(\mathbf{X}) - l(\mathbf{X})]$
- Deficit: The interval shortfall, i.e., $\mathbb{E}[\mathbf{1}\{Y \notin [l(\mathbf{X}), r(\mathbf{X})]\} \cdot \min\{|Y - l(\mathbf{X})|, |Y - r(\mathbf{X})|\}]$
- Excess: The additional width included not needed to capture the true value $Y$, i.e., $\mathbb{E}[\mathbf{1}\{Y \in [l(\mathbf{X}), r(\mathbf{X})]\} \cdot \min\{Y - l(\mathbf{X}), r(\mathbf{X}) - Y\}]$

**Experimental setup.** We use the same underlying predictive model in all comparisons to isolate the effects of the conformal prediction method (since, by definition, a better predictive model can achieve narrower intervals). Our baseline is locally adaptive conformal prediction (denoted **CRF**), where the non-conformity score is the absolute error of the residuals.

In our experiments, we illustrate SSCP using VIME [263] as an example of a potential self-supervision task. We selected VIME as it is regarded as highly performant on tabular data. As a pretext task, from a partially masked input, VIME aims to reconstruct the original sample in addition to recovering the mask vector. We provide additional experiments (different self-supervised tasks, SSCP with CQR), as well as ablations in Appendix E.3. Additional experimental details can be found in Appendix E.2.

### 6.5.1 Synthetic demonstration

We begin by illustrating our method on a synthetic dataset. Although a standard application of self-supervision to improve the predictive model is likely to improve performance, the primary contribution of this chapter is to apply self-supervision to achieve a more adaptive conformal normalization model $\sigma$ (i.e. to improve uncertainty quantification).

We isolate this effect by generating synthetic residuals from a hypothetical predictive model and comparing the performance of a conformal normalization model *with* and *without* access to the errors of a self-supervised task. Therefore, any improvements in this setting are exclusively due to this additional information.

**Data Generation.** In order to visualize the results, we assume that both the inputs $\mathbf{x}$ and the model residuals $r$ are generated as a function of a single, uniformly distributed latent dimension $L \sim \mathscr{U}(0,1)$. We assume that there are discrete regions in which the model has made larger errors and therefore its residuals are distributed as $r \sim \mathscr{N}(\text{step}(L), 0.1)$, where step() denotes a step function defined as

$$\text{step}(x) = \begin{cases} 1.5 & \text{if } x \in (0.2, 0.4) \text{ or } x \in (0.6, 0.8) \\ 0.1 & \text{otherwise.} \end{cases} \tag{6.5}$$

The sign of the residual is chosen at random. A visualization of these residuals (plotted prior to taking their absolute value, which maps them to $\mathbb{R}^+$ as required) is provided in Figure 6.3.

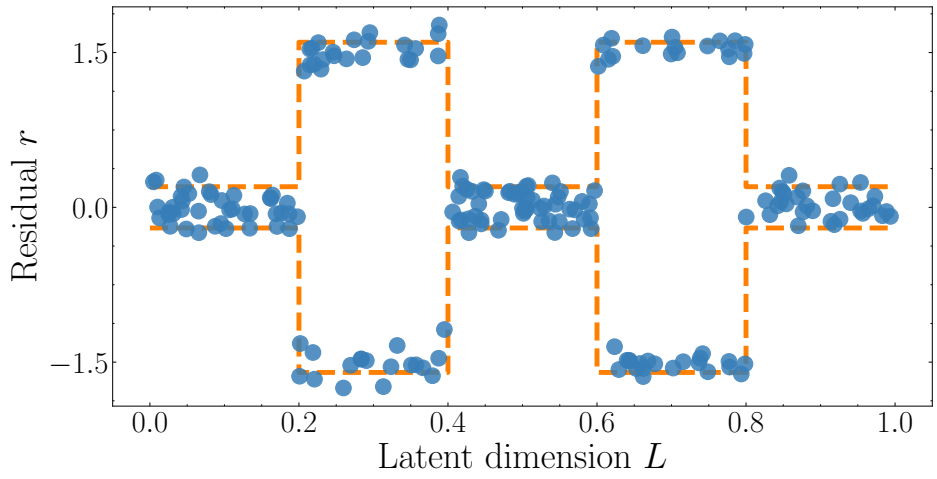



Fig. 6.3 200 observations drawn from our synthetic distribution with the distributions ground truth 90% prediction intervals highlighted with a dashed line.

The input features $\mathbf{x}$ are also sampled as a function of $L$. We consider a 20-dimensional input consisting of 10 uniformly sampled pairs of points along the circumference of a circle with its base at the origin and a radius of $L$. This results in an increasingly sparse input space for larger values of $L$.

**Results.** Predicting $r$ from $\mathbf{x}$ is challenging in the small data setting. However, by applying $\mathbf{x}$, we can better learn the latent process, which in turn can provide useful information for the supervised conformal normalization model. One of the simplest self-supervised tasks is a standard autoencoder that reconstructs the input.

Taking advantage of our knowledge of the data generation process, we consider a single latent unit in the bottleneck layer. Due to the increasing sparsity of inputs with increasing

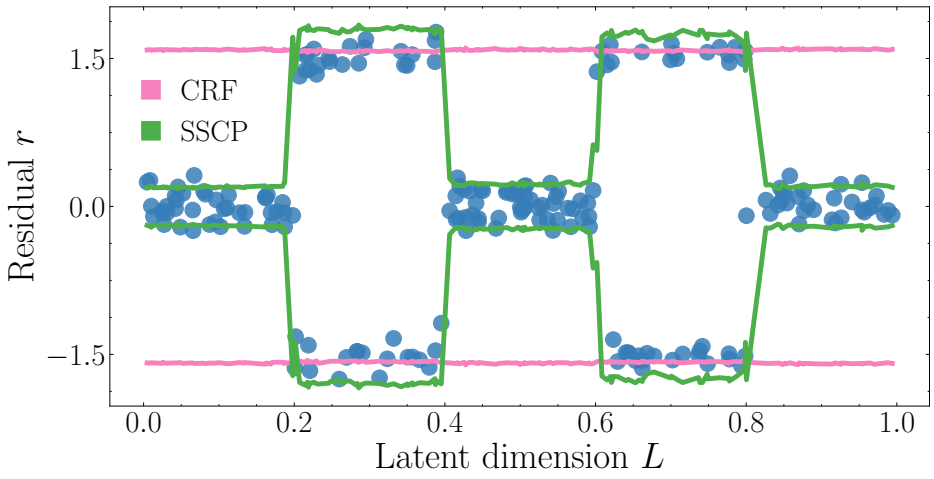



Fig. 6.4 **Conformal prediction intervals.** Standard CRF (pink) fails on this synthetic task due to a complex relationship between the inputs and the latent dimension. By including the errors of a self-supervised task as an input, SSCP (green) is able to learn much more efficient prediction intervals.

$L$, this would result in self-supervised errors that are linearly correlated with $L$ (i.e. the autoencoder could reasonably model the single ground truth latent dimension $L$). Since the residuals are a function of $L$, this self-supervised error provides a useful feature.

In Figure 6.4 we compare the conformal prediction intervals ($\alpha = 0.1$) of a random forest of 1000 trees with and without the additional self-supervised feature, where out-of-bag predictions are used for calibration.

We observe that while both methods achieve the desired level of marginal coverage, the additional information from the self-supervised input results in adaptive intervals. However, the standard approach is unable to model the relationship between inputs and residuals and is forced to default to more conservative intervals to maintain coverage.

This example illustrates that strictly narrower intervals might not be optimal for a given sample. Indeed, for samples with large residuals, CRF achieves narrower intervals than SSCP. However, these intervals are overly tight, and coverage is only maintained over the entire dataset due to overly conservative coverage in low residual regions.

**Takeaway.** Augmenting the conformal normalization model's inputs with a self-supervised loss better models the heteroscedasticity in the predictive model's residuals, which results in more adaptive conformal prediction intervals.

### 6.5.2 Real data experiments

We now assess the performance of SSCP on multiple real-world regression datasets. These datasets have been used in [196]. We assess the utility of using self-supervision in two plausible scenarios.

#### Fully labeled data setting

**Setup.** This experiment evaluates the utility of SSCP if we *only* have access to fully labeled data. We evaluate whether the same labeled data can be repurposed for self-supervision simply by ignoring the label (i.e., we have $\mathscr{D}_{\text{unlabeled}} = \emptyset$). The benefit of this setup is that it is applicable to any supervised learning setting. Formally, $\mathscr{D}_{\text{labeled}}$ is split as

$$\mathscr{D}_{\text{labeled}} = \mathscr{D}_{\text{train}} \sqcup \mathscr{D}_{\text{cal}} \sqcup \mathscr{D}_{\text{res}} \sqcup \mathscr{D}_{\text{test}}.$$

The datasets as described in Chapter 6.4 are used for both the predictive and normalization models. We can use $\pi_{\mathscr{X}}(\mathscr{D}_{\text{train}})$ for the self-supervised task by ignoring the label.

**Analysis.** We assess the aforementioned setup on multiple real-world datasets. Table 6.1 highlights that, indeed, the SSL task on labeled data improves the conformal prediction intervals across all datasets in terms of reduced width and excess, occasionally at the expense of a slightly larger deficit. However, coverage is maintained in all scenarios.

We also note that, of course, the smaller datasets (with respect to samples) see the smallest improvements compared to those datasets with more samples. This is expected, as we simply have fewer data samples on which the SSL task can learn. That said, irrespective of the number of samples, there is still a benefit in repurposing existing labeled data to extract self-supervised information.

**Takeaway.** Self-supervision improves the quality of conformal prediction intervals, even on a fixed dataset without additional data, simply by repurposing the labeled data.

#### Augmented with unlabeled data setting

**Setup.** This experiment highlights the value of self-supervision, coupled with unlabeled data, to improve CRF-based models. We consider the typical setup in self-supervised learning, with $\mathscr{D}_{\text{labeled}}$ and $\mathscr{D}_{\text{unlabeled}}$.

We evaluate the effect for varying proportions $p$ of labeled examples. $\mathscr{D}_{\text{labeled}}$ is then split and used for the predictive and normalization models, as in Chapter 6.5.2. $\mathscr{D}_{unlabeled}$ augments the dataset used to train the self-supervised model (i.e. VIME).

Table 6.1 Assessing the impact of the SSL task on improving CRF, even *without* additional data. We find in all cases, averaged over 5 runs, that the additional SSL information can help to improve width, deficit, and excess (↓ better).

| Dataset | Metric | CRF | SSCP |
|---|---|---|---|
| concrete (n=1030) | Avg.Width | 0.768 | **0.743** |
| | Avg.Deficit | 0.099 | **0.098** |
| | Avg.Excess | 0.263 | **0.253** |
| community (n=1994) | Avg.Width | 2.602 | **2.462** |
| | Avg.Deficit | 0.355 | **0.361** |
| | Avg.Excess | 1.030 | **0.967** |
| star (n=2161) | Avg.Width | 0.293 | **0.263** |
| | Avg.Deficit | 0.036 | **0.031** |
| | Avg.Excess | 0.120 | **0.100** |
| bike (n=10886) | Avg.Width | 0.720 | **0.690** |
| | Avg.Deficit | 0.164 | **0.161** |
| | Avg.Excess | 0.244 | **0.232** |
| Blog data (n=52397) | Avg.Width | 3.474 | **3.360** |
| | Avg.Deficit | **3.084** | 3.155 |
| | Avg.Excess | 1.292 | **1.227** |
| Facebook (n=81311) | Avg.Width | 1.917 | **1.860** |
| | Avg.Deficit | **1.956** | 1.998 |
| | Avg.Excess | 0.584 | **0.554** |

**Analysis.** What happens for varying $p$? For different values of $p$, the inclusion of unlabeled data through self-supervision reduces the width of the prediction intervals (Figure 6.5). Of course, as $p$ increases, so too does the quality of the predictive and normalization models. This explains the natural increase in the relative width of SSCP vs. CRF for increasing $p$, i.e. as expected, with more supervised data to train on (and less unlabeled data), there is a reduced benefit of the unlabeled data. However, even as $p$ increases, we still note the benefit of the self-supervised signal in reducing the width of the intervals relative to CRF.

As an ablation, to quantify the benefit of the unlabeled data, we train the self-supervised task on ONLY the smaller $\mathscr{D}_{labeled}$ rather than the larger combination, which includes $\mathscr{D}_{unlabeled}$. The results indicate that the inclusion of unlabeled data is indeed beneficial with tighter intervals. These results can be found in Appendix E.3.1.

Finally, we expect that the choice and nature of the SSL task is impactful, and specific pretext tasks might provide a superior signal to the normalization model. We discuss this in greater detail in Appendix E.3.4 by assessing alternative modes of self-supervision.

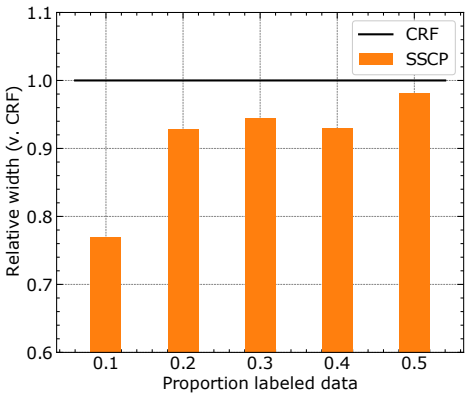
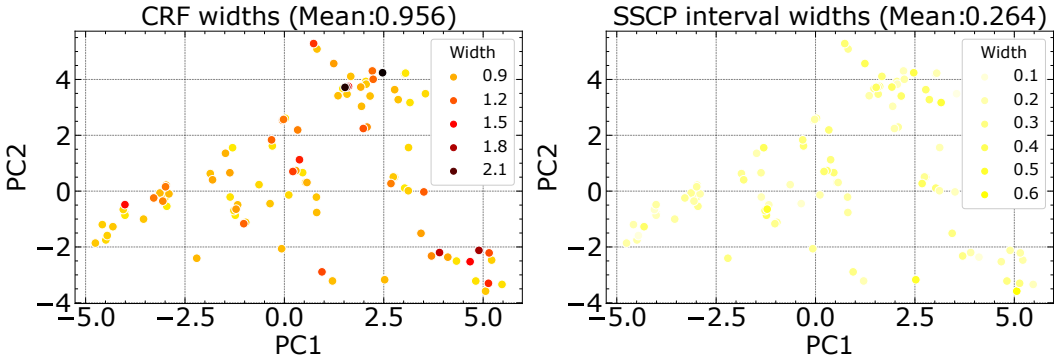



Fig. 6.5 Relative prediction interval widths of SSCP compared to CRF for varying proportions of labeled data (↓ better). Avg. interval width across datasets is lower with the inclusion of unlabeled data across all $p$.

**Takeaway.** Self-supervision using unlabeled data helps to improve the quality of conformal intervals across different proportions of labeled vs. unlabeled data.

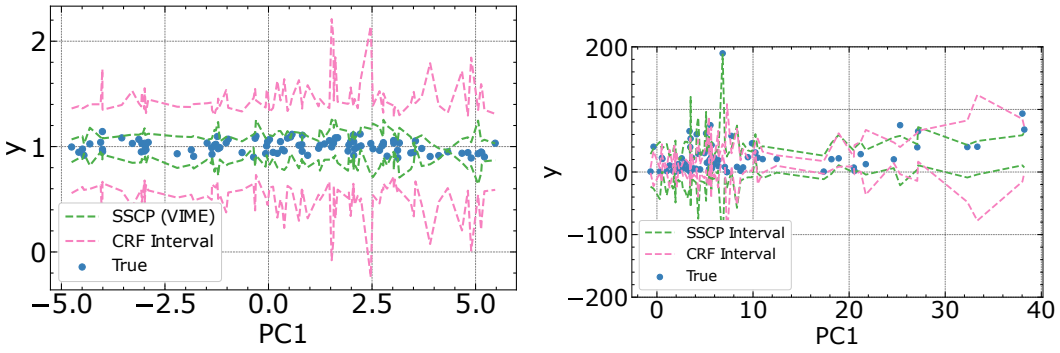
### 6.5.3 Insights

Let us now examine where SSCP helps the most. First, we compare the prediction intervals of samples with CRF vs. the same samples with SSCP. We find that the main source of improvement over CRF is for those samples with the largest interval width. Figure 6.6 highlights that SSCP significantly reduces the interval width for these samples, which is the key source of the reduction in the average width.




(a) CRF. (b) SSCP.

Fig. 6.6 Prediction interval width for most challenging samples on Star. SSCP reduces interval width




(a) Star dataset. (b) Facebook dataset.

Fig. 6.7 Comparing the prediction intervals in different regions. SSCP provides benefits in the sparser regions.

This begs the question of what type of samples/regions SSCP helps. To facilitate visualization, we reduce the dimensionality of the input features $\mathbf{X}$ using principal component analysis (PCA). We plot the first principal component (PC1) vs. output $y$ and compare the prediction intervals for SSCP vs. CRF. In Figure 6.7, we examine two coarse regions of the PC1 space: dense and sparse. In the denser regions, CRF has narrower intervals, similar to SSCP. In the sparser regions, CRF often has much wider prediction intervals compared to SSCP. It is in these sparse regions that SSCP clearly helps.

The rationale is that in the sparser regions, the conformal normalization model $\sigma$ does not perform well. Hence, to satisfy the coverage constraint, the CRF intervals are more conservative and, therefore, more inefficient in these challenging sparser regions.

In contrast, SSCP with the SS signal helps to improve the conformal normalization model (i.e. the residuals are predicted more accurately) and hence by virtue of the improved $\sigma$ it achieves intervals with lower width on the sparser challenging regions. Quantitatively, this improvement to $\sigma$ in MAE is $\pm 14\%$ and $\pm 5\%$ for Figure 6.7 (a) and (b) respectively. Further analysis is found in Appendix E.3.4 and 6.5.3.

**Takeaway.** SSCP improves prediction interval width on the most challenging samples and sparser regions.

### Robustness study

Finally, we wish to assess the robustness of the intervals and their improvements over the different random runs. Consequently, we aim to assess the confidence interval (CI) gains (i.e. reductions) for SSCP (VIME) vs. CRF.

**Analysis.** We assess the 90% CI's of the reduction in interval width (i.e. performance gain) across all datasets averaged across runs. The results are shown in Figure 6.8.

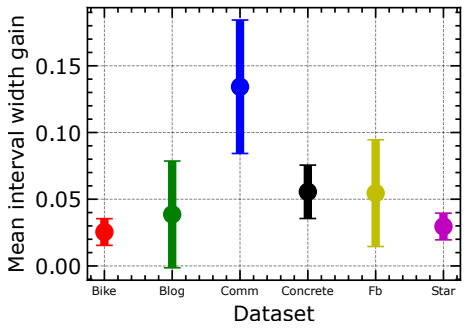



Fig. 6.8 Assessing the 90% confidence interval gains — which implies width reductions across datasets, averaged across runs. We find that SSCP has a net reduction in intervals across all datasets, highlighting the robustness of the approach.

We find that SSCP is robust in the improvements, showing a net gain over CRF (with interval reductions) across all datasets.

**Takeaway.** SSCP shows a performance gain (i.e., interval reduction) across the board, highlighting the robustness of including the self-supervised signal to improve the conformal predictive intervals.

## 6.6 Discussion

Improving the quality of prediction intervals is crucial for more reliable uncertainty quantification- a cornerstone of reliable and trustworthy machine learning. In this chapter, we show the value of self-supervised learning beyond model pretraining and representation learning- and moreover, how data can be used in a novel manner to improve uncertainty quantification.

Specifically, we propose a framework to integrate self-supervised losses to improve conformal prediction intervals. We empirically show with both synthetic and real data that self-supervision improves the prediction intervals of locally adaptive conformal prediction, specifically CRF (but without loss of generality CQR as well). In addition, we provide insights that the major gain is that self-supervision assists in more challenging and sparser data regions.

**Limitations.** SSCP provides a framework to highlight the value of self-supervision for conformal prediction. Our evaluation of pretext tasks (in the main thesis and appendix) are provided as examples and additional tasks could be evaluated using the provided "recipe", inclusive of future work to design a conformal prediction-specific pretext task.

**Societal impact.** Quantifying predictive uncertainty is important for reliable and trustworthy ML, especially in high-stakes settings such as medicine and finance. SSCP seeks to improve this and provide more trustworthy and actionable predictions by providing end users with reliable yet adaptive prediction intervals as a measure of confidence. These uncertainty intervals can then be used by end-users to guide decision making.

**Future work.** The findings of this chapter open up avenues for future research, leveraging the idea that self-supervision can be used to improve conformal prediction. The following are proposed areas of future exploration: (i) assessing alternative self-supervised approaches (e.g. [33, 126]), (ii) development of new conformal-aware self-supervised tasks, (iii) application of self-supervision to improve feature-wise conformal prediction (as per [210], [239]), (iv) the use of self-supervision with other data modalities such as images, where the self-supervised tasks tend to be distinct from the tabular setting.

# Chapter 7

# Conclusion

## 7.1 Broader Perspectives

**Summary.** This thesis advances the emerging research area of data-centric AI, a shift in focus from the de facto model-centric ML paradigm. The research presented in this thesis proposes systematic ML methods and algorithms that enhance ML reliability and trustworthiness from a data-centric perspective. Collectively, these contributions demonstrate that developing new ML approaches to handle data has a significant impact on ensuring AI reliability, rather than simply improving the model. Moreover, this thesis challenges the traditional perspective that larger and more complex models inherently lead to better AI and instead argues that well-characterized, curated, and contextually evaluated data is an equally important foundation of reliable and trustworthy ML.

**Shift from Model-Centric to Data-Centric AI.** For much of ML's history, research and development have been model-centric, with efforts focused on designing more sophisticated architectures, fine-tuning hyperparameters, and leveraging larger-scale computing resources to push the limits of performance. This focus has led to the rapid evolution of deep learning, with models growing increasingly large and powerful, achieving state-of-the-art results across benchmarks. However, despite these advances, many AI systems remain unreliable in real-world settings, particularly in domains such as healthcare, finance, and scientific research, where poor generalization and unpredictable behavior can lead to severe consequences [67].

A fundamental issue underlying these limitations is that most AI failures are not due to inadequacies in model architectures but rather due to deficiencies in the data itself. No matter how sophisticated a model may be, it is ultimately constrained by the quality of the data it learns from. A model trained on biased, inconsistent, or incomplete data will inevitably inherit these flaws, leading to errors in critical decision-making processes. This realization has led to a growing movement toward data-centric AI, which posits that improving data

quality, through better characterization, curation, augmentation, and evaluation, is vital to enhancing ML reliability and trustworthiness.

This thesis advances this area by introducing novel and systematic ML methods. Let us unpack the different data-centric ML methods introduced in this thesis. (1) Data-IQ enables a principled understanding of data subgroups through the lens of uncertainty estimation and sample-level learning dyanmics, thereafter showing how these identified groups could guide data sculpting, feature acquisition, dataset auditing and more. (2) Data-SUITE helps identify incongruous instances at test/deployment time, wherein models could fail, both in terms of poor performance and to guide future data collection. (3) Curated LLM explores how LLM-based synthetic data, along with principled curation, can help address the data logjam plaguing ultra-low data regimes, thereby unlocking the capabilities of ML in these underserved settings. Beyond data quality, this thesis offers a data-centric lens for improving the evaluation and testing of machine learning models. (4) Context-aware testing introduces a new paradigm for auditing data slice failures or underperformance, demonstrating that evaluating models with context-guided search is more valuable and reliable than data-only approaches. Finally, from a deployment perspective, this work tackles the challenges of data-centric uncertainty quantification. (5) Self-supervised conformal prediction demonstrates how data (and the information contained therein) can be re-purposed via self-supervised representations to improve uncertainty quantification.

These findings suggest that future AI systems should prioritize these data-centric aspects, fundamentally reorienting how AI research approaches the challenge of real-world ML reliability and impact.

**Reliable and Trustworthy ML.** Building reliable and trustworthy AI requires more than just high accuracy on benchmark datasets. Real-world applications, particularly in high-stakes environments such as healthcare, finance, and decision-making, demand generalization, calibrated uncertainty quantification, and an understanding of when a model's predictions can or cannot be trusted [67].

This thesis directly addresses these concerns by developing tools that enhance AI trustworthiness through improved data understanding and evaluation. By designing Data-IQ to systematically characterize data heterogeneity, this research highlights the importance of recognizing how different samples within a dataset may exhibit varying levels of learnability and how this impacts model training. The model may well learn better on some subsets, while others introduce greater ambiguity due to noise or label uncertainty. Similarly, Data-SUITE not only flags inconsistent data but also provides an understanding of why failures occur at the data level. In addition, its findings can also influence future data collection to prevent such issues.

Another key aspect of reliability is the ability to evaluate models in a way that reflects real-world complexity and context. The Context-Aware Testing (CAT) paradigm and subsequent SMART Testing framework is a novel approach to model evaluation, leveraging LLMs to integrate domain knowledge and context-guided search into the testing process, allowing for adaptive and contextually aware error analysis. Unlike traditional evaluation methods that rely purely on data alone, SMART Testing dynamically generates hypotheses for where and why a model might fail, ensuring that evaluations are not static but instead tailored to real-world deployment scenarios and reducing false positive and negatives.

Finally, a critical component of trust in AI is the quantification of uncertainty, which is often overlooked. Self-Supervised Conformal Prediction is introduced as a method to enhance uncertainty quantification by utilizing unlabeled data through self-supervised learning to improve conformal prediction intervals and provide more adaptive uncertainty estimates, thereby aiding in informed decision-making.

Together, these contributions establish a more comprehensive framework for ML reliability and trustworthiness. By systematically addressing the data-related challenges, we provide practitioners with a comprehensive toolkit for building ML systems that are not just accurate on average but reliable and trustworthy, towards enabling ML systems that can be deployed with confidence in high-stakes domains such as healthcare, finance etc.

**Connecting the contributions: toward an integrated data-centric pipeline.** Although the methods are presented as standalone contributions, they can be composed into a unified data-centric workflow. First, **Data-IQ** can provide structured *context* for **SMART Testing**: the Ambiguous/Hard partitions identify where training dynamics suggest instability or underspecification, which can be summarized as candidate "risk factors" (e.g., sparse regions, conflicting labels, or feature collision) and supplied to SMART as additional context to guide hypothesis generation toward plausible failure modes. This would reduce reliance on the LLM's generic priors and focus testing on data-supported suspects.

Second, **Data-SUITE** can complement **SSCP** by providing a *data-centric* measure of incongruity at deployment time. Concretely, Data-SUITE's feature-wise prediction interval widths and inconsistency indicators can be used as auxiliary signals for local adaptation in conformal prediction: instances with high incongruity could trigger wider intervals, different calibration buckets, or targeted collection of additional labels for re-calibration. Conversely, SSCP-style self-supervised signals could be explored as richer representations within Data-SUITE's "representer" block to better capture structure in high-dimensional or multimodal data.

These connections point toward a broader research agenda: data-centric systems that continuously (i) characterize training data, (ii) test models using context grounded in these characterizations, and (iii) monitor deployment-time data congruence to adapt uncertainty estimates and guide targeted data acquisition.

## 7.2 Societal Implications

The data-centric lens proposed in this thesis has significant societal implications as it addresses fundamental challenges around training and deployment of ML systems in high-stakes domains. Although, the technical contributions advance the field of AI, their ultimate value lies in how they can improve decision-making systems that impact humans. In the following, we explore the societal implications of each chapter, highlighting both the opportunities they present and the ethical considerations they raise.

**Data-IQ: Implications for Equitable Model Development and Deployment.** Data-IQ's systematic characterization of examples into Easy, Ambiguous, and Hard subgroups has implications for fairness and equity in ML applications. In healthcare settings, for instance, identifying ambiguous examples, where patients with similar features have different outcomes, can highlight subpopulations for which current feature sets are insufficient for accurate prediction. This insight can guide clinicians in collecting additional biomarkers or clinical measurements specifically for that patient subgroup, potentially addressing disparities in model predictions and ultimately care quality.

The feature acquisition guidance provided by Data-IQ also has economic implications, especially in resource-constrained settings. By quantifying which features would most reduce ambiguity, organizations can prioritize their data collection efforts and allocate resources more efficiently. This capability is particularly valuable in developing regions or public health contexts where comprehensive data collection may be prohibitively expensive, yet targeted improvements could yield substantial benefits in model performance.

However, the identification of "Hard or unlearnable" examples raises important ethical considerations. While we have demonstrated that the hard or unlearnable samples are likely mislabeled (and can therefore be either pruned from the dataset or corrected in a targeted manner), there are still potential risks. To mitigate this risk, Data-IQ implementations must include appropriate human oversight (i.e. human-in-the-loop) and domain expert validation.

**Data-SUITE: Implications for Transparency and Data Governance.** Data-SUITE's ability to identify incongruous regions for in-distribution data has significant implications for regulatory compliance and data governance frameworks. As regulations increasingly

mandate transparency in AI systems, Data-SUITE provides a systematic mechanism to identify potential blind spots in models at deployment time.

Furthermore, the prototype-based insights generated by Data-SUITE could also facilitate more effective communication between technical teams and stakeholders. By providing concrete, interpretable examples of where models may be unreliable, Data-SUITE enables non-technical decision-makers to understand model limitations and make more informed deployment decisions, or could guide future targeted data collection to mitigate the gaps.

**CLLM: Implications for Data Equity and Global Access to AI.** CLLM's approach to data augmentation in low-data regimes has immense potential for democratizing access to ML technologies. By enabling effective model training with as few as 20 to 100 examples, CLLM can empower organizations in low-resource settings to develop context-specific AI solutions without requiring massive data collection efforts or computational resources. This is particularly relevant for developing regions and specialized domains where data scarcity has traditionally been a barrier to AI adoption.

The demonstrated ability of CLLM to improve performance for underrepresented subgroups has direct implications for healthcare equity. Medical conditions that disproportionately affect minority populations often suffer from data scarcity due to historical disparities. By enhancing model performance on these underrepresented groups, CLLM could help address disparities in model performance and diagnostic accuracy across demographic groups.

However, the reliance on LLMs for synthetic data generation introduces concerns about embedded biases in these foundation models. While our curation mechanism was shown to mitigate this risk by filtering out potentially harmful or biased synthetic samples, implementations of CLLM must include ongoing monitoring for unexpected biases that may emerge in specific applications. Additionally, transparency about the use of synthetic data in training is crucial for maintaining trust with end-users and ensuring regulatory compliance. Finally, issues surrounding privacy are vital to consider when using synthetic data.

**SMART: Implications for Accountability and Safe Deployment.** SMART Testing represents an advancement for AI model evaluation beyond data-only approaches by identifying meaningful model failures. In safety-critical domains, the ability to proactively identify failure modes that align with real-world context can prevent harmful deployment decisions and inform targeted improvements.

The context-aware nature of SMART has significant implications for regulatory compliance and the development of industry standards. As regulatory bodies increasingly require evidence of thorough testing for AI systems in high-stakes applications, SMART provides a documentable, principled approach that integrates domain knowledge into the testing process. This aligns with the trend toward more rigorous pre-deployment verification. Additionally,

the comprehensive model reports generated by SMART also facilitate more effective communication between model developers, domain experts, organizational decision-makers and regulatory bodies. That said, users should still be cautious as LLMs can introduce biases. Although our self-falsification mechanism has been empirically shown to mitigate these biases, we recommend that human audits of the model reports remain vital in cases where specific testing dimensions are overlooked.

**SSCP: Implications for Responsible Decision Support.** SSCP's improvements to uncertainty quantification has direct implications for responsible decision support systems across various domains. By providing more adaptive and efficient prediction intervals, SSCP enables decision makers to appropriately calibrate their confidence in model predictions, particularly in challenging or sparse data regions where traditional methods produce overly conservative intervals.

In healthcare settings, more precise uncertainty quantification could improve the prioritization of human interventions. For example, in healthcare, when an ML model makes a prediction about a patient, an adaptive measure of uncertainty can help clinicians make more informed decisions about whether to intervene. This has both individual benefits for patient care and system-level benefits for managing healthcare resources.

The ability of SSCP to leverage unlabeled data for improved uncertainty quantification also has benefits. In contexts where labeled data is scarce, SSCP's approach allows for improving prediction reliability using more readily available unlabeled data. This creates new possibilities for developing robust predictive models in sensitive domains without requiring an extensive collection of labeled examples.

**Broader Societal Considerations.**

Across all the methods developed in this thesis, several overarching societal considerations emerge. Firstly, these methods collectively highlight the importance of human oversight (i.e. human-in-the-loop). Rather than replacing human judgment, the approaches developed here are designed to augment human decision-making. Maintaining this human-centered perspective is essential as these methods are deployed in practice, rather than the algorithms being used in a fully automated manner.

Secondly, there is a broader consideration of how these methods might reshape practices around data collection and management. By highlighting the value of targeted, quality-focused data collection over purely large-scale data collection, these approaches could help shift the focus toward thoughtful data practices. This cultural shift may be one of the most significant long-term impacts of the data-centric paradigm.

In conclusion, the data-centric approaches developed in this dissertation have the potential to advance not only the technical capabilities of ML systems, but also their reliable and

trustworthy implementation. By addressing fundamental challenges in data quality, characterization, testing and deployment, these methods contribute to a future where AI systems can be deployed with greater confidence in their reliability, fairness, and trustworthiness across diverse contexts and populations.

## 7.3 Future Agenda

This thesis is just the beginning. The works presented here represent initial steps in a broader agenda for data-centric AI, aiming for reliable and trustworthy ML. For additional avenues not covered in this thesis, see DC-Check [216] or the publications not covered in this thesis, see Chapter 1.5.

While the respective conclusions and discussions in each chapter have already outlined various directions for future work, we outline some particularly promising avenues below.

**Data-Centric AI Across Modalities.** While much of the work in this thesis has focused on structured tabular data and text and image data, real-world AI systems must operate across a wide range of data modalities, including time-series and graph-based data. Each of these modalities presents unique challenges in terms of data characterization, quality assessment, and uncertainty quantification.

For example, in time-series data, data characterization and performance attribution take on a temporal dimension, as challenges might manifest at certain timesteps and not others. Hence, future work should consider how this affects the time series at a global level. Similarly, in graph-based systems, data quality is often tied to structural consistency and relational integrity. Many graph learning models operate under the assumption that node and edge relationships are correct, but real-world graphs—such as directed graphs, social networks, or protein interaction maps—are often incomplete, noisy, or biased by data collection methods. Future research must explore how graph-aware data-centric AI methods can improve graph-based models by refining connectivity structures, detecting anomalies, and identifying spurious relationships before models are trained.

Moreover, the intersection of these modalities in multimodal learning presents even greater challenges, as quality issues may manifest differently across modalities or emerge from their interactions (especially since the modalities do not reside in the same unified latent space). Consequently, understanding how data quality propagates across modalities and developing unified frameworks that can characterize heterogeneous data represents a significant research frontier with wide-ranging utility and applications.

Furthermore, while qualitative examples for image and text data were provided (e.g., in Chapter 2), the methodologies proposed in this thesis have primarily been evaluated on struc-

tured, tabular data. Adapting these methods comprehensively to unstructured data modalities presents unique challenges that require further investigation. For example, ambiguity in tabular data often arises from discrete feature collisions, whereas in images or text, it may stem from semantic overlaps, occlusions, or linguistic nuances. Future research should systematically adapt these data-centric frameworks to unstructured modalities, exploring how metrics like aleatoric uncertainty and incongruity translate to high-dimensional, continuous feature spaces like images, audio, and raw text.

**Integrating Data-Centric Frameworks.** While the methodologies introduced in this thesis have largely been presented as independent solutions addressing specific stages of the ML pipeline, there is significant potential in integrating them to create a unified, end-to-end data-centric ecosystem. For instance, the subsets of "ambiguous" or "hard" examples identified by Data-IQ during training could serve as a rich source of context to automatically generate targeted failure hypotheses for SMART Testing. Similarly, the incongruity scores derived from Data-SUITE at deployment time could be directly integrated into the Self-Supervised Conformal Prediction (SSCP) framework. By weighting the non-conformity scores or self-supervised losses using Data-SUITE's incongruity metrics, the system could provide even more adaptive and robust uncertainty estimates in highly heterogeneous regions. Exploring these intersections represents a promising frontier for developing holistic data-centric AI systems.

**Beyond Prediction: Data-Centric AI for Diverse Tasks.** In this thesis, we have primarily looked at data-centric AI through the lens of predictive tasks and supervised learning. However, many critical real-world applications go beyond classification and regression. In particular, a future agenda could explore data-centric challenges in the context of causal inference and reinforcement learning. For causal inference, understanding the quality and limitations of observational data is paramount, as unmeasured confounding and selection bias can fundamentally undermine causal conclusions. For reinforcement learning, the characteristics of the data collected during exploration directly impact what policies can be learned, suggesting potential benefits from more principled approaches to experience collection and assessment. Alternatively, the data we have on hand might impact areas like off-policy evaluation (i.e. what we can evaluate depends on the data itself). Each of these problem domains presents unique challenges but also opportunities to develop specialized tools that address their particular requirements.

**Adaptive and Self-Improving Data Systems.** A fundamental limitation of current ML pipelines is their reliance on static, manually curated datasets. As ML and AI systems are increasingly deployed in dynamic, real-world environments, they must be capable of autonomously acquiring, assessing and refining their own data. The future of data-centric AI

lies in self-improving AI systems that can continuously update their training distributions, detect shifts in data quality, and adapt their learning objectives accordingly.

One key challenge is designing adaptive data filtering and augmentation pipelines that can identify and incorporate new, high-value training data while discarding outdated or redundant information. This requires moving beyond traditional dataset curation, towards developing AI systems that actively monitor their data sources, detect distributional shifts, and refine their training distributions over time. In areas such as autonomous robotics, AI-driven scientific discovery, and adaptive healthcare monitoring, the ability to self-assess data quality and dynamically adjust learning strategies will be critical.

Another promising avenue is the development of AI models that can guide their own data collection via real-world experimentation. For example, an AI-driven drug discovery system might not only learn from existing biomedical datasets but also suggest new real-world experiments to acquire data that can either improve the system or reduce uncertainty.

This future vision of adaptive, self-improving data-centric AI systems necessitates new methods for quantifying data utility, identifying gaps in training distributions, and making informed decisions about which data to acquire next.

**Human-AI Intersection in Data-Centric Systems.** As AI systems are increasingly used in decision-making processes, understanding the intersection between humans and AI becomes increasingly important. In particular, humans often possess expert domain knowledge or ethical intuitions that complement the quantitative data-centric methods. Hence, while this thesis has focused on improving data from a technical perspective, future work could explore how humans interact with, interpret, and contribute to AI-driven data curation and evaluation.

### 7.3.1 Concluding remarks.

In closing, the hope that comes with this thesis is that, by taking the first step in formalizing and investigating these problems in data-centric AI, the ML community may draw inspiration from (1) the problem classes (and proposed algorithmic solutions) and (2) the fundamental value of the broader data-centric AI agenda. Consequently, we hope that this thesis contributes to achieving the broader goal of developing reliable and trustworthy ML systems that can be deployed with confidence across a wide range of high-stakes domains.

# References


[1] Ade-Ibijola, A. and Okonkwo, C. (2023). Artificial intelligence in africa: Emerging challenges. In *Responsible AI in Africa: Challenges and Opportunities*, pages 101–117. Springer International Publishing Cham.

[2] Alaa, A., van Breugel, B., Saveliev, E., and van der Schaar, M. (2022). How faithful is your synthetic data? sample-level metrics for evaluating and auditing generative models.

[3] Alami, H., Rivard, L., Lehoux, P., Hoffman, S. J., Cadeddu, S. B. M., Savoldelli, M., Samri, M. A., Ag Ahmed, M. A., Fleet, R., and Fortin, J.-P. (2020). Artificial intelligence in health care: laying the foundation for responsible, sustainable, and inclusive innovation in low-and middle-income countries. *Globalization and Health*, 16:1–6.

[4] Algan, G. and Ulusoy, I. (2021). Image classification with deep learning in the presence of noisy labels: A survey. *Knowledge-Based Systems*, 215:106771.

[5] Angwin, J., Larson, J., Kirchner, L., and Mattu, S. (2016). Machine bias. *ProPublica: https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing*.

[6] Arora, S., Du, S., Hu, W., Li, Z., and Wang, R. (2019). Fine-grained analysis of optimization and generalization for overparameterized two-layer neural networks. In *International Conference on Machine Learning*, pages 322–332. PMLR.

[7] Arpit, D., Jastrzębski, S., Ballas, N., Krueger, D., Bengio, E., Kanwal, M. S., Maharaj, T., Fischer, A., Courville, A., Bengio, Y., et al. (2017). A closer look at memorization in deep networks. In *International conference on machine learning*, pages 233–242. PMLR.

[8] Ashmore, R., Calinescu, R., and Paterson, C. (2021). Assuring the machine learning lifecycle: Desiderata, methods, and challenges. *ACM Computing Surveys (CSUR)*, 54(5):1–39.

[9] Asiedu, M. N., Dieng, A., Oppong, A., Nagawa, M., Koyejo, S., and Heller, K. (2023). Globalizing fairness attributes in machine learning: A case study on health in africa. *arXiv preprint arXiv:2304.02190*.

[10] Astorga, N., Liu, T., Seedat, N., and van der Schaar, M. (2024). Partially observable cost-aware active-learning with large language models. *Advances in Neural Information Processing Systems*, 38.

[11] Asuncion, A. and Newman, D. (2007a). Uci machine learning repository.

[12] Asuncion, A. and Newman, D. (2007b). UCI machine learning repository.

[13] Ayres-de Campos, D., Bernardes, J., Garrido, A., Marques-de Sa, J., and Pereira-Leite, L. (2000). Sisporto 2.0: a program for automated analysis of cardiotocograms. *Journal of Maternal-Fetal Medicine*, 9(5):311–318.

[14] Balasubramanian, V., Ho, S.-S., and Vovk, V. (2014). *Conformal prediction for reliable machine learning: theory, adaptations and applications*. Newnes.

[15] Baqui, P., Bica, I., Marra, V., Ercole, A., and van Der Schaar, M. (2020a). Ethnic and regional variations in hospital mortality from covid-19 in brazil: a cross-sectional observational study. *The Lancet Global Health*, 8(8):e1018–e1026.

[16] Baqui, P., Bica, I., Marra, V., Ercole, A., and van Der Schaar, M. (2020b). Ethnic and regional variations in hospital mortality from covid-19 in brazil: a cross-sectional observational study. *The Lancet Global Health*, 8(8):e1018–e1026.

[17] Bedford, T. and Cooke, R. M. (2001). Probability density decomposition for conditionally dependent random variables modeled by vines. *Annals of Mathematics and Artificial intelligence*, 32(1):245–268.

[18] Beede, E., Baylor, E., Hersch, F., Iurchenko, A., Wilcox, L., Ruamviboonsuk, P., and Vardoulakis, L. M. (2020). A human-centered evaluation of a deep learning system deployed in clinics for the detection of diabetic retinopathy. In *Proceedings of the 2020 CHI conference on human factors in computing systems*, pages 1–12.

[19] Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal statistical society: series B (Methodological)*, 57(1):289–300.

[20] Berk, R., Heidari, H., Jabbari, S., Joseph, M., Kearns, M., Morgenstern, J., Neel, S., and Roth, A. (2017). A convex framework for fair regression. *arXiv preprint arXiv:1706.02409*.

[21] Berrevoets, J., Seedat, N., Imrie, F., and Van Der Schaar, M. (2023). Differentiable and transportable structure learning. In *International Conference on Machine Learning*, pages 2206–2233. PMLR.

[22] Biswas, A., Nasim, M., Imran, A., Sejuty, A. T., Fairooz, F., Puppala, S., and Talukder, S. (2023). Generative adversarial networks for data augmentation. *arXiv preprint arXiv:2306.02019*.

[23] Borisov, V., Leemann, T., Seßler, K., Haug, J., Pawelczyk, M., and Kasneci, G. (2021). Deep neural networks and tabular data: A survey. *arXiv preprint arXiv:2110.01889*.

[24] Borisov, V., Sessler, K., Leemann, T., Pawelczyk, M., and Kasneci, G. (2023). Language models are realistic tabular data generators. In *The Eleventh International Conference on Learning Representations*.

[25] Boström, H., Linusson, H., Löfström, T., and Johansson, U. (2016). Evaluation of a variance-based nonconformity measure for regression forests. In *Symposium on Conformal and Probabilistic Prediction with Applications*, pages 75–89. Springer.

[26] Breunig, M. M., Kriegel, H.-P., Ng, R. T., and Sander, J. (2000). Lof: identifying density-based local outliers. In *Proceedings of the 2000 ACM SIGMOD international conference on Management of data*, pages 93–104.

[27] Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33:1877–1901.

[28] Cabrera, A. A., Kahng, M., Hohman, F., Morgenstern, J., and Chau, D. H. (2019). Discovery of intersectional bias in machine learning using automatic subgroup generation. In *ICLR Debugging Machine Learning Models Workshop*.

[29] Calmon, F., Wei, D., Vinzamuri, B., Natesan Ramamurthy, K., and Varshney, K. R. (2017). Optimized pre-processing for discrimination prevention. *Advances in neural information processing systems*, 30.

[30] Chapelle, O., Schölkopf, B., and Zien, A., editors (2006). *Semi-Supervised Learning*. The MIT Press.

[31] Chawla, N. V., Bowyer, K. W., Hall, L. O., and Kegelmeyer, W. P. (2002). Smote: synthetic minority over-sampling technique. *Journal of artificial intelligence research*, 16:321–357.

[32] Chen, I. Y., Pierson, E., Rose, S., Joshi, S., Ferryman, K., and Ghassemi, M. (2021). Ethical machine learning in healthcare. *Annual Review of Biomedical Data Science*, 4:123–144.

[33] Chen, T., Kornblith, S., Norouzi, M., and Hinton, G. (2020). A simple framework for contrastive learning of visual representations. In *International Conference on Machine Learning*, pages 1597–1607. PMLR.

[34] Chernozhukov, V., Wüthrich, K., and Zhu, Y. (2021). Distributional conformal prediction. *Proceedings of the National Academy of Sciences*, 118(48):e2107794118.

[35] Chowdhery, A., Narang, S., Devlin, J., Bosma, M., Mishra, G., Roberts, A., Barham, P., Chung, H. W., Sutton, C., Gehrmann, S., et al. (2022). Palm: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311*.

[36] Chowdhury, R. (2021). Sharing learnings about our image cropping algorithm.

[37] Christakis, M., Eniser, H. F., Hoffmann, J., Singla, A., and Wüstholz, V. (2023). Specifying and testing k-safety properties for machine-learning models. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*, pages 4748–4757.

[38] Chug, S., Kaushal, P., Kumaraguru, P., and Sethi, T. (2021). Statistical learning to operationalize a domain agnostic data quality scoring. *arXiv preprint arXiv:2108.08905*.

[39] Chung, Y., Kraska, T., Polyzotis, N., Tae, K. H., and Whang, S. E. (2019). Slice finder: Automated data slicing for model validation. In *2019 IEEE 35th International Conference on Data Engineering (ICDE)*, pages 1550–1553. IEEE.

[40] Ciecierski-Holmes, T., Singh, R., Axt, M., Brenner, S., and Barteit, S. (2022). Artificial intelligence for strengthening healthcare systems in low-and middle-income countries: a systematic scoping review. *npj Digital Medicine*, 5(1):162.

[41] Collins, G. S., Reitsma, J. B., Altman, D. G., and Moons, K. G. (2015). Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (tripod) the tripod statement. *Circulation*, 131(2):211–219.

[42] Costello, K. and Rimol, M. (2019). Gartner identifies the top strategic technology trends for 2021. *Gartner*.

[43] d'Eon, G., d'Eon, J., Wright, J. R., and Leyton-Brown, K. (2022). The spotlight: A general method for discovering systematic errors in deep learning models. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, pages 1962–1981.

[44] Der Kiureghian, A. and Ditlevsen, O. (2009). Aleatory or epistemic? does it matter? *Structural safety*, 31(2):105–112.

[45] Dong, Q., Li, L., Dai, D., Zheng, C., Wu, Z., Chang, B., Sun, X., Xu, J., and Sui, Z. (2022). A survey for in-context learning. *arXiv preprint arXiv:2301.00234*.

[46] Duggan, M. A., Anderson, W. F., Altekruse, S., Penberthy, L., and Sherman, M. E. (2016a). The surveillance, epidemiology and end results (seer) program and pathology: towards strengthening the critical relationship. *The American journal of surgical pathology*, 40(12):e94.

[47] Duggan, M. A., Anderson, W. F., Altekruse, S., Penberthy, L., and Sherman, M. E. (2016b). The surveillance, epidemiology and end results (SEER) program and pathology: towards strengthening the critical relationship. *The American Journal of Surgical Pathology*, 40(12):e94.

[48] Dwork, C., Hardt, M., Pitassi, T., Reingold, O., and Zemel, R. (2012). Fairness through awareness. In *Proceedings of the 3rd innovations in theoretical computer science conference*, pages 214–226.

[49] El Emam, K., Mosquera, L., and Hoptroff, R. (2020). *Practical synthetic data generation: balancing privacy and the broad availability of data*. O'Reilly Media.

[50] Esteban, C., Hyland, S. L., and Rätsch, G. (2017). Real-valued (medical) time series generation with recurrent conditional gans. *arXiv preprint arXiv:1706.02633*.

[51] Eyuboglu, S., Varma, M., Saab, K., Delbrouck, J.-B., Lee-Messer, C., Dunnmon, J., Zou, J., and Re, C. (2022). Domino: Discovering systematic errors with cross-modal embeddings. *arXiv preprint arXiv:2203.14960*.

[52] Farahani, A., Voghoei, S., Rasheed, K., and Arabnia, H. R. (2021). A brief review of domain adaptation. *Advances in data science and information engineering: proceedings from ICDATA 2020 and IKE 2020*, pages 877–894.

[53] Fehrman, E., Muhammad, A. K., Mirkes, E. M., Egan, V., and Gorban, A. N. (2017). The five factor model of personality and evaluation of drug consumption risk. In *Data science: innovative developments in data analysis and clustering*, pages 231–242. Springer.

[54] Feldman, M., Friedler, S. A., Moeller, J., Scheidegger, C., and Venkatasubramanian, S. (2015). Certifying and removing disparate impact. In *proceedings of the 21th ACM SIGKDD international conference on knowledge discovery and data mining*, pages 259–268.

[55] Finn, C., Abbeel, P., and Levine, S. (2017). Model-agnostic meta-learning for fast adaptation of deep networks. In *International conference on machine learning*, pages 1126–1135. PMLR.

[56] Foygel Barber, R., Candes, E. J., Ramdas, A., and Tibshirani, R. J. (2021). The limits of distribution-free conditional predictive inference. *Information and Inference: A Journal of the IMA*, 10(2):455–482.

[57] Gal, Y. and Ghahramani, Z. (2016). Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *international Conference on Machine Learning*, pages 1050–1059. PMLR.

[58] Gallegos, I. O., Rossi, R. A., Barrow, J., Tanjim, M. M., Kim, S., Dernoncourt, F., Yu, T., Zhang, R., and Ahmed, N. K. (2023). Bias and fairness in large language models: A survey. *arXiv preprint arXiv:2309.00770*.

[59] Gartlehner, G., West, S. L., Mansfield, A. J., Poole, C., Tant, E., Lux, L. J., and Lohr, K. N. (2012). Clinical heterogeneity in systematic reviews and health technology assessments: synthesis of guidance documents and the literature. *International Journal of Technology Assessment in Health Care*, 28(1):36–43.

[60] Gavves, E. (2019). Bayesian deep learning, lecture 10, university of amsterdam.

[61] Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Iii, H. D., and Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM*, 64(12):86–92.

[62] Ghorbani, A. and Zou, J. (2019). Data shapley: Equitable valuation of data for machine learning. In *International Conference on Machine Learning*, pages 2242–2251. PMLR.

[63] Ghosh, S., Yao, J., and Doshi-Velez, F. (2018). Structured variational learning of bayesian neural networks with horseshoe priors. In *International Conference on Machine Learning*, pages 1744–1753. PMLR.

[64] Ghosheh, G. O., Thwaites, C. L., and Zhu, T. (2023). Synthesizing electronic health records for predictive models in low-middle-income countries (lmics). *Biomedicines*, 11(6):1749.

[65] Gianfrancesco, M. A., Tamang, S., Yazdany, J., and Schmajuk, G. (2018). Potential biases in machine learning algorithms using electronic health record data. *JAMA internal medicine*, 178(11):1544–1547.

[66] Goel, K., Gu, A., Li, Y., and Re, C. (2020). Model patching: Closing the subgroup performance gap with data augmentation. In *International Conference on Learning Representations*.

[67] Goetz, L., Seedat, N., Vandersluis, R., and van der Schaar, M. (2024). Generalization—a key challenge for responsible ai in patient-facing clinical applications. *npj Digital Medicine*, 7(1):126.

[68] Goncalves, A., Ray, P., Soper, B., Stevens, J., Coyle, L., and Sales, A. P. (2020). Generation and evaluation of synthetic patient data. *BMC medical research methodology*, 20(1):1–40.

[69] Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., and Bengio, Y. (2014a). Generative adversarial nets. In *Advances in Neural Information Processing Systems 2014*, volume 27.

[70] Goodfellow, I. J., Shlens, J., and Szegedy, C. (2014b). Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572*.

[71] Graves, A. (2011). Practical variational inference for neural networks. *Advances in neural information processing systems*, 24.

[72] Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B., and Smola, A. (2012). A kernel two-sample test. *The Journal of Machine Learning Research*, 13(1):723–773.

[73] Grill, J.-B., Strub, F., Altché, F., Tallec, C., Richemond, P., Buchatskaya, E., Doersch, C., Avila Pires, B., Guo, Z., Gheshlaghi Azar, M., et al. (2020). Bootstrap your own latent A new approach to self-supervised learning. *Advances in Neural Information Processing Systems*, 33:21271–21284.

[74] Grinsztajn, L., Oyallon, E., and Varoquaux, G. (2022). Why do tree-based models still outperform deep learning on typical tabular data? *Advances in neural information processing systems*, 35:507–520.

[75] Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V., and Courville, A. (2017). Improved training of wasserstein gans. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, pages 5769–5779.

[76] Gupta, N., Patel, H., Afzal, S., Panwar, N., Mittal, R. S., Guttula, S., Jain, A., Nagalapatti, L., Mehta, S., Hans, S., et al. (2021). Data quality toolkit: Automatic assessment of data quality and remediation for machine learning datasets. *arXiv preprint arXiv:2108.05935*.

[77] Haeusser, P., Frerix, T., Mordvintsev, A., and Cremers, D. (2017). Associative domain adaptation. In *Proceedings of the IEEE international conference on computer vision*, pages 2765–2773.

[78] Hansen, L., Seedat, N., van der Schaar, M., and Petrovic, A. (2023). Reimagining synthetic tabular data generation through data-centric ai: A comprehensive benchmark. *Advances in Neural Information Processing Systems*, 37:33781–33823.

[79] Hardt, M., Price, E., and Srebro, N. (2016). Equality of opportunity in supervised learning. *Advances in neural information processing systems*, 29.

[80] Harries, M. and Wales, N. S. (1999). Splice-2 comparative evaluation: Electricity pricing.

[81] Hashimoto, T., Srivastava, M., Namkoong, H., and Liang, P. (2018). Fairness without demographics in repeated loss minimization. In *International Conference on Machine Learning*, pages 1929–1938. PMLR.

[82] Hegselmann, S., Buendia, A., Lang, H., Agrawal, M., Jiang, X., and Sontag, D. (2023). Tabllm: Few-shot classification of tabular data with large language models. In *International Conference on Artificial Intelligence and Statistics*, pages 5549–5581. PMLR.

[83] Hendrycks, D. and Dietterich, T. (2018). Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations*.

[84] Heremans, E. R., Seedat, N., Buyse, B., Testelmans, D., van der Schaar, M., and De Vos, M. (2024). U-pass: An uncertainty-guided deep learning pipeline for automated sleep staging. *Computers in Biology and Medicine*, 171:108205.

[85] Heremans, E. R., Van den Bulcke, L., Seedat, N., Devulder, A., Borzée, P., Buyse, B., Testelmans, D., Van Den Bossche, M., van Der Schaar, M., and De Vos, M. (2025). Automated remote sleep monitoring needs uncertainty quantification. *Journal of Sleep Research*, 34(1):e14300.

[86] Ho, T. K. and Basu, M. (2002). Complexity measures of supervised classification problems. *IEEE transactions on pattern analysis and machine intelligence*, 24(3):289–300.

[87] Hollmann, N., Müller, S., Eggensperger, K., and Hutter, F. (2022). Tabpfn: A transformer that solves small tabular classification problems in a second. In *The Eleventh International Conference on Learning Representations*.

[88] Hospedales, T. M., Antoniou, A., Micaelli, P., and Storkey, A. J. (2021). Meta-learning in neural networks: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.

[89] Houlsby, N., Huszár, F., Ghahramani, Z., and Lengyel, M. (2011). Bayesian active learning for classification and preference learning. *arXiv preprint arXiv:1112.5745*.

[90] Hsu, Y.-C., Shen, Y., Jin, H., and Kira, Z. (2020). Generalized odin: Detecting out-of-distribution image without learning from out-of-distribution data. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10951–10960.

[91] Hüllermeier, E. and Waegeman, W. (2021). Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine Learning*, 110:457–506.

[92] Huo, Z., Zhang, L., Khera, R., Huang, S., Qian, X., Wang, Z., and Mortazavi, B. J. (2021). Sparse gated mixture-of-experts to separate and interpret patient heterogeneity in ehr data. In *2021 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, pages 1–4. IEEE.

[93] Huynh, N., Berrevoets, J., Seedat, N., Crabbé, J., Qian, Z., and van der Schaar, M. (2024). Dagnosis: Localized identification of data inconsistencies using structures. In *International Conference on Artificial Intelligence and Statistics*, pages 1864–1872. PMLR.

[94] Izbicki, R., Shimizu, G. T., and Stern, R. B. (2020). Flexible distribution-free conditional predictive bands using density estimators. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 3068–3077. PMLR.

[95] Jain, A., Patel, H., Nagalapatti, L., Gupta, N., Mehta, S., Guttula, S., Mujumdar, S., Afzal, S., Sharma Mittal, R., and Munigala, V. (2020). Overview and importance of data quality for machine learning tasks. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 3561–3562.

[96] Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., de las Casas, D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L., Lavaud, L. R., Lachaux, M.-A., Stock, P., Scao, T. L., Lavril, T., Wang, T., Lacroix, T., and Sayed, W. E. (2023). Mistral 7b.

[97] Jiang, A. Q., Sablayrolles, A., Roux, A., Mensch, A., Savary, B., Bamford, C., Chaplot, D. S., Casas, D. d. l., Hanna, E. B., Bressand, F., et al. (2024). Mixtral of experts. *arXiv preprint arXiv:2401.04088*.

[98] Jiang, W. and Pardos, Z. A. (2021). Towards equity and algorithmic fairness in student grade prediction. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, pages 608–617.

[99] Jin, G., Yi, X., Zhang, L., Zhang, L., Schewe, S., and Huang, X. (2020). How does weight correlation affect generalisation ability of deep neural networks? *Advances in Neural Information Processing Systems*, 33:21346–21356.

[100] Jin, Q. and Ucar, T. (2023). Benchmarking tabular representation models in transfer learning settings. In *NeurIPS 2023 Second Table Representation Learning Workshop*.

[101] Joe, H. (2014). *Dependence modeling with copulas*. CRC press.

[102] Johansson, U., Sönströd, C., and Linusson, H. (2015a). Efficient conformal regressors using bagged neural nets. In *2015 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8.

[103] Johansson, U., Sönströd, C., and Linusson, H. (2015b). Efficient conformal regressors using bagged neural nets. In *2015 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8.

[104] Kaggle (2017). Kaggle machine learning and data science survey.

[105] Kandel, S., Paepcke, A., Hellerstein, J. M., and Heer, J. (2012). Enterprise data analysis and visualization: An interview study. *IEEE Transactions on Visualization and Computer Graphics*, 18(12):2917–2926.

[106] Karim, M. R., Shajalal, M., Graß, A., Döhmen, T., Chala, S. A., Boden, A., Beecks, C., and Decker, S. (2023). Interpreting black-box machine learning models for high dimensional datasets. In *2023 IEEE 10th International Conference on Data Science and Advanced Analytics (DSAA)*, pages 1–10. IEEE.

[107] Kendall, A. and Gal, Y. (2017). What uncertainties do we need in Bayesian deep learning for computer vision? *Advances in Neural Information Processing Systems*, 30.

[108] Kenton, J. D. M.-W. C. and Toutanova, L. K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186.

[109] Kim, J., Lee, C., and Park, N. (2022). Stasy: Score-based tabular data synthesis. In *The Eleventh International Conference on Learning Representations*.

[110] Kim, M. P., Ghorbani, A., and Zou, J. (2019). Multiaccuracy: Black-box post-processing for fairness in classification. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society*, pages 247–254.

[111] Kingma, D. P., Salimans, T., and Welling, M. (2015). Variational dropout and the local reparameterization trick. *Advances in neural information processing systems*, 28:2575–2583.

[112] Kingma, D. P. and Welling, M. (2014). Auto-encoding variational bayes. In *2nd International Conference on Learning Representations*.

[113] Kirsch, A., Van Amersfoort, J., and Gal, Y. (2019). Batchbald: Efficient and diverse batch acquisition for deep bayesian active learning. *Advances in neural information processing systems*, 32.

[114] Knaus, W. A., Harrell, F. E., Lynn, J., Goldman, L., Phillips, R. S., Connors, A. F., Dawson, N. V., Fulkerson, W. J., Califf, R. M., Desbiens, N., et al. (1995). The support prognostic model: Objective estimates of survival for seriously ill hospitalized adults. *Annals of internal medicine*, 122(3):191–203.

[115] Koenker, R. and Bassett Jr, G. (1978). Regression quantiles. *Econometrica: journal of the Econometric Society*, pages 33–50.

[116] Koenker, R. and Hallock, K. F. (2001). Quantile regression. *Journal of economic perspectives*, 15(4):143–156.

[117] Koh, P. W. and Liang, P. (2017). Understanding black-box predictions via influence functions. In *International Conference on Machine Learning*, pages 1885–1894. PMLR.

[118] Koh, P. W., Sagawa, S., Marklund, H., Xie, S. M., Zhang, M., Balsubramani, A., Hu, W., Yasunaga, M., Phillips, R. L., Gao, I., Lee, T., David, E., Stavness, I., Guo, W., Earnshaw, B. A., Haque, I. S., Beery, S., Leskovec, J., Kundaje, A., Pierson, E., Levine, S., Finn, C., and Liang, P. (2021). WILDS: A benchmark of in-the-wild distribution shifts. In *International Conference on Machine Learning*, pages 5637–5664. PMLR.

[119] Kotelnikov, A., Baranchuk, D., Rubachev, I., and Babenko, A. (2022). Tabddpm: Modelling tabular data with diffusion models. *arXiv preprint arXiv:2209.15421*.

[120] Koutroumpis, P., Leiponen, A., and Thomas, L. D. (2020). Markets for data. *Industrial and Corporate Change*, 29(3):645–660.

[121] Krishnan, R. and Tickoo, O. (2020). Improving model calibration with accuracy versus uncertainty optimization. *Advances in Neural Information Processing Systems*, 33:18237–18248.

[122] Kumagai, A. and Iwata, T. (2019). Unsupervised domain adaptation by matching distributions based on the maximum mean discrepancy via unilateral transformations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 4106–4113.

[123] Kuzilek, J., Hlosta, M., and Zdrahal, Z. (2017). Open university learning analytics dataset. *Scientific data*, 4(1):1–8.

[124] Kwon, Y., Won, J.-H., Kim, B. J., and Paik, M. C. (2020). Uncertainty quantification using bayesian neural networks in classification: Application to biomedical image segmentation. *Computational Statistics & Data Analysis*, 142:106816.

[125] Lakshminarayanan, B., Pritzel, A., and Blundell, C. (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in neural information processing systems*, 30.

[126] Lee, C., Imrie, F., and van der Schaar, M. (2022). Self-supervision enhanced feature selection with correlated gates. In *International Conference on Learning Representations*.

[127] Lee, K., Lee, K., Lee, H., and Shin, J. (2018). A simple unified framework for detecting out-of-distribution samples and adversarial attacks. *Advances in neural information processing systems*, 31.

[128] Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R. J., and Wasserman, L. (2018). Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111.

[129] Leibig, C., Allken, V., Ayhan, M. S., Berens, P., and Wahl, S. (2017). Leveraging uncertainty information from deep neural networks for disease detection. *Scientific reports*, 7(1):1–14.

[130] Lemaître, G., Nogueira, F., and Aridas, C. K. (2017). Imbalanced-learn: A python toolbox to tackle the curse of imbalanced datasets in machine learning. *Journal of Machine Learning Research*, 18(17):1–5.

[131] Lemmerich, F. and Becker, M. (2018). pysubgroup: Easy-to-use subgroup discovery in python. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 658–662.

[132] Leslie, D., Mazumder, A., Peppin, A., Wolters, M. K., and Hagerty, A. (2021). Does "ai" stand for augmenting inequality in the era of covid-19 healthcare? *BMJ*, 372.

[133] Levin, R., Cherepanova, V., Schwarzschild, A., Bansal, A., Bruss, C. B., Goldstein, T., Wilson, A. G., and Goldblum, M. (2022). Transfer learning with deep tabular models. *arXiv preprint arXiv:2206.15306*.

[134] Li, Y., Ma, T., and Zhang, H. R. (2020a). Learning over-parametrized two-layer neural networks beyond ntk. In *Conference on learning theory*, pages 2613–2682. PMLR.

[135] Li, Z., Zhao, Y., Botta, N., Ionescu, C., and Hu, X. (2020b). Copod: copula-based outlier detection. In *2020 IEEE International Conference on Data Mining (ICDM)*, pages 1118–1123. IEEE.

[136] Li, Z., Zhao, Y., and Fu, J. (2020c). Sync: A copula based framework for generating synthetic data from aggregated sources. In *2020 International Conference on Data Mining Workshops (ICDMW)*, pages 571–578. IEEE.

[137] Liang, W., Tadesse, G. A., Ho, D., Fei-Fei, L., Zaharia, M., Zhang, C., and Zou, J. (2022). Advances, challenges and opportunities in creating data for trustworthy ai. *Nature Machine Intelligence*, 4(8):669–677.

[138] Liu, E. Z., Haghgoo, B., Chen, A. S., Raghunathan, A., Koh, P. W., Sagawa, S., Liang, P., and Finn, C. (2021a). Just train twice: Improving group robustness without training group information. In *International Conference on Machine Learning*, pages 6781–6792. PMLR.

[139] Liu, F. T., Ting, K. M., and Zhou, Z.-H. (2012). Isolation-based anomaly detection. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 6(1):1–39.

[140] Liu, J., Seedat, N., Cui, P., and van der Schaar, M. (2025). Going beyond static: Understanding shifts with time-series attribution. In *The Thirteenth International Conference on Learning Representations*.

[141] Liu, P., Yuan, W., Fu, J., Jiang, Z., Hayashi, H., and Neubig, G. (2023). Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing. *ACM Computing Surveys*, 55(9):1–35.

[142] Liu, T., Astorga, N., Seedat, N., and van der Schaar, M. (2024). Large language models to enhance bayesian optimization. In *The Twelfth International Conference on Learning Representations*.

[143] Liu, X., Zhang, F., Hou, Z., Mian, L., Wang, Z., Zhang, J., and Tang, J. (2021b). Self-supervised learning: Generative or contrastive. *IEEE Transactions on Knowledge and Data Engineering*.

[144] Long, M., Cao, Y., Wang, J., and Jordan, M. (2015). Learning transferable features with deep adaptation networks. In *International conference on machine learning*, pages 97–105. PMLR.

[145] Lorena, A. C., Garcia, L. P. F., Lehmann, J., Souto, M. C. P., and Ho, T. K. (2019). How complex is your classification problem? a survey on measuring classification complexity. *ACM Comput. Surv.*, 52(5).

[146] Machado, P., Fernandes, B., and Novais, P. (2022). Benchmarking data augmentation techniques for tabular data. In *International Conference on Intelligent Data Engineering and Automated Learning*, pages 104–112. Springer.

[147] Margeloiu, A., Simidjievski, N., Lio, P., and Jamnik, M. (2022). Graph-conditioned mlp for high-dimensional tabular biomedical data. *arXiv preprint arXiv:2211.06302*.

[148] McInnes, L., Healy, J., Saul, N., and Großberger, L. (2018). Umap: Uniform manifold approximation and projection. *Journal of Open Source Software*, 3(29):861.

[149] Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., and Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM computing surveys (CSUR)*, 54(6):1–35.

[150] Meng, Y., Michalski, M., Huang, J., Zhang, Y., Abdelzaher, T., and Han, J. (2023). Tuning language models as training data generators for augmentation-enhanced few-shot learning. In *International Conference on Machine Learning*, pages 24457–24477. PMLR.

[151] Mirchandani, S., Xia, F., Florence, P., Ichter, B., Driess, D., Arenas, M. G., Rao, K., Sadigh, D., and Zeng, A. (2023). Large language models as general pattern machines. *arXiv preprint arXiv:2307.04721*.

[152] Mollura, D. J., Culp, M. P., Pollack, E., Battino, G., Scheel, J. R., Mango, V. L., Elahi, A., Schweitzer, A., and Dako, F. (2020). Artificial intelligence in low-and middle-income countries: innovating global health radiology. *Radiology*, 297(3):513–520.

[153] Mukhoti, J. and Gal, Y. (2018). Evaluating bayesian deep learning methods for semantic segmentation. *arXiv preprint arXiv:1811.12709*.

[154] Mussmann, S. and Liang, P. (2018). On the relationship between data efficiency and error for uncertainty sampling. In *International Conference on Machine Learning*, pages 3674–3682. PMLR.

[155] Navratil, J., Arnold, M., and Elder, B. (2020). Uncertainty prediction for deep sequential regression using meta models. *arXiv preprint arXiv:2007.01350*.

[156] Ng, A., Aroyo, L., Coleman, C., Diamos, G., Reddi, V. J., Vanschoren, J., Wu, C.-J., and Zhou, S. (2021). Neurips data-centric ai workshop.

[157] Nguyen, V.-L., Shaker, M. H., and Hüllermeier, E. (2022). How to measure uncertainty in uncertainty sampling for active learning. *Machine Learning*, 111(1):89–122.

[158] Oakden-Rayner, L., Dunnmon, J., Carneiro, G., and Ré, C. (2020). Hidden stratification causes clinically meaningful failures in machine learning for medical imaging. In *Proceedings of the ACM Conference on Health, Inference, and Learning*, pages 151–159.

[159] Obermeyer, Z., Powers, B., Vogeli, C., and Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464):447–453.

[160] Oh, E. J., Parikh, R. B., Chivers, C., and Chen, J. (2021). Two-stage approaches to accounting for patient heterogeneity in machine learning risk prediction models in oncology. *JCO Clinical Cancer Informatics*, 5:1015–1023.

[161] OpenAI (2023). Gpt-4 technical report.

[162] Oshingbesan, A., Omondi, W. G., Tadesse, G. A., Cintas, C., and Speakman, S. (2022). Beyond protected attributes: Disciplined detection of systematic deviations in data. In *Workshop on Trustworthy and Socially Responsible Machine Learning, NeurIPS 2022*.

[163] Owoyemi, A., Owoyemi, J., Osiyemi, A., and Boyd, A. (2020). Artificial intelligence for healthcare in africa. *Frontiers in Digital Health*, 2:6.

[164] Pan, S. J. and Yang, Q. (2009). A survey on transfer learning. *IEEE Transactions on knowledge and data engineering*.

[165] Papadopoulos, H. (2008). Inductive conformal prediction: Theory and application to neural networks. In *Tools in Artificial Intelligence*. Citeseer.

[166] Papadopoulos, H., Gammerman, A., and Vovk, V. (2008). Normalized nonconformity measures for regression conformal prediction. In *Proceedings of the IASTED International Conference on Artificial Intelligence and Applications (AIA 2008)*, pages 64–69.

[167] Papadopoulos, H. and Haralambous, H. (2010). Neural networks regression inductive conformal predictor and its application to total electron content prediction. In *International Conference on Artificial Neural Networks*, pages 32–41. Springer.

[168] Papadopoulos, H., Vovk, V., and Gammerman, A. (2011). Regression conformal prediction with nearest neighbours. *Journal of Artificial Intelligence Research*, 40:815–840.

[169] Papamakarios, G., Nalisnick, E., Rezende, D. J., Mohamed, S., and Lakshminarayanan, B. (2021). Normalizing flows for probabilistic modeling and inference. *The Journal of Machine Learning Research*, 22(1):2617–2680.

[170] Park, C., Awadalla, A., Kohno, T., and Patel, S. (2021). Reliable and trustworthy machine learning for health using dataset shift detection. *Advances in Neural Information Processing Systems*, 34.

[171] Pastor, E., De Alfaro, L., and Baralis, E. (2021). Looking for trouble: Analyzing classifier behavior via pattern divergence. In *Proceedings of the 2021 International Conference on Management of Data*, pages 1400–1412.

[172] Patel, K., Fogarty, J., Landay, J. A., and Harrison, B. (2008). Investigating statistical machine learning as a tool for software development. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pages 667–676.

[173] Paul, M., Ganguli, S., and Dziugaite, G. K. (2021). Deep learning on a data diet: Finding important examples early in training. *Advances in Neural Information Processing Systems*, 34.

[174] PCUK, P. C. U. (2019). Cutract. *https://prostatecanceruk.org*.

[175] Pianykh, O. S., Langs, G., Dewey, M., Enzmann, D. R., Herold, C. J., Schoenberg, S. O., and Brink, J. A. (2020). Continuous learning AI in radiology: Implementation principles and early applications. *Radiology*, 297(1):6–14.

[176] Platzer, M. and Reutterer, T. (2021). Holdout-based empirical assessment of mixed-type synthetic data. *Frontiers in big Data*, page 43.

[177] Pleiss, G., Zhang, T., Elenberg, E., and Weinberger, K. Q. (2020). Identifying mislabeled data using the area under the margin ranking. *Advances in Neural Information Processing Systems*, 33:17044–17056.

[178] Pocock, S. J., Ariti, C. A., McMurray, J. J., Maggioni, A., Køber, L., Squire, I. B., Swedberg, K., Dobson, J., Poppe, K. K., Whalley, G. A., et al. (2013). Predicting survival in heart failure: a risk score based on 39 372 patients from 30 studies. *European Heart Journal*, 34(19):1404–1413.

[179] Polyzotis, N., Roy, S., Whang, S. E., and Zinkevich, M. (2017). Data management challenges in production machine learning. In *Proceedings of the 2017 ACM International Conference on Management of Data*, pages 1723–1726.

[180] Polyzotis, N. and Zaharia, M. (2021). What can data-centric ai learn from data and ml engineering? *arXiv preprint arXiv:2112.06439*.

[181] Polyzotis, N., Zinkevich, M., Roy, S., Breck, E., and Whang, S. (2019). Data validation for machine learning. *Proceedings of machine learning and systems*, 1:334–347.

[182] Popper, K. R. (1963). Science as falsification. *Conjectures and refutations*, 1(1963):33–39.

[183] Pouplin, T., Jeffares, A., Seedat, N., and Van Der Schaar, M. (2024). Relaxed quantile regression: Prediction intervals for asymmetric noise. In *International Conference on Machine Learning*, pages 40951–40981. PMLR.

[184] Qian, Z., Cebere, B.-C., and van der Schaar, M. (2023). Synthcity: facilitating innovative use cases of synthetic data in different data modalities.

[185] Qiu, L., Jiang, L., Lu, X., Sclar, M., Pyatkin, V., Bhagavatula, C., Wang, B., Kim, Y., Choi, Y., Dziri, N., et al. (2023). Phenomenal yet puzzling: Testing inductive reasoning capabilities of language models with hypothesis refinement. *arXiv preprint arXiv:2310.08559*.

[186] Ranathunga, S., Lee, E.-S. A., Prifti Skenduli, M., Shekhar, R., Alam, M., and Kaur, R. (2023). Neural machine translation for low-resource languages: A survey. *ACM Computing Surveys*, 55(11):1–37.

[187] Raskar, R., Vepakomma, P., Swedish, T., and Sharan, A. (2019). Data markets to support ai for all: Pricing, valuation and governance. *arXiv preprint arXiv:1905.06462*.

[188] Rasouli, M. and Jordan, M. I. (2021). Data sharing markets. *arXiv preprint arXiv:2107.08630*.

[189] Rauba, P., Seedat, N., Kacprzyk, K., and van der Schaar, M. (2024a). Self-healing machine learning: A framework for autonomous adaptation in real-world environments. *Advances in Neural Information Processing Systems*, 37:42225–42267.

[190] Rauba, P., Seedat, N., Ruiz Luyten, M., and van der Schaar, M. (2024b). Context-aware testing: A new paradigm for model testing with large language models. *Advances in Neural Information Processing Systems*, 37:112505–112553.

[191] Recht, B., Roelofs, R., Schmidt, L., and Shankar, V. (2019). Do ImageNet classifiers generalize to ImageNet? In Chaudhuri, K. and Salakhutdinov, R., editors, *International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pages 5389–5400. PMLR.

[192] Ren, M., Zeng, W., Yang, B., and Urtasun, R. (2018). Learning to reweight examples for robust deep learning. In *International Conference on Machine Learning*, pages 4334–4343. PMLR.

[193] Renggli, C., Rimanic, L., Gürel, N. M., Karlas, B., Wu, W., and Zhang, C. (2021). A data quality-driven view of mlops. *IEEE Data Engineering Bulletin*.

[194] Rezende, D. and Mohamed, S. (2015). Variational inference with normalizing flows. In *International conference on machine learning*, pages 1530–1538. PMLR.

[195] Ribeiro, M. T., Wu, T., Guestrin, C., and Singh, S. (2020). Beyond accuracy: Behavioral testing of nlp models with checklist. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912.

[196] Romano, Y., Patterson, E., and Candes, E. (2019). Conformalized quantile regression. *Advances in Neural Information Processing Systems*, 32.

[197] Rostamzadeh, N., Hutchinson, B., Greer, C., and Prabhakaran, V. (2021). Thinking beyond distributions in testing machine learned models. In *NeurIPS 2021 Workshop on Distribution Shifts: Connecting Methods and Applications*.

[198] Röttger, P., Vidgen, B., Nguyen, D., Waseem, Z., Margetts, H., Pierrehumbert, J., et al. (2021). Hatecheck: Functional tests for hate speech detection models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, page 41. Association for Computational Linguistics.

[199] Ruiz, C., Ren, H., Huang, K., and Leskovec, J. (2023). Enabling tabular deep learning when d $\setminus \gg n$ with an auxiliary knowledge graph. *arXiv preprint arXiv:2306.04766*.

[200] Sagadeeva, S. and Boehm, M. (2021). Sliceline: Fast, linear-algebra-based slice finding for ml model debugging. In *Proceedings of the 2021 International Conference on Management of Data*, pages 2290–2299.

[201] Sagawa, S., Koh, P. W., Hashimoto, T. B., and Liang, P. (2019). Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. *arXiv preprint arXiv:1911.08731*.

[202] Sajjadi, M. S., Bachem, O., Lucic, M., Bousquet, O., and Gelly, S. (2018). Assessing generative models via precision and recall. *Advances in neural information processing systems*, 31.

[203] Sambasivan, N., Kapania, S., Highfill, H., Akrong, D., Paritosh, P., and Aroyo, L. M. (2021a). "Everyone wants to do the model work, not the data work": Data cascades in high-stakes AI. In *proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, pages 1–15.

[204] Sambasivan, N., Kapania, S., Highfill, H., Akrong, D., Paritosh, P., and Aroyo, L. M. (2021b). "everyone wants to do the model work, not the data work": Data cascades in high-stakes ai. In *proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, pages 1–15.

[205] Saria, S. and Subbaswamy, A. (2019). Tutorial: safe and reliable machine learning. *ACM Conference on Fairness, Accountability, and Transparency*.

[206] Saveliev, E., Liu, J., Seedat, N., Boyd, A., and van der Schaar, M. (2025). Towards human-guided, data-centric llm co-pilots. *arXiv preprint arXiv:2501.10321*.

[207] Seedat, N. (2020). Mcu-net: A framework towards uncertainty representations for decision support system patient referrals in healthcare contexts. *arXiv preprint arXiv:2007.03995*.

[208] Seedat, N., Crabbé, J., Bica, I., and van der Schaar, M. (2022a). Data-iq: Characterizing subgroups with heterogeneous outcomes in tabular data. *Advances in Neural Information Processing Systems*, 35:23660–23674.

[209] Seedat, N., Crabbé, J., Qian, Z., and van der Schaar, M. (2023a). Triage: Characterizing and auditing training data for improved regression. In *Thirty-seventh Conference on Neural Information Processing Systems*.

[210] Seedat, N., Crabbé, J., and van der Schaar, M. (2022b). Data-SUITE: Data-centric identification of in-distribution incongruous examples. In *International Conference on Machine Learning*, pages 19467–19496. PMLR.

[211] Seedat, N., Huynh, N., Imrie, F., and van der Schaar, M. (2024a). You can't handle the (dirty) truth: Data-centric insights improve pseudo-labeling. *Journal of Data-centric Machine Learning Research*.

[212] Seedat, N., Huynh, N., van Breugel, B., and van der Schaar, M. (2024b). Curated LLM: Synergy of LLMs and data curation for tabular augmentation in low-data regimes. In *Forty-first International Conference on Machine Learning*.

[213] Seedat, N., Imrie, F., Bellot, A., Qian, Z., and van der Schaar, M. (2022c). Continuous-time modeling of counterfactual outcomes using neural controlled differential equations. In *International Conference on Machine Learning*, pages 19497–19521. PMLR.

[214] Seedat, N., Imrie, F., and van der Schaar, M. (2022d). Dc-check: A data-centric ai checklist to guide the development of reliable machine learning systems. *arXiv preprint arXiv:2211.05764*.

[215] Seedat, N., Imrie, F., and van der Schaar, M. (2023b). Dissecting sample hardness: Fine-grained analysis of hardness characterization methods. In *The Twelfth International Conference on Learning Representations*.

[216] Seedat, N., Imrie, F., and van der Schaar, M. (2023c). Navigating data-centric artificial intelligence with DC-Check: Advances, challenges, and opportunities. *IEEE Transactions on Artificial Intelligence*.

[217] Seedat, N., Jeffares, A., Imrie, F., and van der Schaar, M. (2023d). Improving adaptive conformal prediction using self-supervised learning. In *International Conference on Artificial Intelligence and Statistics*, pages 10160–10177. PMLR.

[218] Seedat, N. and Kanan, C. (2019). Towards calibrated and scalable uncertainty representations for neural networks. *arXiv preprint arXiv:1911.00104*.

[219] Seedat, N., Tozzi, C., Ardiaca, A. H., van der Schaar, M., Weatherall, J., and Taylor, A. (2024c). Unlocking historical clinical trial data with align: A compositional large language model system for medical coding. *arXiv preprint arXiv:2411.13163*.

[220] Seedat, N. and van der Schaar, M. (2025). Bootstrapping self-improvement of language model programs for zero-shot schema matching. In *Forty-second International Conference on Machine Learning*.

[221] Sesia, M. and Romano, Y. (2021). Conformal prediction using conditional histograms. *Advances in Neural Information Processing Systems*, 34:6304–6315.

[222] Settles, B. (2009). Active learning literature survey. Technical report, University of Wisconsin-Madison.

[223] Shafer, G. and Vovk, V. (2008). A tutorial on conformal prediction. *Journal of Machine Learning Research*, 9(3).

[224] Shankar, S., Fawaz, L., Gyllstrom, K., and Parameswaran, A. (2023). Automatic and precise data validation for machine learning. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, pages 2198–2207.

[225] Sharma, A. and Wehrheim, H. (2020). Higher income, larger loan? monotonicity testing of machine learning models. In *Proceedings of the 29th ACM SIGSOFT International Symposium on Software Testing and Analysis*, pages 200–210.

[226] Shwartz-Ziv, R. and Armon, A. (2022). Tabular data: Deep learning is not all you need. *Information Fusion*, 81:84–90.

[227] Singhal, K., Azizi, S., Tu, T., Mahdavi, S. S., Wei, J., Chung, H. W., Scales, N., Tanwani, A., Cole-Lewis, H., Pfohl, S., et al. (2023). Large language models encode clinical knowledge. *Nature*, pages 1–9.

[228] Singla, S., Nushi, B., Shah, S., Kamar, E., and Horvitz, E. (2021). Understanding failures of deep networks via robust feature extraction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12853–12862.

[229] Sklar, A. (1959). Fonctions de répartition à n dimensions et leurs marges. *Publications de l'Institut de Statistique de l'Université de Paris*, 8:229–231.

[230] Sohoni, N., Dunnmon, J., Angus, G., Gu, A., and Ré, C. (2020). No subclass left behind: Fine-grained robustness in coarse-grained classification problems. *Advances in Neural Information Processing Systems*, 33:19339–19352.

[231] Song, H., Kim, M., Park, D., Shin, Y., and Lee, J.-G. (2020). Learning from noisy labels with deep neural networks: A survey. *arXiv preprint arXiv:2007.08199*.

[232] Srivastava, A., Valkov, L., Russell, C., Gutmann, M. U., and Sutton, C. (2017). Veegan: Reducing mode collapse in gans using implicit variational learning. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, pages 3310–3320.

[233] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., and Salakhutdinov, R. (2014). Dropout: a simple way to prevent neural networks from overfitting. *The journal of machine learning research*, 15(1):1929–1958.

[234] Sun, H., Chan, A. J., Seedat, N., Hüyük, A., and van der Schaar, M. (2024). When is off-policy evaluation (reward modeling) useful in contextual bandits? a data-centric perspective. *Journal of Data-centric Machine Learning Research*.

[235] Sun, H., van Breugel, B., Crabbé, J., Seedat, N., and van der Schaar, M. (2023). What is flagged in uncertainty quantification? latent density models for uncertainty categorization. *Advances in Neural Information Processing Systems*, 36:4664–4684.

[236] Suresh, H., Gong, J. J., and Guttag, J. V. (2018). Learning tasks for multitask learning: Heterogenous patient populations in the ICU. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 802–810.

[237] Swayamdipta, S., Schwartz, R., Lourie, N., Wang, Y., Hajishirzi, H., Smith, N. A., and Choi, Y. (2020). Dataset cartography: Mapping and diagnosing datasets with training dynamics. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9275–9293.

[238] Tanaka, F. H. K. D. S. and Aranha, C. (2019). Data augmentation using gans. *arXiv preprint arXiv:1904.09135*.

[239] Teng, J., Wen, C., Zhang, D., Bengio, Y., Gao, Y., and Yuan, Y. (2022). Predictive inference with feature conformal prediction. *arXiv preprint arXiv:2210.00173*.

[240] Tibshirani, R. J., Foygel Barber, R., Candes, E., and Ramdas, A. (2019). Conformal prediction under covariate shift. *Advances in Neural Information Processing Systems*, 32.

[241] Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., Rodriguez, A., Joulin, A., Grave, E., and Lample, G. (2023). Llama: Open and efficient foundation language models.

[242] van Breugel, B., Seedat, N., Imrie, F., and van der Schaar, M. (2024). Can you rely on your model evaluation? improving model evaluation with synthetic test data. *Advances in Neural Information Processing Systems*, 36.

[243] van Engelen, J. E. and Hoos, H. H. (2019). A survey on semi-supervised learning. *Machine Learning*, 109:373–440.

[244] Vanschoren, J., van Rijn, J. N., Bischl, B., and Torgo, L. (2013). Openml: Networked science in machine learning. *SIGKDD Explorations*, 15(2):49–60.

[245] Vargas, D. V., Takano, H., and Murata, J. (2013). Contingency training. In *The SICE Annual Conference 2013*, pages 1361–1366. IEEE.

[246] Varshney, K. R. (2020). On mismatched detection and safe, trustworthy machine learning. In *2020 54th Annual Conference on Information Sciences and Systems (CISS)*, pages 1–4. IEEE.

[247] Vincent, J. (2020). Ai camera operator repeatedly confuses bald head for soccer ball during live stream.

[248] Vovk, V. (2012). Conditional validity of inductive conformal predictors. In *Asian Conference on Machine Learning*, pages 475–490. PMLR.

[249] Vovk, V. (2013). Transductive conformal predictors. In *IFIP International Conference on Artificial Intelligence Applications and Innovations*, pages 348–360. Springer.

[250] Vovk, V., Gammerman, A., and Shafer, G. (2005a). *Algorithmic learning in a random world*. Springer Science & Business Media.

[251] Vovk, V., Gammerman, A., and Shafer, G. (2005b). Conformal prediction. *Algorithmic learning in a random world*, pages 17–51.

[252] Wang, W. and Pai, T.-W. (2023). Enhancing small tabular clinical trial dataset through hybrid data augmentation: Combining smote and wcgan-gp. *Data*, 8(9):135.

[253] Wang, Y., Yao, Q., Kwok, J. T., and Ni, L. M. (2020). Generalizing from a few examples: A survey on few-shot learning. *ACM Computing Surveys (CSUR)*.

[254] Wasserman, L. (2004). *All of statistics: a concise course in statistical inference*, volume 26. Springer.

[255] Wei, J., Wei, J., Tay, Y., Tran, D., Webson, A., Lu, Y., Chen, X., Liu, H., Huang, D., Zhou, D., et al. (2023). Larger language models do in-context learning differently. *arXiv preprint arXiv:2303.03846*.

[256] Whiteson, D. (2014). HIGGS. UCI Machine Learning Repository. DOI: https://doi.org/10.24432/C5V312.

[257] Williams, C. K. and Rasmussen, C. E. (2006). *Gaussian processes for machine learning*, volume 2. MIT press Cambridge, MA.

[258] Xia, T., Han, J., Qendro, L., Dang, T., and Mascolo, C. (2021). Uncertainty-aware covid-19 detection from imbalanced sound data. *arXiv preprint arXiv:2104.02005*.

[259] Xu, L., Skoularidou, M., Cuesta-Infante, A., and Veeramachaneni, K. (2019). Modeling tabular data using conditional GAN. *Advances in Neural Information Processing Systems*, 32.

[260] Yan, H., Ding, Y., Li, P., Wang, Q., Xu, Y., and Zuo, W. (2017). Mind the class weight bias: Weighted maximum mean discrepancy for unsupervised domain adaptation. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 945–954.

[261] Yang, C., Wang, X., Lu, Y., Liu, H., Le, Q. V., Zhou, D., and Chen, X. (2023). Large language models as optimizers. *arXiv preprint arXiv:2309.03409*.

[262] Yoon, J., Davtyan, C., and van der Schaar, M. (2016). Discovery and clinical decision support for personalized healthcare. *IEEE journal of biomedical and health informatics*, 21(4):1133–1145.

[263] Yoon, J., Zhang, Y., Jordon, J., and van der Schaar, M. (2020). Vime: Extending the success of self-and semi-supervised learning to tabular domain. *Advances in Neural Information Processing Systems*, 33.

[264] Zha, D., Bhat, Z. P., Lai, K.-H., Yang, F., Jiang, Z., Zhong, S., and Hu, X. (2023). Data-centric artificial intelligence: A survey. *arXiv preprint arXiv:2303.10158*.

[265] Zhang, B. H., Lemoine, B., and Mitchell, M. (2018). Mitigating unwanted biases with adversarial learning. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society*, pages 335–340.

[266] Zhang, L., Goldstein, M., and Ranganath, R. (2021). Understanding failures in out-of-distribution detection with deep generative models. In *International Conference on Machine Learning*, pages 12427–12436. PMLR.

[267] Zhang, Y., Zame, W., and van der Schaar, M. (2020). AutoCP: Automated pipelines for accurate prediction intervals. *arXiv preprint arXiv:2006.14099*.

[268] Zhao, Y., Hu, X., Cheng, C., Wang, C., Wan, C., Wang, W., Yang, J., Bai, H., Li, Z., Xiao, C., et al. (2021). Suod: Accelerating large-scale unsupervised heterogeneous outlier detection. *Proceedings of Machine Learning and Systems*, 3.

[269] Zhuang, F., Qi, Z., Duan, K., Xi, D., Zhu, Y., Zhu, H., Xiong, H., and He, Q. (2020). A comprehensive survey on transfer learning. *Proceedings of the IEEE*, 109(1):43–76.

[270] Zliobaite, I. (2013). How good is the electricity benchmark for evaluating concept drift adaptation. *arXiv preprint arXiv:1301.3524*.

# Appendix A

# Appendix for Chapter 2

## A.1 Additional Details

### A.1.1 Extended Related Work

We present a comparison of our data characterization framework, Data-IQ, and provide further contrast to the related work. Table A.1 highlights that the related methods do not satisfy all the properties (P1- P4). Furthermore, even for cases where it is satisfied, the properties are not naturally satisfied; rather, they require our framework in order to enable the use case. *P2* around data collection and selection has been largely unaddressed by prior methods. In the case of *P3*, the related methods focus on training time without taking into account end-to-end deployment scenarios. Finally, these prior methods are typically applicable only to neural networks, which limits the broad applicability related to *P4*, where in high-stakes settings such as healthcare and finance, which often have tabular data, practitioners typically use non-neural network methods, such as Gradient Boosting or XGBoost.

Table A.1 Comparison of related work, highlights that *ONLY* Data-IQ naturally address all desiderata. (♡): satisfied naturally by construction, (♢): needs our framework

| | Assessment metric | (P1) Data characterization consistent across models | (P2) Principled data collection & selection | (P3) Applicable at deployment time | (P4) Applicable to *any* ML model |
|---|---|---|---|---|---|
| Data-IQ (Ours) | Aleatoric uncertainty | V. High | ✓ | ✓(♡) | ✓(♡) |
| Data Maps [237] | Training variability | Medium | ✗ | ✓(♢) | ✓(♢) |
| AUM [177] | Logit difference | High | ✗ | ✗ | ✓ |
| GraNd [173] | Gradient Norm | Medium | ✗ | ✗ | ✗ |
| JTT [138] | Training errors | Low | ✗ | ✓(♢) | ✓ |

The properties of *P1* were evaluated in Chapter 2.2. We delve into each subsequent property below and show that the related works have different goals and hence do not satisfy the properties:

- *P2*: Data Maps [237] does not fulfill this property, as was shown experimentally in Chapter 2.4.2. The method is shown to be insufficiently sensitive for the task. AUM [177] uses the differences between logits to flag mislabeled instances. By virtue, AUM solves a different task of label errors, whereas the task of improving the data by collecting better features, etc., is focused on the inputs rather than the labels. GraNd [173] looks at the gradient norm to identify important examples. Their task, rather than feature acquisition, is to assist in the pruning of a dataset. JTT [138] looks at training errors to detect samples with spurious correlations. Not only is the task different, but flagged errors could also link to both ambiguity (needing more features) or hardness (an example being difficult to learn). Therefore, it is not sufficiently fine-grained to be leveraged for feature acquisition. In the best case, we might want to augment the dataset with such flagged examples.

- *P3*: Data Maps naturally operates only at training time (as per the original work). However, if it leverages Data-IQ's representation space, then it could be extended to be used at test time. AUM operates to flag mislabeled data (i.e. outcome labels). These do not exist at test time, hence AUM is a training-time ONLY method. GraNd uses gradients to flag examples. Hence, it is only applicable at training time and not at test time since we do not compute gradients at test time (as we do not have access to labels). JTT surfaces training errors and hence, we would not know test time errors as we do not have labels. However, if JTT leverages Data-IQ's representation space idea, then it could also be extended to be used at test time by flagging examples relative to a training time embedding.

- *P4*: Data Maps was only studied for usage with neural networks. For use with *any* ML model, it can be extended using our framework (see Chapter 2.3.4 and Appendix A.1.2). AUM could, of course, be used with any ML model with output logits. GraNd does not facilitate this, as only ML models where gradients can be computed are usable. Finally, since JTT simply captures model errors, it is easily usable with any ML model.

**Tangential related work.** We wish to highlight that while methods such as Influence functions [117] and Data Shapley [62] may seem related, they have fundamentally different goals around capturing the contributing role of examples to the model's decision boundary and predictions. Hence, they are considered as tangential to our work of characterizing data into subgroups. Additionally, these methods assign an instance importance related to its contribution to the decision boundary of a specific model. This is different from our task of characterizing the properties inherent in the data and then using them to stratify the examples into subgroups.

Another key difference lies in the computational requirements, where both Influence functions and Data Shapley are highly computationally intensive, on top of the model training. By contrast, Data-IQ's characterization can be obtained easily, simply by training an ML model.

### A.1.2 Data-IQ adapted to any model

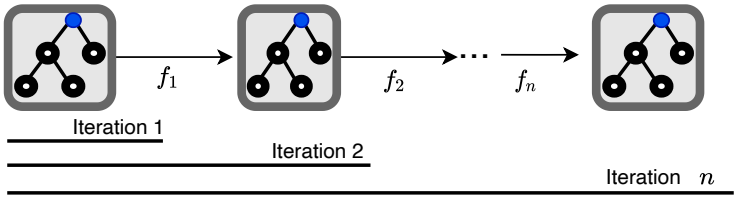



Fig. A.1 Pseudo-ensemble for GBDTs for usage with Data-IQ, where each sub-model is a checkpoint

Data-IQ's formulation allows us to use it with *any* ML model training in stages (i.e., iterative learning) such as gradient boosting decision trees (GBDTs) methods, such as XGBoost, LightGBM, etc. We simply need a set of checkpoints through training.

This property is incredibly important for practical utility since methods such as GBDTs or XGBoost are widely used by practitioners due to their high performance on tabular data (often outperforming neural networks). GBDTs iteratively combine weak models and have shown great success on tabular data, often outperforming neural networks. Hence, having our method applicable for GBDTs in addition to neural networks adds to the broad utility.

Before outlining why methods such as GBDTs fit the Data-IQ paradigm, we provide a brief overview.

Formally, given a dataset $D_{\text{train}}$, GBDT iteratively constructs a model $F : X \to \mathbb{R}$ to minimise the empirical risk. At each iteration $e$, the model is updated as

$$F^{(e)}(x) = F^{(e-1)}(x) + \eta\, h^{(e)}(x), \tag{A.1}$$

where $\eta \in (0, 1]$ is the learning rate and $h^{(e)} : X \to \mathbb{R}$ is the weak learner (typically a regression tree) fitted at iteration $e$ to the current pseudo-residuals.

We now provide guidance on how to apply Data-IQ to such methods. Naively, we could construct the checkpoints as an ensemble of multiple independent GBDTs. However, this is inefficient as the space and time complexity scales with $E$ models. To avoid increasing the overhead of a single model (from a practitioner perspective), we create a pseudo-ensemble using a single GBDT; see Figure A.1.

Similar to neural networks, the iterative learning nature of a GBDT means that the sequential submodels can be considered as checkpoints. Formally, each sub-model has parameters $\theta^{(i)}$, hence the ensemble of checkpoints can be described as $\Theta = \{\theta^{(i)}, 1 \leq i \leq E\}$.

We then apply Data-IQ as normal to the checkpoints $(\theta_1, \theta_2 ... \theta_E)$. The flexibility of this approach is that it applies both to training a new model, but interestingly, we can also apply this to an *ALREADY* trained model by looping through the structure to create the pseudo-ensemble.

### A.1.3 Why does Data-IQ produce a bell-shape?

We now theoretically analyze why Data-IQ has a bell-shaped curve. We show that if the x-axis is aleatoric uncertainty and if the y-axis is predictive confidence, we always obtain a bell-shaped curve.

Recall two terms from our formulation: (1) $P(x, \theta)$: the model's predictive confidence for an input $x$ applied to a model with parameters $\theta$, (2) $\bar{P}(x)$: the model's average confidence for the ground-truth class. Now assume $P(x, \theta)$ is peaked around $\bar{P}(x)$, as shown in Figure A.2 below.

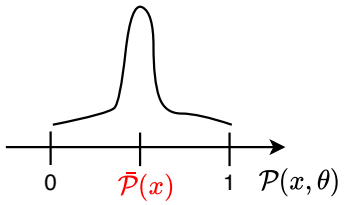



Fig. A.2 $\mathscr{P}(x, \theta)$ peaked around $\bar{\mathscr{P}}(x)$

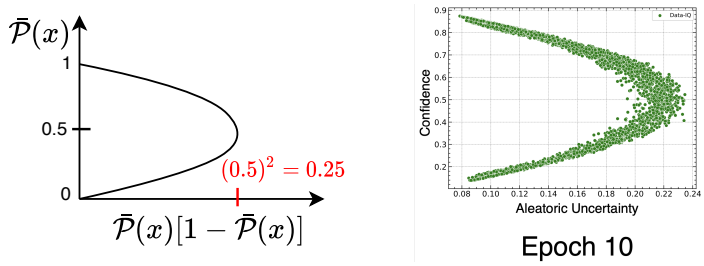
Based on this, we compute Data-IQ and obtain the following axes. This results in Figure A.3 (a) with the bell-shape, which matches what we observe in practice (Figure A.3 (b))

- y-axis: $1/E \sum_{e=1}^{E} P(x, \theta_e) \approx \bar{P}(x)$
- x-axis: $\frac{1}{E} \sum_{e=1}^{E} P(x, \theta_e)(1 - P(x, \theta_e)) \approx \bar{P}(x)[1 - \bar{P}(x)]$

### A.1.4 Evaluating Data-IQ and Data Maps over stages

**Goal.** Recall that we utilize *any* ML model trained in stages (e.g. epochs or iterations) for both Data-IQ and Data Maps. We train our models to convergence, with early stopping on a validation set. As is common practice, early stopping is done to prevent the model from overfitting. Once early stopping kicks in and we stop model training, we then compute the Data-IQ and Data Maps, with scores averaged over all previously checkpointed models.

This brings two questions to mind, which we seek to answer:




(a) Theoretical bell-shape (b) Real bell-shape

Fig. A.3 The bell-shapes derived theoretically and practically match.

1. How do Data-IQ and Data Maps differ at various stages (epochs)?

2. What happens if we don't do early stopping and let the model overfit?

Using Figure A.4, we answer both questions. For this, we train a neural network, and around 10 epochs, early stopping would normally kick in. However, instead of using early stopping, we continue training the neural network until 20 epochs, thereby letting the model overfit.

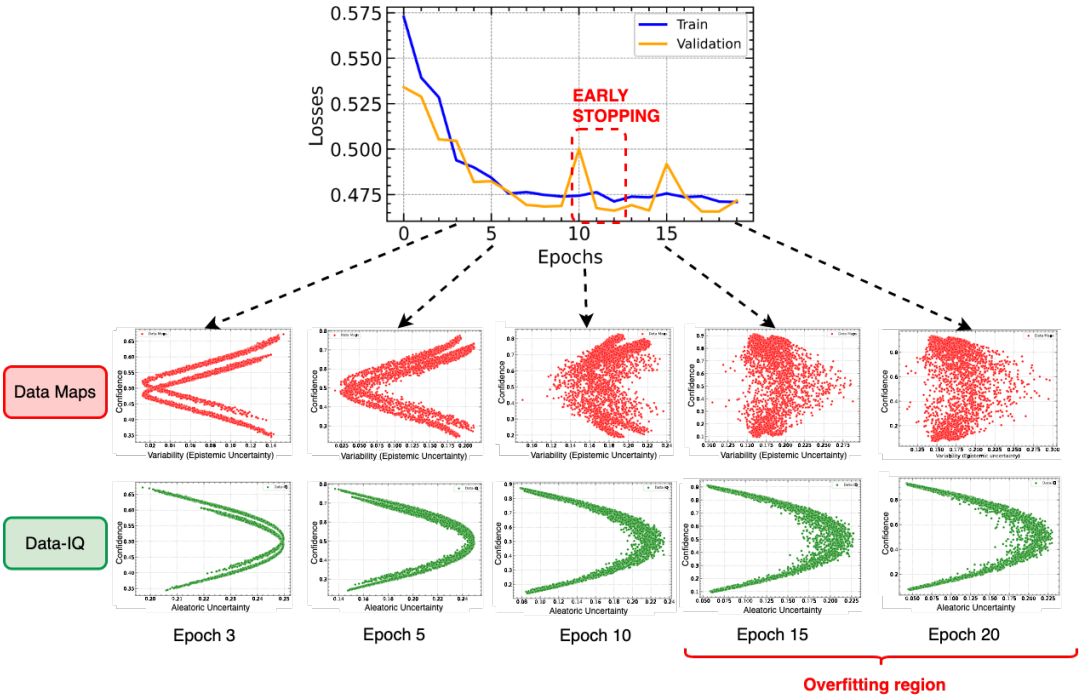



Fig. A.4 Evaluation of Data-IQ and Data Maps over various epochs (stages), even letting the model overfit. Data-IQ is more stable, whereas Data Maps changes shape after overfitting, which is undesirable.

**Takeaway.**

1. **Differences across epochs:** Data Maps change significantly over the course of training. This is due to its sensitivity, as the model parameters are changed/updated through training. This sensitivity is reflected in the variability (epistemic uncertainty) on the x-axis. In contrast, Data-IQ is less sensitive and rapidly converges. The rationale is that Data-IQ models the inherent uncertainty in the data (aleatoric), thereby being more stable and less sensitive.

2. **Model overfits:** An interesting phenomenon occurs after the early stopping phase and into the overfitting stage. Data-IQ remains consistent/stable, as expected. It is during this overfitting stage that the Data Maps shape changes and begins to look similar to Data-IQ. The reason for this is that after $\approx$10 epochs (when the model begins to overfit), the model parameters do not get updated much (the loss doesn't change). The small changes of model parameters are then reflected by the epistemic/model uncertainty not changing much (x-axis of Data Maps). The minimal change means that when we average at the end of a longer training, the effects get washed out. This washing out of the effect causes the change in shape by Epoch 20. We will delve into this point in greater detail in the next sub-section as it explains the difference between Data Maps in our setting when compared to other settings.

### A.1.5 Data Maps: difference in our setting

**Goal.** We now take a deep dive, based on the interesting findings at the model overfitting stage, where the Data Maps shape changes. We believe that the difference of our setting from the original NLP setting of Data Maps provides a clear explanation for why the shape we observe for Data Maps is different from the shape observed in the original paper [237].

In typical NLP settings, the models are larger and highly parameterized. Hence, the models are trained for a large number of epochs (sometimes hundreds). However, in the tabular setting, models need to train for significantly fewer (as is evident by the early stopping around $\approx$ 10 epochs. Further, our models are much shallower compared to highly parameterized NLP models, motivating why our setting requires early stopping to prevent overfitting.

**Takeaway.** We now begin our deep dive. As we saw in Figure A.4, after around 10 epochs, the training loss plateaus. This means that subsequently, the model parameters do not get updated much over training. The result of this is that the epistemic uncertainty would have

much less variability after 10 epochs. Thus, when we average the epistemic uncertainty for longer epochs, these minimal changes would wash out effects from earlier in training.

This effect over longer training explains why the shape of Data Maps is different in the NLP settings, which are often trained for a large number of epochs, and, assuming, we let the model overfit in our setting.

What implications does this have? We assess the evolution of three examples where we know their ground truth subgroup (*Easy*, *Ambiguous*, *Hard*).

First, we assess Data Maps, which use epistemic uncertainty. Figure A.5 shows the evolution of epistemic uncertainty (Data Maps) for the three subgroup examples during training. We see that during model training (before early stopping), the *Easy* and *Hard* examples, in fact, have higher epistemic uncertainty than the *Ambiguous* example. This would lead us to the incorrect conclusion that the *Ambiguous* example is Easy.

In practice, we do not want to let our model knowingly overfit. However, we can only obtain the correct characterization for Data Maps if we let the model overfit and wash out the earlier effects. This highlights a limitation of Data Maps vs Data-IQ.

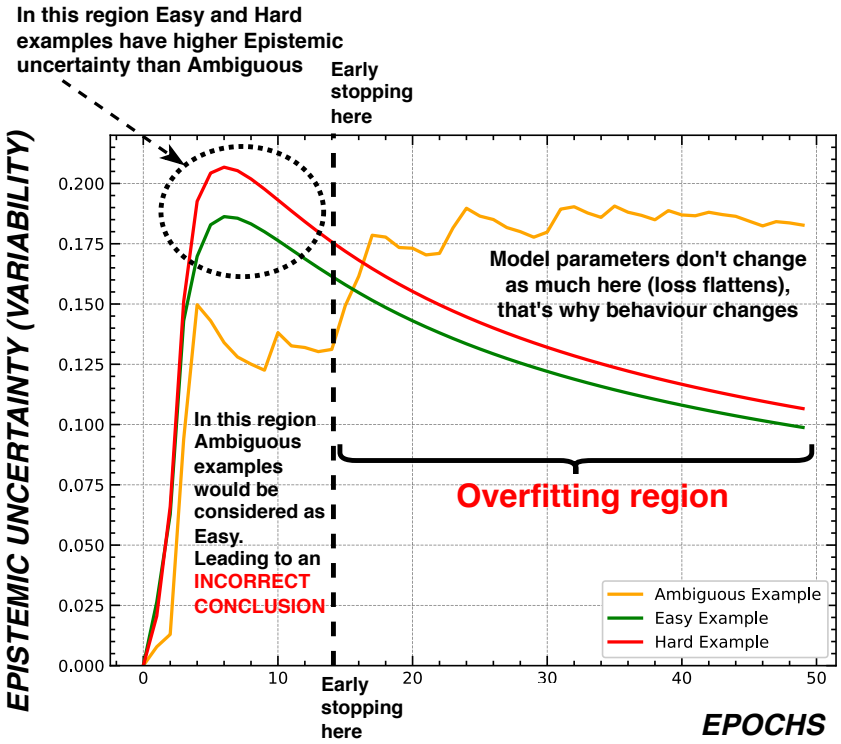



Fig. A.5 Data Maps: Epistemic Uncertainty (variability) evolution over epochs per subgroup

We contrast this with Data-IQ, which uses Aleatoric uncertainty. We not only have consistent behavior for the *Ambiguous* examples, but also, irrespective of whether we use early stopping or the model overfits, we obtain the correct characterization of samples.

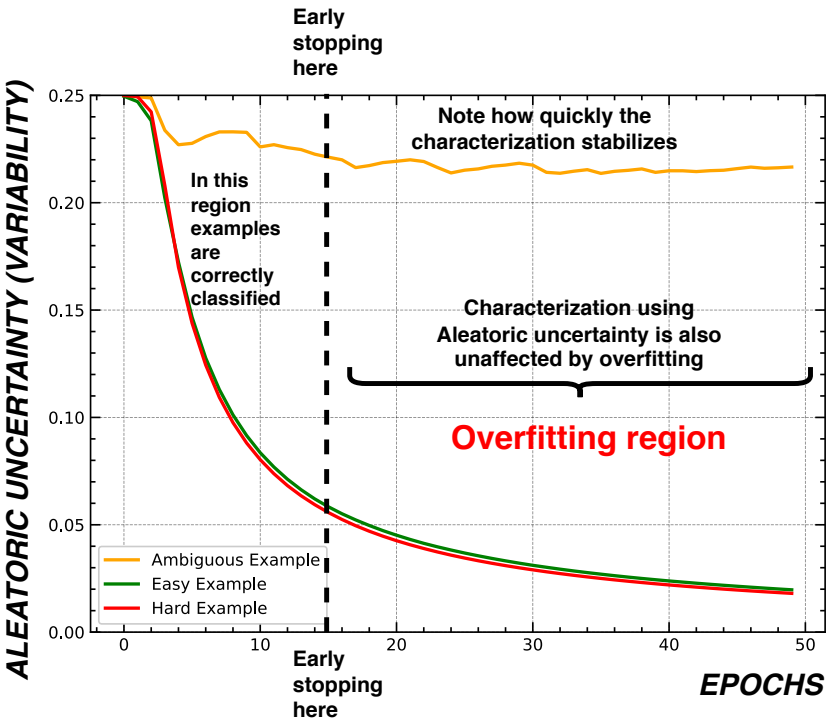
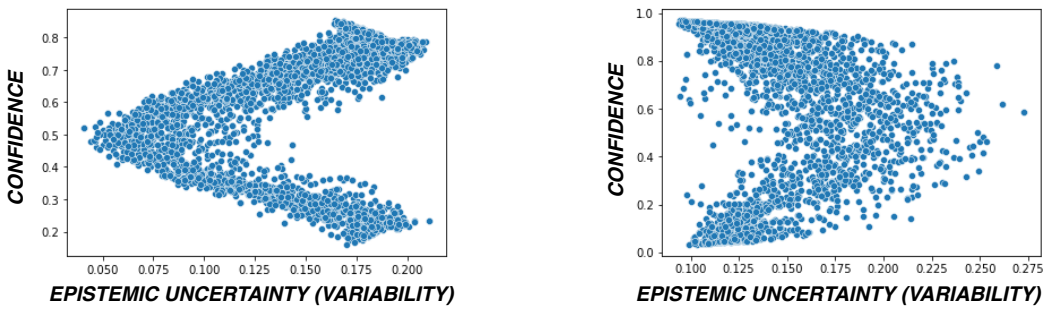



Fig. A.6 Data-IQ: Aleatoric Uncertainty evolution over epochs per subgroup

Finally, we look at the Data Maps shapes at the point where we perform early stopping (Figure A.7(a)) and at the end of training if we let the model severely overfit (Figure A.7(b)).




(a) Data Maps @ 10 epochs (Correct model)

(b) Data Maps @ 30 epochs (OVERFIT model)

Fig. A.7 Comparison of Data Maps when the model is just right and overfits, illustrating how the shape changes

Figure A.7 shows we can only obtain the expected shape from the original paper [237] if we let the model overfit. This is, of course, not an ideal solution, as in practice, we do not want to overfit our models by training for extra epochs. This is especially important in our tabular setting with less parameterized models and on smaller tabular datasets.

This further motivates why we require Data-IQ and why its characterization of examples based on aleatoric uncertainty is more useful and principled.

### A.1.6 Data-IQ thresholds

As outlined in Chapter 2.3.3, we stratify samples based on the Aleatoric uncertainty and predictive confidence.

Typically, we set the percentile $Q_n = 50$th percentile for aleatoric uncertainty. The rationale for this is that any example where the aleatoric uncertainty is greater than the 50th percentile should be considered *Ambiguous*. This is based on the definition of *Ambiguous* examples. Of course, this threshold can be updated by practitioners.

Typically, we set the quantile $Q_{50}$ for aleatoric uncertainty (i.e., the 50th percentile). The rationale for this is that any example where the aleatoric uncertainty is greater than $Q_{50}$ should be considered *Ambiguous*. This is based on the definition of *Ambiguous* examples. Of course, this threshold can be updated by practitioners.

This means that the examples below $Q_{50}$ of the aleatoric uncertainty are then *Easy* or *Hard*.

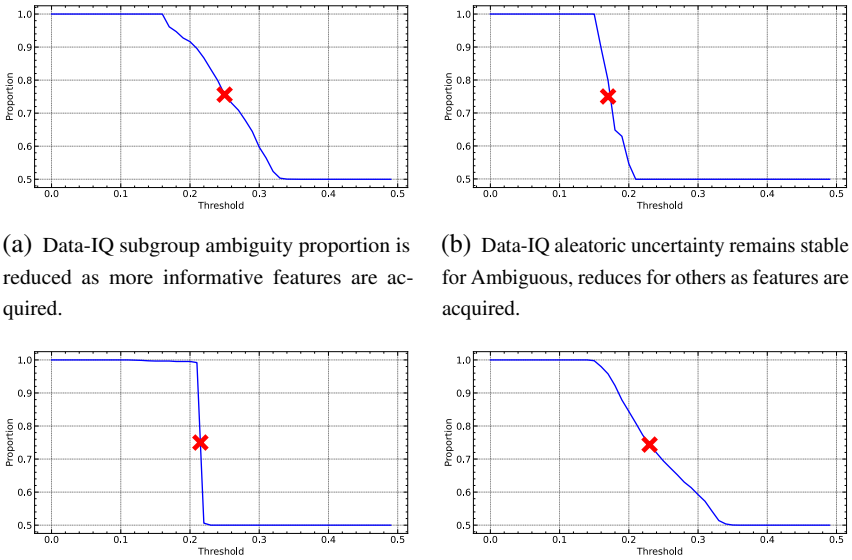
This means that the examples below $Q_{50}$ of the aleatoric uncertainty are then *Easy* or *Hard*. To differentiate the two, we look at the predictive confidence. We consider examples with high confidence on the true label as *Easy* and those with low confidence on the true label as *Hard*. This means that we need to practically set $C_{\text{up}}$ and $C_{\text{low}}$.

We define a threshold $thresh \in [0, 0.5]$ such that $C_{\text{up}} = 1 - thresh$ and $C_{\text{low}} = thresh$. An obvious threshold=0.25 such that $C_{\text{up}} = 0.75$ and $C_{\text{low}} = 0.25$. However, we present a practical method that could be applied to determine this threshold. We note that from our characterizations, examples with high aleatoric uncertainty are often uncertain about their predictive confidence. Hence, they never have very high or low uncertainty, rather the uncertainty is around 0.5. However, since it is never exactly 0.5, this motivates a band, namely $C_{\text{up}}$ and $C_{\text{low}}$.

Now, assume for any dataset, we train a model and apply Data-IQ. We can then sweep $thresh \in \{0, 0.5\}$ and assess the proportion of examples in each group.

We show results in Figure A.8 below as we sweep the threshold. For low threshold values, the proportions are, of course, stable (selecting the most confident samples). However, there is a threshold value beyond which the curve rises until a threshold knick point. After

that knick-point, there is stability once again. We propose that this knick-point threshold of stability be used as the threshold in practice. The rationale is that this will allow for better consistency of data characterization.




(a) Data-IQ subgroup ambiguity proportion is reduced as more informative features are acquired.

(b) Data-IQ aleatoric uncertainty remains stable for Ambiguous, reduces for others as features are acquired.

(c) Data Maps subgroup proportions largely unaffected as features acquired.

(d) Data Maps variability increases across subgroups as features acquired.

Fig. A.8 SUPPORT Quantifying the value of feature acquisition based on change in ambiguity. Only Data-IQ captures this relationship.

### A.1.7 Data-IQ subgroup characterization

With Data-IQ, we can obtain a plot of the data where the x-axis represents aleatoric uncertainty and the y-axis represents predictive confidence for the true label. As discussed, this means we obtain a bell-shaped characterization. Our goal is to highlight where on the plot each subgroup lies. i.e. illustrate the location of *Easy*, *Ambiguous* and *Hard* on the plot.

We highlight in Figure A.9 that *Ambiguous* examples are more likely to lie on the right side with higher aleatoric uncertainty. We note that based on our characterization, this region of the plot never has high and low predictive uncertainty. Rather, it typically only has uncertainty approximately around 0.5.

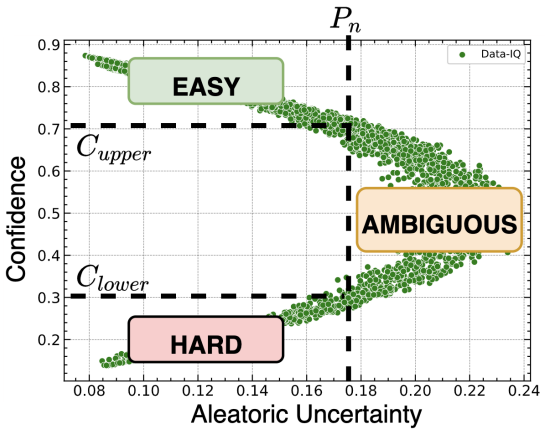



Fig. A.9 $\mathscr{P}(x,\theta)$ peaked around $\bar{\mathscr{P}}(x)$

## A.2 Experimental Details

### A.2.1 Code details

Code for this work can be found at: https://github.com/seedatnabeel/Data-IQ

### A.2.2 Benchmarks, Datasets and Implementation details

#### Data-IQ

The description of Data-IQ is detailed in the main part of the thesis, where we stratify examples based on aleatoric uncertainty and predictive confidence.

**Implementation details.** We implement Data-IQ via a class that is versatile and can plug into any framework, including Pytorch, Tensorflow/Keras, Skorch, and Scikit-learn style APIs. The rationale is that since Data-IQ can be used with any model, the functionality should be equally plug-and-play from a usability perspective. Note that for libraries including Pytorch, Tensorflow/Keras, and Skorch, we can easily include Data-IQ simply by adding it to the training callbacks of the respective training loops.

#### Data-IQ algorithm

Algorithm 2 below presents the Data-IQ algorithm in more detail.

**Algorithm 2:** Data-IQ Algorithm

**Data:** Dataset $\mathscr{D}$

**Result:** Subgroup assignment $g$ per sample $(x,y)$, i.e. $g((x,y),\mathscr{D}_{\text{train}})$

**Inputs:** $C_{\text{up}} = 0.75$, $C_{\text{low}} = 0.25$, $Q_{50}$ = 50th percentile

**Training:** Train model $(f)$ for $E$ epochs. At each epoch $e$, compute $P(x,\theta_e)$ for datapoint $x$, where $\theta_e$ is the model parameters at epoch $e$.

**Characterize samples into subgroups** $(g((x_i,y_i),\mathscr{D}_{\textbf{train}}))$**: for** $(x_i,y_i)$ *in* $\mathscr{D}_{train}$ **do**

  **Metrics:** $\bar{P}(x_i,y_i) = {}^1/_E \sum_{e=1}^{E} P(x_i,\theta_e)$;

  $v_{\text{al}}(x_i,y_i) = \frac{1}{E}\sum_{e=1}^{E} P(x_i,\theta_e)(1-P(x_i,\theta_e))$

  **if** $\bar{P}(x_i,y_i) \geq C_{\text{up}} \wedge v_{\text{al}}(x_i,y_i) < Q_{50}\left[v_{\text{al}}(\mathscr{D}_{train})\right]$ **then**

    Assign *Easy* subgroup

  **else if** $\bar{P}(x_i,y_i) \leq C_{\text{low}} \wedge v_{\text{al}}(x_i,y_i) < Q_{50}\left[v_{\text{al}}(\mathscr{D}_{train})\right]$ **then**

    Assign *Hard* subgroup

  **else**

    Assign *Ambiguous* subgroup

  **end**

**end**

$$g((x,y),\mathscr{D}_{\text{train}}) = \begin{cases} \text{Easy} & \text{if } \bar{P}(x,y) \geq C_{\text{up}} \ \wedge \ v_{\text{al}}(x,y) < Q_{50}\left[v_{\text{al}}(\mathscr{D}_{\text{train}})\right] \\ \text{Hard} & \text{if } \bar{P}(x,y) \leq C_{\text{low}} \ \wedge \ v_{\text{al}}(x,y) < Q_{50}\left[v_{\text{al}}(\mathscr{D}_{\text{train}})\right] \\ \text{Ambiguous} & \text{otherwise} \end{cases} \tag{A.2}$$

### Data Maps

Similar to Data-IQ, we have previously detailed Data Maps.

**Implementation details.** Our implementation of Data Maps can be used with the flexible approach we built for Data-IQ. We adapt the implementation of Data Maps from [1].

### AUM

The Area-under-the-Margin (AUM) metric first computes the margin for data $x, y$ at epoch $t$.

$$M^{(t)}(x,y) = \overbrace{z_y^{(t)}(x)}^{\text{assigned logit}} - \overbrace{\max_{i \neq y} z_i^{(t)}(x)}^{\text{largest other logit}}. \tag{A.3}$$

Over all epochs, the AUM is then computed as the average of these margin calculations.

$$AUM(x,y) = \frac{1}{T}\sum_{t=1}^{T} M^{(t)}(x,y), \tag{A.4}$$

**Implementation details.** The benchmark is based on [177] and we use the implementation from [2].

### GraNd

The gradient norm at epoch $t$ for an input $x, y$ is computed as :
$\chi_t(x,y) = \mathbb{E}\left\|\sum_{k=1}^{K} \nabla_{f^{(k)}} \ell\left(f_t(x),y\right)^T \psi_t^{(k)}(x)\right\|_2$
where $\psi_t^{(k)}(x) = \nabla_{\mathbf{w}_t} f_t^{(k)}(x)$.

**Implementation details.** The benchmark is based on [173]. We adapt the Jax implementation from [3] to Pytorch with the help of [4].

[1] https://github.com/allenai/cartography
[2] https://github.com/asappresearch/aum
[3] https://github.com/mansheej/data_diet
[4] https://github.com/cybertronai/autograd-lib

### JTT

The Just-Train-Twice (JTT) method is a two-stage method. For a model, trained via ERM, it first identifies errors in the training examples during training $i.e. y \neq \hat{y}$.

The second step is to train a final model, where the error samples are increased via weighting.

$$L_{\text{up-ERM}}(\theta, E) = (\lambda \sum_{(x,y)\in E} \ell(x,y;\theta) + \sum_{(x,y)\notin E} \ell(x,y;\theta)), \tag{A.5}$$

where $\lambda$ is the weighting factor and $E$ errors.

The goal is that the weighting factor will upsample the samples with errors and by training on this augmented dataset, it will help improve worst-group performance.

**Implementation details.** Our benchmark is based on [138] and we adapt the implementation from [5].

## A.2.3 Datasets

We describe the four real-world medical datasets in greater detail.

**SEER Dataset** The SEER dataset is a publicly available dataset consisting of 240,486 patients enrolled in the American SEER program [46]. The dataset comprises features used to characterize prostate cancer, including age, PSA (severity score), Gleason score, clinical stage, treatments, etc. A summary of the covariate features can be found in Table A.2. The classification task is to predict patient mortality, which is a binary label $\in \{0,1\}$.

The dataset is highly imbalanced, where 94% of patients survive. Hence, we extract a balanced subset of 20,000 patients (i.e. 10,000 with label=0 and 10,000 with label=1).

Table A.2 Summary of features for the SEER Dataset [46]

| Feature | Range |
|---|---|
| Age | $37-95$ |
| PSA | $0-98$ |
| Comorbidities | $0,1,2,\geq 3$ |
| Treatment | Hormone Therapy (PHT), Radical Therapy - RDx (RT-RDx),Radical Therapy -Sx (RT-Sx), CM |
| Grade | $1,2,3,4,5$ |
| Stage | $1,2,3,4$ |
| Primary Gleason | $1,2,3,4,5$ |
| Secondary Gleason | $1,2,3,4,5$ |

[5] https://github.com/anniesch/jtt

**CUTRACT Dataset** The CUTRACT dataset is a private dataset consisting of 10,086 patients enrolled in the British Prostate Cancer UK program [174]. Similar to the SEER dataset, it consists of the same features to characterize prostate cancer. In addition, it has the same task to predict mortality. A summary of the covariate features can be found in Table A.3. Once again, the dataset is highly imbalanced, hence we then extract a balanced subset of 2,000 patients (i.e. 1000 with label=0 and 1000 with label=1).

Table A.3 Summary of features for the CUTRACT Dataset [174]

| Feature | Range |
|---|---|
| Age | $44-95$ |
| PSA | $1-100$ |
| Comorbidities | $0,1,2,\geq 3$ |
| Treatment | Hormone Therapy (PHT), Radical Therapy - RDx (RT-RDx),Radical Therapy -Sx (RT-Sx), CM |
| Grade | $1,2,3,4,5$ |
| Stage | $1,2,3,4$ |
| Primary Gleason | $1,2,3,4,5$ |
| Secondary Gleason | $1,2,3,4,5$ |

**Fetal Dataset** The Fetal dataset [13] is a publicly available dataset consisting of 2126 patients who underwent fetal cardiotograms (CTG). The patients had specific diagnostic features measured and the CTGs were classified by three expert obstetricians (with a consensus label) to assign three fetal states (normal, suspect and pathologic). Our goal is to predict these three fetal states, hence the task is a multi-class classification problem. A summary of the covariate features can be found in Table A.4.

Table A.4 Summary of features for the Fetal Dataset [13]

| Feature | Range |
|---|---|
| LB (FHR baseline) | $106-160$ |
| AC (Acceleration/sec) | $0-260$ |
| FM (fetal movements/sec) | $0-564$ |
| UC (uterine contractions/sec) | $0-23$ |
| ASTV (% of time with abnormal short term variability) | $12-87$ |
| ALTV (% of time with abnormal long term variability) | $0-91$ |
| MLTV (mean value of long term variability) | $0-51$ |
| DS (severe deceleration/sec) | $0-1$ |
| DP (prolonged deceleration/sec) | $0-4$ |
| Nzeros (n histogram zeros) | $0-10$ |
| Variance (histogram variance) | $0-269$ |
| Tendency (histogram tendency) | $0-1$ |
| NSF (fetal class code) | $1(Normal),2(suspect),3(pathologic)$ |

**Covid-19 Dataset** The Covid-19 dataset [15] consists of Covid patients from Brazil. The dataset is publicly available as it is based on SIVEP-Gripe data. The dataset consists of 6882 patients from Brazil recorded between Februrary 27-May 4 2020. The dataset captures risk factors, including comorbidities, symptoms, and demographic characteristics. There is a mortality label from COVID-19, making it a binary classification task. A summary of the characteristics of the covariates can be found in Table A.5.

Table A.5 Summary of features for the Covid-19 Dataset [15]

| Feature | Range |
|---|---|
| Sex | 0 (Female), 1(Male) |
| Age | $1-104$ |
| Fever | $0,1$ |
| Cough | $0,1$ |
| Sore throad | $0,1$ |
| Shortness of breath | $0,1$ |
| Respiratory discomfort | $0,1$ |
| SPO2 | $0-1$ |
| Diharea | $0,1$ |
| Vomitting | $0,1$ |
| Cardiovascular | $0,1$ |
| Asthma | $0,1$ |
| Diabetes | $0,1$ |
| Pulmonary | $0,1$ |
| Immunosuppresion | $0,1$ |
| Obesity | $0,1$ |
| Liver | $0,1$ |
| Neurologic | $0,1$ |
| Branca (Region) | $0,1$ |
| Preta (Region) | $0,1$ |
| Amarela (Region) | $0,1$ |
| Parda (Region) | $0,1$ |
| Indigena (Region) | $0,1$ |

**Support Dataset** The Support [114] is a private dataset that examines outcomes for seriously ill hospitalized patients, consisting of 9105 patients. The features include demographics, medical history, clinical variables, and laboratory variables. Our goal is to predict patient mortality. A summary of the covariate features can be found in Table A.6.

### A.2.4 Additional experiment details

We note that all experiments were performed using a single Nvidia Tesla P100 GPU.

**Robustness to variation in experiment details.**

We assess robustness to variation across different models and/or parameterizations. The rationale for this is that these are model changes or perturbations that different practitioners might easily make. However, we desire consistency and stability in the metrics such that the data insights remain consistent. For layers, we evaluate a 3-layer MLP, 4-layer MLP, and 5-layer MLP. For the number of hidden units, we define two types: type 1 has 64 units in layer one, which is reduced by half per layer, and type 2 has 256 units in layer one, which is reduced by half per layer. Finally, we evaluate these combinations with Adam as the optimizer during training. These models perform at a similar level (i.e. accuracy). We then assess the consistency of the metrics for all examples using these different models/parameterizations based on Spearman rank correlation.

**Data insights from subgroups experiment details.**

For the data insights experiment, we perform clustering on each subgroup identified by Data-IQ. We use Gaussian Mixture Models (GMM), similar to [230]. The difference is the sub-space in which the clustering is applied. In Data-IQ's case, we cluster within each subgroup identified by Data-IQ. For each supergroup, we search over $k \in 2, \ldots, 10$. We choose the $k$ value that yields the highest average Silhouette score. Our implementation is based on [6]

**Principled feature acquisition experiment details.**

We assess the potential for principled feature acquisition using a semi-synthetic experiment. We assess how valuable each feature is based on the Pearson correlation of the feature with the target. We then rank-sort the correlations from lowest to highest. This ordering is then used to determine the order in which to "acquire" and assess the features.

**Principled dataset comparison experiment details.**

We compare two synthetic datasets using two models representing two synthetic data vendors, namely CTGAN and Gaussian Copula. We use the implementations of [7].

[6] https://scikit-learn.org [7] https://github.com/sdv-dev/SDV

**Less is more experiment details.**

For this experiment, we note that we train multiple baseline models to evaluate the generalization based on dataset sculpting. We evaluated three different XGBoost models with 100,150 and 200 estimators each. Our XGBoost implementation uses the Python version from [8].

**Group-DRO experiment details.**

Group DRO uses training group annotations to directly minimize the worst-group error on the training set. Hence, we compare different methods to obtain the group annotations. This experiment compares Group-DRO applied to groups identified by Data-IQ and as a baseline, George [230]. We also compare this to Just-Train-Twice (JTT) [138]. We use the implementation of Group-DRO from [9], which is based on [201]. Our George benchmark is based on the implementation from [10] and the JTT benchmark is based on the implementation from [11]. All code is in Pytorch.

The group DRO objective can then be written as:

$$L_{\text{group-DRO}}(\theta) = \max_{g \in G} \frac{1}{n_g} \sum_{i|g_i=g} \ell(x_i, y_i; \theta) \tag{A.6}$$

where $n_g$ is the number of training points with group $g_i = g$.

**Subgroup informed usage of uncertainty estimation experiment details.**

We obtain uncertainty estimates using a Bayesian Neural Network based on [63]. Specifically, we train a 5-layer MLP model. A Gaussian prior is placed over the weights and we optimize the KL divergence during training. Our benchmark is based on [63] and we use the implementation from [12].

[8] https://github.com/dmlc/xgboost
[9] https://github.com/HazyResearch/hidden-stratification
[10] https://github.com/HazyResearch/hidden-stratification
[11] https://github.com/anniesch/jtt
[12] https://github.com/IBM/UQ360

Table A.6 Summary of features for the Support dataset [114]

| Feature | Range |
|---|---|
| Sex | 0 (Female), 1(Male) |
| Age | 18 – 102 |
| Number of comorbities | 0 – 9 |
| ARF/MOSF w/Sepsis | 0, 1 |
| COPD | 0, 1 |
| CHF | 0, 1 |
| Cirrhosis | 0, 1 |
| Coma | 0, 1 |
| Colon Cancer | 0, 1 |
| Lung cancer | 0, 1 |
| ARF/MOSF w/Malig | 0, 1 |
| Cancer | 0, 1 |
| under $11k | 0, 1 |
| $11k-$25k | 0, 1 |
| $25k-$50k | 0, 1 |
| >$50k | 0, 1 |
| white | 0, 1 |
| black | 0, 1 |
| asian | 0, 1 |
| hispanic | 0, 1 |
| yeas of education | 0 – 31 |
| avg TISS | 1 – 83 |
| diabetes | 0, 1 |
| dementia | 0, 1 |
| mean bp | 0 – 195 |
| white blood cell count | 0 – 200 |
| heart rate | 0 – 300 |
| respiratory rate | 0 – 90 |
| temperature | 31 – 42 |
| pafi | 12 – 890 |
| albumnin | 0.39 – 29 |
| bilirubin | 0.09 – 63 |
| creatinine | 0.09 – 63 |
| sodium | 6 – 181 |
| ph | 6.8 – 7.81 |
| glucose | 0 – 1092 |
| bun | 1 – 300 |
| urine output | 0 – 9000 |
| adlp | 0 – 7 |
| adls | 0 – 7 |

## A.3 Additional Experiments

### A.3.1 Understanding the representation space

**Goal.** At deployment/testing time, we stratify the examples into subgroups using the representation space of the training examples. Our goal is to illustrate the points made in Chapter 2.3.3 with respect to the representation space.

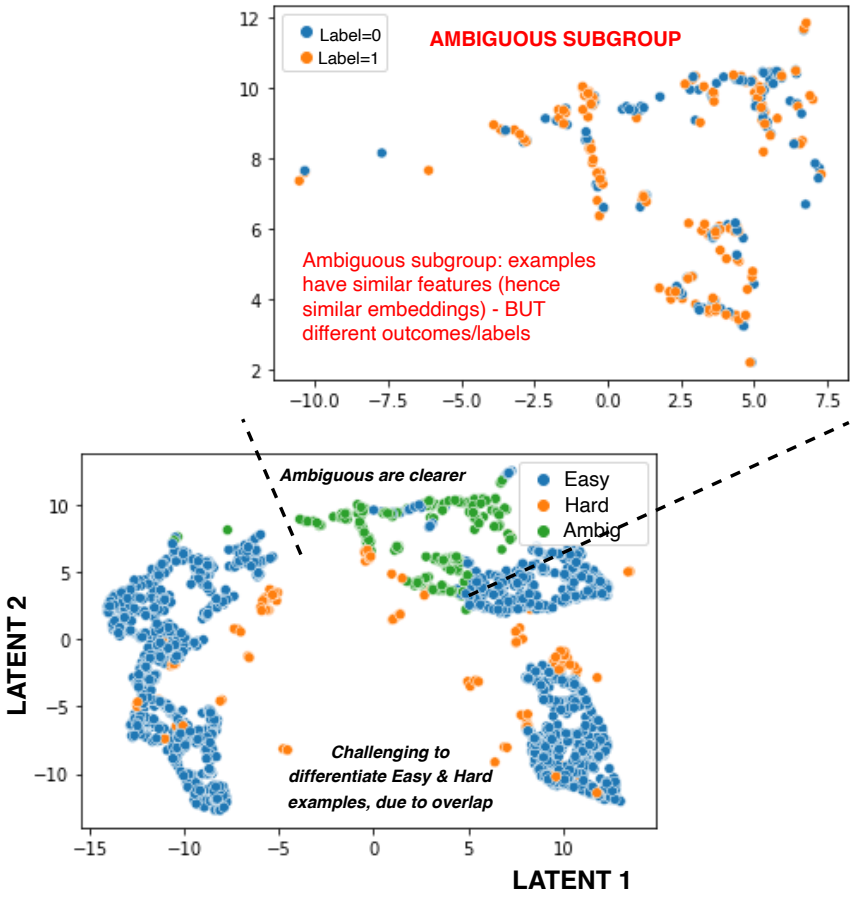



Fig. A.10 Representation space of training examples broken down by sub-group. We then zoom into the Ambiguous subgroup and illustrate the heterogeneous outcomes, where examples with similar features can have different outcomes.

**Takeaway.** We recall the two main points about the representation space and illustrate them using the representation space (i.e. UMAP embedding) shown in Figure A.10.

1. *Ambiguous* examples have distinctive features and are clustered in embedding space. Thus, we can distinguish the *Ambiguous* examples using the embedding. We can clearly see this with the green points in Figure A.10.

2. It is not possible to reliably distinguish *Easy* examples from *Hard* examples based on the embedding because *Hard* examples are a minority with outcome randomness that have similar features as the *Easy* examples. Naturally, since the Easy and Hard example embeddings are similar, they also have similar features. We can clearly see this where the orange (Hard) examples are randomly scattered with the Blue (Easy) examples in Figure A.10. This makes these two groups difficult to distinguish.

We also zoom in to the *Ambiguous* region (sub-figure of Figure A.10). This illustrates the specific issue of outcome heterogeneity, where similar patients (with similar features) have different outcomes. This is illustrated by the different colors, which represent the different outcomes (i.e. label=0 is survival and label=1 is death).

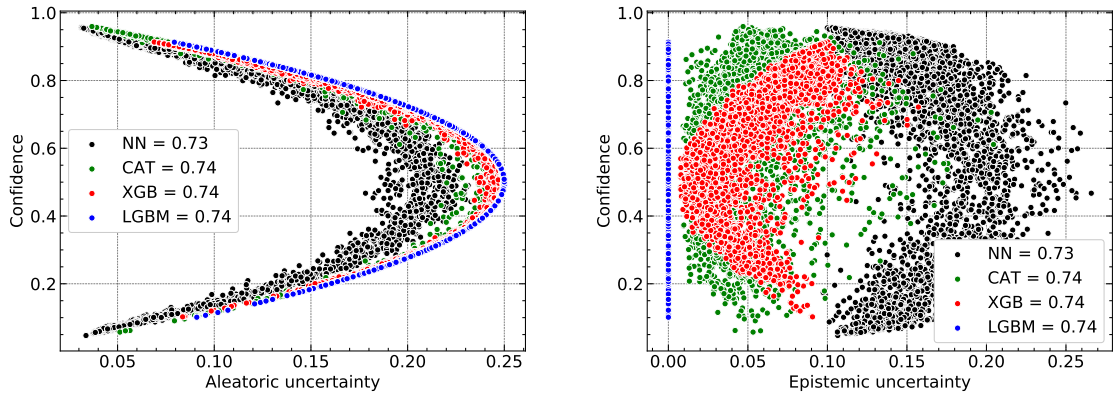
### A.3.2 Data-IQ: Neural Networks vs Other Model Classes

**Goal.** As discussed in the main text and Appendix A.1.2, Data-IQ can be used with *any* ML model trained in stages. Methods such as XGBoost, LightGBM, and CatBoost models are widely used by practitioners on tabular data, often more so than neural networks [23]. While Data-IQ can naturally handle any such model, as long as there are checkpoints (over iterations), Data Maps naturally cannot (or at least it was never formulated to do so). However, to enable comparison, we can use Data Maps within our framework so that it can be applied to a more general class of models.

Ideally, based on *P1*, we desire that the characterization of examples be consistent for similar performing models, irrespective of whether the model is a neural network or, for example, an XGBoost model.

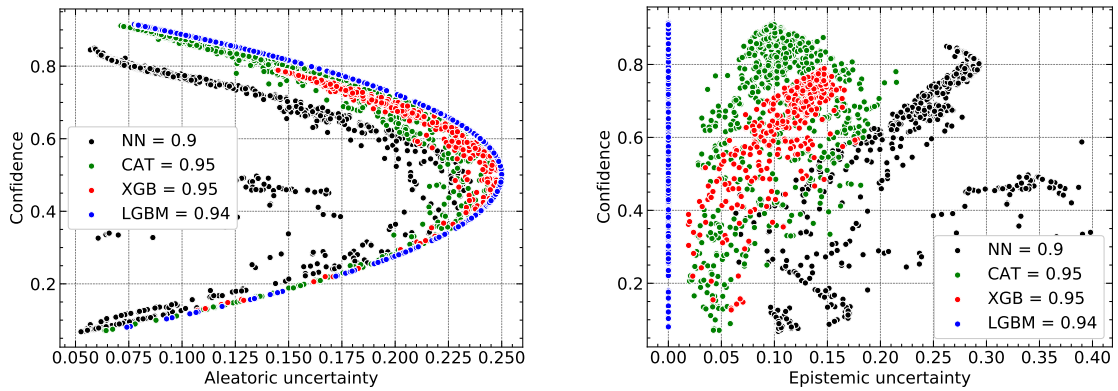
To assess the robustness of both Data-IQ and Data Maps, we train neural network, XGBoost, LightGBM, and CatBoost models to achieve the same performance and then perform the characterization for all models. The evaluation of all data sets is shown in Figures A.11-A.14.

**Takeaway** .We can clearly see that Data-IQ has a similar characterization across all four models. Contrastingly, for Data Maps, the characterizations are significantly different for the different model classes. The key difference is Data-IQ's use of Aleatoric uncertainty, rather than the Data Maps usage of Epistemic uncertainty. The implication of this result is that Data-IQ captures the uncertainty inherent to the data (aleatoric uncertainty) and this leads to a more consistent and stable characterization of the data itself. In particular, this highlights that Data-IQ characterizes the data in a manner that is not as sensitive to the choice of model when compared to Data Maps.




(a) Data-IQ (b) Data Maps

Fig. A.11 NN vs XGBoost comparison on the Covid-19 dataset, showing Data-IQ is more consistent

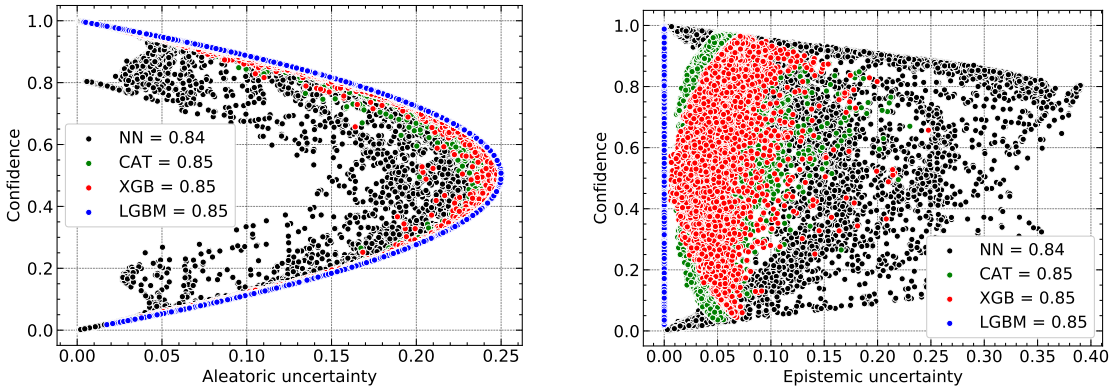



(a) Data-IQ (b) Data Maps

Fig. A.12 NN vs XGBoost comparison on the Fetal dataset, showing Data-IQ is more consistent

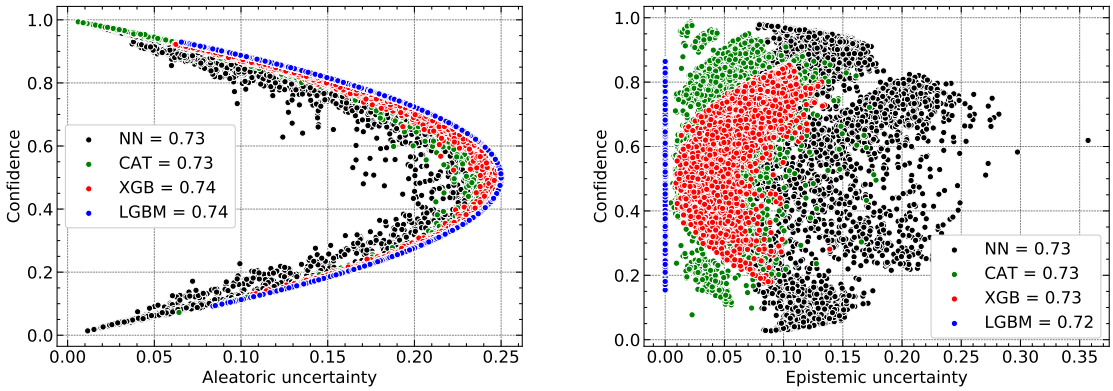
To further augment these results, we quantitatively assess the consistency across model types. Similar to Chapter 4.1, we have compared the Spearman rank correlation between the metrics of all model combinations (neural networks, XGBoost, LightGBM, CatBoost). Note that all four models are trained to perform similarly on a held-out dataset.

Below, we present Table A.7 showing the average and standard deviation. We see that Data-IQ has higher scores for all datasets, highlighting better consistency compared to Data Maps.




(a) Data-IQ (b) Data Maps

Fig. A.13 NN vs XGBoost comparison on the Prostate dataset, showing Data-IQ is more consistent




(a) Data-IQ (b) Data Maps

Fig. A.14 NN vs XGBoost comparison on the Support dataset, showing Data-IQ is more consistent

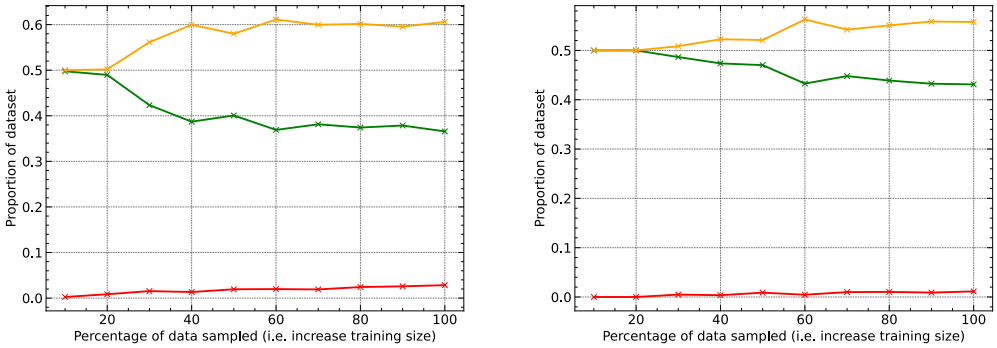
Table A.7 Comparison of robustness/consistency across different models on the basis of the Spearman rank correlation

| Dataset | (Ours) Data-IQ | Data Maps |
|---|---|---|
| Covid | $0.85 \pm 0.05$ | $0.53 \pm 0.07$ |
| Support | $0.84 \pm 0.06$ | $0.48 \pm 0.04$ |
| Prostate | $0.84 \pm 0.11$ | $0.44 \pm 0.17$ |
| Fetal | $0.72 \pm 0.07$ | $0.45 \pm 0.06$ |

### A.3.3 Collecting more data - it might make things worse

**Goal.** Typically, it is assumed that simply collecting more data will mitigate model problems. We aim to assess the influence of adding more data samples, in contrast to experiments in the main thesis, which focus on acquiring features. Primarily, we want to show how, in the case of heterogeneous outcomes, *blindly* adding more data might not always be the solution.

**Experiment.** We sub-sample proportions of the dataset from 0-100% of the original size of the dataset. We then recompute the Data-IQ subgroups on each proportion as more data examples are added. We show two examples in Figure A.15, in which the subgroups evolve as more data is added.




(a) Covid more samples (b) Support more samples

Fig. A.15 The *Ambiguous* subgroup proportion in fact increases as the dataset size increases, i.e. more examples are added. This is due to the increased probability of feature collisions.

**Takeaway.** As the number of data examples increases, the proportion covered by the *Ambiguous* subgroup increases. This is due to the increased probability of "feature collisions" as more examples are added. Therefore, the result shows that the simple solution, in which we naively collect more data, can sometimes do more harm.

### A.3.4 Identifying subsets in an outcome-driven manner: assessing quality

**Goal.** Data-IQ identifies subgroups in an outcome-driven manner, compared to the conventional unsupervised manner. We compare the cluster quality on the outcome-driven subgroups obtained by Data-IQ compared to other benchmarks, namely: George [230], clustering in the latent space, clustering in a PCA space, and clustering the raw feature space. All methods use a Gaussian Mixture model for consistency.

We evaluated these methods based on two well-known cluster quality metrics, namely, (1) Silhouette score (SIL) and (2) Davies-Bouldin (DB) score. The results for two example datasets can be seen in Table A.8.

Table A.8 Clustering results

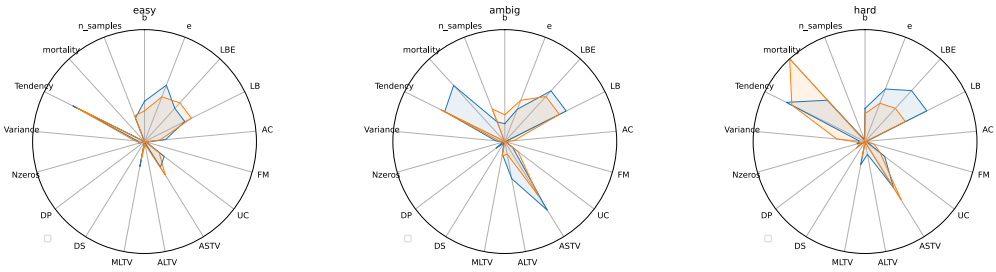
| Dataset | Metric | Data-IQ (Ours) | Outcome latent space (George) | Latent space | PCA | Feature space |
|---|---|---|---|---|---|---|
| Covid-19 | SIL (↑) | 0.68 | 0.48 | 0.47 | 0.51 | 0.02 |
| | DB (↓) | 0.42 | 0.87 | 0.78 | 0.81 | 5.7 |
| Prostate | SIL (↑) | 0.80 | 0.6 | 0.56 | 0.57 | 0.54 |
| | DB (↓) | 0.3 | 0.6 | 0.63 | 0.57 | 0.61 |

**Takeaway.** We can clearly see that Data-IQ has the highest SIL score and the lowest DB score. These results demonstrate that the outcome-driven clustering approach taken by Data-IQ, where the clusters are within the Data-IQ subgroups, results in higher-quality clusters.

### A.3.5 Data Insights: additional results

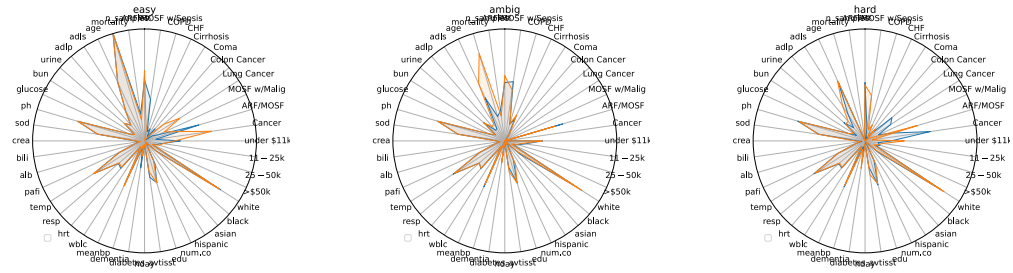
**Goal.** The main text (Chapter 2.4.1) illustrated the data insights that can be obtained for a single dataset (Prostate). Here, we include the results for all other datasets (using the same setup). See Figures A.16-A.18.

**Takeaway.** In general, for each dataset, the subgroups represent: (1) *Easy*: Severe patients with a death outcome and less severe patients with a survival outcome, (2) *Ambiguous*: Patients with similar features but different outcomes. This could suggest that the features we have at hand are insufficient to separate the differences in outcomes and (3) *Hard*: Severe patients with a survival outcome and less severe patients with a death outcome. i.e. opposite outcomes as expected due to randomness in the outcomes.



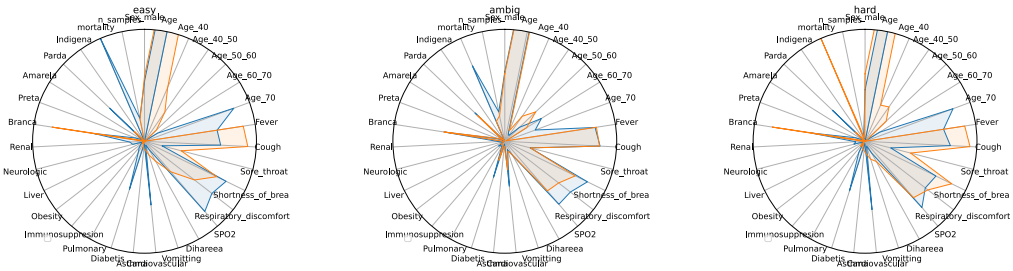

(a) Easy (b) Ambiguous (c) Hard

Fig. A.16 Fetal subgroups identified by Data-IQ (descriptions above). Colors represent the GMM clusters




(a) Easy (b) Ambiguous (c) Hard

Fig. A.17 Support subgroups identified by Data-IQ (descriptions above). Colors represent the GMM clusters



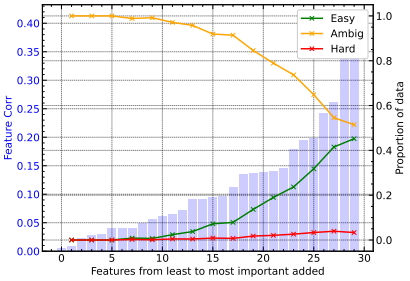

(a) Easy (b) Ambiguous (c) Hard

Fig. A.18 Covid subgroups identified by Data-IQ (descriptions above). Colors represent the GMM clusters

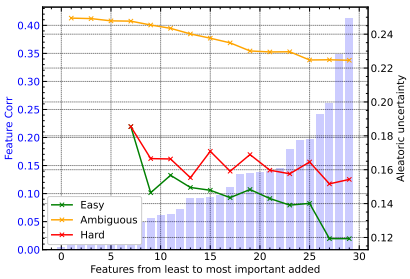
### A.3.6 Principled feature acquisition (P2): additional results

**Goal.** The main text (Chapter 2.4.2) illustrated principled feature acquisition using Data-IQ for a single dataset. Here, we include the results for all other datasets (using the same semi-synthetic setup, based on the correlation of the feature with the target). Recall our formulation, that a valuable feature should decrease the ambiguity of an example, overall decreasing the proportion of *Ambiguous* examples in the dataset. This permits a principled approach to feature acquisition.

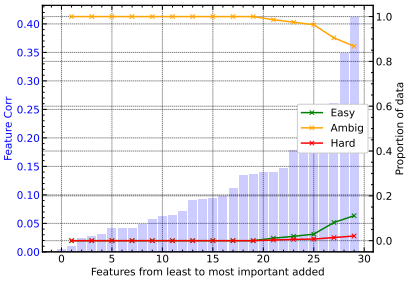
**Takeaway.** We see similar results to the main text for all datasets. For Data-IQ, we see that as we acquire "valuable" features, the proportion of the *Ambiguous* subgroup drops, while the proportion of the *Easy* subgroup increases. There are significant changes in the important features. This shows that Data-IQ's subgroup characterization can be used to quantify a feature's value by its ability to decrease ambiguity. In contrast, Data Maps shows minimal response to feature acquisition, suggesting that it may not be sensitive enough to capture the feature's value.



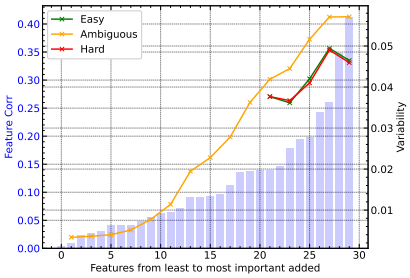

(a) Data-IQ subgroup ambiguity proportion is reduced as more informative features are acquired.




(b) Data-IQ aleatoric uncertainty remains stable for Ambiguous, reduces for others as features are acquired.

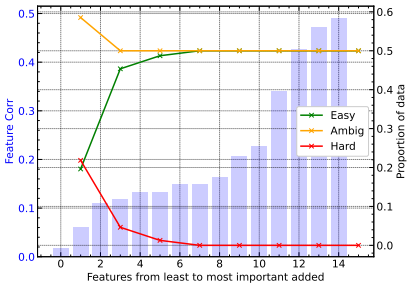



(c) Data Maps subgroup proportions largely unaffected as features acquired.

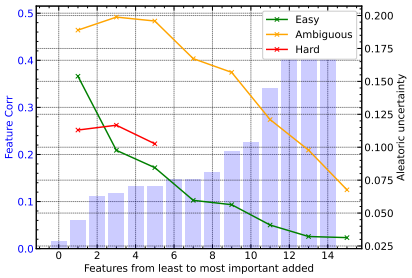



(d) Data Maps variability increases across subgroups as features acquired.

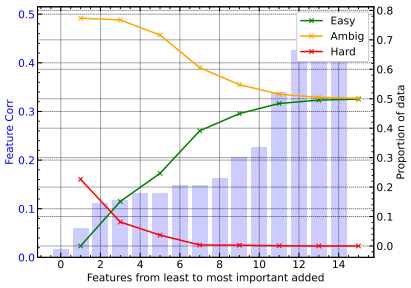
Fig. A.19 COVID Quantifying the value of feature acquisition based on change in ambiguity. Only Data-IQ captures this relationship.



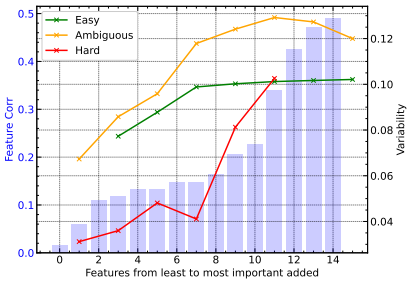

(a) Data-IQ subgroup ambiguity proportion is reduced as more informative features are acquired.




(b) Data-IQ aleatoric uncertainty remains stable for Ambiguous, reduces for others as features are acquired.

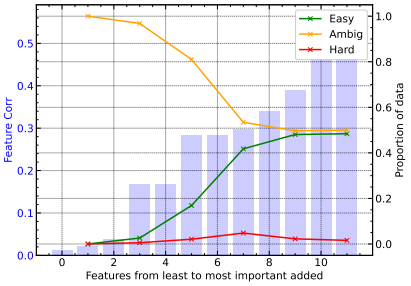



(c) Data Maps subgroup proportions largely unaffected as features acquired.

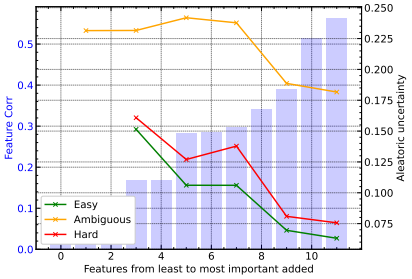



(d) Data Maps variability increases across subgroups as features acquired.

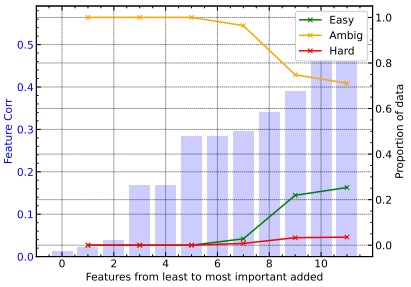
Fig. A.20 FETAL Quantifying the value of feature acquisition based on change in ambiguity. Only Data-IQ captures this relationship.



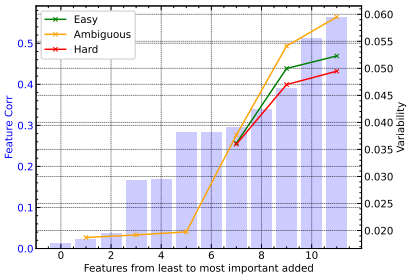

(a) Data-IQ subgroup ambiguity proportion is reduced as more informative features are acquired.




(b) Data-IQ aleatoric uncertainty remains stable for Ambiguous, reduces for others as features are acquired.

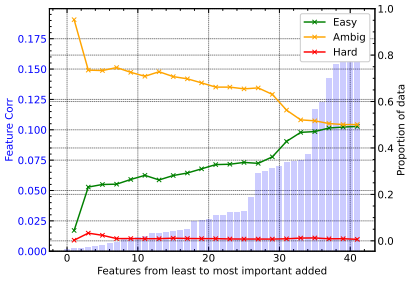



(c) Data Maps subgroup proportions largely unaffected as features acquired.

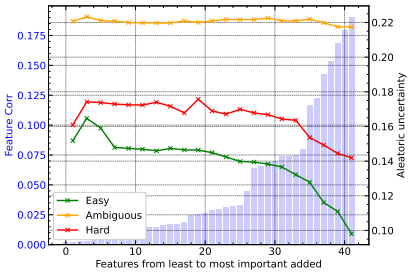



(d) Data Maps variability increases across subgroups as features acquired.

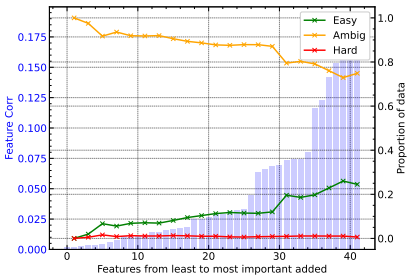
Fig. A.21 Prostate Quantifying the value of feature acquisition based on change in ambiguity. Only Data-IQ captures this relationship.



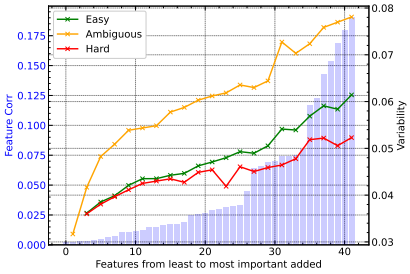

(a) Data-IQ subgroup ambiguity proportion is reduced as more informative features are acquired.




(b) Data-IQ aleatoric uncertainty remains stable for Ambiguous, reduces for others as features are acquired.

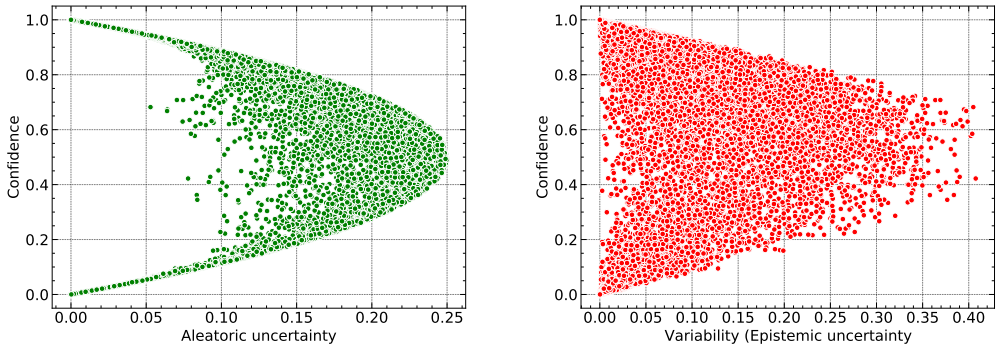



(c) Data Maps subgroup proportions largely unaffected as features acquired.




(d) Data Maps variability increases across subgroups as features acquired.

Fig. A.22 Support Quantifying the value of feature acquisition based on change in ambiguity. Only Data-IQ captures this relationship.




(a) Data-IQ: TEXT

(b) Data Maps: TEXT

(c) Data-IQ: IMAGE

(d) Data Maps: IMAGE

Fig. A.23 Data-IQ vs Data Maps compared on text and image modalities

### A.3.7 Effect of averaging over the uncertainty metric

**Goal.** We wish to assess the impact of averaging/taking an expectation over the metric, such as aleatoric uncertainty over training. Specifically, we assess the case where an example has high variance at the beginning of training and flattens with time, compared to an example that remains at a medium level throughout training. These represent the EASY/HARD (high variance in the beginning and then flattening) and AMBIGUOUS (medium level throughout) subgroups, respectively. The question is, by taking an average, will the uncertainties be similar despite the training dynamics being different? If the average uncertainties are not similar, that means we can distinguish the groups with different training dynamics.

**Experiment.** We conduct an experiment to assess this by sampling 100 examples from each of these groups and plot the following three metrics per sample:

- Training dynamic (i.e. the aleatoric uncertainty over training epochs).
- The distribution over aleatoric uncertainty scores across all training steps.
- The "average" aleatoric uncertainty.

**Takeaway.** The result, as shown in Figure A.24 below, shows that even by taking an average, there is still a distinct separation to be able to distinguish between the two types of samples with different training dynamics. This suggests that averaging over the uncertainty is indeed a useful metric to evaluate the uncertainty of an individual data sample.

### A.3.8 Effect of additional sources of randomness

**Goal.** We wish to assess the impact of incorporating additional sources of randomness, for example, dropout.

**Experiment.** We tested the model parameterizations but additionally trained an additional 3 models with the same architecture, yet with the inclusion of different dropout masks with dropout probability: $p = 0.1, p = 0.2, p = 0.3$.

We then assess the Spearman correlation of the metrics across all model combinations, as was done in the experiment on robustness to variation. A higher correlation implies better robustness to variations.

**Takeaway.** Data-IQ remains most robust to variation, even with this additional randomness. We highlight the results on all four datasets in Table A.9 below.

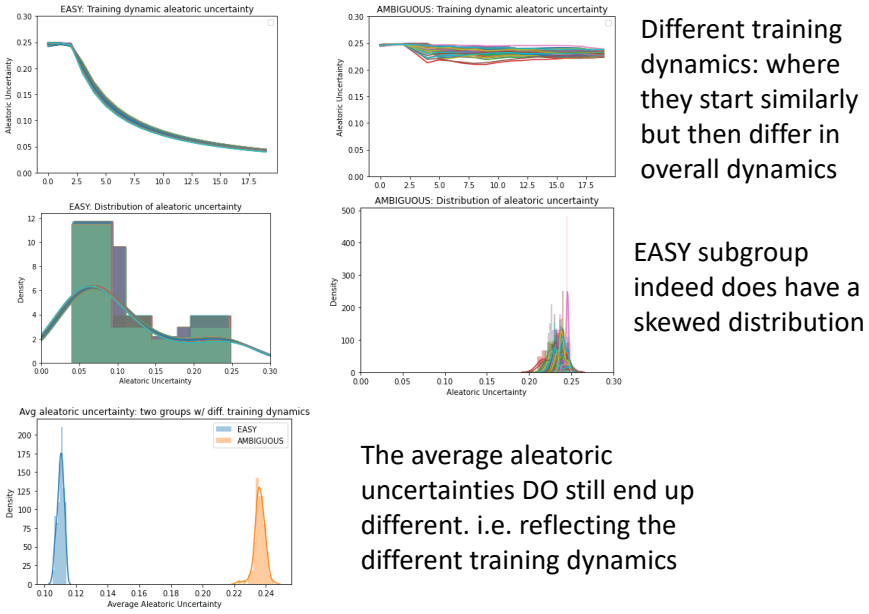



Fig. A.24 Assessment of whether we can distinguish groups with different training dynamics when we average over the metric

Table A.9 Comparison of robustness/consistency across different models on the basis of the Spearman rank correlation

| Dataset | (Ours) | Data-IQ | Data Maps |
|---|---|---|---|
| Covid | $0.93 \pm 0.07$ | $0.66 \pm 0.16$ |
| Support | $0.92 \pm 0.02$ | $0.81 \pm 0.11$ |
| Prostate | $0.97 \pm 0.01$ | $0.96 \pm 0.01$ |
| Fetal | $0.86 \pm 0.06$ | $0.81 \pm 0.13$ |

### A.3.9 Effect of hyper-parameters on the recovery of estimated groups

**Goal.** We wish to assess how model choices/parameterizations affect the recovery of the different subgroups. i.e. we desire that if a sample is categorized as EASY with one parameterization, that similarly it should be categorized as EASY for another parameterization. Similarly, for the other subgroups, we desire that subgroup characterization remains the same across different model parameterizations.

**Experiment.** We use the same sets of model parameterizations as assessed previously (Robustness to variation experiment). We then compute the overlap of sample identifiers for each subgroup (*Easy*, *Ambiguous*, *Hard*) between the different parameterizations. This can be thought of as sample-level stability for assignment to a specific subgroup.

**Takeaway.** We present the results across all four datasets below in Table A.10, which suggests that the recovery of subgroups is stable, with more than 90% of samples being assigned to the same subgroup between models.

Table A.10 Assessment of stability in group assignment across different parameterizations. The score reflects the average overlap of id characterization.

| Dataset | (Ours) Data-IQ |
|---|---|
| Covid | $0.94 \pm 0.03$ |
| Support | $0.91 \pm 0.01$ |
| Prostate | $0.95 \pm 0.01$ |
| Fetal | $0.77 \pm 0.14$ |

### A.3.10 Data sculpting: Absolute number of samples

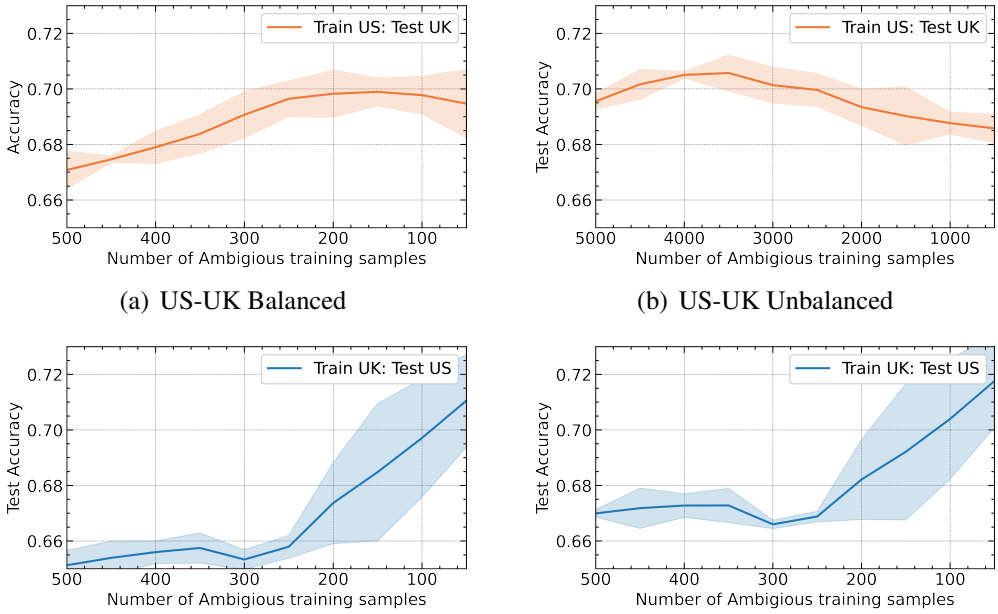
**Goal.** When performing data sculpting, we desire to know the absolute number of samples removed in order to quantify the importance of the effect of removing such data points on model performance.

**Experiment.** We present this analysis for two scenarios:

- Balanced SEER-CUTRACT: 1000 samples each
- Unbalanced SEER-CUTRACT: 10000 samples in SEER and 1000 in CUTRACT

**Takeaway.** For the balanced experiment, the full Ambiguous subgroup (i.e. proportion=1) has a total of 500 samples. Hence, when we sweep the x-axis representing the proportions of Ambiguous samples; $p = \{0, 0.2, 0.4, 0.6, 0.8, 1\}$, this is reflective of $\{0, 100, 200, 300, 400, 500\}$ absolute samples. This result shows that the number of samples that can be excluded is fairly substantial.

We compare this to the unbalanced experiment. The results are in Figure A.25. It shows that a substantial number of Ambiguous samples can be removed to improve performance. However, in the case of (b), when there are a lot of samples, removing too many can also be harmful.




(a) US-UK Balanced

(b) US-UK Unbalanced

(c) UK-US Balanced

(d) UK-US Unbalanced

Fig. A.25 Data sculpting - absolute number of *Ambiguous* samples removed vs test accuracy

### A.3.11 Correlation of weights through training

**Goal.** We wish to assess the correlation of weights through model training.

**Experiment.** To provide this assessment, we follow Jin et al [99], where weight correlation is defined as the average cosine similarity between weight vectors of neurons. We conduct an experiment to assess the correlation of model weights over epochs $e \in \{1,2...E\}$. i.e. we assess the average correlation of model weights $(e_1,e_2),(e_2,e_3)$ and so on.

**Takeaway.** We note that [99], shows that LOW correlation is less than 0.2. Our validation shows that the weights have LOW correlation, with an average weight correlation across the 4 datasets of $0.1 \pm 0.02$. Thus, even though our goal is different from a typical sampling setting (i.e. we just want to measure the dispersion of observed variables) - we still wish to highlight the low correlation of weights between epochs.

However, as a recommendation for practitioners, if this is a concern, a practical solution is that the practitioner could decide to simply increase the intervals for which the dynamics are sampled, thereby decreasing this correlation even further.

# Appendix B

# Appendix for Chapter 3

## B.1 Additional Details

### B.1.1 Extended Related Work

We present a comparison of our framework Data-SUITE and contrast it to related work of uncertainty quantification and learning with noisy labels. Table B.1 highlights both similarities and differences across multiple dimensions. We highlight three key features that distinguish Data-SUITE: (1) Data-centric uncertainty is a novel paradigm compared to the predominant model-centric approaches, (2) Our method offers increased flexibility, as it can be used independent of task-specific predictive model. Hence, any conclusions that we draw from Data-SUITE are not model-specific. (3) Our method provides theoretical guarantees concerning the validity of coverage.

Table B.1 Comparison of related work

| | Data-centric uncertainty | Model-centric uncertainty | Task Model independent | No noise assumptions | Coverage guarantees |
|---|---|---|---|---|---|
| Data-SUITE (Ours) | ✓ | × | ✓ | ✓ | ✓ |
| Uncertainty Quantification | × | ✓ | × | ✓ | × |
| Noisy labels | × | ✓ | × | × | × |

### B.1.2 Data-SUITE Details

We next have in-depth discussions on both the generator and conformal predictor. We outline motivations as well as technical details not covered in the main thesis.

#### Generator: Copulas

**Motivation.** Recall that in our formulation, we have a set of training instances $\mathscr{D}_{\text{train}}$ and we wish to learn the dependency between features $\mathscr{X}$. Hence, we model the multivariate joint distribution of $\mathscr{D}_{\text{train}}$ to capture the dependence structure between random variables. A significant challenge is modeling distributions in high dimensions. Parametric approaches such as Kernel Density prediction (KDE), often using Gaussian distributions, are largely inflexible.

On the other hand, non-parametric versions are typically infeasible with complex data distributions and the curse of dimensionality. Additionally, while Variational Autoencoders (VAEs) [112] and Generative Adversarial Networks (GANs) [69] can model and learn the joint distributions, they both have limitations for our setting. VAEs make strong distributional assumptions [194], while GANs involve training multiple models, which leads to associated difficulties [232, 75] (both from training, computational burden and inherent to GANs - e.g., mode collapse).

An attractive approach, particularly for tabular data, is Copulas; which flexibly couple the marginal distributions of the different random variables into a joint distribution. One important reason lies in the following theorem:

**Property B.1.1** (Sklar's theorem)**.** *A d-dimensional random vector* $\boldsymbol{X} = (X_1, ..., X_d)$ *with joint distribution* $F$ *and marginal distributions* $F_1, ..., F_d$ *can be expressed as* $F(X_1, ..., X_d) = C_\theta\{F_1(X_1)...F_d(X_d)\}$*, where* $C_\theta : [0,1]^d \to [0,1]$ *is a Copula. This is attractive in high dimensions, as it separates learning of univariate marginal distributions from learning of the coupled multivariate dependence structure.*

Parametric copulas have limited expressivity; hence, we use pair copula constructions (vine copulas) [17], which are hierarchical and express a multivariate copula as a cascade of bivariate copulas.

**Copula Details.** To learn the copula, we factorize the d-dimensional copula density into a product of $d(d-1)/2$, bivariate conditional densities by means of the chain rule.

The graphical model has edges being each bivariate copula that encodes the (conditional) dependence between variables. The graphical model additionally consists of levels (where

there are as many levels as features of the dataset). Each node will be a variable, and edges are couplings between variables based on bivariate copulas. As each level is constructed, the number of nodes decreases per level. The product of all pair-copula densities is then taken to define the joint copula.

Once we have learned the copula density, we sample the copula to obtain an augmented dataset of pseudo/synthetic samples. The copula samples are then easily transformed back to the natural data scale using the inverse probability integral transform.

Specifically, assume we have $\mathbf{U} = (U_1, U_2 ... U_d)$ random variable $\sim U(0,1)$. We can then use the Copula $C_\theta$ to define variables $\mathbf{S} = (S_1, S_2 ... S_d)$, where $S_1 = C^{-1}(U_1), S_2 = C^{-1}(U_2|U_1) ... S_d = C^{-1}(U_d|U_1, U_2 ... U_{d-1})$. This means that $\mathbf{S}$ is the inverse Rosenblatt transform of $\mathbf{U}$ and hence, $\mathbf{S} \sim C$, which allows us to simulate synthetic/pseudo samples. For more on Copulas in general, we refer the reader to [101].

**Complexity.** As we go through each tree, there is a decreasing number of pair copulas. i.e. ($T_1 = d-1$, $T_2 = d-2 ... T_{d-1} = 1$). Hence, the complexity of this algorithm is $O(n) \times d \times$ truncation level), which for all purposes is $O(n)$.

### Conformal predictor

**Motivation.** Conformal prediction allows us to transform any underlying point predictor into a valid interval predictor. We will not discuss the generalized framework of conformal prediction (*Transductive Conformal prediction*), which requires model training to be redone for every data point. This has a huge computational burden for modern datasets with many data points.

We instead only discuss *Inductive Conformal prediction*, which is used in Data-SUITE. The inductive method splits the two processes needed: 1. the training of the underlying model and 2. computing the conformal estimates.

**Conformal Prediction Details.** Practically, we split the training set ($|\mathscr{D}^+_{\text{train}}| = n$) into two disjoint sets, namely the proper training set and calibration set: $\mathscr{D}^+_{\text{train}} = \mathscr{D}^+_{\text{train2}} \sqcup \mathscr{D}^+_{\text{cal}}$, where $|\mathscr{D}^+_{\text{train2}}| = m$ and ($|\mathscr{D}^+_{\text{cal}}| = n - m$).

We use the proper training set to create our prediction rules for the feature-wise regressor ($g$).The calibration set is used for "conformalization", i.e. for computing the non-conformity scores and p-values.

The non-conformity score $\mu_i$ of each example is a function that computes the disagreement (i.e. non-conformance) between the prediction and true value. Note that we only compute non-conformity scores on the calibration set.

To obtain our intervals, we need to determine the critical value $\varepsilon$ based on the nonconformity scores. We first sort them in descending order. The critical value $\varepsilon$ is then the $\lceil(|\mathscr{D}^{+}_{\text{cal}}|+1)(1-\alpha)\rceil$-th smallest residual [249]. Consequently, for any confidence level (1-$\alpha$), we can use the critical value to find $p(x) > \alpha$, which corresponds to the maximum and minimum values with p-values larger than $\alpha$. As a consequence, we can obtain the maximum and minimum values of our predictive intervals.

The whole process is detailed in Algorithm 3. Normalization to obtain adaptive instance-wise intervals can be done using a normalization function $\sigma_i$, as described previously.

**Remarks on theoretical guarantees.** A motivation for conformal prediction in high-stakes settings is the theoretical guarantees on PI coverage validity (See Property B.1.2).

i.e., at a confidence level $(1-\alpha)$ of 95% ($\alpha = 0.05$), the true value will be within the CIs in at least 95% of the cases. The framework is non-parametric and only makes the exchangeability assumption (which we detail next), and the guarantees of validity of coverage hold for any choice of dataset, underlying model, or nonconformity measure, which makes it a versatile option.

**Property B.1.2** (Validity)**.** *Under the exchangeability assumption, the conformal predictor will return an interval,* $\mathbb{P}(X_i \in [l_i(X), r_i(X)]) \geq 1-\alpha$*, i.e., error* $\leq \alpha$*.*

The validity of the PI holds if the data is independent and identically distributed (IID). In practice, the weaker assumption of exchangeability (see Assumption B.1.1) also guarantees validity [128]. This means that we aren't required to impose any additional requirements for the validity of the PI since the aforementioned assumptions on the underlying data are typically made for any ML model.

We also highlight that the validity is maintained even in the case of normalization, as long as the proper training set and calibration set are disjoint.

**Assumption B.1.1** (Exchangeability)**.** *In a dataset of n observations, the data points do not follow any particular order, i.e., all n permutations are equiprobable. Exchangeability is weaker than IID observations; however, IID observations satisfy exchangeability.*

**Algorithm 3:** General Inductive Conformal Prediction

function CE $(\alpha, \mathscr{D}^{+}_{\text{train2}}, \mathscr{D}^{+}_{\text{cal}}, \mu)$;
**Input :** Significance $\alpha$, nonconformity measure $\mu$, proper training set, $\mathscr{D}^{+}_{\text{train2}}$ and calibration set, $\mathscr{D}^{+}_{\text{cal}}$
**Output :** Interval estimate

Train the underlying model $g$ on $\mathscr{D}^{+}_{\text{train2}}$;
Compute non-conformity scores on the calibration set $\mathscr{D}^{+}_{\text{cal}}$:
$P = \{\}$ ;
**foreach** *feature* $i \in \{1, \ldots, d\}$ **do**
  $\mu_i(\mathbf{x}) = |x_i - (g_i \circ f)(\mathbf{x})|$ ;
  Add $\mu_i(\mathbf{x})$ to $P$ ;
**end**
**CONFORMALIZATION.**
Sort $P$ in descending order to obtain scores $S$ ;
Determine the critical value of $\varepsilon \leftarrow \lceil (|\mathscr{D}^{+}_{\text{cal}}| + 1)(1-\alpha) \rceil$-th smallest residual in $S$.
Construct the interval predictor for each new value:;

**procedure** $\Gamma^{\alpha}(\mathbf{x} : \mathscr{X})$
  **foreach** $\mathbf{x} \in \mathscr{D}_{test}$ **do**
    Apply $x_m = (g_i \circ f)(\mathbf{x})$
  **end**
  **return** $[l_i(\mathbf{x}), r_i(\mathbf{x})] = [x_m - \varepsilon, x_m + \varepsilon]$;

## B.2 Experimental Details

### B.2.1 Benchmarks & Implementations

### B.2.2 Code details

Code for this work can be found at: https://github.com/seedatnabeel/Data-SUITE

#### Data-SUITE

**Implementation details.** Data-SUITE adopts a pipeline-based approach to constructing CIs, leveraging copula modeling, representation learning, and conformal estimation.

We break down each of these below:

*Copula Modeling*: We use the Copula to estimate a multivariate distribution with univariate marginal distributions. We make use of vine copulas [17] for this task.

Specifically, we use Direct-Vines (D-Vines), which impose constraints on the edges such that we only learn vines of the structure given in Figure B.1 below.

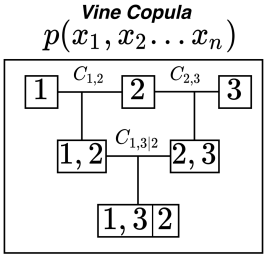



Fig. B.1 D-Vine structure

When fitting each tree, we select the "best" base copula based on likelihood from Gaussian, Frank, Clayton or Gumbel copulas.

Finally, when we sample from the copula, we specifically sample $n_{samples} = |\mathscr{D}_{\text{train}}|$.

*Representation learning*: For the representer, we specifically make use of Principal Components Analysis (PCA). That said, we can simply replace this block with any alternative, such as an autoencoder.

We pre-process all data prior to the representer by standardizing the data, such that each feature has zero mean and unit variance (see EquationB.1). Note that if the features were indeed categorical, we would perform one-hot encoding of the features.

$$z = \frac{(x - \mu)}{\sigma} \tag{B.1}$$

When applying PCA to learn the latent representation, we halve the dimensionality of $\mathscr{D}_{\text{train}}$, i.e., $d_X/2$.

*Conformal Prediction*: The two most important design decisions for conformal estimation are selecting the underlying feature-wise regressor ($g_i$) and the non-conformity score ($\mu_i$).

We selected $g_i$, as in conventional machine learning, by grid-search over different possible underlying models, evaluated on a validation set. We evaluate an MLP, KNN, Decision Trees and Random Forests. We ultimately selected a Decision Tree to serve as the base model for all $n = |d_X|$ feature-wise regressors. We use the following parameters max depth=None, min samples split=2, min samples leaf=5.

It is interesting that a simpler base model proved to be more effective and outperformed more parameterized models, which would require more computing. This is an additional advantage of Data-SUITE. We motivate that the simpler model might be directly related to the fact that we build feature-wise regressors. Hence, the mapping function is easier to approximate.

Finally, we use the absolute error non-conformity score. In practice, we can use any non-conformity score; however, intuitively, for our application, where we want to approximate the true value as closely as possible, the absolute error makes the most sense. Our non-conformity score is given by Equation B.2 below.

$$\mu_i(x) = |x_i - (g_i \circ f)(x)| \tag{B.2}$$

**Bayesian Neural Network (BNN)**

Bayesian modelling, when applied to neural networks, involves a likelihood function $p(Y|X,\theta)$, where the parameters $\theta$ are estimated by a neural network. In contrast to conventional neural networks, which have point estimates for the parameters, BNNs aim to learn distributions over parameters. However, modern neural networks have many parameters and weights. This makes exact inference largely intractable.

Hence, approximate Bayesian methods such as variational inference are often used instead. By this, we mean we do not compute the posterior in the conventional manner. Rather, the variational approximation means we replace the posterior $p(\theta|\mathscr{D})$ with a more tractable variational distribution $q(\theta;\lambda)$. The Kullback-Leibler (KL) divergence between the true distribution and the variational distribution is then minimized as the loss function, which equates to maximizing the evidence lower bound (ELBO); see EquationB.3. We note, however, that the reparameterization trick [111] is needed to make backpropagation possible.

$$\mathscr{L}_{\mathrm{VI}}(\mathscr{D};\lambda) := D_{\mathrm{KL}}\big(q(\theta;\lambda)\,||\,p(\theta)\big) - \mathbb{E}\big[\log p(\mathscr{D}\,|\,\theta)\big] \tag{B.3}$$

**Implementation details.** We train a 5-layer MLP model. A Gaussian prior is placed over the weights and we optimize the KL divergence during training. Our implementation is based on [63] and we use the implementation from [1].

### Deep Ensembles (ENSEMBLE)

Deep Ensembles [125] is widely regarded as the state-of-the-art non-Bayesian uncertainty estimation method. The rationale is that training multiple randomly initialized models allows more robust predictions. Uncertainty can be computed as the variance of the different model predictions. We highlight some important features of Deep Ensembles: (1) optimization based on a *proper scoring rule* such that the loss has a unique minimum, encouraging the model to approximate the true probability distribution. However, the proper nature of the scoring rule introduces a distributional assumption. (2) Deep Ensembles use adversarial perturbations based on the Fast Gradient Sign Method [70] and given by Equation B.4.

$$\mathbf{x}' := \mathbf{x} + \eta \odot \mathrm{sgn}\big(\nabla_{\mathbf{x}}\mathscr{L}(\mathbf{x},y;\theta)\big)\,, \tag{B.4}$$

where $\odot$ denotes element-wise multiplication.

This modifies the loss function used for gradient descent:

$$\mathscr{L}_{\mathrm{tot}}(\mathbf{X},\mathbf{y};\theta) := \mathscr{L}(\mathbf{X},\mathbf{y};\theta) + \mathscr{L}(\mathbf{X}',\mathbf{y};\theta)\,, \tag{B.5}$$

To construct prediction intervals with an uncertainty estimate, an assumption of a conditionally normal distribution is assumed, and the intervals with uncertainty estimates are given as per Equation B.6.

$$\Gamma(\mathbf{x}^*) := \left[\mathbb{E}[y^*\,|\,\mathbf{x}^*] - \sqrt{\mathrm{Var}[y^*\,|\,\mathbf{x}^*]}, \mathbb{E}[y^*\,|\,\mathbf{x}^*] + \sqrt{\mathrm{Var}[y^*\,|\,\mathbf{x}^*]}\right]. \tag{B.6}$$

**Implementation details.** We have 5 models in the ensemble, all randomly initialized. Each model is a 3-layer MLP, which we train for 10 epochs. The learning rate was empirically determined based on a validation set. To compute uncertainty estimates at test time, we

[1] https://github.com/IBM/UQ360

obtain predictions from each model in the ensemble. The prediction interval is computed as per Equation B.6.

### Gaussian Process (GP)

Gaussian processes (GPs) are fully characterized by the mean and covariance functions $\mu$ and $\sum$. The inference step can be performed exactly as marginalizing the multivariate normal distributions can be written in closed form. See EquationB.7, where by conditioning on dataset $\mathscr{D}$, we can compute the posterior.

$$(y^* \mid \mathbf{x}^*, \mathscr{D}) \sim \mathscr{N}\left(\mu^* + (\Sigma^*)^t \Sigma^{-1}(\mathbf{y} - \mu), \Sigma^{**} - (\Sigma^*)^t \Sigma^{-1} \Sigma^*\right). \tag{B.7}$$

We note the two assumptions made by GPs: (1) data is conditionally normally distributed and (2) the covariance selection is correct. That said, violations of these assumptions can severely impact performance. In addition, GPs have an issue of computational complexity due to matrix inversion of $O(n^3)$. For high-dimensional data, stochastic approximations are often used instead.

**Implementation details.** The GP is fit with a radial basis function (RBF) kernel, and we make use of the Scikit-learn [2] implementation of GPs, which is based on [257].

### Monte-Carlo Dropout (MCD)

Dropout layers are typically used as a regularizer at training time [233]. As shown by [57], dropout networks can be used at test time (Monte-Carlo Dropout (MCD)) and are a variational approximation to deep Gaussian processes. Note that this induces the assumption of normality similar to GPs.

A key component of MCD is that dropout at test time effectively allows us to obtain an ensemble of different models without having to retrain the model itself. For this, we do multiple stochastic forward passes at test time.

When making predictions, we compute a conditional mean. This is approximated by Monte-Carlo integration, i.e. mean of multiple forward passes. The prediction interval to characterize uncertainty is given by Equation B.8:

$$\Gamma(\mathbf{x}^*) := \left[\mathrm{E}[y^* \mid \mathbf{x}^*] - \sqrt{var[y^* \mid \mathbf{x}^*]}, \mathrm{E}[y^* \mid \mathbf{x}^*] + \sqrt{var[y^* \mid \mathbf{x}^*]}\right]. \tag{B.8}$$

[2] https://scikit-learn.org

**Implementation details.** We train a 3-layer MLP with dropout ($p = 0.1$)for 10 epochs. The learning rate was empirically determined based on a validation set. To compute uncertainty estimates at test time, we perform 20 Monte-Carlo Samples (forward passes at test time). The prediction interval is computed as per Equation B.8.

#### Quantile Regression (QR)

Quantile regressors estimate the conditional quantiles of a distribution (rather than the conditional mean). Consequently, there is a natural measure of the underlying distribution's spread.

Typically, the MSE loss is replaced by the pinball loss, otherwise known as quantile loss, given by EquationB.9, which balances the number of points above and below the quantiles [116]. Neural networks are easily applied and optimize the loss function with the appropriate number of output heads.

$$\mathscr{L}_{\text{pinball}}(\mathscr{T}) := \sum_{(\mathbf{x},y)\in\mathscr{T}} \max\Big((1-\alpha)(\hat{q}_\alpha(\mathbf{x})-y), \alpha(y-\hat{q}_\alpha(\mathbf{x}))\Big). \tag{B.9}$$

**Implementation details.** The base model for predicting the quantiles is a Gradient Boosting Regressor, with 10 estimators, max depth=5, minimum samples per leaf=5, minimum samples for split=10. We use the implementation from [3].

### B.2.3 Synthetic Experiment Details

#### Dataset & Experiment Configurations

The synthetic data $\mathbf{X} = [X_1, X_2, X_3]$ is drawn IID from a multivariate Gaussian distribution, parameterized by mean vector $\mu$ and a positive definite covariance matrix $\sum$.

$$\mu = \begin{bmatrix} 5.0 \\ 0.0 \\ 10.0 \end{bmatrix}$$

$$\sum = \begin{bmatrix} 3.40 & -2.75 & -2.00 \\ -2.75 & 5.50 & 1.50 \\ -2.00 & 1.50 & 1.25 \end{bmatrix}$$

[3] https://github.com/IBM/UQ360

We encode inconsistency and uncertainty into the features of the test set $\mathscr{D}_{\text{test}}$ using a multivariate additive model $\hat{\mathbf{X}} = \mathbf{X} + \mathbf{Z}$, where $\mathbf{Z} \in \mathbb{R}^{M \times d_X}$, is the perturbation matrix (with $d_X = 3$).

We conduct three experiments with different configurations $(D_a, D_b, D_c)$, see Table B.2

Table B.2 Different configurations of the synthetic data

| | Noise distribution | Perturbation Proportions | Perturbation variance |
|---|---|---|---|
| $D_a$ | $\{Normal\}$ | $\{0.1, 0.25, 0.5, 0.75\}$ | $\{2\}$ |
| $D_b$ | $\{Normal\}$ | $\{0.5\}$ | $\{1, 2, 3\}$ |
| $D_c$ | $\{Beta, Gamma, Normal, Weibull\}$ | $\{0.5\}$ | $\{2\}$ |

### Downstream Model

In the main text, we have a downstream task wherein we compute the MSE for different models. The base model that we train using $\mathscr{D}_{\text{train}}$ is a linear regression model.

## B.2.4 Real-Data Experiment Details

### Datasets

**SEER Dataset** The SEER dataset consists of 240,486 patients enrolled in the American SEER program [46]. The dataset consists of features used to characterize **prostate cancer**: including age, PSA (severity score), Gleason score, clinical stage, treatments etc. A summary of the covariate features can be found in Table B.3. The classification task is to predict patient mortality, which is binary label $\in \{0, 1\}$.

The dataset is highly imbalanced, where 94% of patients survive. Hence, we extract a balanced subset of 20,000 patients (i.e., 10,000 with label=0 and 10,000 with label=1).

Table B.3 Summary of features for the SEER Dataset [46]

| Feature | Range |
|---|---|
| Age | $37 - 95$ |
| PSA | $0 - 98$ |
| Comorbidities | $0, 1, 2, \geq 3$ |
| Treatment | Hormone Therapy (PHT), Radical Therapy - RDx (RT-RDx),Radical Therapy -Sx (RT-Sx), CM |
| Grade | $1, 2, 3, 4, 5$ |
| Stage | $1, 2, 3, 4$ |
| Primary Gleason | $1, 2, 3, 4, 5$ |
| Secondary Gleason | $1, 2, 3, 4, 5$ |

**CUTRACT Dataset** The CUTRACT dataset is a private dataset consisting of 10,086 patients enrolled in the British Prostate Cancer UK program [**?** ]. Similar to the SEER dataset, it consists of the same features to characterize prostate cancer. Additionally, it has the same task to predict mortality. A summary of the covariate features can be found in Table B.4.

Once again, the dataset is highly imbalanced, hence we then choose to extract a balanced subset of 2,000 patients (i.e. 1000 with label=0 and 1000 with label=1).

Table B.4 Summary of features for the CUTRACT Dataset [174]

| Feature | Range |
|---|---|
| Age | $44-95$ |
| PSA | $1-100$ |
| Comorbidities | $0,1,2,\geq 3$ |
| Treatment | Hormone Therapy (PHT), Radical Therapy - RDx (RT-RDx),Radical Therapy -Sx (RT-Sx), CM |
| Grade | $1,2,3,4,5$ |
| Stage | $1,2,3,4$ |
| Primary Gleason | $1,2,3,4,5$ |
| Secondary Gleason | $1,2,3,4,5$ |

**ADULT Dataset** The ADULT dataset [11] has 32,561 instances with a total of 13 attributes capturing demographic (age, gender, race), personal (marital status) and financial (income) features amongst others. The classification task predicts whether a person earns over \$50K or not. We encode the features (e.g. race, sex, gender etc) and a summary can be found in Table B.5.

There is a known bias between gender and income in the dataset. We perform a train-test split such that $\mathscr{D}_{\text{train}}$ and $\mathscr{D}_{\text{test}}$ have approximately equal sizes, with 15,378 and 14,784 samples, respectively. In particular, to highlight the data exploration use case.

Note that there is an imbalance across certain features; however, these are amongst the sensitive attributes. Thus, we do not want to balance the datasets based on this, as we want to show both for the data exploration and model deployment experiments that we can identify these biases in the datasets. Balancing might eliminate these biases.

**ELECTRICITY Dataset.** The Electricity dataset [80], represents energy pricing in Australia, over the period of May-1996 to December 1998, with recordings every 30 minutes giving 45312 samples. The dataset records energy price and demand for New South Wales and Victoria and the amount of power transferred between the two states. The goal is to predict whether the transfer price increases or decreases.

Table B.5 Summary of features for the ADULT Dataset [11]

| Feature | Range |
|---|---|
| Age | $17-90$ |
| education-num | $1-16$ |
| marital-status | $0,1$ |
| relationship | $0,1,2,3,4$ |
| race | $0,1,2,3,4$ |
| sex | $0,1$ |
| capital-gain | $0,1$ |
| capital-loss | $0,1$ |
| hours-per-week | $1-99$ |
| country | $0,1$ |
| employment-type | $0,1,2,3$ |
| salary | $0,1$ |

The covariates outlined in Table B.6 are normalized to the interval $[0,1]$

We temporally partition the dataset where $\mathscr{D}_{\text{train}}$ is Mid-1996 to early-1997 and $\mathscr{D}_{\text{test}}$ is early-1997 to 1998. We note the dataset has been characterized as having concept shift for some features over the test period (however, without an explicit timepoint or label). This could be due to behavioural changes or consumption pattern changes [270]. Hence, $\mathscr{D}_{\text{test}}$ consists of data that is both congruous and incongruous with $\mathscr{D}_{\text{train}}$.

Table B.6 Summary of features for the ELECTRICITY Dataset [80]

| Feature | Range |
|---|---|
| data | $0-1$ |
| period | $0-1$ |
| nswprice | $0-1$ |
| nswdemand | $0-1$ |
| vicprice | $0-1$ |
| vicdemand | $0-1$ |
| transfer | $0-1$ |
| class | $0-1$ |

**Downstream Model**

In the main text, we have a downstream task for each of the three datasets. For all the datasets, our base model which we train using $\mathscr{D}_{\text{train}}$ is a Random Forest Classification model with 100 estimators in the ensemble and splits are based on the 'Gini' criterion.

## B.3 Additional Experiments

This appendix presents additional experiments, validating further properties of Data-SUITE, conducting further comparisons or deep-dives into the regions identified by Data-SUITE, and what insights can be garnered.

### B.3.1 Regions identified by Data-SUITE as NOT OOD

In this experiment, we address the question of whether the regions identified by Data-SUITE as uncertain or inconsistent are, in fact, OOD. We benchmark four widely used methods (with different detection mechanisms), which have been applied in the literature for OOD and outlier detection:

- Mahalanobis distance [127]
- SUOD: Accelerating Large-**S**cale **U**nsupervised Heterogeneous **O**utlier **D**etection [268]
- COPOD: **Cop**ula-Based **O**utlier **D**etection [135]
- Isolation Forest [139]

We note for SUOD, much like the original paper, we make use of an ensemble of base estimators namely: Local outlier factor (LOF) [26], COPOD, Isolation Forest.

For each dataset, we have the instance IDs identified by Data-SUITE for various proportions. We then apply each of the aforementioned methods and compute the overlap between the predicted OOD/Outlier instances and our identified uncertain and inconsistent instances.

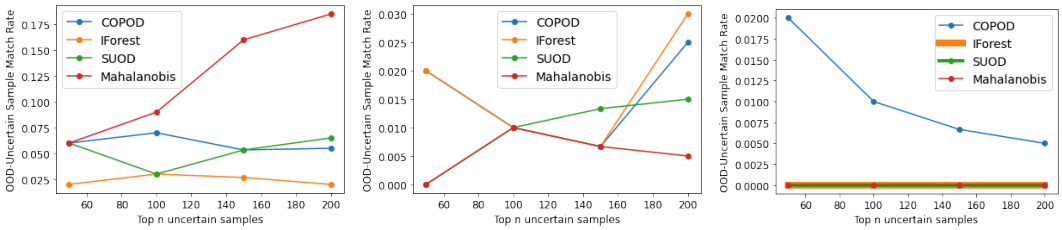
The results for the overlap of uncertain instances can be found in Figure B.2. We see minimal overlap across methods, ranging between 2-18%.

We additionally evaluate the confidence scores (for those methods that provide confidence scores as outputs). The goal is to see if an instance is predicted as OOD/outlier and then with what confidence the detection method ascribes to the instance. The results in Figure B.3

suggest that the methods were often unconfident, with average confidence scores ranging between 5-50%. This suggests that the identified uncertain instances are unlikely OOD.

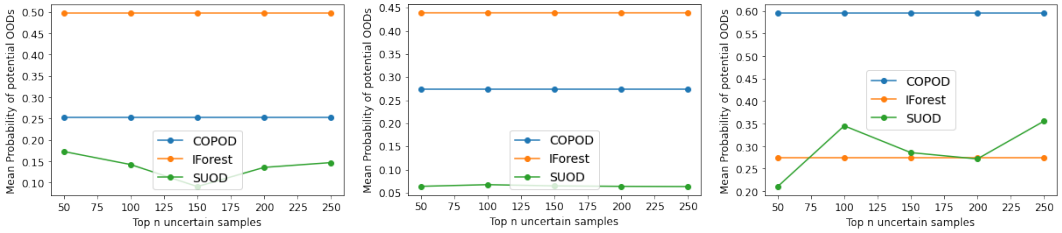
A similar question might be asked for the inconsistent instances. The results are similar to the uncertain instances, as shown in Figure B.4. This result, similar to uncertain instances, suggests the identified inconsistent instances are unlikely OOD.

**Takeaway:** The Data-SUITE identified *uncertain* and *inconsistent* instances are unlikely OOD. The reason is the limited overlap of predicted OOD and both *uncertain* and *inconsistent* instances, coupled with the unconfident (low probability) predictions for OOD.



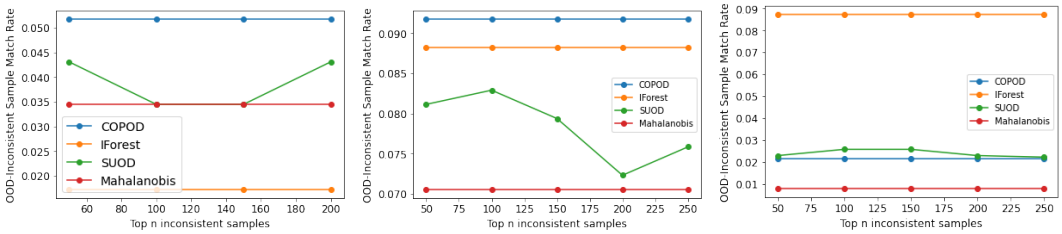

(a) SEER-CUTRACT (b) Adult (c) Electricity

Fig. B.2 OOD-Uncertain Instances Match Rate or overlap




(a) SEER-CUTRACT (b) Adult (c) Electricity

Fig. B.3 Mean probability of predicted OOD, i.e. confidence




(a) SEER-CUTRACT (b) Adult (c) Electricity

Fig. B.4 OOD-Inconsistent Instances Match Rate or overlap

### B.3.2 EDA & digestable prototype insights

As discussed in the main thesis, we now conduct a more detailed analysis of the regions identified by Data-SUITE, as well as the digestible average prototypes. This illustrates the full potential of detail that practitioners can get from Data-SUITE and how they can practically use Data-SUITE to garner insights about the data. We present the diagram again for easy reference (see Figure B.6).

***SEER-CUTRACT.*** Data-SUITE identifies distinct *certain* (Green) and *uncertain* (Red) regions of ID data, as shown in Figure B.6 (i). Beyond average prototypes for $\mathscr{D}_{\text{test}}$ regions i.e., CUTRACT (UK), we also find the nearest neighbor SEER (USA) prototypes for each instance in the identified regions. Comparing these prototypes assists us in teasing out the differences between the two geographic sites.

We note three specific insights that can assist end-users. (1) *certain* instances represent less severe patients than the *uncertain* instances (see PSA values). (2) *certain* instances: CUTRACT (UK) and SEER (USA) prototypes are similar in all feature values. (3) *uncertain* instances: CUTRACT (UK) and SEER (USA) prototypes have differences in certain feature values (PSA, more comorbidities, different treatment, staging score).

We conduct an extended deep dive below, highlighting features such as PSA that show no difference on a population level or for *certain* instances. However, when teasing out *uncertain* instances and their prototypes, we see this difference. This example illustrates the full potential of how Data-SUITE can be used by practitioners to uncover differences across sites (to benefit clinical practice), whilst also identifying patients where model performance would either be reliable or sub-standard (independent of the model).

We highlight this in Figure B.5, which shows no PSA difference between the USA and the UK on a population level or for certain instances, but when teasing out *uncertain* instances and their prototypes, we see this difference.

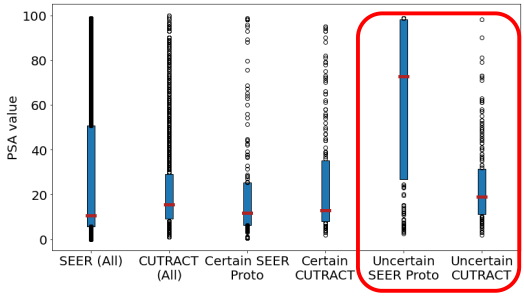
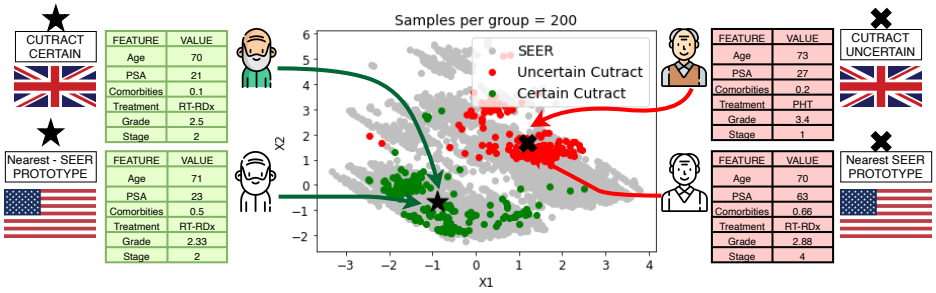
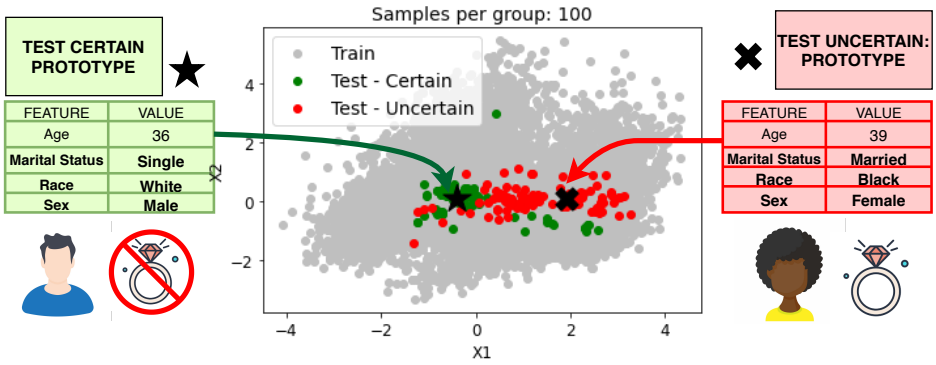


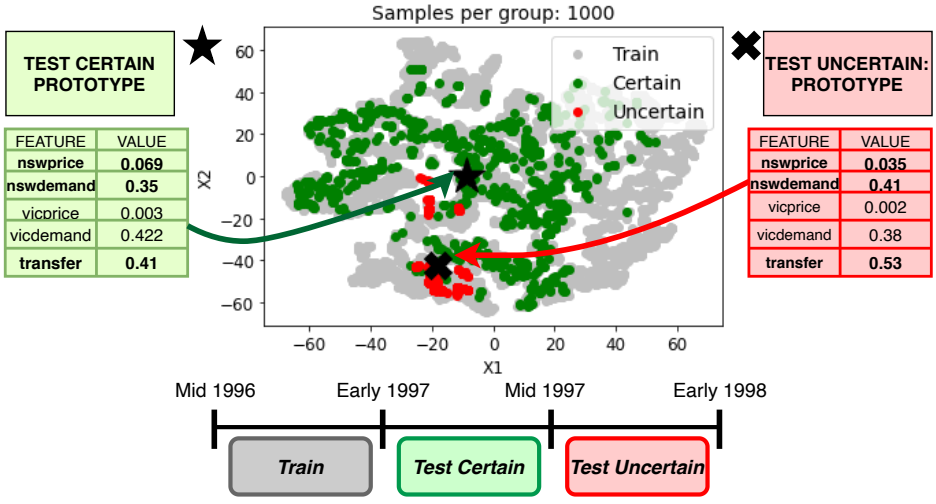

Fig. B.5 PSA Deep-Dive, which highlights how the difference is not evident at the population level or for certain instances. Only the uncertain instances highlight the heterogeneity. However, it does highlight common support based on the whiskers of the box plot.




(a) i. SEER-CUTRACT: CUTRACT *certain* instances are similar to their SEER nearest prototypes, whilst CUTRACT *uncertain* instances are different from their nearest SEER prototypes (e.g., PSA).




(b) ii. Adult: The *certain* and *uncertain* instances represent two different demographics, aligning with the known dataset biases toward females. The *uncertain* instances specifically highlight a subgroup of black females.




(c) iii. Electricity: The *certain* instances are similar to the training set in features and time. The *uncertain* instances identified represent a later time period, wherein concept drift has likely occurred.

Fig. B.6 Insights of prototypes identified by Data-SUITE. Tables describe the average prototypes for *certain* and *uncertain* instances.

We extend this analysis to further illustrate the full potential of the detail that practitioners can get from Data-SUITE. We do this by comparing the Earth Movers Distance (EMD) - see Figure B.7, which is a common metric to flag drift. We see no PSA difference between the USA and the UK on a population level or for certain instances. However, when teasing out *uncertain* instances and their prototypes, we see this difference.

**Takeaway:** We highlight that while not evident at population-level, the heterogeneous groups can be teased out by using Data-SUITE. This fully demonstrates the added capability of what Data-SUITE offers to practitioners.

### B.3.3 Inconsistent instances $\lambda$ sensitivity

As described in Chapter 3.3.3, an instance $x$ is *inconsistent* if the fraction of inconsistent features is above a predetermined threshold $\lambda \in [0,1]$: $\nu(x) > \lambda$. In our implementation, we use $\lambda = 0.5$, i.e., if more than 50% of the features are inconsistent, then the sample is considered inconsistent.

For completeness, we conduct an analysis of the sensitivity to the values of $\lambda \in [0,1]$. We show the accuracy score as a function of $\lambda$, as well as the number of instances that would be classified as inconsistent for that value of $\lambda$. The results are shown in Figure B.8. The flat steady performance for small $\lambda$ and then drop off approximately around $\lambda = 0.5$ across all datasets, suggests that our chosen value of $\lambda = 0.5$ is indeed a sensible choice.

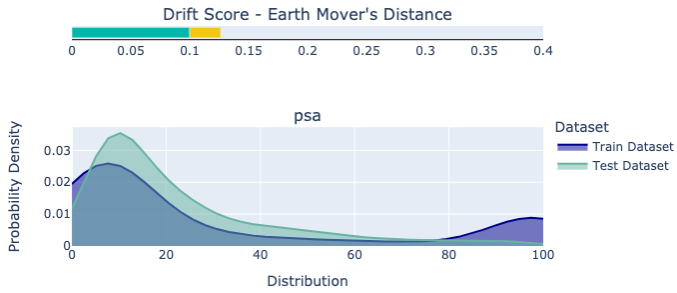
**Takeaway**: Based on the sweep across all three datasets, we see that $\lambda = 0.5$ is a sensible choice in practice.

### B.3.4 Computation time comparison

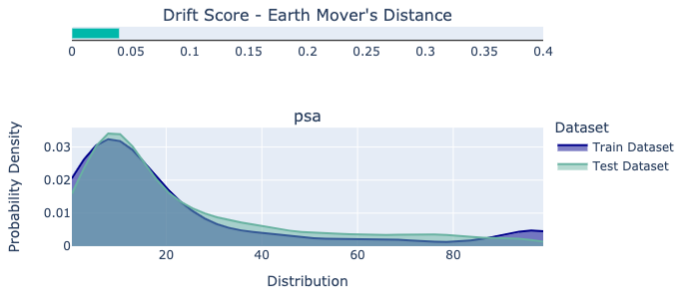
All experiments were run on CPU on a MacBook Pro with an Intel Core i5 and 16GB RAM. Besides actual performance values, practitioners are often interested in the computation time associated with different methods. This is especially true if the algorithms are applied to very large datasets with many instances.

We hence conduct a comparison of the computation time needed by each method to both train and to construct the predictive intervals for all instances in $\mathscr{D}_{\text{test}}$. We present the results in Table B.7, with the recorded computation time provided to the nearest minute.

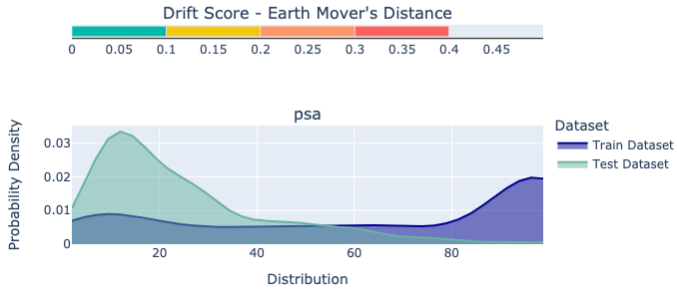
**Takeaway**: Data-SUITE does not have the fastest computation time (due to the interconnected framework). However, the computation times are neither too dissimilar nor prohibitive for practitioners to use.




(a) Earth Movers Distance on the FULL Dataset (Population): Train (SEER) & Test (CUTRACT) - No Difference

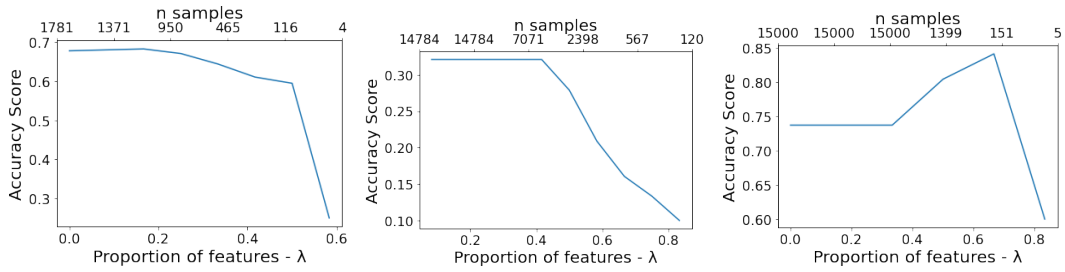



(b) Earth Movers Distance: *CERTAIN*: Train (SEER Nearest Neighbors) & Test (CUTRACT) - No Difference




(c) Earth Movers Distance: *UNCERTAIN*: Train (SEER Nearest Neighbors) & Test (CUTRACT)- Clear Difference

Fig. B.7 Earth Movers Distance Deep-Dive into PSA, which highlights how the difference is not evident at the population level or for certain instances. Only the uncertain instances highlight the heterogeneity.




(a) SEER-CUTRACT (b) Adult (c) Electricity

Fig. B.8 $\lambda$ sweep to flag inconsistent instances

Table B.7 Comparison of computation time across methods to the nearest minute

| | SEER-CUTRACT | Adult | Electricity |
|---|---|---|---|
| **Data-SUITE** | 4 | 1 | 1 |
| BNN | 2 | 1 | 1 |
| CONFORMAL | 2 | 1 | 2 |
| ENSEMBLE | 28 | 3 | 3 |
| GP | 22 | 15 | 24 |
| MCD | 19 | 13 | 19 |
| QR | 1 | 1 | 1 |

### B.3.5 Comparison to domain adaptation

In the paradigm of domain adaptation, a popular metric is the maximum mean discrepancy (MMD) [72]. For example, [122, 144, 260, 77] have used MMD as the metric to compare distributions and subsequently, optimize to minimize the distributional difference of representations.

The MMD metric is a distance-based metric that compares the mean embeddings of two probability distributions, source $S$ and target $T$, in a reproducing kernel Hilbert space $H_k$. This is given by EquationB.10

$$MMD(S,T) = ||\mu_k(S) - \mu_k(T)||_{H_k} \tag{B.10}$$

We can compute unbiased estimates of samples from the two distributions after we apply the kernel trick (in this case, we use a Radial Basis Function Kernel).

In the domain adaptation literature, the goal is to minimize the latent feature divergence by optimizing the MMD. Hence, MMD is a fundamental component of domain adaptation and is used to identify instances with the discrepancy between source and target domains.

As a comparison to our approach (Data-SUITE), we cast the problem as a potential domain adaptation problem and apply MMD, where the source distribution ($S \sim \mathscr{D}_{\text{train}}$) and target distribution ($T \sim \mathscr{D}_{\text{test}}$). We then use the computed MMD to stratify instances, where lower MMD means *certain* and large MMD means *uncertain*.

The mean performance improvement (MPI) is then computed for instances stratified based on MMD, similar to the experiment in Chapter 3.4.3. Table B.8 contrasts the results using MMD vs Data-SUITE to stratify instances.

**Takeaway.** The approach of Data-SUITE achieves greater average performance improvement based on MPI across all three datasets when compared to framing the problem in the context of domain adaptation (using the MMD metric to identify and stratify instances).

Table B.8 Comparison of average performance improvement for Data-SUITE vs MMD (Domain Adaptation)

| | **Data-SUITE** | MMD |
|---|---|---|
| SEER-CUTRACT | **0.11** | 0.088 |
| Adult | **0.64** | -0.06 |
| Electricity | **0.26** | -0.17 |

# Appendix C

# Appendix for Chapter 4

## C.1 Additional Details & Extended Related Work

This chapter primarily engages with the work on data augmentation when we have limited data, where our primary goal is synthetic data generation to augment the dataset. Generating synthetic datasets not only helps improve downstream performance but also provides a flexible solution, as it does not tie the data consumer to any particular downstream model. That said, beyond the major difference of synthetic data generation, for completeness, we contrast our setting of learning from limited data with other seemingly similar settings and highlight their differences.

**Contrasting learning w/ limited data vs other settings.** The challenge of learning from limited data, while seemingly related to several other learning paradigms, presents distinct differences and unique intricacies that warrant dedicated study.

*Transfer learning* [164], *domain adaptation* [52], and *few-shot learning* [253] employ additional data resources or rely on specific task-related assumptions to improve learning performance. These methods exploit large labeled data from a source domain, unlabeled data in a target domain, or leverage knowledge from related tasks, respectively. For example, [133] and [100] use models trained on labeled data from a source domain, while [199] and [147] leverage knowledge graphs. This is in contrast to our setting, i.e. learning with limited data, which must function with whatever scarce labeled data it has, without making any assumptions about the availability of additional data or tasks.

**Contrast of CLLM and transfer learning / meta-learning / few-shot learning.**

In addition to the above, we emphasize three dimensions along which CLLM differs from the transfer learning and meta-learning literature (which permits few-shot learning). Specifically, we highlight why transfer learning and meta-learning generally cannot apply to the setting considered in CLLM. We provide empirical evidence on why the following

dimensions are important, notably the point regarding the choice of the downstream backbone. Additionally, we compare CLLM with TabPFN [87], a few-shot learning method designed for small tabular problems (see Chapter C.3.6).

1. *Access to external datasets*: our problem setting in CLLM assumes access to a single small training set $D_{\text{train}}$, without access to any external/additional datasets. This mirrors the unique characteristics of low-to-middle income countries (LMICs), where data scarcity may be pervasive, hence making it unrealistic to assume that external datasets are available to a practitioner. In contrast, transfer learning and meta-learning make more stringent assumptions about the data requirements. (i) *Transfer learning* (Definition 3 in [269]): one typically assumes access to at least one additional source dataset $D_{\text{source}}$, usually bigger than $D_{\text{train}}$. This external $D_{\text{source}}$ can then be used to pretrain a model, which is adapted using $D_{\text{train}}$. (ii) *Meta-learning* [88]: one assumes access to a set of $m$ tasks, which define a set of source datasets $\{D_{\text{source}}^{(i)} \mid i \in [m]\}$. It is also worth noting that a common assumption in meta-learning is that the source datasets and $D_{\text{train}}$ share the same feature space, restricting its applicability. To summarize, both of these learning paradigms often rely on external data, an assumption not required in CLLM. We acknowledge that CLLM can be viewed as a form of transfer in its broadest definition, as it leverages prior knowledge with the LLM. However, a key point is that it does not require external/additional datasets.

2. *Flexibility of the backbone model*: since CLLM is a data augmentation method for tabular data, it enables the practitioner to use any downstream model backbone, such as neural networks or tree-based methods (XGBoost, Random forest, CatBoost). This flexibility is important as it allows the practitioner to choose the best-suited model for the task at hand. For example, one can use any tree-based method with CLLM, which is appealing given that tree-based methods are often preferred over neural networks for tabular data [74], [226]. In contrast, most approaches to transfer learning and meta-learning are not flexible as they traditionally require neural networks as backbone models. This stems from their methodology, where fine-tuning a pre-trained model on $D_{\text{train}}$ is a prevailing approach [55].

3. *Ease of use*: CLLM is distinct in that it is easy to use, without the need for costly or complex operations, such as fine-tuning large pre-trained models or using them at inference time, which may be impossible in LMICs, due to the cost associated with these operations.

*Active learning* [222] and *semi-supervised learning* [243, 30] also operate under the premise of having access to plentiful unlabeled data and the capacity to interactively query

labels. However, in our setting, which is considered learning with limited data, it does not inherently assume such capabilities. Rather, the focus is instead on limited labeled data only. Furthermore, active learning primarily focuses on the iterative process of selecting data samples that, when labeled, are expected to most significantly improve the model's performance. This selection is typically based on criteria such as uncertainty sampling, which focuses on **epistemic uncertainty** [154, 89, 113, 157]. The primary objective is to minimize labeling effort while maximizing the model's learning efficiency. Additionally, active learning aims to label instances based on epistemic uncertainty where the model struggles to make accurate predictions, yet the samples themselves are correct. In contrast, CLLM leverages training dynamics based on **aleatoric uncertainty** and confidence and is designed to discard samples that might jeopardize the downstream accuracy. These samples can be considered to have inherent issues or are erroneous, such as being "mislabeled". To summarize, in active learning, epistemic uncertainty is used to identify data points that, if labeled, would yield the most significant insights for model training. In our approach, they serve to identify and exclude/filter data points that could potentially deteriorate the model's performance.

*Self-supervised learning* [143] leverages large amounts of unlabeled data to learn useful representations for downstream tasks. However, in our setting, learning with limited data does not inherently assume such access to vast amounts of unlabeled data.

**Data-centric AI.** Ensuring high data quality is a critical but often overlooked problem in ML, where the focus is optimizing models [204]. Even when it is considered, the process of evaluating data sets is ad hoc or artisanal [214]. However, the recent push of data-centric AI [137, 180, 264, 215] aims to develop systematic tools to curate existing datasets. Our work contributes to this body of work – presenting CLLM, which, to the best of our knowledge, is the first systematic data-centric framework looking at how we can tailor synthetic datasets (rather than real datasets) for downstream task use with data curation.

**Why Data Augmentation?** Data augmentation is a flexible approach to address the low-data regime. An alternative might be to resort to a pre-trained black-box model for classification, which, for example, could be via in-context learning for classification [45]. However, such a solution is inadequate for several reasons, many of which would prevent real-world utility (e.g. in LMICs):

► *Not economical over the long term:* While using an LLM like GPT for classification may seem attractive due to its few-shot capabilities, it is likely not economically viable in real-world settings, especially in LMICs. The reason is that classifying each sample will incur a cost to call the LLM, hence it scales linearly with the number of test samples. Over time, the cumulative cost of these calls will surpass the one-off fixed cost associated

with generating data. With data augmentation, once the dataset is augmented, there are no additional deployment time costs associated with the LLM. Indeed, the downstream models, e.g., a random forest or XGBoost, have negligible inference costs.

▶ *Control, interpretability and auditability:* Relying on a large, pre-trained LLM as a black-box classifier raises several concerns. (1) We have no control over our downstream classifier and its architecture. (2) There is a lack of interpretability and auditability of the LLM when issuing predictions. In contrast, training a downstream model on augmented data maintains the ability to understand and explain how the model is making decisions (e.g. feature importance). This is especially crucial in contexts where accountability, transparency, and validation of machine learning processes are paramount.

▶ *Independence and self-sufficiency:* Relying on third-party services for continuous classification means being dependent on their availability, pricing models, and potential changes in the LLM version. By augmenting data and training a downstream classifier on the augmented dataset, we ensure that no external dependencies such as increasing costs or reduced performance with LLM version updates.

▶ *Hardware and financial constraints:* Even if we opt for an open-source LLM (e.g. LLaMA [241]), deploying and running it locally demands significant computational resources. Typically, these models require GPUs with high amounts of VRAM for optimal performance (e.g. needing around 40 GB, hence requiring an A100 GPU for LLaMA-2 65B). Such high-end GPUs are expensive and are likely to be inaccessible in an LMIC setting. Furthermore, renting hardware by the hour can quickly become prohibitively expensive. Data augmentation, on the other hand, can often be performed on modest hardware, and once the augmented dataset is created, many classifiers can be trained without the need for high-end GPUs, making the entire process more financially accessible.

In conclusion, while large language models offer vast knowledge, data augmentation provides a more cost-effective, controllable, and interpretable solution for building robust classifiers in low-data settings of low-income countries.

## C.2 Experimental Details

We provide details on our datasets used, as well as other experimental specifics, including generation, curation, downstream model, and prompt template.

### C.2.1 Code details

Code for this work can be found at: https://github.com/seedatnabeel/CLLM

### C.2.2 Datasets

We summarize the different datasets we use in Table C.1. The datasets vary in the number of samples, number of features, and domain.

Table C.1 Summary of the datasets used. * Denotes private/proprietary datasets.

| Name | $n$ samples | $n$ features | Domain |
|---|---|---|---|
| Adult Income [11] | 30k | 12 | Finance |
| Compas [5] | 5k | 13 | Criminal justice |
| *Covid-19 [16] | 7k | 29 | Healthcare/Medicine |
| *CUTRACT Prostate [174] | 2k | 12 | Healthcare/Medicine |
| Drug [53] | 2k | 27 | Healthcare/Medicine |
| *MAGGIC [178] | 41k | 29 | Healthcare/Medicine |
| *SEER Prostate [47] | 20k | 12 | Healthcare/Medicine |

The private datasets are de-identified and used in accordance with the guidance of the respective data providers. We follow recommendations to use the Azure OpenAI service when using GPT-4 and GPT-3.5 models, where via the agreement, we ensure the medical data is not sent for human review or stored, hence respecting the guidelines given by the dataset providers.

We detail the dataset splits used in Chapter 4.3.1. For each dataset and number of samples $n \in \{20, 40, 100, 200\}$, we sample a training set $\mathscr{D}_{\text{train}}$ such that $|\mathscr{D}_{\text{train}}| = n$, and each target class has the same number of samples. We then split the remaining samples into two non-overlapping datasets, $D_{\text{oracle}}$ and $\mathscr{D}_{\text{test}}$, which have the same cardinality. This procedure is repeated $n_{\text{seed}} = 10$ times, thus leading to different training and test sets. Note that the different generative models use the same $\mathscr{D}_{\text{train}}$ and $\mathscr{D}_{\text{test}}$ for a given seed.

**Motivation for the choice of datasets.**

1. **Open-source:** Adult, Drug and Compas are widely used open-source datasets used in the tabular data literature. Adult and Drug are both UCI datasets that have been used in many papers, while Compas is part of OpenML [244]. Our reason for selecting them is that, despite them being open-source, they are highly reflective of domains in which we might be unable to collect many samples.Hence, in reality, we would often be in a low-data regime.

2. **Private datasets:** We wanted to disentangle the possible role of memorization in the strong performance of the LLM. To ensure the datasets are not in the LLMs training corpus, we selected 4 private medical datasets that require an authorization process to access. Hence, these datasets would not be part of the LLMs training corpus, given their proprietary nature, and hence would be unseen to the LLM. While the private and unseen aspect is the primary motivation, we also wish to highlight that these are real-world medical datasets. Consequently, this allows us to test a highly realistic problem setting.

### C.2.3 Data generation.

**GPT-4 and GPT-3.5** We access GPT-4 [161] and GPT-3.5-Turbo [27] through the API. We use a temperature of 0.9.

**GReaT.** GReaT [24] is a generative model that fine-tunes an LLM based on a training set. We use the implementation provided by the authors.

**Generative model-based augmentation.** For the other baselines used in 4.3.1, we use the library SynthCity [184], using the defaults. We detail each next.

- TVAE: this is a conditional Variational Auto Encoder (VAE) for tabular data and is based on Xu et al. [259]

- CTGAN: A conditional generative adversarial network that can handle tabular data and is based on Xu et al. [259]

- NFLOW: Normalizing Flows are generative models that produce tractable distributions where both sampling and density evaluation can be efficient and exact.

- TabDDPM: A diffusion model that can be universally applied to any tabular dataset, handles any type of feature and is based on Kotelnikov et al. [119]

**Traditional Data Augmentation.** We use SMOTE [31], which augments data by considering nearest neighbors and performing linear interpolations. We use the implementation provided by [130] and set the number of neighbors $k$ to 5.

### C.2.4 Data curation

**Learning dynamics computation** We train an XGBoost with 100 estimators on $\mathscr{D}_{\text{train}}$. We then compute predictive confidence and aleatoric uncertainty for the samples in $D_{\text{syn}}$. The motivation for the choice of an XGBoost backbone is that we cannot expect good performance by choosing "any" curation model, but rather, we require a curation model with enough capacity and generalization properties, where boosting methods like XGBoost used in our work have been shown to achieve best performance on tabular data. This leads to our guideline for the curation step: the model used for curation should be **at least as flexible** as the model that the practitioner intends to use for the downstream task.

**Learning dynamics thresholds** Recall that CLLM has two thresholds $\tau_{\text{conf}}$ and $\tau_{\text{al}}$ on the predictive confidence and aleatoric uncertainty, respectively, as defined in Chapter 4.2.2. We set $\tau_{\text{conf}} = 0.2$, in order to select high-confidence samples. We adopt an adaptive threshold for $\tau_{\text{al}}$ based on the dataset, such that $\tau_{\text{al}} = 0.75 \cdot (\max \text{val}_{\mathscr{F}}(\mathscr{D}_{\text{syn}}) - \min \text{val}_{\mathscr{F}}(\mathscr{D}_{\text{syn}}))$, where $\text{val}_{\mathscr{F}}(\mathscr{D}_{\text{syn}}) := \{\text{val}_{\mathscr{F}}(x,y) \mid (x,y) \in \mathscr{D}_{\text{syn}}\}$. Note that by definition $\text{val}_{\mathscr{F}}(x,y)$ is bounded between 0 and 0.25.

**Example of learning dynamics** We include examples of learning dynamics computed for 20 samples in Figure C.1.

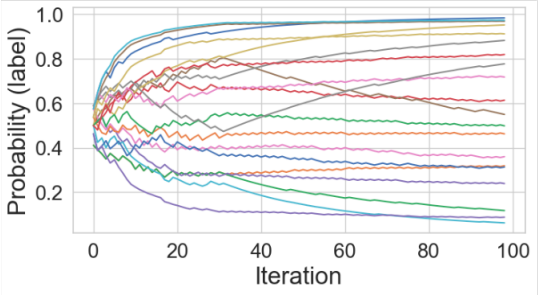



Fig. C.1 Learning dynamics computed for 20 samples

### C.2.5 Downstream task

We compute downstream performance in Chapter 4.3.1 using four different downstream models: XGBoost, Random Forest, Decision Tree, and Logistic Regression.

### C.2.6 Prompt example

We include the template of the prompts used. We show how we include (1) in-context examples (demonstrations), (2) contextual information including dataset background and feature information and (3) the instruction.

```
System role: 'You are a tabular synthetic data generation model.'

You are a synthetic data generator.
Your goal is to produce data which mirrors \
the given examples in causal structure and feature and label distributions \
but also produce as diverse samples as possible.

I will give you real examples first.

Context: Leverage your medical knowledge about covid and Brazil to generate 1000 realistic but diverse samples.

example data: {data}

The output should be a markdown code snippet formatted in the following schema:

"Sex_male": string  // feature column
"Age": string  // feature column
"Age_40": string  // feature column
"Age_40_50": string  // feature column
"Age_50_60": string  // feature column
"Age_60_70": string  // feature column
"Age_70": string  // feature column
"Fever": string  // feature column
"Cough": string  // feature column
"Sore_throat": string  // feature column
"Shortness_of_breath": string  // feature column
"Respiratory_discomfort": string  // feature column
"SPO2": string  // feature column
"Diharea": string  // feature column
"Vomiting": string  // feature column
"Cardiovascular": string  // feature column
"Asthma": string  // feature column
"Diabetes": string  // feature column
"Pulmonary": string  // feature column
"Immunosuppression": string  // feature column
```

```
    "Obesity": string  // feature column
    "Liver": string  // feature column
    "Neurologic": string  // feature column
    "Renal": string  // feature column
    "Branca": string  // feature column
    "Preta": string  // feature column
    "Amarela": string  // feature column
    "Parda": string  // feature column
    "Indigena": string  // feature column
    "is_dead": string  // label if patient dead or not, is_dead

    DO NOT COPY THE EXAMPLES but generate realistic but new and
   diverse samples which have the correct label conditioned on the
   features.
```

Listing C.1 Template of the prompt

## C.3 Additional Experiments

### C.3.1 Curation mechanism also improves open-source LLM generated data

In this subsection, we demonstrate the wide applicability of the CLLM framework. Different factors may affect the choice of the LLM backbone, such as operational costs and the desired parameterization of the LLM (affecting the quality of generation). In addition to GPT-4 and GPT-3.5, we use the open source models Mistral-7b [96] and LLaMa-13b (4-bit quantized) [241], LLaMa-70b [241] and Mixtral-8x7b [97] to generate augmented datasets [1]. We then compute the downstream performance when training a model on both the uncurated and curated data. As can be seen in Table C.2, downstream performance with uncurated is lower for these open-source models compared to GPT-4 — which is expected given their significantly smaller size (i.e. parameter count). However, the curation mechanism, which is the key contribution of CLLM, almost always improves downstream performance for all LLM backbones investigated. Overall, this demonstrates the versatility and wide applicability of CLLM for tabular data augmentation in low-data regimes.

**Practical tip:** The size of the LLM plays a role: the larger models (with more parameters) outperform those with fewer parameters (e.g. GPT-4 vs Mixtral-8x7b vs Mistral-7b). Hence, while curation helps improve all LLM-generated data, ideally, the best LLM possible should be used.

[1] For LLaMa-70b and Mixtral-8x7b, we only run open-source datasets due to data sharing restrictions with the endpoints for those two models.

Table C.2 AUC averaged over 4 downstream models on $\mathscr{D}_{\text{test}}$ for GPT-4, GPT-3.5, Mistral-7b, LLAMA-13b, LLAMA-70b and Mixtral-8x7b (Curated and Uncurated)

| Dataset | Open AI | | | | Open source | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | GPT-4 | | GPT-3.5 | | Mistral-7b | | LLAMA-13b | | LLAMA-70b | | Mixtral | |
| | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. |
| covid (n=20) | 73.78 | **73.87** ↑ | 69.85 | **71.41** ↑ | 71.47 | **72.80** ↑ | 63.25 | **67.24** ↑ | N.A. | N.A. | N.A. | N.A. |
| cutract (n=20) | 71.15 | **72.50** ↑ | 69.97 | **71.54** ↑ | 69.60 | **71.34** ↑ | 67.84 | 66.71 | N.A. | N.A. | N.A. | N.A. |
| maggic (n=20) | 60.70 | **61.48** ↑ | 57.54 | **58.69** ↑ | 53.64 | 52.06 | 53.30 | **53.96** ↑ | N.A. | N.A. | N.A. | N.A. |
| seer (n=20) | 84.53 | **84.82** ↑ | 83.34 | **83.71** ↑ | 83.60 | **85.18** ↑ | 80.94 | **82.84** ↑ | N.A. | N.A. | N.A. | N.A. |
| compas (n=20) | 68.01 | 67.91 | 62.07 | **64.43** ↑ | 56.26 | **59.95** ↑ | 60.08 | **60.34** ↑ | 56.10 | **60.68** ↑ | 54.34 | **61.66** ↑ |
| adult (n=20) | 50.39 | **71.48** ↑ | 49.23 | **72.37** ↑ | 47.68 | **65.84** ↑ | 48.82 | **66.00** ↑ | 51.96 | **68.40** ↑ | 57.78 | **76.84** ↑ |
| drug (n=20) | 75.08 | **75.29** ↑ | 71.68 | **72.14** ↑ | 74.14 | **75.21** ↑ | 67.96 | 67.87 | 66.77 | **69.12** ↑ | 72.45 | 71.63 |
| covid (n=40) | 73.40 | **73.95** ↑ | 70.42 | **71.93** ↑ | 69.61 | **71.47** ↑ | 61.32 | **65.57** ↑ | N.A. | N.A. | N.A. | N.A. |
| cutract (n=40) | 69.87 | **71.72** ↑ | 68.47 | **69.56** ↑ | 68.74 | **72.36** ↑ | 64.34 | **67.46** ↑ | N.A. | N.A. | N.A. | N.A. |
| maggic (n=40) | 59.29 | **60.77** ↑ | 57.50 | **59.15** ↑ | 52.43 | **53.79** ↑ | 52.61 | **53.45** ↑ | N.A. | N.A. | N.A. | N.A. |
| seer (n=40) | 84.29 | **84.93** ↑ | 83.46 | **84.44** ↑ | 83.88 | **85.21** ↑ | 78.82 | **82.65** ↑ | N.A. | N.A. | N.A. | N.A. |
| compas (n=40) | 67.57 | **67.85** ↑ | 61.34 | **62.84** ↑ | 57.07 | **60.87** ↑ | 59.52 | **61.46** ↑ | 57.48 | **61.20**↑ | 58.90 | **63.32**↑ |
| adult (n=40) | 48.31 | **73.82** ↑ | 49.21 | **74.27** ↑ | 48.80 | **74.13** ↑ | 54.34 | **69.31** ↑ | 64.44 | **74.83**↑ | 56.59 | **78.30** ↑ |
| drug (n=40) | 74.30 | **75.79** ↑ | 71.33 | **72.76** ↑ | 73.12 | **74.12** ↑ | 69.84 | **72.50** ↑ | 64.14 | **68.14** ↑ | 74.34 | **76.07** ↑ |
| covid (n=100) | 73.77 | **74.71** ↑ | 70.71 | **72.76** ↑ | 71.02 | **73.37** ↑ | 63.76 | **70.68** ↑ | N.A. | N.A. | N.A. | N.A. |
| cutract (n=100) | 70.20 | **72.51** ↑ | 69.97 | **71.94** ↑ | 68.92 | **71.17** ↑ | 64.81 | **69.85** ↑ | N.A. | N.A. | N.A. | N.A. |
| maggic (n=100) | 58.98 | **61.32** ↑ | 55.71 | **58.90** ↑ | 52.53 | **53.36** ↑ | 54.27 | 53.65 | N.A. | N.A. | N.A. | N.A. |
| seer (n=100) | 84.45 | **85.37** ↑ | 83.92 | **85.08** ↑ | 82.23 | **84.36** ↑ | 81.00 | **81.99** ↑ | N.A. | N.A. | N.A. | N.A. |
| compas (n=100) | 68.02 | **68.19** ↑ | 60.10 | **62.47** ↑ | 53.74 | **61.28** ↑ | 57.90 | **61.79** ↑ | 60.96 | **63.36** ↑ | 59.07 | **63.13** ↑ |
| adult (n=100) | 46.09 | **74.57** ↑ | 47.56 | **73.97** ↑ | 40.51 | **71.08** ↑ | 48.85 | **72.91** ↑ | 54.45 | **76.67** ↑ | 54.27 | **77.61** ↑ |
| drug (n=100) | 76.24 | **76.74** ↑ | 69.46 | **71.05** ↑ | 74.02 | **76.55** ↑ | 66.86 | **75.31** ↑ | 71.47 | **75.00** ↑ | 73.22 | **75.83** ↑ |
| covid (n=200) | 73.40 | **74.62** ↑ | 70.70 | **73.12** ↑ | 70.81 | **73.26** ↑ | 62.53 | **70.67** ↑ | N.A. | N.A. | N.A. | N.A. |
| cutract (n=200) | 71.39 | **73.01** ↑ | 70.28 | **72.39** ↑ | 67.69 | **70.29** ↑ | 65.77 | **69.11** ↑ | N.A. | N.A. | N.A. | N.A. |
| maggic (n=200) | 58.92 | **61.41** ↑ | 57.33 | **60.16** ↑ | 52.78 | 52.40 | 52.56 | **52.90** ↑ | N.A. | N.A. | N.A. | N.A. |
| seer (n=200) | 84.39 | **85.56** ↑ | 83.48 | **84.80** ↑ | 83.15 | **84.18** ↑ | 80.88 | **82.74** ↑ | N.A. | N.A. | N.A. | N.A. |
| compas (n=200) | 67.02 | **68.15** ↑ | 60.48 | **63.39** ↑ | 53.96 | **59.22** ↑ | 57.56 | **59.97** ↑ | 61.82 | **63.53** ↑ | 60.99 | **64.72** ↑ |
| adult (n=200) | 40.96 | **75.84** ↑ | 49.89 | **76.11** ↑ | 44.13 | **74.23** ↑ | 48.25 | **78.35** ↑ | 54.42 | **76.23** ↑ | 42.42 | **76.06** ↑ |
| drug (n=200) | 75.58 | **76.06** ↑ | 70.66 | **72.81** ↑ | 71.89 | **76.54** ↑ | 70.88 | **75.46** ↑ | 69.12 | **74.22** ↑ | 73.43 | **76.31** ↑ |

### C.3.2 Ablation for contextual information on Compas

We conduct a similar experiment as in Table 4.2, and use the dataset Compas. We report the results in Table C.3. These results highlight the importance of incorporating contextual information in the prompt, as it enables us to exploit the prior knowledge of the LLM.

Table C.3 Including contextual information in the prompt improves precision (P), recall (R), and utility (U) in low-sample settings (results shown for Compas)

| $n_{\text{samples}}$ in $\mathscr{D}_{\text{train}}$ | GPT-4 w/ context | | | GPT-4 no context | | | TVAE | | |
|---|---|---|---|---|---|---|---|---|---|
| | P | R | U | P | R | U | P | R | U |
| 20 | $\mathbf{0.69}_{(\mathbf{0.02})}$ | $\mathbf{0.88}_{(\mathbf{0.02})}$ | $\mathbf{0.69}_{(\mathbf{0.02})}$ | $0.27_{(0.03)}$ | $\mathbf{0.89}_{(\mathbf{0.03})}$ | $0.60_{(0.03)}$ | $0.43_{(0.02)}$ | $0.43_{(0.05)}$ | $0.55_{(0.04)}$ |
| 40 | $\mathbf{0.70}_{(\mathbf{0.0})}$ | $\mathbf{0.92}_{(\mathbf{0.01})}$ | $\mathbf{0.65}_{(\mathbf{0.03})}$ | $0.31_{(0.06)}$ | $0.84_{(0.03)}$ | $0.57_{(0.01)}$ | $0.54_{(0.02)}$ | $0.80_{(0.02)}$ | $0.50_{(0.04)}$ |
| 100 | $\mathbf{0.69}_{(\mathbf{0.02})}$ | $\mathbf{0.89}_{(\mathbf{0.02})}$ | $\mathbf{0.69}_{(\mathbf{0.01})}$ | $0.34_{(0.1)}$ | $0.85_{(0.05)}$ | $0.62_{(0.01)}$ | $0.60_{(0.03)}$ | $0.86_{(0.02)}$ | $0.59_{(0.03)}$ |
| 200 | $\mathbf{0.70}_{(\mathbf{0.01})}$ | $\mathbf{0.89}_{(\mathbf{0.02})}$ | $\mathbf{0.69}_{(\mathbf{0.01})}$ | $0.31_{(0.05)}$ | $0.87_{(0.03)}$ | $0.58_{(0.05)}$ | $0.65_{(0.02)}$ | $0.88_{(0.01)}$ | $0.63_{(0.01)}$ |

### C.3.3 Comparison to random noise baseline

We compare CLLM to a random noise baseline, where we augment the dataset with random additive Gaussian noise. In order to capture the correlations between the different features, we fit a Kernel Density Estimator with a Gaussian kernel and bandwidth given by Scott's rule. We then sample 1000 points to create an augmented dataset $D_{\text{syn}}$. We report the performance gap between CLLM and this baseline (with and without curation) for the Covid and Compas datasets in Figure C.2. We observe that the random noise baseline does not match the performance of CLLM (i.e., it has a performance gap), although the baseline naturally improves as the dataset $D_{\text{train}}$ grows in size.

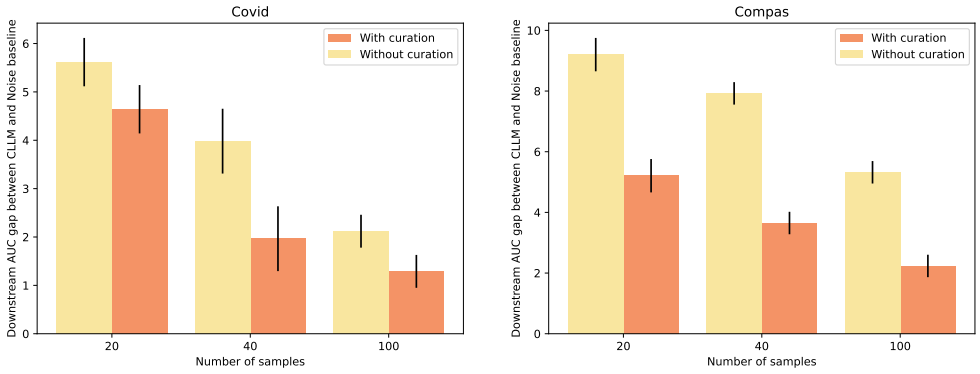



Fig. C.2 The random noise baseline does not match the performance of CLLM

### C.3.4 Detailed results for Chapter 4.3.1

We report additional results for Chapter 4.3.1, showing the AUC for each downstream model (XGBoost, Random forest, Logistic regression, Decision tree). As we can see, the conclusion that curation helps improve downstream performance holds for each of these various downstream models, as is indicated by the green arrows in Tables C.4, C.5, C.6, C.7.

Table C.4 AUC for the RF model on $\mathscr{D}_{\text{test}}$ where curation improves performance for all methods across all sample sizes $n$, as indicated by ↑.

| | Real data | | CLLM (OURS) | | | | Baselines | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | GPT-4 | | GPT-3.5 | | CTGAN | | TabDDPM | | GReaT | | NFLOW | | SMOTE | | TVAE | |
| Dataset | $\mathscr{D}_{\text{oracle}}$ | $\mathscr{D}_{\text{train}}$ | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. |
| covid (n=20) | 76.11 | 72.52 | **75.67** | 75.59 | 72.37 | 73.32 ↑ | 61.66 | 65.24 ↑ | 70.22 | 70.13 | 57.78 | 68.24 ↑ | 65.54 | 70.72 ↑ | 71.40 | 71.42 ↑ | 63.90 | 67.69 ↑ |
| cutract (n=20) | 74.16 | 73.34 | 73.45 | **74.24** ↑ | 72.22 | 73.51 ↑ | 66.24 | 70.45 ↑ | 69.73 | 70.49 ↑ | 52.61 | 68.11 ↑ | 66.63 | 72.39 ↑ | 71.42 | 72.34 ↑ | 71.18 | 71.89 ↑ |
| maggic (n=20) | 71.28 | 59.18 | 62.92 | **63.97** ↑ | 60.10 | 61.25 ↑ | 53.21 | 55.14 ↑ | 55.61 | 56.50 ↑ | 50.92 | 57.23 ↑ | 55.91 | 59.07 ↑ | 58.20 | 58.55 ↑ | 55.43 | 57.71 ↑ |
| seer (n=20) | 90.09 | 85.22 | 86.30 | **86.82** ↑ | 85.86 | 85.73 | 78.11 | 80.52 ↑ | 82.61 | 82.74 ↑ | 47.57 | 75.70 ↑ | 77.35 | 81.45 ↑ | 82.09 | 83.00 ↑ | 77.63 | 81.16 ↑ |
| compas (n=20) | 66.79 | 64.47 | **68.40** | 67.85 | 61.93 | 64.29 ↑ | 56.87 | 63.21 ↑ | 58.56 | 61.26 ↑ | 52.23 | 63.30 ↑ | 59.60 | 64.79 ↑ | 61.05 | 62.18 ↑ | 59.31 | 63.37 ↑ |
| adult (n=20) | 85.83 | 83.31 | 51.35 | 73.45 ↑ | 49.17 | 74.27 ↑ | 75.63 | 78.99 ↑ | 77.27 | 77.64 ↑ | 69.35 | 78.79 ↑ | 68.32 | 78.27 ↑ | 77.75 | 78.34 ↑ | 76.13 | 79.02 ↑ |
| drug (n=20) | 83.12 | 77.07 | **79.23** | 78.90 | 77.11 | 76.95 | 72.81 | 77.24 ↑ | 72.00 | 74.03 ↑ | 63.34 | 73.29 ↑ | 65.47 | 70.19 ↑ | 75.48 | 75.36 | 72.22 | 74.32 ↑ |
| covid (n=40) | 76.11 | 75.21 | 75.63 | **75.70** ↑ | 72.27 | 73.59 ↑ | 66.64 | 70.75 ↑ | 75.56 | 75.39 | 56.91 | 70.88 ↑ | 70.47 | 73.15 ↑ | 73.59 | 73.63 ↑ | 64.27 | 69.71 ↑ |
| cutract (n=40) | 74.16 | 72.31 | 71.67 | **73.20** ↑ | 69.92 | 71.01 ↑ | 64.67 | 69.21 ↑ | 69.16 | 69.67 ↑ | 53.41 | 68.93 ↑ | 61.23 | 69.21 ↑ | 70.82 | 71.21 ↑ | 59.99 | 67.56 ↑ |
| maggic (n=40) | 71.28 | 60.91 | 61.80 | **63.11** ↑ | 59.38 | 61.50 ↑ | 56.49 | 58.17 ↑ | 56.50 | 58.21 ↑ | 48.43 | 57.84 ↑ | 55.47 | 59.94 ↑ | 59.82 | 60.11 ↑ | 56.76 | 58.92 ↑ |
| seer (n=40) | 90.09 | 86.96 | 86.01 | 86.29 ↑ | 86.73 | **87.09** ↑ | 83.46 | 86.62 ↑ | 85.86 | 85.19 | 54.71 | 83.22 ↑ | 83.23 | 86.81 ↑ | 83.76 | 84.82 ↑ | 80.34 | 85.82 ↑ |
| compas (n=40) | 66.79 | 62.73 | 68.06 | **68.13** ↑ | 61.17 | 62.27 ↑ | 56.05 | 60.92 ↑ | 59.76 | 61.03 ↑ | 57.82 | 64.25 ↑ | 58.89 | 63.79 ↑ | 61.20 | 61.25 ↑ | 55.89 | 60.54 ↑ |
| adult (n=40) | 85.83 | 83.61 | 50.26 | 75.55 ↑ | 46.68 | 76.63 ↑ | 75.13 | 81.56 ↑ | 76.10 | 80.40 ↑ | 68.04 | 81.08 ↑ | 73.29 | 81.46 ↑ | 80.35 | 81.10 ↑ | 75.81 | 82.11 ↑ |
| drug (n=40) | 83.12 | 78.81 | 78.35 | **79.54** ↑ | 77.33 | 77.94 ↑ | 74.68 | 76.64 ↑ | 74.50 | 77.18 ↑ | 70.25 | 77.77 ↑ | 65.86 | 74.55 ↑ | 74.84 | 75.26 ↑ | 71.29 | 75.98 ↑ |
| covid (n=100) | 76.11 | 75.78 | 75.86 | **76.33** ↑ | 73.02 | 74.40 ↑ | 72.00 | 74.74 ↑ | 74.64 | 75.74 ↑ | 65.76 | 74.69 ↑ | 67.05 | 75.47 ↑ | 74.00 | 74.09 ↑ | 72.11 | 74.44 ↑ |
| cutract (n=100) | 74.16 | 73.93 | 72.92 | 74.73 ↑ | 72.51 | 73.93 ↑ | 70.18 | 74.26 ↑ | 71.71 | 73.69 ↑ | 55.48 | 70.79 ↑ | 69.79 | **74.75** ↑ | 71.18 | 72.47 ↑ | 68.47 | 73.78 ↑ |
| maggic (n=100) | 71.28 | 63.06 | 60.99 | **63.36** ↑ | 57.97 | 60.86 ↑ | 58.98 | 60.76 ↑ | 58.88 | 60.13 ↑ | 49.66 | 59.57 ↑ | 57.82 | 62.24 ↑ | 61.55 | 61.90 ↑ | 57.46 | 60.55 ↑ |
| seer (n=100) | 90.00 | 87.53 | 86.33 | 87.31 ↑ | 86.40 | 87.06 ↑ | 84.59 | 87.27 ↑ | 85.89 | 87.00 ↑ | 70.32 | 85.97 ↑ | 84.30 | 87.52 ↑ | 85.12 | 85.82 ↑ | 81.92 | 86.41 ↑ |
| compas (n=100) | 66.79 | 63.18 | 68.44 | **68.67** ↑ | 59.41 | 62.00 ↑ | 60.28 | 64.34 ↑ | 60.17 | 62.63 ↑ | 59.32 | 63.98 ↑ | 60.10 | 65.19 ↑ | 61.34 | 61.37 ↑ | 59.58 | 63.37 ↑ |
| adult (n=100) | 85.83 | 84.38 | 46.37 | 76.29 ↑ | 47.30 | 75.74 ↑ | 77.45 | 82.63 ↑ | 80.88 | 81.69 ↑ | 79.23 | 83.33 ↑ | 73.52 | 83.38 ↑ | 82.54 | 83.05 ↑ | 76.60 | 82.28 ↑ |
| drug (n=100) | 83.12 | 80.31 | 79.29 | 79.75 ↑ | 74.55 | 76.46 ↑ | 74.54 | 78.01 ↑ | 78.34 | 79.88 ↑ | 73.19 | 79.72 ↑ | 67.01 | 76.46 ↑ | 76.92 | 77.57 ↑ | 74.53 | 78.82 ↑ |
| covid (n=200) | 76.11 | 76.08 | 75.23 | 75.84 ↑ | 72.62 | 74.99 ↑ | 74.04 | 76.25 ↑ | 75.28 | **76.82** ↑ | 67.08 | 75.62 ↑ | 67.73 | 75.67 ↑ | 74.82 | 75.27 ↑ | 70.64 | 75.03 ↑ |
| cutract (n=200) | 74.16 | 74.25 | 73.48 | 75.28 ↑ | 72.01 | 74.37 ↑ | 71.99 | 74.93 ↑ | 74.33 | **76.19** ↑ | 68.05 | 74.68 ↑ | 71.47 | 75.67 ↑ | 72.44 | 72.85 ↑ | 68.55 | 74.43 ↑ |
| maggic (n=200) | 71.28 | 64.77 | 61.56 | 63.78 ↑ | 59.76 | 62.53 ↑ | 60.85 | 63.57 ↑ | 58.23 | 58.85 ↑ | 50.84 | 61.49 ↑ | 57.52 | 63.00 ↑ | 63.72 | 64.01 ↑ | 58.92 | 62.08 ↑ |
| seer (n=200) | 90.09 | 88.13 | 86.96 | 87.72 ↑ | 85.28 | 86.70 ↑ | 85.05 | 87.83 ↑ | 87.14 | **88.80** ↑ | 84.86 | 88.14 ↑ | 80.80 | 87.46 ↑ | 86.57 | 87.11 ↑ | 82.30 | 87.59 ↑ |
| compas (n=200) | 66.79 | 63.61 | 67.68 | **68.80** ↑ | 60.67 | 63.43 ↑ | 61.20 | 65.11 ↑ | 60.88 | 63.87 ↑ | 60.10 | 64.91 ↑ | 56.94 | 64.76 ↑ | 61.57 | 62.76 ↑ | 59.95 | 64.14 ↑ |
| adult (n=200) | 85.83 | 85.16 | 40.18 | 77.61 ↑ | 46.74 | 78.71 ↑ | 80.92 | 84.26 ↑ | 84.50 | **85.44** ↑ | 82.73 | 85.05 ↑ | 77.48 | 84.72 ↑ | 84.26 | 84.33 ↑ | 78.12 | 83.94 ↑ |
| drug (n=200) | 83.12 | 81.47 | 78.70 | 78.98 ↑ | 75.87 | 77.59 ↑ | 77.19 | 80.03 ↑ | 73.25 | 75.75 ↑ | 78.50 | 81.13 ↑ | 67.83 | 78.81 ↑ | 79.50 | 80.16 ↑ | 74.81 | 79.45 ↑ |

Table C.5 AUC for the XGB model on $\mathscr{D}_{\text{test}}$ where curation improves performance for all methods across all sample sizes $n$, as indicated by ↑.

| | Real data | | CLLM (OURS) | | | | Baselines | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | GPT-4 | | GPT-3.5 | | CTGAN | | TabDDPM | | GReaT | | NFLOW | | SMOTE | | TVAE | |
| Dataset | $\mathscr{D}_{\text{oracle}}$ | $\mathscr{D}_{\text{train}}$ | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. |
| covid (n=20) | 77.59 | 69.70 | **75.59** | 75.49 | 71.27 | 72.25 ↑ | 61.40 | 65.96 ↑ | 69.39 | 68.21 | 55.99 | 67.06 ↑ | 62.28 | 69.67 ↑ | 68.49 | 68.43 | 62.49 | 67.16 ↑ |
| cutract (n=20) | 73.14 | 71.04 | 73.26 | **74.30** ↑ | 71.62 | 73.04 ↑ | 65.13 | 69.67 ↑ | 68.46 | 68.87 ↑ | 51.99 | 68.31 ↑ | 62.93 | 70.87 ↑ | 68.28 | 70.72 ↑ | 69.96 | 71.98 ↑ |
| maggic (n=20) | 71.38 | 57.98 | 61.54 | **62.81** ↑ | 58.76 | 60.01 ↑ | 53.62 | 55.44 ↑ | 55.97 | 57.06 ↑ | 50.14 | 56.96 ↑ | 54.91 | 58.22 ↑ | 56.79 | 57.15 ↑ | 55.41 | 57.72 ↑ |
| seer (n=20) | 91.80 | 80.41 | **87.00** | 86.56 | 85.67 | 85.42 | 76.84 | 80.00 ↑ | 81.79 | 80.21 | 45.06 | 75.39 ↑ | 76.25 | 81.45 ↑ | 78.57 | 79.78 ↑ | 75.69 | 81.01 ↑ |
| compas (n=20) | 69.17 | 65.49 | **69.71** | 68.79 | 62.55 | 64.68 ↑ | 56.34 | 63.92 ↑ | 58.19 | 61.42 ↑ | 51.73 | 63.51 ↑ | 59.01 | 64.84 ↑ | 62.35 | 63.46 ↑ | 59.18 | 63.62 ↑ |
| adult (n=20) | 88.64 | 79.21 | 52.12 | 74.28 ↑ | 47.70 | 72.91 ↑ | 71.70 | 78.26 ↑ | 76.50 | 77.22 ↑ | 67.79 | 78.67 ↑ | 66.79 | 77.77 ↑ | 77.11 | 77.97 ↑ | 75.46 | **79.35** ↑ |
| drug (n=20) | 81.06 | 69.26 | **78.17** | 77.28 | 75.23 | 75.71 ↑ | 70.60 | 74.31 ↑ | 70.29 | 71.25 ↑ | 57.18 | 69.67 ↑ | 62.96 | 69.12 ↑ | 70.84 | 72.30 ↑ | 69.78 | 72.45 ↑ |
| covid (n=40) | 77.59 | 71.57 | 75.23 | **75.36** ↑ | 71.55 | 73.37 ↑ | 66.17 | 70.85 ↑ | 72.21 | 72.68 ↑ | 56.03 | 70.80 ↑ | 67.39 | 71.82 ↑ | 70.02 | 69.96 | 61.19 | 69.77 ↑ |
| cutract (n=40) | 73.14 | 69.50 | 71.26 | **73.57** ↑ | 69.66 | 71.34 ↑ | 62.52 | 68.69 ↑ | 64.50 | 68.05 ↑ | 51.23 | 68.23 ↑ | 60.74 | 68.91 ↑ | 69.42 | 69.19 | 59.34 | 67.80 ↑ |
| maggic (n=40) | 71.38 | 58.76 | 60.27 | **61.88** ↑ | 57.87 | 60.29 ↑ | 55.87 | 58.47 ↑ | 55.38 | 58.03 ↑ | 48.50 | 57.02 ↑ | 54.53 | 59.01 ↑ | 58.12 | 58.29 ↑ | 56.19 | 58.33 ↑ |
| seer (n=40) | 91.80 | 84.51 | 86.63 | **87.40** ↑ | 86.27 | 86.80 ↑ | 82.48 | 86.55 ↑ | 81.62 | 81.58 | 52.80 | 83.03 ↑ | 80.04 | 85.84 ↑ | 82.53 | 83.62 ↑ | 79.35 | 85.81 ↑ |
| compas (n=40) | 69.17 | 62.90 | **69.24** | 68.90 | 61.74 | 63.21 ↑ | 56.63 | 62.55 ↑ | 59.07 | 59.38 ↑ | 58.01 | 64.95 ↑ | 58.44 | 64.44 ↑ | 61.81 | 61.89 ↑ | 56.60 | 62.26 ↑ |
| adult (n=40) | 88.64 | 82.16 | 48.51 | 76.59 ↑ | 45.92 | 76.92 ↑ | 73.38 | 82.53 ↑ | 74.03 | 79.84 ↑ | 66.60 | 80.98 ↑ | 69.47 | 81.56 ↑ | 80.01 | 80.51 ↑ | 74.20 | **82.53** ↑ |
| drug (n=40) | 81.06 | 72.72 | 77.03 | **78.85** ↑ | 76.37 | 76.48 ↑ | 70.99 | 74.48 ↑ | 74.65 | 76.53 ↑ | 64.52 | 75.09 ↑ | 62.58 | 73.14 ↑ | 71.80 | 72.15 ↑ | 67.61 | 74.04 ↑ |
| covid (n=100) | 77.59 | 73.16 | 75.55 | **76.56** ↑ | 72.69 | 74.36 ↑ | 70.54 | 73.99 ↑ | 72.99 | 74.50 ↑ | 63.40 | 73.98 ↑ | 63.50 | 73.79 ↑ | 72.64 | 73.35 ↑ | 70.45 | 73.76 ↑ |
| cutract (n=100) | 73.14 | 70.49 | 72.29 | 74.01 ↑ | 71.51 | 73.48 ↑ | 68.64 | 73.63 ↑ | 71.32 | **74.02** ↑ | 54.74 | 69.69 ↑ | 66.97 | 72.98 ↑ | 67.71 | 69.42 ↑ | 66.44 | 72.32 ↑ |
| maggic (n=100) | 71.38 | 60.55 | 60.04 | **62.57** ↑ | 56.98 | 60.49 ↑ | 58.21 | 60.42 ↑ | 58.11 | 59.42 ↑ | 49.71 | 58.77 ↑ | 55.36 | 60.58 ↑ | 60.02 | 60.15 ↑ | 56.47 | 60.28 ↑ |
| seer (n=100) | 91.80 | 85.52 | 86.81 | **87.64** ↑ | 86.39 | 87.21 ↑ | 83.74 | 87.29 ↑ | 84.66 | 86.85 ↑ | 68.20 | 85.49 ↑ | 82.37 | 86.80 ↑ | 83.18 | 83.87 ↑ | 80.45 | 86.49 ↑ |
| compas (n=100) | 69.17 | 62.71 | **69.37** | 69.37 | 59.83 | 62.80 ↑ | 60.56 | 64.90 ↑ | 58.20 | 62.00 ↑ | 58.51 | 64.60 ↑ | 59.76 | 64.68 ↑ | 61.33 | 62.03 ↑ | 60.37 | 63.42 ↑ |
| adult (n=100) | 88.64 | 82.78 | 44.60 | 76.75 ↑ | 46.34 | 76.83 ↑ | 74.97 | 82.89 ↑ | 77.38 | 81.42 ↑ | 78.84 | **83.96** ↑ | 68.31 | 82.87 ↑ | 82.58 | 83.24 ↑ | 76.15 | 82.47 ↑ |
| drug (n=100) | 81.06 | 75.49 | 79.12 | **79.81** ↑ | 72.52 | 74.78 ↑ | 68.89 | 75.91 ↑ | 76.88 | 78.33 ↑ | 66.38 | 77.02 ↑ | 62.27 | 75.13 ↑ | 74.40 | 74.49 ↑ | 71.02 | 76.85 ↑ |
| covid (n=200) | 77.59 | 73.06 | 75.12 | **76.64** ↑ | 71.74 | 74.66 ↑ | 72.57 | 75.50 ↑ | 73.56 | 75.88 ↑ | 66.02 | 74.51 ↑ | 63.29 | 74.62 ↑ | 73.34 | 73.89 ↑ | 68.14 | 74.21 ↑ |
| cutract (n=200) | 73.14 | 71.68 | 73.43 | 75.14 ↑ | 71.76 | 74.31 ↑ | 69.35 | 73.80 ↑ | 72.76 | **75.33** ↑ | 65.83 | 73.43 ↑ | 68.05 | 74.27 ↑ | 70.06 | 71.35 ↑ | 66.65 | 73.34 ↑ |
| maggic (n=200) | 71.38 | 62.40 | 59.46 | **63.15** ↑ | 58.05 | 61.68 ↑ | 59.04 | 62.58 ↑ | 57.63 | 59.23 ↑ | 50.63 | 60.09 ↑ | 55.55 | 61.36 ↑ | 61.73 | 61.88 ↑ | 57.09 | 61.15 ↑ |
| seer (n=200) | 91.80 | 85.87 | 86.75 | 88.08 ↑ | 85.74 | 86.97 ↑ | 83.97 | 87.64 ↑ | 86.21 | **88.46** ↑ | 83.21 | 87.65 ↑ | 76.91 | 87.16 ↑ | 84.70 | 85.59 ↑ | 81.45 | 87.46 ↑ |
| compas (n=200) | 69.17 | 63.14 | 68.80 | **69.54** ↑ | 61.62 | 64.59 ↑ | 60.65 | 65.67 ↑ | 60.97 | 65.12 ↑ | 60.66 | 65.67 ↑ | 56.49 | 64.34 ↑ | 61.68 | 62.93 ↑ | 60.32 | 64.18 ↑ |
| adult (n=200) | 88.64 | 84.12 | 40.38 | 77.39 ↑ | 47.40 | 79.40 ↑ | 80.00 | 84.61 ↑ | 83.73 | 85.73 ↑ | 81.92 | **85.76** ↑ | 74.90 | 85.13 ↑ | 84.78 | 84.45 | 75.54 | 84.39 ↑ |
| drug (n=200) | 81.06 | 78.41 | 78.25 | 79.09 ↑ | 73.31 | 76.76 ↑ | 72.82 | 79.00 ↑ | 72.62 | 74.89 ↑ | 74.30 | **79.75** ↑ | 63.91 | 77.60 ↑ | 77.64 | 77.92 ↑ | 70.16 | 77.89 ↑ |

Table C.6 AUC for the DT model on $\mathscr{D}_{\text{test}}$ where curation improve performance for all methods across all sample sizes $n$, as indicated by ↑.

| | Real data | | CLLM (OURS) | | | | Baselines | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | GPT-4 | | GPT-3.5 | | CTGAN | | TabDDPM | | GReaT | | NFLOW | | SMOTE | | TVAE | |
| Dataset | $\mathscr{D}_{\text{oracle}}$ | $\mathscr{D}_{\text{train}}$ | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. |
| covid (n=20) | 63.46 | 61.92 | 67.52 | **67.67** ↑ | 61.07 | 64.89 ↑ | 52.94 | 56.62 ↑ | 58.95 | 59.19 ↑ | 51.83 | 58.73 ↑ | 54.37 | 61.53 ↑ | 59.56 | 59.01 | 56.46 | 59.49 ↑ |
| cutract (n=20) | 62.49 | 64.62 | 62.40 | **66.03** ↑ | 62.10 | 65.15 ↑ | 55.67 | 61.73 ↑ | 57.91 | 58.97 ↑ | 51.89 | 62.76 ↑ | 57.12 | 64.85 ↑ | 62.85 | 62.38 | 60.25 | 62.34 ↑ |
| maggic (n=20) | 56.53 | 54.80 | 54.67 | **55.03** ↑ | 52.54 | 53.54 ↑ | 50.90 | 52.08 ↑ | 52.39 | 52.70 ↑ | 49.70 | 53.08 ↑ | 51.52 | 53.23 ↑ | 52.55 | 52.44 | 50.91 | 51.52 ↑ |
| seer (n=20) | 78.86 | 71.09 | 76.17 | **77.17** ↑ | 73.92 | 75.58 ↑ | 64.18 | 70.69 ↑ | 74.78 | 74.72 | 46.60 | 68.79 ↑ | 65.77 | 72.25 ↑ | 70.54 | 71.19 ↑ | 64.47 | 70.11 ↑ |
| compas (n=20) | 61.05 | 58.73 | 62.07 | **63.91** ↑ | 57.17 | 60.85 ↑ | 53.83 | 58.83 ↑ | 51.77 | 56.11 ↑ | 52.39 | 60.96 ↑ | 54.64 | 60.27 ↑ | 58.16 | 57.57 | 53.52 | 58.97 ↑ |
| adult (n=20) | 73.84 | 68.03 | 52.80 | 67.10 ↑ | 49.15 | 66.52 ↑ | 61.57 | 68.93 ↑ | 70.46 | 71.19 ↑ | 62.04 | **71.87** ↑ | 59.07 | 69.16 ↑ | 65.52 | 63.33 | 64.40 | 70.41 ↑ |
| drug (n=20) | 65.66 | 63.49 | 64.40 | **66.87** ↑ | 62.69 | 64.21 ↑ | 56.76 | 64.18 ↑ | 61.04 | 62.48 ↑ | 52.22 | 62.62 ↑ | 53.50 | 61.30 ↑ | 62.34 | 60.69 | 55.00 | 58.89 ↑ |
| covid (n=40) | 63.46 | 65.10 | 66.55 | **67.85** ↑ | 62.54 | 64.64 ↑ | 57.16 | 63.41 ↑ | 63.15 | 60.93 | 51.66 | 61.81 ↑ | 57.04 | 63.13 ↑ | 62.19 | 60.49 | 54.51 | 60.32 ↑ |
| cutract (n=40) | 62.49 | 63.01 | 62.49 | **65.29** ↑ | 61.61 | 63.01 ↑ | 56.40 | 62.63 ↑ | 58.18 | 58.97 ↑ | 51.41 | 62.50 ↑ | 55.64 | 62.24 ↑ | 60.93 | 61.29 ↑ | 51.94 | 60.24 ↑ |
| maggic (n=40) | 56.53 | 53.74 | 53.20 | **54.71** ↑ | 52.33 | 52.97 ↑ | 51.85 | 53.85 ↑ | 52.48 | 53.45 ↑ | 49.61 | 52.87 ↑ | 51.24 | 54.45 ↑ | 53.69 | 53.37 | 50.60 | 52.66 ↑ |
| seer (n=40) | 78.86 | 72.40 | 76.02 | 77.13 ↑ | 75.18 | **78.26** ↑ | 68.31 | 73.89 ↑ | 77.94 | 73.24 | 51.45 | 72.32 ↑ | 67.78 | 75.95 ↑ | 70.32 | 72.81 ↑ | 68.79 | 74.48 ↑ |
| compas (n=40) | 61.05 | 58.23 | 62.00 | **63.43** ↑ | 56.73 | 59.19 ↑ | 52.23 | 56.84 ↑ | 56.67 | 57.55 ↑ | 54.52 | 60.09 ↑ | 54.47 | 58.92 ↑ | 56.59 | 56.47 | 52.62 | 57.20 ↑ |
| adult (n=40) | 74.84 | 69.75 | 49.47 | 68.90 ↑ | 47.85 | 66.79 ↑ | 60.55 | 70.14 ↑ | 64.06 | 70.56 ↑ | 58.77 | 71.63 ↑ | 60.26 | 71.50 ↑ | 69.43 | 70.58 ↑ | 59.76 | **72.41** ↑ |
| drug (n=40) | 65.66 | 64.18 | 62.86 | **65.90** ↑ | 60.09 | 63.43 ↑ | 60.84 | 65.40 ↑ | 63.05 | 64.90 ↑ | 54.80 | 65.25 ↑ | 54.68 | 62.40 ↑ | 61.21 | 61.98 ↑ | 53.77 | 60.23 ↑ |
| covid (n=100) | 63.46 | 63.17 | 67.14 | **68.60** ↑ | 62.40 | 66.02 ↑ | 61.34 | 64.63 ↑ | 63.33 | 66.42 ↑ | 55.86 | 64.53 ↑ | 55.44 | 64.91 ↑ | 61.96 | 63.40 ↑ | 59.75 | 64.47 ↑ |
| cutract (n=100) | 62.49 | 62.69 | 60.96 | 65.60 ↑ | 62.12 | 65.95 ↑ | 59.06 | **66.20** ↑ | 63.38 | 65.72 ↑ | 52.62 | 61.73 ↑ | 58.11 | 65.08 ↑ | 60.15 | 61.34 ↑ | 58.82 | 63.76 ↑ |
| maggic (n=100) | 56.53 | 54.83 | 53.58 | **55.70** ↑ | 50.72 | 53.98 ↑ | 52.29 | 55.06 ↑ | 53.93 | 54.61 ↑ | 49.71 | 53.34 ↑ | 51.82 | 54.53 ↑ | 54.11 | 54.34 ↑ | 51.31 | 53.33 ↑ |
| seer (n=100) | 78.86 | 73.83 | 76.50 | **77.84** ↑ | 74.81 | 77.80 ↑ | 69.61 | 76.91 ↑ | 74.07 | 77.21 ↑ | 59.62 | 75.40 ↑ | 69.51 | 76.54 ↑ | 71.04 | 72.63 ↑ | 66.35 | 76.44 ↑ |
| compas (n=100) | 61.05 | 56.61 | 63.04 | **63.39** ↑ | 55.57 | 57.40 ↑ | 55.07 | 58.95 ↑ | 53.38 | 55.69 ↑ | 54.61 | 59.31 ↑ | 54.38 | 57.92 ↑ | 56.69 | 57.01 ↑ | 54.34 | 56.72 ↑ |
| adult (n=100) | 73.84 | 72.19 | 46.68 | 66.58 ↑ | 47.94 | 67.37 ↑ | 62.42 | 71.78 ↑ | 68.06 | 70.77 ↑ | 66.29 | **74.36** ↑ | 58.28 | 72.14 ↑ | 71.51 | 72.68 ↑ | 62.11 | 72.18 ↑ |
| drug (n=100) | 65.66 | 64.20 | 66.18 | **67.41** ↑ | 60.77 | 61.21 ↑ | 57.67 | 63.41 ↑ | 63.26 | 64.55 ↑ | 54.89 | 65.72 ↑ | 52.06 | 63.38 ↑ | 61.21 | 62.30 ↑ | 55.40 | 63.32 ↑ |
| covid (n=200) | 63.46 | 63.45 | 67.19 | **68.74** ↑ | 63.25 | 65.73 ↑ | 61.58 | 66.27 ↑ | 63.55 | 66.98 ↑ | 57.20 | 65.54 ↑ | 55.08 | 63.77 ↑ | 63.60 | 63.40 | 60.01 | 64.27 ↑ |
| cutract (n=200) | 62.49 | 63.53 | 63.36 | 65.33 ↑ | 63.01 | 65.47 ↑ | 60.51 | 65.01 ↑ | 62.55 | **66.94** ↑ | 57.40 | 65.14 ↑ | 59.13 | 65.20 ↑ | 61.66 | 62.40 ↑ | 58.29 | 64.85 ↑ |
| maggic (n=200) | 56.53 | 55.06 | 52.76 | 54.41 ↑ | 52.41 | 54.34 ↑ | 52.47 | **55.21** ↑ | 52.40 | 53.39 ↑ | 50.43 | 53.66 ↑ | 51.58 | 54.82 ↑ | 54.64 | 54.36 | 51.78 | 54.22 ↑ |
| seer (n=200) | 78.86 | 74.51 | 75.20 | 77.26 ↑ | 75.33 | 77.41 ↑ | 70.96 | 77.08 ↑ | 74.51 | **79.20** ↑ | 71.44 | 78.66 ↑ | 65.84 | 76.60 ↑ | 73.11 | 74.77 ↑ | 66.37 | 76.36 ↑ |
| compas (n=200) | 61.05 | 57.20 | 61.09 | **62.89** ↑ | 56.15 | 59.02 ↑ | 54.20 | 57.78 ↑ | 55.72 | 58.68 ↑ | 55.83 | 59.81 ↑ | 52.13 | 57.25 ↑ | 56.55 | 57.34 ↑ | 54.07 | 58.66 ↑ |
| adult (n=200) | 73.84 | 72.26 | 44.41 | 67.99 ↑ | 48.46 | 67.24 ↑ | 66.56 | 74.12 ↑ | 71.61 | 74.89 ↑ | 69.91 | **76.52** ↑ | 60.70 | 74.05 ↑ | 73.11 | 74.08 ↑ | 63.10 | 73.82 ↑ |
| drug (n=200) | 65.66 | 67.04 | 66.29 | 66.57 ↑ | 61.88 | 64.34 ↑ | 59.10 | 66.06 ↑ | 63.59 | 63.40 | 59.11 | **69.04** ↑ | 53.78 | 64.55 ↑ | 64.56 | 65.35 ↑ | 56.82 | 63.02 ↑ |

Table C.7 AUC for the LR model on $\mathscr{D}_{\text{test}}$ where curation improves performance for all methods across all sample sizes $n$, as indicated by ↑.

| | Real data | | CLLM (OURS) | | | | Baselines | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | GPT-4 | | GPT-3.5 | | CTGAN | | TabDDPM | | GReaT | | NFLOW | | SMOTE | | TVAE | |
| Dataset | $\mathscr{D}_{\text{oracle}}$ | $\mathscr{D}_{\text{train}}$ | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. |
| covid (n=20) | 80.47 | 69.85 | 76.35 | **76.73** ↑ | 74.69 | 75.18 ↑ | 59.99 | 66.87 ↑ | 68.78 | 69.84 ↑ | 63.94 | 71.81 ↑ | 69.28 | 72.32 ↑ | 68.36 | 68.44 ↑ | 63.92 | 70.12 ↑ |
| cutract (n=20) | 79.12 | 71.47 | **75.50** | 75.41 | 73.93 | 74.47 ↑ | 69.01 | 70.06 ↑ | 68.11 | 68.02 | 53.04 | 68.90 ↑ | 71.09 | 73.55 ↑ | 71.09 | 71.53 ↑ | 74.37 | 74.69 ↑ |
| maggic (n=20) | 70.46 | 56.55 | 63.67 | **64.12** ↑ | 58.74 | 59.98 ↑ | 53.27 | 55.37 ↑ | 54.40 | 55.28 ↑ | 50.39 | 55.27 ↑ | 56.52 | 59.00 ↑ | 55.83 | 56.45 ↑ | 54.59 | 57.82 ↑ |
| seer (n=20) | 90.91 | 85.96 | 88.63 | **88.74** ↑ | 87.90 | 88.11 ↑ | 78.22 | 83.69 ↑ | 83.17 | 84.74 ↑ | 51.07 | 77.84 ↑ | 84.86 | 84.75 | 85.72 | 86.11 ↑ | 80.33 | 82.63 ↑ |
| compas (n=20) | 73.02 | 63.74 | **71.88** | 71.10 | 66.62 | 67.89 ↑ | 55.62 | 64.29 ↑ | 62.15 | 64.70 ↑ | 56.97 | 66.59 ↑ | 64.71 | 68.59 ↑ | 62.68 | 63.16 ↑ | 61.19 | 64.36 ↑ |
| adult (n=20) | 88.37 | 79.24 | 45.28 | 71.07 ↑ | 50.89 | 75.77 ↑ | 80.03 | **81.27** ↑ | 73.19 | 74.10 ↑ | 68.84 | 79.66 ↑ | 75.65 | 80.74 ↑ | 74.61 | 75.04 ↑ | 76.82 | 78.84 ↑ |
| drug (n=20) | 81.41 | 73.56 | **78.54** | 78.10 | 71.69 | 71.68 | 73.07 | 74.88 ↑ | 69.14 | 70.97 ↑ | 62.38 | 69.99 ↑ | 66.57 | 70.37 ↑ | 71.99 | 72.28 ↑ | 69.39 | 71.08 ↑ |
| covid (n=40) | 80.47 | 71.20 | 76.17 | **76.91** ↑ | 75.32 | 76.13 ↑ | 64.56 | 68.82 ↑ | 71.08 | 72.77 ↑ | 61.40 | 71.21 ↑ | 70.74 | 73.84 ↑ | 68.85 | 69.65 ↑ | 64.13 | 69.58 ↑ |
| cutract (n=40) | 79.12 | 71.90 | 74.05 | **74.82** ↑ | 72.69 | 72.89 ↑ | 68.43 | 70.97 ↑ | 70.67 | 72.39 ↑ | 61.50 | 74.10 ↑ | 67.98 | 71.55 ↑ | 70.26 | 70.12 | 67.87 | 70.88 ↑ |
| maggic (n=40) | 70.46 | 59.65 | 61.89 | **63.39** ↑ | 60.41 | 61.84 ↑ | 55.81 | 56.64 ↑ | 56.58 | 58.08 ↑ | 48.68 | 58.82 ↑ | 57.50 | 60.90 ↑ | 57.98 | 58.00 ↑ | 56.61 | 59.42 ↑ |
| seer (n=40) | 90.91 | 87.86 | 88.50 | **88.89** ↑ | 85.66 | 85.62 | 85.95 | 87.62 ↑ | 84.92 | 85.45 ↑ | 60.75 | 85.84 ↑ | 88.46 | 88.85 ↑ | 86.56 | 87.58 ↑ | 86.30 | 88.36 ↑ |
| compas (n=40) | 73.02 | 65.50 | **70.96** | 70.93 | 65.70 | 66.72 ↑ | 60.24 | 63.78 ↑ | 59.91 | 62.49 ↑ | 65.17 | 68.19 ↑ | 62.63 | 67.00 ↑ | 63.70 | 64.18 ↑ | 58.63 | 64.17 ↑ |
| adult (n=40) | 88.37 | 82.23 | 44.99 | 74.25 ↑ | 56.38 | 76.76 ↑ | 78.24 | 82.22 ↑ | 71.83 | 81.15 ↑ | 73.65 | 81.54 ↑ | 81.49 | **84.34** ↑ | 81.80 | 83.16 ↑ | 80.54 | 83.02 ↑ |
| drug (n=40) | 81.41 | 71.74 | **78.98** | 78.88 | 71.54 | 73.21 ↑ | 71.34 | 74.44 ↑ | 72.12 | 73.68 ↑ | 70.00 | 76.45 ↑ | 66.91 | 73.78 ↑ | 69.09 | 69.75 ↑ | 68.21 | 70.94 ↑ |
| covid (n=100) | 80.47 | 74.19 | 76.53 | **77.34** ↑ | 74.74 | 76.27 ↑ | 72.30 | 75.15 ↑ | 75.45 | 76.20 ↑ | 69.05 | 74.94 ↑ | 71.01 | 76.38 ↑ | 71.71 | 72.28 ↑ | 73.90 | 75.17 ↑ |
| cutract (n=100) | 79.12 | 76.73 | 74.62 | 75.70 ↑ | 73.75 | 74.40 ↑ | 73.89 | 75.61 ↑ | 75.70 | 74.49 | 60.03 | 74.35 ↑ | 75.49 | **76.86** ↑ | 76.13 | 75.47 | 73.85 | 76.21 ↑ |
| maggic (n=100) | 70.46 | 60.14 | 61.31 | **63.64** ↑ | 57.17 | 60.28 ↑ | 59.32 | 61.13 ↑ | 58.12 | 58.97 ↑ | 49.10 | 59.95 ↑ | 60.43 | 63.06 ↑ | 59.87 | 59.59 | 59.43 | 61.28 ↑ |
| seer (n=100) | 90.91 | 88.92 | 88.16 | 88.69 ↑ | 88.09 | 88.25 ↑ | 88.45 | 89.11 ↑ | 87.53 | 88.25 ↑ | 83.13 | 88.44 ↑ | 88.46 | **89.27** ↑ | 87.95 | 87.65 | 86.81 | 88.67 ↑ |
| compas (n=100) | 73.02 | 67.75 | 71.21 | **71.34** ↑ | 65.61 | 67.67 ↑ | 64.14 | 66.72 ↑ | 61.55 | 65.03 ↑ | 67.43 | 68.88 ↑ | 65.84 | 68.39 ↑ | 66.39 | 66.53 ↑ | 65.61 | 67.77 ↑ |
| adult (n=100) | 88.37 | 85.59 | 46.71 | 78.68 ↑ | 48.64 | 75.94 ↑ | 82.30 | 84.48 ↑ | 77.41 | 78.98 ↑ | 83.99 | 84.99 ↑ | 82.67 | **85.77** ↑ | 85.63 | 85.43 | 81.32 | 83.97 ↑ |
| drug (n=100) | 81.41 | 74.34 | **80.35** | 80.01 | 70.00 | 71.76 ↑ | 71.66 | 75.80 ↑ | 71.26 | 72.40 ↑ | 74.57 | 78.65 ↑ | 69.34 | 77.50 ↑ | 71.04 | 71.75 ↑ | 71.91 | 75.38 ↑ |
| covid (n=200) | 80.47 | 76.73 | 76.06 | 77.25 ↑ | 75.18 | 77.09 ↑ | 76.11 | 77.54 ↑ | 77.51 | **78.09** ↑ | 71.92 | 76.62 ↑ | 74.06 | 77.53 ↑ | 74.94 | 74.94 | 72.75 | 76.00 ↑ |
| cutract (n=200) | 79.12 | 77.53 | 75.28 | 76.30 ↑ | 74.36 | 75.38 ↑ | 75.25 | 75.88 ↑ | **77.66** | 77.66 | 75.37 | 76.69 ↑ | 76.45 | 77.50 ↑ | 76.77 | 76.59 | 72.94 | 74.86 ↑ |
| maggic (n=200) | 70.46 | 63.32 | 61.89 | **64.30** ↑ | 59.10 | 62.10 ↑ | 61.57 | 63.98 ↑ | 56.77 | 57.35 ↑ | 51.05 | 63.17 ↑ | 59.17 | 63.84 ↑ | 62.86 | 62.86 | 60.94 | 63.47 ↑ |
| seer (n=200) | 90.91 | 90.01 | 88.65 | 89.20 ↑ | 87.57 | 88.14 ↑ | 88.20 | 88.80 ↑ | 89.69 | 89.81 ↑ | 89.10 | 89.68 ↑ | 87.35 | 89.54 ↑ | 89.13 | 89.14 ↑ | 88.74 | 89.65 ↑ |
| compas (n=200) | 73.02 | 69.14 | 70.53 | **71.36** ↑ | 63.49 | 66.50 ↑ | 66.26 | 68.73 ↑ | 64.83 | 66.40 ↑ | 67.85 | 69.93 ↑ | 60.78 | 68.07 ↑ | 68.18 | 68.18 | 66.26 | 68.99 ↑ |
| adult (n=200) | 88.37 | 86.94 | 38.89 | 80.35 ↑ | 56.97 | 79.07 ↑ | 85.25 | 86.29 ↑ | 86.81 | 86.63 | 85.70 | 85.94 ↑ | 84.18 | 86.65 ↑ | 86.88 | 86.70 | 84.09 | 85.91 ↑ |
| drug (n=200) | 81.41 | 77.47 | 79.06 | 79.61 ↑ | 71.56 | 72.54 ↑ | 76.15 | 78.84 ↑ | 69.00 | 73.14 ↑ | 77.51 | 79.72 ↑ | 75.47 | **80.07** ↑ | 75.82 | 75.69 | 71.76 | 76.95 ↑ |

### C.3.5 Full results for performance evaluation

We report full results with standard deviation for the results from the main thesis. The performance is AUC averaged over XGBoost, Random Forest, Logistic Regression, and Decision Tree.

Table C.8 AUC averaged over four downstream models on $\mathscr{D}_{\text{test}}$ where curation improves performance for all methods across all sample sizes $n$, as indicated by ↑. CLLM w/ GPT-4 (Curated) dataset provides the strongest performance for both private/proprietary datasets and public datasets

| | Real data | | CLLM (OURS) | | | | Baselines | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | GPT-4 | | GPT-3.5 | | CTGAN | | TabDDPM | | GReaT | | NFLOW | | SMOTE | | TVAE | |
| Dataset | $\mathscr{D}_{\text{oracle}}$ | $\mathscr{D}_{\text{train}}$ | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. | Uncur. | Cur. |
| covid (n=20) | $74.41_{(0.11)}$ | $68.50_{(1.57)}$ | $73.78_{(0.31)}$ | **$73.87_{(0.50)}$** | $69.85_{(0.75)}$ | $71.41_{(0.92)}$ | $59.00_{(2.25)}$ | $63.67_{(2.51)}$ | $66.84_{(1.66)}$ | $66.85_{(1.56)}$ | $57.38_{(1.47)}$ | $66.46_{(0.80)}$ | $62.87_{(0.98)}$ | $68.56_{(1.07)}$ | $66.95_{(1.66)}$ | $66.82_{(1.89)}$ | $61.69_{(2.72)}$ | $66.11_{(2.79)}$ |
| cutract (n=20) | $72.23_{(0.65)}$ | $70.12_{(1.16)}$ | $71.15_{(0.46)}$ | **$72.50_{(0.76)}$** | $69.97_{(1.13)}$ | $71.54_{(1.38)}$ | $64.01_{(2.26)}$ | $67.98_{(1.63)}$ | $66.05_{(1.14)}$ | $66.59_{(1.49)}$ | $52.38_{(1.78)}$ | $67.02_{(0.97)}$ | $64.44_{(1.05)}$ | $70.42_{(1.25)}$ | $68.41_{(1.34)}$ | $69.24_{(1.25)}$ | $68.94_{(1.38)}$ | $70.22_{(1.12)}$ |
| maggic (n=20) | $67.41_{(0.06)}$ | $57.13_{(0.85)}$ | $60.70_{(0.38)}$ | **$61.48_{(0.49)}$** | $57.54_{(0.83)}$ | $58.69_{(0.76)}$ | $52.75_{(1.34)}$ | $54.51_{(1.40)}$ | $54.59_{(1.02)}$ | $55.39_{(1.07)}$ | $50.29_{(0.52)}$ | $55.64_{(0.43)}$ | $54.72_{(1.38)}$ | $57.38_{(1.06)}$ | $55.84_{(1.21)}$ | $56.15_{(1.20)}$ | $54.08_{(1.05)}$ | $56.19_{(0.76)}$ |
| seer (n=20) | $87.92_{(0.08)}$ | $80.67_{(1.67)}$ | $84.53_{(0.33)}$ | **$84.82_{(0.47)}$** | $83.34_{(0.95)}$ | $83.71_{(0.52)}$ | $74.34_{(4.11)}$ | $78.73_{(3.13)}$ | $80.59_{(1.32)}$ | $80.60_{(1.32)}$ | $47.57_{(2.51)}$ | $74.43_{(3.36)}$ | $76.06_{(1.68)}$ | $79.98_{(1.65)}$ | $79.23_{(2.17)}$ | $80.02_{(2.08)}$ | $74.53_{(1.37)}$ | $78.73_{(1.24)}$ |
| compas (n=20) | $67.51_{(0.03)}$ | $63.11_{(0.96)}$ | **$68.01_{(0.44)}$** | $67.91_{(0.55)}$ | $62.07_{(1.51)}$ | $64.43_{(1.38)}$ | $55.67_{(1.60)}$ | $62.56_{(1.61)}$ | $57.67_{(1.90)}$ | $60.87_{(1.33)}$ | $53.33_{(1.85)}$ | $63.59_{(1.30)}$ | $59.49_{(1.72)}$ | $64.62_{(1.14)}$ | $61.06_{(1.92)}$ | $61.59_{(2.15)}$ | $58.30_{(1.77)}$ | $62.58_{(1.87)}$ |
| adult (n=20) | $84.17_{(0.10)}$ | $77.45_{(1.25)}$ | $50.39_{(3.99)}$ | $71.48_{(2.34)}$ | $49.23_{(2.57)}$ | $72.37_{(2.26)}$ | $72.23_{(1.26)}$ | $76.86_{(1.25)}$ | $74.35_{(1.48)}$ | $75.04_{(1.82)}$ | $67.00_{(4.72)}$ | $77.25_{(1.49)}$ | $67.46_{(3.76)}$ | $76.48_{(1.77)}$ | $73.75_{(1.64)}$ | $73.67_{(1.46)}$ | $73.20_{(1.51)}$ | $76.90_{(1.64)}$ |
| drug (n=20) | $77.81_{(0.55)}$ | $70.84_{(2.25)}$ | $75.08_{(1.17)}$ | **$75.29_{(1.11)}$** | $71.68_{(2.25)}$ | $72.14_{(2.64)}$ | $68.31_{(2.81)}$ | $72.65_{(2.00)}$ | $68.12_{(2.38)}$ | $69.68_{(2.41)}$ | $58.78_{(4.26)}$ | $68.89_{(3.57)}$ | $62.13_{(4.94)}$ | $67.75_{(3.72)}$ | $70.16_{(1.87)}$ | $70.16_{(1.75)}$ | $66.60_{(2.95)}$ | $69.18_{(2.59)}$ |
| covid (n=40) | $74.41_{(0.11)}$ | $70.77_{(0.96)}$ | $73.40_{(0.61)}$ | **$73.95_{(0.67)}$** | $70.42_{(0.92)}$ | $71.93_{(0.60)}$ | $63.63_{(1.15)}$ | $68.46_{(0.93)}$ | $70.50_{(1.49)}$ | $70.44_{(1.37)}$ | $56.50_{(0.83)}$ | $68.68_{(1.39)}$ | $66.41_{(2.51)}$ | $70.48_{(1.48)}$ | $68.66_{(1.48)}$ | $68.44_{(1.28)}$ | $61.03_{(0.96)}$ | $67.35_{(0.60)}$ |
| cutract (n=40) | $72.23_{(0.65)}$ | $69.18_{(0.65)}$ | $69.87_{(0.62)}$ | **$71.72_{(0.46)}$** | $68.47_{(0.59)}$ | $69.56_{(0.50)}$ | $63.01_{(2.33)}$ | $67.87_{(1.36)}$ | $65.63_{(2.71)}$ | $67.27_{(2.38)}$ | $54.39_{(1.59)}$ | $68.44_{(0.41)}$ | $61.40_{(1.95)}$ | $67.98_{(1.06)}$ | $67.86_{(0.72)}$ | $67.95_{(0.59)}$ | $59.79_{(1.85)}$ | $66.62_{(1.06)}$ |
| maggic (n=40) | $67.50_{(0.04)}$ | $58.26_{(0.55)}$ | $59.29_{(0.50)}$ | **$60.77_{(0.36)}$** | $57.50_{(0.69)}$ | $59.15_{(0.48)}$ | $55.00_{(1.27)}$ | $56.78_{(1.17)}$ | $55.24_{(0.63)}$ | $56.94_{(0.50)}$ | $48.81_{(0.73)}$ | $56.64_{(0.63)}$ | $54.68_{(0.53)}$ | $58.58_{(0.54)}$ | $57.40_{(0.57)}$ | $57.44_{(0.67)}$ | $55.04_{(1.14)}$ | $57.33_{(1.06)}$ |
| seer (n=40) | $87.92_{(0.08)}$ | $82.93_{(0.55)}$ | $84.29_{(0.39)}$ | **$84.93_{(0.46)}$** | $83.46_{(1.20)}$ | $84.44_{(0.69)}$ | $80.05_{(0.93)}$ | $83.67_{(0.51)}$ | $82.59_{(1.48)}$ | $81.37_{(1.11)}$ | $54.93_{(2.06)}$ | $81.11_{(1.24)}$ | $79.88_{(0.53)}$ | $84.36_{(0.62)}$ | $80.79_{(0.67)}$ | $82.21_{(0.67)}$ | $78.69_{(2.32)}$ | $83.62_{(1.06)}$ |
| compas (n=40) | $67.51_{(0.03)}$ | $62.34_{(0.79)}$ | $67.57_{(0.40)}$ | **$67.85_{(0.37)}$** | $61.34_{(1.67)}$ | $62.84_{(1.21)}$ | $56.29_{(1.96)}$ | $61.02_{(1.69)}$ | $58.85_{(1.28)}$ | $60.11_{(1.22)}$ | $58.88_{(1.04)}$ | $64.37_{(0.91)}$ | $58.61_{(1.35)}$ | $63.54_{(1.11)}$ | $60.83_{(1.19)}$ | $60.95_{(1.22)}$ | $55.94_{(1.50)}$ | $61.04_{(1.56)}$ |
| adult (n=40) | $84.17_{(0.10)}$ | $79.44_{(1.03)}$ | $48.31_{(3.58)}$ | $73.82_{(2.19)}$ | $49.21_{(1.78)}$ | $74.27_{(1.49)}$ | $71.82_{(1.22)}$ | $79.11_{(0.71)}$ | $71.51_{(1.85)}$ | $77.99_{(0.65)}$ | $66.77_{(3.32)}$ | $78.81_{(1.36)}$ | $71.13_{(2.27)}$ | $79.71_{(1.02)}$ | $77.90_{(1.07)}$ | $78.84_{(1.19)}$ | $72.58_{(1.34)}$ | **$80.02_{(0.76)}$** |
| drug (n=40) | $77.81_{(0.55)}$ | $71.86_{(1.07)}$ | $74.30_{(0.59)}$ | **$75.79_{(0.39)}$** | $71.33_{(0.88)}$ | $72.76_{(0.97)}$ | $69.46_{(2.23)}$ | $72.74_{(1.70)}$ | $71.08_{(1.72)}$ | $73.07_{(1.03)}$ | $64.89_{(1.39)}$ | $73.64_{(0.87)}$ | $62.51_{(3.03)}$ | $70.97_{(1.91)}$ | $69.23_{(1.68)}$ | $69.78_{(1.46)}$ | $65.22_{(1.37)}$ | $70.30_{(1.04)}$ |
| covid (n=100) | $74.41_{(0.11)}$ | $71.57_{(0.48)}$ | $73.77_{(0.27)}$ | **$74.71_{(0.34)}$** | $70.71_{(0.46)}$ | $72.76_{(0.44)}$ | $69.05_{(0.96)}$ | $72.13_{(0.66)}$ | $71.60_{(0.59)}$ | $73.22_{(0.46)}$ | $63.52_{(1.29)}$ | $72.04_{(0.57)}$ | $64.25_{(1.51)}$ | $72.64_{(0.64)}$ | $70.08_{(0.67)}$ | $70.78_{(0.59)}$ | $69.05_{(0.48)}$ | $71.96_{(0.49)}$ |
| cutract (n=100) | $72.23_{(0.65)}$ | $70.96_{(0.68)}$ | $70.20_{(0.45)}$ | **$72.51_{(0.55)}$** | $69.97_{(0.97)}$ | $71.94_{(0.89)}$ | $67.94_{(1.01)}$ | $72.42_{(0.66)}$ | $70.53_{(1.51)}$ | $71.98_{(1.39)}$ | $55.72_{(2.04)}$ | $69.14_{(0.93)}$ | $67.59_{(0.71)}$ | $72.42_{(0.54)}$ | $68.79_{(0.83)}$ | $69.68_{(0.76)}$ | $66.89_{(1.03)}$ | $71.52_{(0.70)}$ |
| maggic (n=100) | $67.50_{(0.04)}$ | $59.65_{(0.50)}$ | $58.98_{(0.23)}$ | **$61.32_{(0.42)}$** | $55.71_{(0.83)}$ | $58.90_{(0.72)}$ | $57.20_{(0.91)}$ | $59.34_{(0.64)}$ | $57.26_{(0.50)}$ | $58.28_{(0.46)}$ | $49.54_{(0.71)}$ | $57.91_{(0.74)}$ | $56.36_{(0.54)}$ | $60.11_{(0.54)}$ | $58.89_{(0.51)}$ | $58.99_{(0.42)}$ | $56.17_{(0.68)}$ | $58.86_{(0.69)}$ |
| seer (n=100) | $87.92_{(0.08)}$ | $83.95_{(0.32)}$ | $84.45_{(0.38)}$ | **$85.37_{(0.47)}$** | $83.92_{(0.81)}$ | $85.08_{(0.32)}$ | $81.60_{(0.73)}$ | $85.14_{(0.36)}$ | $83.04_{(0.78)}$ | $84.83_{(0.49)}$ | $70.32_{(2.52)}$ | $83.83_{(0.39)}$ | $81.16_{(1.03)}$ | $85.03_{(0.39)}$ | $81.82_{(0.45)}$ | $82.49_{(0.46)}$ | $78.88_{(0.71)}$ | $84.50_{(0.44)}$ |
| compas (n=100) | $67.51_{(0.03)}$ | $62.56_{(0.72)}$ | $68.02_{(0.29)}$ | **$68.19_{(0.37)}$** | $60.10_{(1.60)}$ | $62.47_{(1.11)}$ | $60.01_{(1.16)}$ | $63.73_{(0.99)}$ | $58.32_{(0.91)}$ | $61.34_{(0.94)}$ | $59.97_{(0.82)}$ | $64.19_{(0.83)}$ | $60.02_{(0.63)}$ | $64.04_{(0.60)}$ | $61.44_{(0.84)}$ | $61.73_{(0.81)}$ | $59.97_{(1.00)}$ | $62.82_{(1.05)}$ |
| adult (n=100) | $84.17_{(0.10)}$ | $81.24_{(0.48)}$ | $46.09_{(1.86)}$ | $74.57_{(1.74)}$ | $47.56_{(3.43)}$ | $73.97_{(1.57)}$ | $74.29_{(1.23)}$ | $80.45_{(0.81)}$ | $75.93_{(1.45)}$ | $78.22_{(1.22)}$ | $77.09_{(0.74)}$ | **$81.66_{(0.53)}$** | $70.70_{(1.02)}$ | $81.04_{(0.37)}$ | $80.56_{(0.53)}$ | $81.10_{(0.44)}$ | $74.04_{(1.44)}$ | $80.23_{(0.98)}$ |
| drug (n=100) | $77.81_{(0.55)}$ | $73.58_{(0.72)}$ | $76.24_{(0.64)}$ | **$76.74_{(0.51)}$** | $69.46_{(2.43)}$ | $71.05_{(2.30)}$ | $68.19_{(2.06)}$ | $73.28_{(1.41)}$ | $72.43_{(0.96)}$ | $73.79_{(0.65)}$ | $67.26_{(1.22)}$ | $75.28_{(0.46)}$ | $62.67_{(2.23)}$ | $73.12_{(0.97)}$ | $70.90_{(1.08)}$ | $71.53_{(1.13)}$ | $68.22_{(1.60)}$ | $73.59_{(0.97)}$ |
| covid (n=200) | $74.41_{(0.11)}$ | $72.33_{(0.52)}$ | $73.40_{(0.25)}$ | **$74.62_{(0.13)}$** | $70.70_{(0.73)}$ | $73.12_{(0.48)}$ | $71.07_{(0.35)}$ | $73.89_{(0.40)}$ | $72.47_{(0.50)}$ | $74.44_{(0.44)}$ | $65.55_{(0.67)}$ | $73.07_{(0.38)}$ | $65.04_{(0.69)}$ | $72.90_{(0.37)}$ | $71.68_{(0.49)}$ | $71.87_{(0.49)}$ | $67.89_{(0.56)}$ | $72.38_{(0.42)}$ |
| cutract (n=200) | $72.23_{(0.65)}$ | $71.75_{(0.71)}$ | $71.39_{(0.76)}$ | $73.01_{(0.85)}$ | $70.28_{(0.67)}$ | $72.39_{(0.84)}$ | $69.28_{(0.55)}$ | $72.41_{(0.65)}$ | $71.83_{(0.65)}$ | **$74.03_{(0.69)}$** | $66.66_{(0.94)}$ | $72.49_{(0.71)}$ | $68.77_{(0.71)}$ | $73.16_{(0.68)}$ | $70.23_{(0.83)}$ | $70.80_{(0.79)}$ | $66.61_{(0.76)}$ | $71.87_{(0.71)}$ |
| maggic (n=200) | $67.50_{(0.04)}$ | $61.39_{(0.44)}$ | $58.92_{(0.47)}$ | **$61.41_{(0.39)}$** | $57.33_{(0.51)}$ | $60.16_{(0.41)}$ | $58.48_{(0.59)}$ | $61.33_{(0.57)}$ | $56.26_{(0.82)}$ | $57.20_{(0.91)}$ | $50.74_{(0.74)}$ | $59.60_{(0.55)}$ | $55.95_{(0.72)}$ | $60.75_{(0.39)}$ | $60.73_{(0.44)}$ | $60.78_{(0.49)}$ | $57.18_{(0.56)}$ | $60.23_{(0.44)}$ |
| seer (n=200) | $87.92_{(0.08)}$ | $84.63_{(0.35)}$ | $84.39_{(0.19)}$ | $85.56_{(0.30)}$ | $83.48_{(0.41)}$ | $84.80_{(0.34)}$ | $82.04_{(0.70)}$ | $85.34_{(0.36)}$ | $84.39_{(0.44)}$ | **$86.57_{(0.27)}$** | $82.15_{(0.43)}$ | $86.03_{(0.23)}$ | $77.73_{(1.65)}$ | $85.19_{(0.22)}$ | $83.38_{(0.32)}$ | $84.15_{(0.42)}$ | $79.71_{(0.58)}$ | $85.26_{(0.25)}$ |
| compas (n=200) | $67.51_{(0.03)}$ | $63.27_{(0.60)}$ | $67.02_{(0.53)}$ | **$68.15_{(0.40)}$** | $60.48_{(1.86)}$ | $63.39_{(1.28)}$ | $60.58_{(0.83)}$ | $64.32_{(0.78)}$ | $60.60_{(0.90)}$ | $63.52_{(1.04)}$ | $61.11_{(0.63)}$ | $65.08_{(0.53)}$ | $56.58_{(1.03)}$ | $63.60_{(0.83)}$ | $61.99_{(0.64)}$ | $62.80_{(0.61)}$ | $60.15_{(1.14)}$ | $63.99_{(0.71)}$ |
| adult (n=200) | $84.17_{(0.10)}$ | $82.12_{(0.41)}$ | $40.96_{(2.47)}$ | $75.84_{(1.43)}$ | $49.89_{(3.22)}$ | $76.11_{(1.43)}$ | $78.18_{(0.26)}$ | $82.32_{(0.29)}$ | $81.66_{(0.19)}$ | $83.17_{(0.20)}$ | $80.06_{(0.53)}$ | **$83.32_{(0.35)}$** | $74.31_{(0.81)}$ | $82.64_{(0.33)}$ | $82.26_{(0.38)}$ | $82.39_{(0.33)}$ | $75.21_{(0.69)}$ | $82.02_{(0.25)}$ |
| drug (n=200) | $77.81_{(0.55)}$ | $76.10_{(0.45)}$ | $75.58_{(0.55)}$ | $76.06_{(0.42)}$ | $70.66_{(1.56)}$ | $72.81_{(1.18)}$ | $71.31_{(1.17)}$ | $75.98_{(0.76)}$ | $69.61_{(2.76)}$ | $71.79_{(1.99)}$ | $72.35_{(0.61)}$ | **$77.41_{(0.56)}$** | $65.25_{(2.47)}$ | $75.26_{(0.77)}$ | $74.38_{(0.50)}$ | $74.78_{(0.54)}$ | $68.39_{(1.26)}$ | $74.33_{(0.61)}$ |

### C.3.6 Comparing CLLM vs TabPFN

We outlined in Appendix C.1 various dimensions on which CLLM and transfer learning / meta-learning / few-shot learning differ. To provide further empirical evidence on why the above dimensions are important (notably the point on the choice of downstream backbone), we compare CLLM with TabPFN [87], a few-shot learning method designed for small tabular problems. We chose TabPFN because it meets our criteria, as it does not require access to external datasets. The model is pre-trained on an extensive set of synthetic tabular datasets and can perform few-shot learning with its transformer backbone.

We use the pre-trained model released by the authors at https://github.com/automl/TabPFN and show the results in Table C.9.

**Takeaways:** We see that CLLM outperforms TabPFN on $6/7$ datasets for the different $n$. The performance gains are especially noticeable in the ultra low-sample regime ($n = 20$).

Finally, we acknowledge that CLLM is not a one-size-fits-all approach. When the assumptions underpinning transfer learning or meta-learning hold (e.g. availability of external

Table C.9 Comparison of CLLM vs TabPFN. We report the AUC averaged over 4 downstream models on $\mathscr{D}_{\text{test}}$.

| Dataset | Uncurated | Curated (CLLM) | TabPFN |
|---|---|---|---|
| covid (n=20) | 73.78 | 73.87 | 66.31 |
| cutract (n=20) | 71.15 | 72.50 | 63.86 |
| maggic (n=20) | 60.70 | 61.48 | 55.49 |
| seer (n=20) | 84.53 | 84.82 | 75.30 |
| compas (n=20) | 68.01 | 67.91 | 57.04 |
| adult (n=20) | 50.39 | 71.48 | 71.70 |
| drug (n=20) | 75.08 | 75.29 | 69.18 |
| covid (n=40) | 73.40 | 73.95 | 67.26 |
| cutract (n=40) | 69.87 | 71.72 | 65.93 |
| maggic (n=40) | 59.29 | 60.77 | 57.75 |
| seer (n=40) | 84.29 | 84.93 | 79.48 |
| compas (n=40) | 67.57 | 67.85 | 57.72 |
| adult (n=40) | 48.31 | 73.82 | 75.36 |
| drug (n=40) | 74.30 | 75.79 | 71.32 |
| covid (n=100) | 73.77 | 74.71 | 69.50 |
| cutract (n=100) | 70.20 | 72.51 | 70.01 |
| maggic (n=100) | 58.98 | 61.32 | 59.07 |
| seer (n=100) | 84.45 | 85.37 | 80.96 |
| compas (n=100) | 68.02 | 68.19 | 63.42 |
| adult (n=100) | 46.09 | 74.57 | 77.36 |
| drug (n=100) | 76.24 | 76.74 | 72.68 |
| covid (n=200) | 73.40 | 74.62 | 70.82 |
| cutract (n=200) | 71.39 | 73.01 | 71.59 |
| maggic (n=200) | 58.92 | 61.41 | 60.67 |
| seer (n=200) | 84.39 | 85.56 | 82.05 |
| compas (n=200) | 67.02 | 68.15 | 65.51 |
| adult (n=200) | 40.96 | 75.84 | 79.18 |
| drug (n=200) | 75.58 | 76.06 | 74.29 |

datasets), combining ideas from these learning paradigms with the augmentation methodology of CLLM could constitute an interesting research direction, but this falls beyond the scope of our current work.

### C.3.7 CLLM with context in low-resource languages

We investigate how the language of tabular column names affects the performance of CLLM. To do so, for a given dataset, we translate the feature names from English to 2 low-resource languages [186], i.e. to (i) *Swahili* and (ii) *Hausa*. Changing the language of the features alters the contextual information provided to the LLM to generate synthetic data.

We report the results in Table C.10. As expected, we observe for both Swahili and Hausa a performance drop compared to using English feature names, thereby mirroring the results in Chapter 4.2.1 on the importance of contextual information in the prompt. However, we note that CLLM for both Swahili and Hausa remains highly competitive compared to other baselines, notably thanks to the curation mechanism.

Table C.10 Test AUC in settings where the names of features have been translated to Swahili and Hausa, with and without curation

| Dataset | Uncurated *English* | Curated *English* | Uncurated *Swahili* | Curated *Swahili* | Uncurated *Hausa* | Curated *Hausa* |
|---|---|---|---|---|---|---|
| covid ($n = 20$) | 73.78 | 73.87 | 71.32 | 73.06 | 68.75 | 70.01 |
| seer ($n = 20$) | 84.53 | 84.82 | 83.92 | 85.47 | 81.57 | 82.73 |
| compas ($n = 20$) | 68.01 | 67.91 | 64.42 | 61.61 | 62.85 | 63.06 |
| covid $n = 40$ | 73.40 | 73.95 | 71.53 | 73.20 | 70.15 | 72.13 |
| seer $n = 40$ | 84.29 | 84.93 | 79.71 | 84.25 | 81.47 | 84.66 |
| compas $n = 40$ | 67.57 | 67.85 | 64.92 | 65.78 | 62.31 | 63.20 |

**Takeaways:**

1. CLLM can provide valuable augmentation and performance gains even with feature names in low-resource languages.
2. For optimal performance, we recommend using CLLM with English feature names or translating the feature names to English before applying CLLM.

### C.3.8 Comparison of curation mechanism vs Data-IQ

We compare CLLM's curation mechanism with the approach taken by Data-IQ [210].

They differ along four dimensions:

1. **Problem setting** (data augmentation vs data understanding): CLLM focuses on the problem of data augmentation in low-data regimes by curating synthetic samples. In contrast, Data-IQ aims to understand and characterize subgroups within a given real dataset.

2. **Conceptually**: CLLM curates a large synthetic dataset $D_{\text{syn}}$ with respect to a very small gold standard real dataset $D_{\text{train}}$. In contrast, Data-IQ aims to characterize subgroups of learnable samples in a single large real dataset $D$ (e.g. to find the hard samples in $D$).

3. **Technically**: CLLM trains the curation model only on the small but gold standard $D_{\text{train}}$. The learning dynamics of the synthetic samples in $D_{\text{syn}}$ are then assessed with respect to this curation model. In contrast, Data-IQ computes learning dynamics for real samples in $D$ using a curator model trained on the same $D$ it assesses.

4. **Empirical performance**: In CLLM, the curation aims at discarding synthetic samples that contradict the learning signal obtained from the real data. If we were to perform the curation like in Data-IQ, this implies that we would instead have to merge $D_{\text{train}}$ and $D_{\text{syn}}$, and train a curator model on $D_{\text{merged}} = D_{\text{train}} \cup D_{\text{syn}}$ to assess $D_{\text{syn}}$. Intuitively, with such an approach, the signal from the small $D_{\text{train}}$ would be overshadowed by $D_{\text{syn}}$, as the latter is a lot larger in size (1000 samples), thus making the curation irrelevant.

We now show empirical evidence regarding the last point by comparing CLLM with a baseline where we train the curator on $D_{\text{merged}} = D_{\text{train}} \cup D_{\text{syn}}$. The results are in Table C.11. We observe that:

1. CLLM's curation outperforms Data-IQ curation on downstream performance across all datasets and $n$.

2. Data-IQ curation does not always improve upon the uncurated baseline, which aligns with our intuition that $D_{\text{syn}}$ overshadows $D_{\text{train}}$ because of its size.

Table C.11 Comparing the CLLM curation mechanism to Data-IQ curation [210]. We report the AUC averaged over 4 downstream models on $\mathscr{D}_{\text{test}}$.

| Dataset | Uncurated | Curated (CLLM) - Ours | Curated (Data-IQ) |
|---|---|---|---|
| covid (n=20) | 73.78 | 73.87 | 73.45 |
| cutract (n=20) | 71.15 | 72.50 | 70.84 |
| maggic (n=20) | 60.70 | 61.48 | 60.68 |
| seer (n=20) | 84.53 | 84.82 | 83.97 |
| compas (n=20) | 68.01 | 67.91 | 67.45 |
| adult (n=20) | 50.39 | 71.48 | 43.59 |
| drug (n=20) | 75.08 | 75.29 | 74.23 |
| covid (n=40) | 73.40 | 73.95 | 73.36 |
| cutract (n=40) | 69.87 | 71.72 | 69.57 |
| maggic (n=40) | 59.29 | 60.77 | 59.65 |
| seer (n=40) | 84.29 | 84.93 | 84.47 |
| compas (n=40) | 67.57 | 67.85 | 67.07 |
| adult (n=40) | 48.31 | 73.82 | 48.92 |
| drug (n=40) | 74.30 | 75.79 | 74.45 |
| covid (n=100) | 73.77 | 74.71 | 73.77 |
| cutract (n=100) | 70.20 | 72.51 | 70.29 |
| maggic (n=100) | 58.98 | 61.32 | 58.66 |
| seer (n=100) | 84.45 | 85.37 | 84.33 |
| compas (n=100) | 68.02 | 68.19 | 67.82 |
| adult (n=100) | 46.09 | 74.57 | 45.13 |
| drug (n=100) | 76.24 | 76.74 | 76.11 |
| covid (n=200) | 73.40 | 74.62 | 73.27 |
| cutract (n=200) | 71.39 | 73.01 | 71.43 |
| maggic (n=200) | 58.92 | 61.41 | 58.71 |
| seer (n=200) | 84.39 | 85.56 | 84.49 |
| compas (n=200) | 67.02 | 68.15 | 67.26 |
| adult (n=200) | 40.96 | 75.84 | 41.01 |
| drug (n=200) | 75.58 | 76.06 | 75.39 |

### C.3.9 CLLM in specialized domains.

We conduct an experiment with an additional dataset, namely Higgs [256]. We chose this dataset because it comes from a specialized domain (physics), where the dataset consists of kinematic properties of particles measured by the particle detectors of an accelerator, which is likely under-represented in the LLMs training corpus.

We note that the contextual information for this dataset is quite specific, as can be seen from the names of the features, which include for example "m_bb", "m_jj", "m_jjj". This particular contextual information, which is not as semantically meaningful as other datasets,

makes it interesting to compare the performance of CLLM with the traditional baselines. We show the results in Table C.12. As we can see, the performance of CLLM is good for the smaller $n$. As $n$ increases, the downstream task benefits more from the use of the other baselines. We highlight that in this case, our curation mechanism still benefits both CLLM and the baselines.

Table C.12 CLLM performance on the Higgs dataset

| Dataset | C | LLM | CTGAN | | TVAE | | NFLOW | |
|---|---|---|---|---|---|---|---|---|
| | Uncur | Cur | Uncur | Cur | Uncur | Cur | Uncur | Cur |
| higgs (n=20) | 65.82 | 70.25 | 62.73 | 67.31 | 59.14 | 69.89 | 56.45 | 69.63 |
| higgs (n=40) | 65.38 | 71.01 | 63.07 | 70.41 | 59.29 | 67.61 | 60.27 | 71.54 |
| higgs (n=100) | 68.62 | 73.42 | 68.46 | 75.59 | 68.90 | 76.05 | 59.73 | 75.86 |
| higgs (n=200) | 66.39 | 75.18 | 74.27 | 79.75 | 73.65 | 79.32 | 60.99 | 79.42 |

## C.3.10 Behavior beyond the low-data regime.

We examine the behavior of baselines beyond $n = 200$ and the low-data regime. We provide a plot in Figure C.3 for three datasets examining the behavior of CTGAN, TVAE and NFLOW for increasing $n$. We fix the CLLM method at $n = 200$ (as we do not see too much of an additional performance increase due to the LLM context window, which limits the number of in-context samples we can provide).

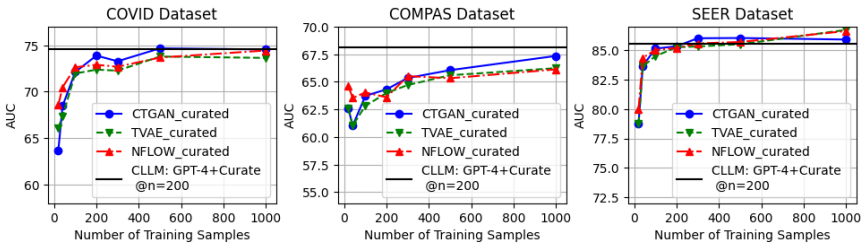



Fig. C.3 Behavior of baselines at high $n$, i.e. beyond the low data regime.

**Takeaways**:

1. Other baselines initially improve with more real samples. However, their performance gains tend to plateau out at around $n \in \{200, 400\}$ samples. This suggests that we only achieve minimal gains as the number of samples added to the baselines increases.

2. CLLM either outperforms the baselines even for very high $n$ (Compas, Covid) or remains competitive (SEER), even when the number of samples used by CTGAN, TVAE and NFLOW is twice or thrice bigger.

### C.3.11 Choice of thresholds

In CLLM, we have thresholds that are used as part of the curation mechanism. The intuition is that our choice of thresholds should discard hard samples, that is, samples for which the confidence is low while the aleatoric uncertainty is also low. Given this intuition, we set an adaptive threshold on the aleatoric uncertainty, with $\tau_{\text{al}} = 0.75 \cdot (\max \text{val}_{\mathscr{F}}(\mathscr{D}_{\text{syn}}) - \min \text{val}_{\mathscr{F}}(\mathscr{D}_{\text{syn}}))$, where $\text{val}_{\mathscr{F}}(\mathscr{D}_{\text{syn}}) := \{\text{val}_{\mathscr{F}}(x,y) \mid (x,y) \in \mathscr{D}_{\text{syn}}\}$ denotes the set of aleatoric uncertainties for the samples in $\mathscr{D}_{\text{syn}}$. Hence, this threshold is adaptive and depends on the dataset at hand, which reduces the number of degrees of freedom to 1, namely, the choice of the confidence threshold. For the latter, we set $\tau_{\text{conf}} = 0.2$. While we do not claim that this value is optimal for all datasets, we find in practice that it is robust for datasets across different domains.

We explore other combinations of thresholds to confirm our intuition. In addition to the choice of thresholds used in our main experiments, we consider two alternatives:

- Aggressive filtering: we set $\tau_{conf} = 0.95$ and $\tau_{al} = 0.2$.
- Permissive filtering: we set $\tau_{conf} = 0.5$ and $\tau_{al} = 0.05$

We then evaluate the test AUC for $n = 20$, with an XGBoost as the downstream model and GPT-3.5 as the LLM backbone. The results are shown in Figure C.4.

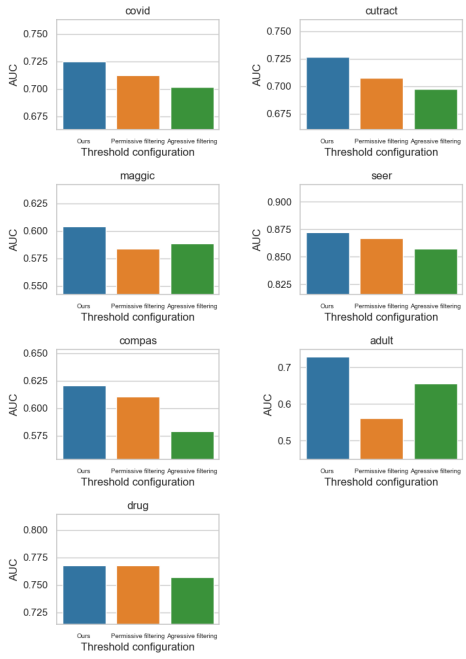



Fig. C.4 Assessment of the effects of different thresholds in CLLM

As we can see, our configuration strikes a good balance between the aggressive filtering and permissive filtering baselines across the 7 datasets.

In an ideal scenario, access to an external validation set could help determine optimal thresholds. However, given our focus on the low-sample regime ($n \leq 100$), we prioritized a versatile configuration that performs consistently across datasets and sample sizes.

# Appendix D

# Appendix for Chapter 5

## D.1 Extended related work

### D.1.1 Enhanced overview of relevant literature

This work primarily engages with works on ML model testing. Consequently, we detail different paradigms of ML model testing next, which we summarize in Table D.1 and Figure D.1 showing that none of the existing paradigms satisfy all the properties for ML testing neither in terms of automation nor from the perspective of relevance as they are unable to incorporate context and/or requirements.

Table D.1 Comparison of ML testing paradigms in terms of how tests are defined, the type of search space (i.e. relevance based on context and requirements), sensitivity to multiple testing and automation to enable scalability.

| | | | Search space | | | | |
|---|---|---|---|---|---|---|---|
| Paradigm | Objective | Test definition | Context-aware | Integrate Requirements | Automated testing | Outputs | Examples |
| Average testing | Overall performance | Data split | N.A. | N.A. . | ✓ | Single score | |
| Behavioral | Test expected behavior | Pre-defined (Human) | N.A. | N.A. | ✗ | Multi-dimensional score | [195, 198, 242] |
| Data-only | Identify divergent groups | Search | ✗ | ✗ | ✓ | Multi-dimensional score | [162, 131, 171, 39, 200] |
| **SMART (Ours)** | Probe for model failure | Search | ✓ | ✓ | ✓ | Multi-dimensional score justifications, report | |

**Existing paradigms in ML model testing.** The ML community has predominantly approached ML model evaluation/testing through the use of held-out test datasets. A single performance metric (accuracy, AUC etc) is then computed on the basis of the test dataset. This single average evaluation may mask nuances of the model's performance along various dimensions. One approach to address this is to create better benchmark datasets when evaluating models on common benchmarking tasks. For example, manual corruptions like Imagenet-C [83] or collecting additional real data, such as the Wilds benchmark [118].

Benchmark datasets are labor-intensive to collect and their utility is limited to the specific benchmark tasks.

What if we want to evaluate models not confined to benchmarking tasks? To test in cases beyond benchmark tasks, the community has proposed trying to find slices or regions wherein the model fails (i.e. via stress tests). As mentioned in Chapter 5.1, this could be ▶ **Behavioral testing**: which requires human expertise and intuition to define the test scenarios (e.g. Checklist [195], HateCheck [198] or 3S-Testing [242]) — which is not automated and does not scale. Moreover, these approaches run a high risk of overlooking critical weaknesses due to human cognitive biases. ▶ **Data-only testing**: which does not account for context or requirements and searches exhaustively (e.g. Autostrat [162], SliceFinder [39] or DivExplorer [171]). This may result in slices focused on arbitrary, less important, or unrealistic and implausible scenarios that are unlikely to occur in reality. Moreover, we run the risk of the multiple testing problem, where due to the large number of tests evaluated, we might discover a divergent group by chance.

**Hypothesis-driven ML model testing.** In contrast, a hypothesis-driven approach to ML model testing brings about the falsifiability approach widely adopted in science. It begins with the formulation of specific, testable hypotheses based on theoretical understanding, context/domain knowledge, and the intended application of the model (i.e. requirements). The concept of hypothesis-driven ML model testing is deeply rooted in work by Popper [182], who proposed the principle of falsifiability as a driver of scientific progress. The progression of knowledge hinges on the formulation and rigorous testing of hypotheses, which can either be falsified or supported by empirical evidence. In the context of hypotheses in ML model testing, testable statements about the model's performance under various conditions, fairness, and robustness can improve our understanding of the model's performance. By rigorously testing these hypotheses, we can uncover the strengths and limitations of ML models.

**Contrast to data-only methods on unstructured data.**

This work contrasts with data-only methods for slice discovery or blind spot discovery applicable to structured data. Specifically, we focus on tabular data, where metadata, in the form of column names, is explicitly encoded into the data. The data-only approaches covered in this work directly search over the raw feature space to identify slices with similar attributes, wherein the model exhibits underperforming predictions.

For completeness, we contrast with data-only approaches, which are often applicable to data without explicit structure and metadata (images, text). There have been numerous methods including [43, 110, 228, 51] to identify slices of unstructured data with systematic failures. They all follow a similar pattern: (1) the data is embedded (often with a pre-trained) model into a representation space and (2) the underperforming slices are identified

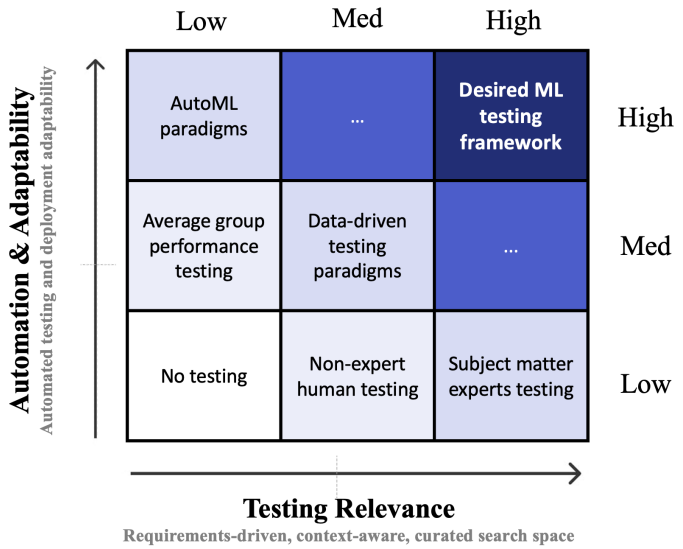



Fig. D.1 Contrasting paradigms of ML model testing along (i) Testing relevance, which accounts for requirements and/or context, and (ii) Degree of automation and adaptability when carrying out testing. We desire a new paradigm of ML testing to address both.

by clustering on the raw embedding space or after dimensionality reduction. We then need to interpret the clusters post-hoc to understand what they represent. This contrasts with the tabular data setting, where we find cohesive groups of features that provide an explicit interpretation. Moreover, in tabular regimes, we do not have access to pre-trained models in the same way as for domains such as images or text.

Specifically, in context-aware testing, we also note that tabular data inherently includes context through interpretable/meaningful feature names and metadata. This context is not naturally present in images (which are tensors of pixel intensities). Hence, extending CAT to other domains like images or text would require incorporating metadata to provide the necessary context, which is often unavailable. Additionally, one would need to develop new methods for operationalizing hypotheses on unstructured data.

**Contrast to software testing.**

Going beyond ML model testing, the idea of testing is also prevalent in software systems. For example, unit tests of functions in a codebase. We highlight that software testing of functional input-output correctness is also an area proposed in software testing, which could theoretically be applied to ML systems as well. That said, we contrast SMART and these paradigms in Table D.2 below, demonstrating that we tackle a different testing problem.

Additionally, a further orthogonal area to ML model testing is that of data validation [224, 181]. In contrast to testing an ML model for failures, data validation aims to test input data pipelines for data quality problems or drift.

Table D.2 Comparison of SMART testing with software testing works

| **Criteria** | SMART | Christakis et al. [37] | Sharma et al [225] | **How is SMART different?** |
|---|---|---|---|---|
| Test case generation | Automatic LLM-based test cases | Pre-Specified dimensions of functional correctness (k-safety properties) | Approximates the black-box model to test with a white-box decision tree model. | SMART uses the LLM to automatically hypothesize the test cases & execute them |
| What is the focus of testing | Identify model failures | Assess I/O functional correctness | SMT solvers find violations to monotonicity properties. i.e. input increase!=output increase | SMART failures encompass a broader range of possibilities |
| Are tests context-aware? | Yes | No | No | SMART uniquely incorporates context awareness into testing, leading to more realistic assessments. |

### D.1.2 Components of the ML Testing pipeline

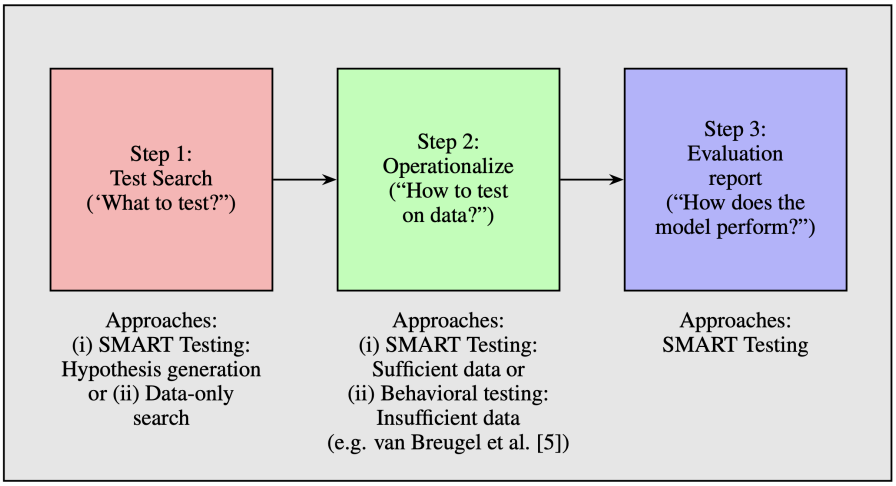



Fig. D.2 Components of ML Testing — (i) Test search, (ii) Operationalize, (iii) Evaluation report, with example approaches for each component.

- Test search: First, we need to decide "what to test". This is a search problem aimed at identifying test dimensions. This could either be done via SMART, which is targeted (via context and requirements), or data-only methods (where the space is larger). Alternatively, if human experts are available, humans could define the test dimensions.

- Operationalize: Second, we need to operationalize the test and address the challenge of "how to carry out this test on data". If we have sufficient data, SMART could operationalize the tests on the data via an interpreter. Alternatively, if we don't have sufficient data to run the test, then SMART could be augmented by behavioral testing approaches such as van Breugel et al. [242], which, once the test has been defined, uses synthetic data to augment small subgroups/slices.
- Evaluation report: Third, we carry out the test and evaluate the model to answer the question, "how does the model perform'. SMART can be used to produce a comprehensive report of failures and justifications in an automated manner.

### D.1.3 Comparison of the features of slice discovery methods

Even as we zoom in on the task of discovering slices where the model might underperform, we observe that SMART has features that are not supported by most other discovery methods. We exemplify some of these features in Table D.3.

Table D.3 Comparison of slice discovery methods

| Criteria | SliceFinder | Pysubgroup | DivExplorer | Autostrat | SMART Testing |
|---|---|---|---|---|---|
| Integrates custom domain knowledge | ✗ | ✗ | ✗ | ✗ | ✓ |
| Constant slice discovery time | ✗ | ✗ | ✗ | ✗ | ✓ |
| Performance is resistant to irrelevant data | ✗ | ✗ | ✗ | ✗ | ✓ |
| Can capture rare slices | ✗ | ✗ | ✗ | ✗ | ✓ |
| Inherently supports logical ORs | ✗ | ✗ | ✗ | ✓ | ✓ |
| Resistant to overfitting the training set | ✗ | ✗ | ✗ | ✗ | ✓ |

## D.2 SMART Details

We present a block diagram of the key components of SMART Testing in Figure D.3. Additionally, for each component, we provide further motivations and technical details not previously covered. We also provide additional technical information on certain implementation details.

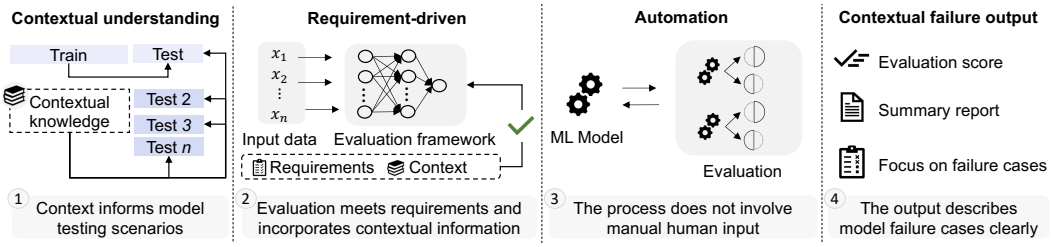



Fig. D.3 A strong machine learning testing framework should incorporate textual understanding during testing, should meet requirements, should be automated, and should provide contextual reports with an emphasis on model failures.

### D.2.1 Contextual understanding

A core component of SMART is leveraging LLMs to strategically identify *what to test for* in tabular ML models. This process is anchored on the premise that LLMs can effectively navigate the space of potential model failure slices, harnessing their contextual understanding to pinpoint slices where the model is most susceptible to failure.

The operation of LLMs within the SMART framework involves critical inputs from the context space $\mathscr{C} = \mathscr{C}_e \times \mathscr{C}_D$:

1. **External Context** ($\mathscr{C}_e$): A string describing the overarching scenario, domain knowledge, and the task at hand.

2. **Dataset Context** ($\mathscr{C}_D$): Extracted from the training data, this includes a description of the observations where the model did not fail ($\hat{y} = y$), or where it failed ($\hat{y} \neq y$). This could be characterized as a string with a description of the covariates of each group (e.g., mean, median, mode, and a textual description of the distribution).

Utilizing these inputs, the LLM generates a set of hypotheses ($\mathscr{H}$) and corresponding justifications ($\mathscr{J}$) regarding potential model failures. The framework also incorporates a *self-refine mechanism* to enhance hypothesis generation. This mechanism iteratively refines hypotheses based on the observed or provided external context ($\mathscr{C}_e$) and dataset context ($\mathscr{C}_D$), re-ranking them by their likelihood.

### D.2.2 Operationalizing variables

Once the hypotheses have been proposed, it is important to operationalize them. Recall from the main text that we define an interpreter function $I : \mathscr{H} \to (\mathscr{X} \to \{0,1\})$ that maps each natural language hypothesis $H_i$ to a corresponding binary-valued indicator function $g_i(\mathbf{x})$. This operationalization to define $g_i$ can be achieved through one of two methods:

1. **LLM-based Operationalization**: Hypothesizing possible operationalizations (e.g., $age > 70$ as a way to operationalize "elderly people"). In this case, the previous interactions and information are provided to the LLM, together with the relevant description of the data. It is then asked to provide possible ways to operationalize a specific hypothesis. This is done by passing an "operationalization prompt" which contains the aforementioned information.

   The following is an example of an operationalization prompt in Python:

```
        f"""
        The following are hypotheses about which people within a dataset the
    model might underperform on.
        Propose specific ranges for each hypothesis. Hypotheses: {hypotheses}.

        Dataset information: {context}. {context_target}

        The dataset contains {len(unique_values)} columns. The columns are {',
    '.join(unique_values.keys())}. The values are {str(unique_values.items())}

        TASK: Propose specific variable ranges for each hypothesis such that
    they are clearly operationalizable and defined. Use this format: Hypothesis:
     <>; Operationalization: <>.
        """

```

   Listing D.1 Operationalization (LLM knowledge): General template

   This is then converted into a specific operationalization for each hypothesis using an external compiler that maps the strings to a function.

2. **Data-driven Operationalization**: Utilizing training data to identify optimal splits for given covariates (e.g., $age >= 82.32$ as a condition to split data based on the "age" covariate). This method receives the covariate (or set of covariates) as input and returns the optimal split. The optimal split is defined as the split that can identify two groups with the largest absolute difference in accuracies given some requirements (e.g. a minimum or maximum group size). This splitting can be done by *any black box splitting function*, which takes in a set of inputs and returns a splitting mechanism.

**Black-box splitting function for data-driven operationalization**. While any black-box splitting function could be applied to the data, we provide details on the specific function used in SMART.

The function implemented is based on a decision tree model, which can be either a regressor or a classifier, depending on the nature of the outcome variable. In all our experiments, we perform classification-based tasks.

The function operates by fitting a decision tree to the data and recursively traversing the tree to find the split that yields the largest absolute difference in the outcome variable's mean value between two slices. The split must also satisfy group size constraints, specified as minimum and maximum group sizes. The traversal process evaluates each potential split, calculating the mean outcome for each slice and the discrepancy between these means. The optimal split is the one that maximizes this discrepancy while adhering to the group size requirements.

The function returns a query string that represents this optimal split. This string can then be used to segment the dataset into the identified slices for further analysis or testing. The following is the pseudocode for the splitting algorithm.

This splitting mechanism is used for continuous features. A separate black-box splitting mechanism is developed for categorical features, based on iterating through different permutations of these features and evaluating them, which aids in variable selection.

In practice, we employ the LLM-based operationalization for the SMART ablation and the data-driven operationalization within the original SMART framework. However, we highlight that this is a design choice that we have found works in practice; there is nothing stopping us from using any operationalization framework within the main SMART framework.

**Algorithm 4:** Optimal splitting mechanism within the SMART framework

**Data:** *dataframe*, *features*, *outcome*, *min_group_size*, *max_group_size*

**Result:** Optimal query string for data split

**Function** `GetOptimalSplitQuery`(*dataframe, features, outcome, min_group_size, max_group_size*)**:**

Validate input features and outcome in *dataframe*

Determine the type of decision tree model based on *outcome* type

Fit the decision tree model to *dataframe* using *features* and *outcome*

Initialize an empty list *conditions* for tracking split conditions

**return** `TraverseTree`(*root,* 0*, conditions*)

**Function** `TraverseTree`(*node, depth, conditions*)**:**

**if** *node is a leaf* **then**

Calculate the discrepancy in *outcome* between the two slices

**return** the condition and discrepancy if group size constraints are met

**end**

Determine left and right conditions based on the threshold at *node*

**return** `TraverseTree`($node.left$, $depth+1$, $conditions \cup \{left_condition\}$)

`// or the right conditions, depending on which has the greater discrepancy`

### D.2.3 Feasibility checks

SMART is built on many modules that can be toggled on or off. One of such modules is a module called "feasibility check" which evaluates whether there are any hypotheses that should be tested in the first place. The experiments presented in Chapter 5.5.1 highlight the importance of being able to identify when no relationship between covariates exists.

The feasibility check contains three steps. First, an LLM is queried to evaluate whether any relationships could exist between the covariates and an outcome variable. For instance, this could be whether a relationship exists between loan default and the annual rainfall in a given region. Second, the answer is self-refinement. This helps to evaluate feasibility because, in practice, initial responses tend to be over-optimistic (such as hypothesizing that annual rainfall is associated with loan default via a geographic proxy). This self-refinement helps to critically evaluate the previous answer. The number of steps in the self-refinement process is a hyperparameter. Lastly, the answer (whether or not a relationship could exist between the two variables and, hence, should be inspected) is converted to a Boolean value via an external function.

The following is pseudocode that implements the feasibility module.

**Algorithm 5:** Feasibility for evaluating slices

**Input:** *unique_values*, *context*, *context_target*, *system_message*, *n_refine*

**Output:** *feasibility_boolean_response*

**Function** `FeasibilityCheck(`*unique_values, context, context_target, system_message, n_refine*`)`**:**

```
// Construct the feasibility task prompt
task ← "Evaluate subgroups for model performance"
task ← task∪ "Context: " ∪ context ∪ "Target: " ∪ context_target
task ← task∪ "Columns: " ∪ join(unique_values.keys())
// Get initial feasibility response
feasibility_response ← GetLLMResponse(task, system_message)
// Refine the answer
feasibility_response ← SelfRefine(unique_values, context, context_target,
  feasibility_response, system_message, n_refine)
// Convert to boolean
boolean_task ← "Based on analysis, provide yes/no answer"
boolean_task ← boolean_task∪ "Analysis: " ∪ feasibility_response
feasibility_boolean_response ← GetLLMResponse(boolean_task)
```

### D.2.4 Data adjustment queries

Given that SMART operates with an LLM, sometimes the framework outputs proposals that do not operationalize well on the data. For instance, even if a column "age" is a categorical variable, SMART might propose to operationalize the hypothesis "elderly people" as $age > 72$, which would cause an error.

To avoid this, we implement an additional data adjustment module that can handle such cases. It catches the error and re-prompts the LLM to find a group that could be operationalized given the data structure and does so iteratively until such a group is found.

The following is a pseudocode function that explains how data adjustment is performed.

**Algorithm 6:** Pseudocode for adjusting subgroup/slice queries

```
Data: X (DataFrame), n_subgroups (integer)
Result: Adjusted subgroup queries in the dataset
Function AdjustSubgroupQuery(X, n_subgroups):
    for each subgroup condition do
        if CheckQueryExistence(X, condition) then
            UpdateSubgroupCondition(condition)
        else
            // Condition yields no rows, adjust it
            adjusted_condition ← GetLLMResponse(condition)
            // Update the condition in the subgroup
            UpdateSubgroupCondition(adjusted_condition)
        end
    end
return
```

### D.2.5 Requirements, automation, and outputs

**Requirements**. SMART can natively integrate user requirements into its framework. This is done by inputting requirements as a string and concatenating it together with the context. Some requirements are directly integrated into the framework itself (e.g. functionalities for determining the minimum or maximum sample size of a data split).

**Automation**. Figure 5.1 showcases the pipeline of SMART and which components are automated. SMART is developed using an "sklearn" style fit, predict framework. The fit method automatically performs a feasibility check, generates hypotheses and justifications, operationalizes them, performs *self-falsification* using empirical data, re-ranks the hypotheses, and saves all intermediate results. The method can then be automatically applied to any piece of data to evaluate whether it underperforms in the groups that have been identified as underperforming.

**Outputs**. In addition to SMART outputting subgroups/slices or a scalar number, it can output a model report. An example report is provided in Appendix D.4.7. We note that this report is simply an example that employs both the findings of the fitting procedure and an additional LLM to summarize the outputs. More fine-grained outputs can be constructed.

### D.2.6 Moving outside of IID data

We also propose that SMART can move outside of IID data and generalize better in the presence of covariate shift (refer to Appendix 5.5.5). We highlight that this is achieved by performing an analysis on the original data and using an LLM to propose possible hypotheses

and splits for a different target domain where no data is available (hence, operationalizing only with the LLM, which has access to previous operationalizations). The following is an example prompt designed to achieve this.

```
f"""
        You have access to the following information.

        Dataset information: {context}. {context_target}

        The dataset contains {len(unique_values)} columns. The columns are {', '.join
    (unique_values.keys())}. The values are {str(unique_values.items())}

        However, you are no longer working with the same data as just described.
    Rather, this is the context: {new_context}.

        These are the hypotheses: {self._updated_hypotheses}.

        TASK: Propose specific variable ranges for each hypothesis such that they are
     clearly operationalizable and defined. Use this format: Hypothesis: <>;
    Operationalization: <>.
        """
```

Listing D.2 Operationalization (LLM knowledge): General template

### D.2.7 SMART and multiple testing

The reason why SMART performs testing via hypothesis generation is that testing for all slices is equivalent to generating and testing a hypothesis on the data. Here, we outline in greater detail how hypothesis generation is connected to multiple testing.

As outlined in the main thesis, searching for $f$ failures may bring up the challenge of multiple hypothesis testing. Specifically, when we evaluate the failure rate of model $f$ across different slices $\mathscr{S}_i \subseteq \mathscr{D}$, we are testing the null hypothesis $H_0^{(i)} : \mu_{\mathscr{S}_i} = \mu_{\mathbb{P}}$ against the alternative $H_1^{(i)} : \mu_{\mathscr{S}_i} \neq \mu_{\mathbb{P}}$, where $\mu_{\mathbb{P}}$ is the population-level expected loss. Empirically, this is evaluated using the sample slice loss $\widehat{\mu}_{\mathscr{S}_i}$ compared against the full empirical dataset loss $\widehat{\mu}_{\mathscr{D}}$. Then, the probability of making a Type I error increases with each test. This drastically inflates the family-wise error rate (FWER). For instance, assuming that each slice is independent, the probability of making one or more Type I errors across all tests is given by $1-(1-\alpha)^m$, where $m$ is the total number of slices tested. While this can be addressed by adjusting for multiple testing, we run into the trade-off between the FWER control and statistical power. As we employ statistical correction methods to account for Type I errors, we increase the probability of Type II errors.

## D.3 Experimental Details

We now summarize all experimental details, datasets and benchmarks.

### D.3.1 Code details

Code for this work can be found at: https://github.com/vanderschaarlab/SMART_Testing

### D.3.2 Datasets

We summarize the different datasets we use in Table D.4. The datasets vary in the number of samples, number of features, and domain.

Table D.4 Summary of the datasets used.

| Name | $n$ samples | $n$ features | Domain | Task |
|---|---|---|---|---|
| Loan | 5000 | 8 | Finance | Loan default |
| Breast | 116 | 8 | Healthcare/Medicine | Breast cancer |
| Compas [5] | 4743 | 8 | Criminal justice | Criminal recidivism |
| OULAD [123] | 5000 | 8 | Education | Exam score |
| Diabetes | 768 | 8 | Healthcare/Medicine | Diabetes label |
| SEER [174] | 2000 | 12 | Healthcare/Medicine | Prostate cancer morality |
| CUTRACT [174] | 20000 | 12 | Healthcare/Medicine | Prostate cancer morality |

Note that, depending on the experiment, we might vary the sample size or features used. This can be because of the added synthetic variables or certain slice discovery algorithms taking too long to run.

### D.3.3 Baselines

We compare SMART testing with the following data-only baselines in Chapter 5.5.

- Autostrat [162]. We use the implementation from [1]
- Pysubgroup [131]. We use the implementation from [2]
- Divexplorer [171]. We use the implementation from [3]
- Slicefinder [39]. We use the implementation from [4]

[1] https://github.com/Adebayo-Oshingbesan/BPS_Autostrat_Code/
[2] https://github.com/flemmerich/pysubgroup [3] https://github.com/elianap/divexplorer
[4] https://github.com/yeounoh/slicefinder

- Sliceline [200]. We use the implementation from [5]

### D.3.4 LLM prompts

#### Generic templates.

```
"""Constructs the task description for the LLM."""
task = f"""Your task is to propose possible hypotheses as to which subgroups within
    the dataset might have worse predictive performance than on average because of
    societal bias in the dataset, insufficient data, other relationships, or others.
    The subgroups might be based on any of the provided characteristics, as well as
    on any combination of such characteristics.

Dataset information: {context}. {context_target}

The dataset contains {len(unique_values)} columns. The columns are {', '.join(
    unique_values.keys())}.

Task: Create {n} hypotheses as to which subgroups within the dataset the model will
    perform worse than on average because of societal biases or other reasons.
    Important: Your hypothesis can contain either one variable or two variables in
    the condition. Therefore, your goal is to find discrepancies in the model's
    performance, not the underlying data outcomes. Justify why you think that for
    each of the {n} hypotheses. Format of the output: Hypothesis: <>; Justification:
    <>.

"""
```

Listing D.3 Generic hypothesis prompt

```
"""
The following are hypotheses about which people within a dataset the model might
    underperform on.
Propose specific ranges for each hypothesis. Hypotheses: {hypotheses}.

TASK: return a dictionary that contains an index number as the key and the column
    value as the value. If there are multiple columns in that hypothesis, return them
     in a list. There are the column names: {', '.join(unique_values.keys())}.
"""
```

Listing D.4 Generic operationalization prompt

#### Example prompts: OULAD Education.

```
"""
----------INPUT TEXT --------------
Your task is to propose possible hypotheses as to which subgroups within the dataset
    might have worse predictive performance than on average because of societal bias
    in the dataset, insufficient data, other relationships, or others. The subgroups
    might be based on any of the provided characteristics, as well as on any
    combination of such characteristics.
```

[5] https://github.com/DataDome/sliceline

```
4
5 Dataset information:
6 Open University Learning Analytics Dataset (OULAD) contains data about courses,
      students and their interactions with Virtual Learning Environment (VLE) for seven
       selected courses (called modules). Presentations of courses start in February
      and October - they are marked by B and J respectively. The dataset consists of
      tables connected using unique identifiers. All tables are stored in the csv
      format.
7
8 The task is to predict the students who are likely to fail or withdraw from a course.
9
10
11 The dataset contains 14 columns. The columns are gender, region, highest_education,
      imd_band, age_band, num_of_prev_attempts, studied_credits, disability, test,
      group_0, group_1, group_2, group_3, group_4. The values are dict_items([('gender
      ', ['M', 'F']), ('region', ['South Region', 'Scotland', 'Yorkshire Region', 'West
       Midlands Region', 'East Anglian Region', 'East Midlands Region', 'South East
      Region', 'Wales', 'North Western Region', 'London Region', 'South West Region', '
      Ireland', 'North Region']), ('highest_education', ['Lower Than A Level', 'A Level
       or Equivalent', 'HE Qualification', 'No Formal quals', 'Post Graduate
      Qualification']), ('imd_band', ['70-80%', '40-50%', '0-10%', '10-20', '50-60%',
      '20-30%', '60-70%', '80-90%', '90-100%', '30-40%', nan]), ('age_band', ['0-35',
      '35-55', '55<=']), ('num_of_prev_attempts', [0, 1, 2, 3, 5, 4, 6]), ('
      studied_credits', {'min': 30, 'mean': 79.77122804326149, 'max': 655}), ('
      disability', ['N', 'Y']), ('test', [False, True]), ('group_0', [False, True]), ('
      group_1', [True, False]), ('group_2', [True, False]), ('group_3', [False, True]),
       ('group_4', [False, True])])
12
13 Task: Create 5 hypotheses as to which subgroups within the dataset the model will
      perform worse than on average because of societal biases or other reasons.
      Therefore, your goal is to find discrepancies in the model's performance, not the
       underlying data outcomes. Justify why you think that. Format of the output:
      Hypothesis: <>; Justification: <>.
14 """
15
16 ----------OUTPUT TEXT --------------
17 Hypothesis 1: Students with disabilities will have worse predictive performance than
      average.
18 Justification: Students with disabilities may face additional challenges in their
      learning process, which could affect their performance in the course. They may
      require additional support or accommodations that are not adequately captured in
      the dataset, leading to lower predictive performance for this subgroup.
19
20 Hypothesis 2: Students from lower socioeconomic backgrounds (indicated by imd_band)
      will have worse predictive performance than average.
21 Justification: Students from lower socioeconomic backgrounds may have limited access
      to resources and support systems that are necessary for successful completion of
      the course. This lack of resources and support may not be adequately captured in
      the dataset, leading to lower predictive performance for this subgroup.
22
23 Hypothesis 3: Students with lower levels of education (indicated by highest_education
      ) will have worse predictive performance than average.
24 Justification: Students with lower levels of education may have less prior knowledge
      and skills that are necessary for success in the course. This lack of
```

```
    foundational knowledge may not be adequately captured in the dataset, leading to
    lower predictive performance for this subgroup.

Hypothesis 4: Students who have attempted the course multiple times (indicated by
    num_of_prev_attempts) will have worse predictive performance than average.
Justification: Students who have attempted the course multiple times may have
    encountered difficulties or challenges in their previous attempts that hindered
    their success. These challenges may not be adequately captured in the dataset,
    leading to lower predictive performance for this subgroup.

Hypothesis 5: Students in certain regions (indicated by region) will have worse
    predictive performance than average.
Justification: Students in certain regions may have different educational systems,
    resources, and support networks that can impact their performance in the course.
    These regional differences may not be adequately captured in the dataset, leading
     to lower predictive performance for subgroups based on region.
```

Listing D.5 Hypothesis generation: OULAD Dataset

```
"""
----------INPUT TEXT --------------

The following are hypotheses about which people within a dataset the model might
    underperform on.
Propose specific ranges for each hypothesis. Hypotheses: Hypothesis 1: Students with
    disabilities will have worse predictive performance than average.

Justification: Students with disabilities may face additional challenges in their
    learning process, which could affect their performance in the course. They may
    require additional support or accommodations that are not adequately captured in
    the dataset, leading to lower predictive performance for this subgroup.

Hypothesis 2: Students from lower socioeconomic backgrounds (indicated by imd_band)
    will have worse predictive performance than average.
Justification: Students from lower socioeconomic backgrounds may have limited access
    to resources and support systems that are necessary for successful completion of
    the course. This lack of resources and support may not be adequately captured in
    the dataset, leading to lower predictive performance for this subgroup.

Hypothesis 3: Students with lower levels of education (indicated by highest_education
    ) will have worse predictive performance than average.
Justification: Students with lower levels of education may have less prior knowledge
    and skills that are necessary for success in the course. This lack of
    foundational knowledge may not be adequately captured in the dataset, leading to
    lower predictive performance for this subgroup.

Hypothesis 4: Students who have attempted the course multiple times (indicated by
    num_of_prev_attempts) will have worse predictive performance than average.
Justification: Students who have attempted the course multiple times may have
    encountered difficulties or challenges in their previous attempts that hindered
    their success. These challenges may not be adequately captured in the dataset,
    leading to lower predictive performance for this subgroup.

Hypothesis 5: Students in certain regions (indicated by region) will have worse
    predictive performance than average.
```

```
Justification: Students in certain regions may have different educational systems, resources, and support networks that can impact their performance in the course. These regional differences may not be adequately captured in the dataset, leading to lower predictive performance for subgroups based on region..

Dataset information:
Open University Learning Analytics Dataset (OULAD) contains data about courses, students and their interactions with Virtual Learning Environment (VLE) for seven selected courses (called modules). Presentations of courses start in February and October - they are marked by B and J respectively. The dataset consists of tables connected using unique identifiers. All tables are stored in the csv format.
.
The task is to predict the students who are likely to fail or withdraw from a course.


The dataset contains 14 columns. The columns are gender, region, highest_education, imd_band, age_band, num_of_prev_attempts, studied_credits, disability, test, group_0, group_1, group_2, group_3, group_4. The values are dict_items([('gender', ['M', 'F']), ('region', ['South Region', 'Scotland', 'Yorkshire Region', 'West Midlands Region', 'East Anglian Region', 'East Midlands Region', 'South East Region', 'Wales', 'North Western Region', 'London Region', 'South West Region', 'Ireland', 'North Region']), ('highest_education', ['Lower Than A Level', 'A Level or Equivalent', 'HE Qualification', 'No Formal quals', 'Post Graduate Qualification']), ('imd_band', ['70-80%', '40-50%', '0-10%', '10-20', '50-60%', '20-30%', '60-70%', '80-90%', '90-100%', '30-40%', nan]), ('age_band', ['0-35', '35-55', '55<=']), ('num_of_prev_attempts', [0, 1, 2, 3, 5, 4, 6]), ('studied_credits', {'min': 30, 'mean': 79.77122804326149, 'max': 655}), ('disability', ['N', 'Y']), ('test', [False, True]), ('group_0', [False, True]), ('group_1', [True, False]), ('group_2', [True, False]), ('group_3', [False, True]), ('group_4', [False, True])])

TASK: Propose specific variable ranges for each hypothesis such that they are clearly operationalizable and defined. Use this format: Hypothesis: <>; Operationalization: <>.
"""

----------OUTPUT TEXT--------------
Hypothesis 1: Students with disabilities will have worse predictive performance than average.
Operationalization: Students with disabilities can be identified by the "disability" variable in the dataset. To test this hypothesis, we can define the range for students with disabilities as those who have a value of "Y" in the "disability" column.

Hypothesis 2: Students from lower socioeconomic backgrounds (indicated by imd_band) will have worse predictive performance than average.
Operationalization: Socioeconomic background can be measured using the "imd_band" variable in the dataset. To test this hypothesis, we can define the range for students from lower socioeconomic backgrounds as those who have an "imd_band" value in the range of "0-30%".

Hypothesis 3: Students with lower levels of education (indicated by highest_education) will have worse predictive performance than average.
```

```
Operationalization: Education level can be determined using the "highest_education" variable in the dataset. To test this hypothesis, we can define the range for students with lower levels of education as those who have a value of "Lower Than A Level" or "No Formal quals" in the "highest_education" column.

Hypothesis 4: Students who have attempted the course multiple times (indicated by num_of_prev_attempts) will have worse predictive performance than average.
Operationalization: Previous attempts can be measured using the "num_of_prev_attempts" variable in the dataset. To test this hypothesis, we can define the range for students who have attempted the course multiple times as those who have a value greater than 1 in the "num_of_prev_attempts" column.

Hypothesis 5: Students in certain regions (indicated by region) will have worse predictive performance than average.
Operationalization: Region can be determined using the "region" variable in the dataset. To test this hypothesis, we can define the range for students in certain regions as those who belong to the regions of "North Region" or "Wales".
```

Listing D.6 Operationalization (LLM knowledge): OULAD Dataset

```
"""
----------INPUT TEXT --------------

The following are groups that are defined based on the dataset. Convert them into a Python dictionary format. Each group should be represented as a key-value pair in the dictionary, where the key is an index (0 to 4), and the value is a string representing the group using Python syntax and logical operators. For multiple conditions, use Python's logical 'and' ('&&') or 'or' ('||'). Ensure the format is a valid Python dictionary.

Examples:
- Single Condition: {0: 'X > 45'}
- Multiple Conditions: {1: '(X > 45) and (Y < 20)'}

Groups to summarize: Hypothesis 1: Students with disabilities will have worse predictive performance than average.

Operationalization: Students with disabilities can be identified by the "disability" variable in the dataset. To test this hypothesis, we can define the range for students with disabilities as those who have a value of "Y" in the "disability" column.

Hypothesis 2: Students from lower socioeconomic backgrounds (indicated by imd_band) will have worse predictive performance than average.
Operationalization: Socioeconomic background can be measured using the "imd_band" variable in the dataset. To test this hypothesis, we can define the range for students from lower socioeconomic backgrounds as those who have an "imd_band" value in the range of "0-30%".

Hypothesis 3: Students with lower levels of education (indicated by highest_education) will have worse predictive performance than average.
Operationalization: Education level can be determined using the "highest_education" variable in the dataset. To test this hypothesis, we can define the range for students with lower levels of education as those who have a value of "Lower Than A Level" or "No Formal quals" in the "highest_education" column.
```

```
19
20 Hypothesis 4: Students who have attempted the course multiple times (indicated by
       num_of_prev_attempts) will have worse predictive performance than average.
21 Operationalization: Previous attempts can be measured using the "num_of_prev_attempts
       " variable in the dataset. To test this hypothesis, we can define the range for
       students who have attempted the course multiple times as those who have a value
       greater than 1 in the "num_of_prev_attempts" column.
22
23 Hypothesis 5: Students in certain regions (indicated by region) will have worse
       predictive performance than average.
24 Operationalization: Region can be determined using the "region" variable in the
       dataset. To test this hypothesis, we can define the range for students in certain
        regions as those who belong to the regions of "North Region" or "Wales".
25 Column names: gender, region, highest_education, imd_band, age_band,
       num_of_prev_attempts, studied_credits, disability, test, group_0, group_1,
       group_2, group_3, group_4
26 Column values: dict_items([('gender', ['M', 'F']), ('region', ['South Region', '
       Scotland', 'Yorkshire Region', 'West Midlands Region', 'East Anglian Region', '
       East Midlands Region', 'South East Region', 'Wales', 'North Western Region', '
       London Region', 'South West Region', 'Ireland', 'North Region']), ('
       highest_education', ['Lower Than A Level', 'A Level or Equivalent', 'HE
       Qualification', 'No Formal quals', 'Post Graduate Qualification']), ('imd_band',
       ['70-80%', '40-50%', '0-10%', '10-20', '50-60%', '20-30%', '60-70%', '80-90%',
       '90-100%', '30-40%', nan]), ('age_band', ['0-35', '35-55', '55<=']), ('
       num_of_prev_attempts', [0, 1, 2, 3, 5, 4, 6]), ('studied_credits', {'min': 30, '
       mean': 79.77122804326149, 'max': 655}), ('disability', ['N', 'Y']), ('test', [
       False, True]), ('group_0', [False, True]), ('group_1', [True, False]), ('group_2
       ', [True, False]), ('group_3', [False, True]), ('group_4', [False, True])])
27 """
28
29 ----------OUTPUT TEXT--------------
30 {
31     0: 'disability == "Y"',
32     1: 'imd_band in ["0-30%"]',
33     2: 'highest_education in ["Lower Than A Level", "No Formal quals"]',
34     3: 'num_of_prev_attempts > 1',
35     4: 'region in ["North Region", "Wales"]'
36 }
```

Listing D.7 Interpreter: OULAD Dataset

### D.3.5 Specific Experimental details

We note that all experiments were performed using two compute resources: a server with NVIDIA RTX A4000 GPU and 18-Core Intel Core i9-10980XE, as well as an Apple M1 Pro 32GB RAM. We exemplify SMART Testing using GPT-4 [**?** ] as the LLM but run further experiments to test the sensitivity to the type of language model in Appendix D.4.1.

### Context-aware testing

**Goal.** We aim to underscore the role of context in ML model testing to prevent false positives, especially when dealing with tabular data where data may contain many irrelevant or uninformative features [106], persisting even post-feature selection [245, 74]. We contrast SMART, which explicitly accounts for context, to data-only approaches, which are context-unaware and only operate on the data.

**Setup.** We fit a predictive model to the training dataset, varying the number of irrelevant, synthetically generated features contained in the dataset, where irrelevant features are drawn from different distributions. We then quantify the proportion of conditions in the identified slices that falsely include the irrelevant synthetic features.

Because different methods are sensitive to different types of irrelevant features, we developed a data-generating process that encompasses many types of variables. Over many runs, different data-only methods pick up on some of these variables, showcasing that all methods are susceptible to randomly sampled irrelevant features in the dataset.

**Sampling mechanism.** To evaluate the impact of irrelevant features, we enrich the dataset by adding synthetic categorical variables. The number of new variables is equal to the number of existing features in the dataset. For each new variable $x_i$, we determine its type by sampling from a Bernoulli distribution with a probability of 0.5. If the sampled value is 0, $x_i$ is a Bernoulli variable with success probability 0.1; otherwise, $x_i$ is a categorical variable with four categories, following a predefined probability distribution (e.g., {0.1, 0.3, 0.4, 0.2}):

$$\begin{aligned} \text{Type}(x_i) &\sim \text{Bernoulli}(0.5), \\ x_i &\sim \begin{cases} \text{Bernoulli}(0.1) & \text{if Type}(x_i) = 0, \\ \text{Categorical}(0.1, 0.3, 0.4, 0.2) & \text{if Type}(x_i) = 1. \end{cases} \end{aligned}$$

We note that there are synthetic data-generating processes that completely break other data-only methods. As an example, creating a unique ID column for each sample breaks the Autostrat algorithm, as all the subgroups/slices identified are unique IDs. The data-generating process employed in our experiment reflects a broad variety of commonly encountered DGPs.

### Requirements-constrained testing

We test whether each of the methods can fulfill three requirements.

The first requirement involved the use of the variable "age" in each detected slice. This was passed as an input to SMART. The other methods do not accept context as input, and therefore, it was not possible to fulfill these requirements. The numbers provided for other methods are simply how often they fulfilled the requirements by chance.

The second and third requirements involved obtaining a minimum and maximum sample size. This was passed as an input to SMART and the ablated SMART version. Based on this, SMART changed its hyperparameter within its function, which asks to indicate a minimum and maximum sample size for the discovered slices. As with the previous experiments, this was not adjusted for the other groups because they do not take textual input.

#### Targeting model failures

To evaluate the targeting of model failures, we test four different tabular models with pre-specified hyperparameters. We find discrepant slices on the training dataset and evaluate them on the testing dataset.

#### Adaptive testing for a deployment environment

The goal of the adaptive testing experiment is to understand the extent to which SMART, as well as other data-only methods, can use data in a source domain to generalize to a new target domain where a covariate shift has been detected. To this end, the datasets provided have a known covariate shift and can be evaluated.

Each method was trained on the UK dataset, and the discovered slices were evaluated on the US dataset. No additional context was provided to data-only methods, as they do not accept any text or context as input.

In contrast, we have provided SMART with the previously discovered slices and hypotheses and have asked to re-evaluate these hypotheses in the context of the US dataset. Specifically, SMART re-hypothesized possible model failures for the US market but used the UK data to operationalize the variables. The ablated version, $SMART_{NSF}$, achieved the best overall performance on the US market. The ablated version (i) did not have access to the UK data and the failures of the models, and (ii) operationalized each covariate using the LLM alone (refer to Chapter D.2.2 for a discussion on operationalizations with different SMART versions). This provides evidence that, in the presence of covariate shift, utilizing inductive knowledge or domain expertise may be more effective in identifying meaningful model failures.

**Discovery of societally important groups and failure understanding**

As discussed previously, SMART provides both possible hypotheses and justifications for model failures, which can be evaluated using a simple "fit" method. Furthermore, SMART prioritizes meaningful data slices that are of societal importance. Such slices can be inspected for any data input.

# D.4 Additional experiments

## D.4.1 Effects of LLMs

In this experiment, we provide additional experimental details that compare the effectiveness of two GPT models, GPT-3.5 and GPT -4. **We highlight that the goal is not to exhaustively test the framework with every LLM**. Rather, the goal is to showcase that SMART is feasible with at least the capabilities of GPT-4. We provide this experiment as a way to measure the sensitivity of the model's performance with smaller LLMs, but highlight that we *do not* recommend using it with smaller LLMs, especially LLMs with fewer than 7B parameters.

### Comparison over identified divergent slices

The following table reproduces the experiment from Appendix 5.5.5 by directly comparing two models - GPT-3.5 and GPT-4. The setup is the same as in the original experiment.

Table D.5 Number of slices identified (out of a maximum of 10) that had significantly divergent performance from average (higher is better). $S_\alpha$ counts the number of significantly divergent groups at $\alpha = 0.05$; $S_{\alpha/n}$ applies the Bonferroni correction.

| | $\mathscr{D}^{\text{UK}}_{\text{train}}$ | | $\mathscr{D}^{\text{UK}}_{\text{test}}$ | | $\mathscr{D}^{\text{US}}_{\text{test}}$ | |
|---|---|---|---|---|---|---|
| | $S_\alpha$ | $S_{\alpha/n}$ | $S_\alpha$ | $S_{\alpha/n}$ | $S_\alpha$ | $S_{\alpha/n}$ |
| Autostrat | $1.00 \pm 0.00$ | $1.00 \pm 0.00$ | $1.00 \pm 0.00$ | $1.00 \pm 0.00$ | $0.95 \pm 0.22$ | $0.95 \pm 0.22$ |
| $\text{PSG}_B$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $1.75 \pm 0.43$ | $1.50 \pm 0.50$ |
| $\text{PSG}_A$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $2.00 \pm 0.00$ | $1.75 \pm 0.43$ | $1.45 \pm 0.50$ |
| Divexplorer | $1.65 \pm 1.88$ | $0.45 \pm 0.92$ | $2.00 \pm 2.28$ | $0.85 \pm 1.53$ | $3.90 \pm 2.57$ | $2.75 \pm 2.21$ |
| Slicefinder | $3.65 \pm 0.96$ | $2.75 \pm 0.62$ | $3.85 \pm 1.11$ | $2.70 \pm 0.46$ | $6.80 \pm 1.03$ | $5.95 \pm 1.02$ |
| Sliceline | $1.00 \pm 0.00$ | $1.00 \pm 0.00$ | $1.35 \pm 0.48$ | $1.35 \pm 0.48$ | $1.35 \pm 0.48$ | $1.35 \pm 0.48$ |
| *SMART_NSF_GPT* $-4$ | $8.30 \pm 0.46$ | $8.00 \pm 0.00$ | $8.45 \pm 0.50$ | $8.00 \pm 0.00$ | $9.20 \pm 0.60$ | $8.35 \pm 0.48$ |
| *SMART_GPT* $-4$ | $9.60 \pm 0.49$ | $9.25 \pm 0.54$ | $9.45 \pm 0.50$ | $8.85 \pm 0.57$ | $8.85 \pm 0.79$ | $8.35 \pm 0.79$ |
| *SMART_NSF_GPT* $-3.5$ | $8.20 \pm 0.60$ | $7.15 \pm 0.65$ | $8.20 \pm 0.75$ | $6.95 \pm 0.67$ | $9.25 \pm 0.62$ | $8.25 \pm 0.43$ |
| *SMART_GPT* $-3.5$ | $10.00 \pm 0.00$ | $9.85 \pm 0.36$ | $10.00 \pm 0.00$ | $9.75 \pm 0.43$ | $7.85 \pm 0.48$ | $7.15 \pm 0.57$ |

The table provides a measure of the model's performance on the training dataset from the same environment ($\mathscr{D}^{\text{UK}}_{\text{train}}$), the testing dataset from the same environment ($\mathscr{D}^{\text{UK}}_{\text{test}}$), and a different deployment environment ($\mathscr{D}^{\text{US}}_{\text{test}}$).

**Takeaway**. Both GPT-3.5 and GPT-4 provide strong increases over benchmark methods with little variability between the two LLMs. One possible reason is that the hypothesis space of potential model failures is somewhat limited. This is evident in the similar hypotheses generated by both GPT models.

### Performance across different models

In this experiment, we vary different tabular machine learning model types and identify how well the ablated and original SMART, identified with GPT-3.5 and GPT-4, can identify slices with large performance discrepancies.

Table D.6 The differences in accuracies between the top slice identified for each method on a testing dataset. The p-value is computed to determine the significance of the difference in accuracy. For the accuracy, higher values imply a greater ability to detect divergent slices (hence, higher is better). For the p-value, the lower value is better. Averages +- standard deviations are shown across 5 runs with random seeds and data splits.

| | Logistic Regression | | SVM | | XGBoost | | Multi-layer Perceptron | |
|---|---|---|---|---|---|---|---|---|
| | $\lvert\Delta Acc\rvert$ | p-value | $\lvert\Delta Acc\rvert$ | p-value | $\lvert\Delta Acc\rvert$ | p-value | $\lvert\Delta Acc\rvert$ | p-value |
| SMART_NSF_GPT-3.5 | $0.23 \pm 0.03$ | $0.00 \pm 0.00$ | $0.23 \pm 0.03$ | $0.00 \pm 0.00$ | $0.12 \pm 0.06$ | $0.14 \pm 0.27$ | $0.23 \pm 0.03$ | $0.00 \pm 0.00$ |
| SMART_GPT-3.5 | $0.34 \pm 0.07$ | $0.00 \pm 0.00$ | $0.34 \pm 0.07$ | $0.00 \pm 0.00$ | $0.28 \pm 0.04$ | $0.00 \pm 0.00$ | $0.34 \pm 0.07$ | $0.00 \pm 0.00$ |
| SMART_NSF_GPT-4 | $0.10 \pm 0.01$ | $0.00 \pm 0.00$ | $0.10 \pm 0.01$ | $0.00 \pm 0.00$ | $0.05 \pm 0.04$ | $0.25 \pm 0.42$ | $0.10 \pm 0.01$ | $0.00 \pm 0.00$ |
| **SMART_GPT-4** | $\mathbf{0.40 \pm 0.02}$ | $\mathbf{0.00 \pm 0.00}$ | $\mathbf{0.40 \pm 0.02}$ | $\mathbf{0.00 \pm 0.00}$ | $\mathbf{0.29 \pm 0.06}$ | $\mathbf{0.00 \pm 0.00}$ | $\mathbf{0.40 \pm 0.02}$ | $\mathbf{0.00 \pm 0.00}$ |

**SMART with deep learning models.** SMART's targeted sampling of hypotheses is entirely independent of the downstream model used. i.e. SMART's context-guided slice sampling mechanism is used to generate hypotheses independently of the downstream model.

We extend our analysis with Logistic Regression, SVM, XGBoost, and MLP to further include two tabular deep learning methods: TabPFN and TabNet. As shown in Table D.7, across all models, SMART is the best at finding subgroups where the models are least reliable.

Table D.7 Identifying slices with the highest performance discrepancies. We demonstrate differences in accuracy ($\lvert\Delta Acc\rvert$) between the top divergent slice identified and average performance across two state-of-the-art deep learning classifiers (over five runs) on the SEER dataset. ↑ is better. 0.00 implies the evaluation method does not support the model.

| Classifier | Evaluation Method | | | | | | |
|---|---|---|---|---|---|---|---|
| | Autostrat | PSG_B | PSG_A | Divexplorer | Slicefinder | Sliceline | SMART |
| TabPFNClassifier | $0.20 \pm 0.10$ | $0.19 \pm 0.05$ | $0.18 \pm 0.05$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.23 \pm 0.05$ | $\mathbf{0.28 \pm 0.17}$ |
| TabNet | $0.10 \pm 0.09$ | $0.10 \pm 0.04$ | $0.10 \pm 0.04$ | $0.02 \pm 0.04$ | $0.00 \pm 0.00$ | $0.17 \pm 0.08$ | $\mathbf{0.17 \pm 0.12}$ |

**Takeaway.** GPT-4 adds additional value when it comes to identifying slices with performance discrepancies. However, GPT-3.5 is still able to find significant performance discrepancies across divergent slices, competitive and in many cases exceeding the performance of other methods (refer to Appendix D.4.2).

### Hypothesis generation

A key component that determines the quality of the SMART method is the ability to navigate the search space by generating hypotheses. Therefore, we provide an overview of the types of hypotheses generated and how they differ based on the type of LLM used.

The table below provides an example of the top 10 hypotheses and justifications generated by GPT-3.5 and GPT-4, for the prostate cancer example in the UK, where the goal is to find specific slices where the model is likely to fail and propose justifications for them.

Table D.8 Comparison of Hypotheses and Justifications generated by GPT-4 and GPT-3.5 for a breast cancer prediction task. The hypothesis targets likely model failures.

| | GPT-4 | | GPT-3.5 | |
|---|---|---|---|---|
| | Hypothesis | Justification | Hypothesis | Justification |
| 1 | Patients with lower age | The model might perform worse for younger patients as prostate cancer is more common in older men, and there might be insufficient data for younger patients. | Subgroup of patients with older age (above 80) will have worse predictive performance due to a higher likelihood of comorbidities and weaker immune systems, leading to higher mortality rates. | Older age is often associated with an increased risk of comorbidities and weaker immune systems, which can negatively impact the prognosis of prostate cancer. |
| 2 | Patients with higher comorbidities | The model might perform worse for patients with multiple comorbidities as the complexity of their health condition might not be fully captured by the model. | Higher Gleason scores indicate more aggressive and advanced prostate cancer, which is generally associated with poorer prognosis and higher mortality rates. | Subgroup of patients with higher Gleason scores (4 or 5) will have worse predictive performance due to more aggressive and advanced cancer, leading to higher mortality rates. |
| 3 | Patients who underwent conservative management | The model might perform worse for these patients as the effectiveness of conservative management might vary greatly among individuals, and this variability might not be well captured by the model. | Subgroup of patients with stage 4 cancer will have worse predictive performance due to the advanced stage of the disease, indicating metastasis and limited treatment options. | Stage 4 prostate cancer is characterized by the spread of cancer to distant organs, indicating a more advanced and aggressive disease. The limited treatment options available for stage 4 cancer can result in worse predictive performance. |
| 4 | Patients with lower prostate-specific antigen levels | The model might perform worse for these patients as lower PSA levels might be associated with early-stage prostate cancer, which might be harder to predict due to lack of symptoms. | Patients with multiple comorbidities often have complex health conditions that can impact the effectiveness of prostate cancer treatment and management, leading to worse predictive performance. | Subgroup of patients with multiple comorbidities (e.g., 3 or 4) will have worse predictive performance due to the presence of other health conditions that may complicate the treatment and management of prostate cancer. |
| 5 | Patients with higher Gleason scores | The model might perform worse for these patients as higher Gleason scores indicate more aggressive cancer, which might progress unpredictably. | Subgroup of patients who received conservative management as their primary treatment option will have worse predictive performance due to the potential delay in receiving more aggressive treatments. | Conservative management may involve watchful waiting or active surveillance, which can result in delayed initiation of more aggressive treatments. This delay in treatment may lead to worse predictive performance. |
| 6 | Patients in stage 4 | The model might perform worse for these patients as stage 4 prostate cancer is advanced and might progress unpredictably. | Primary hormone therapy is often used to control the growth of prostate cancer by reducing the levels of male hormones (androgens). Patients who do not receive this therapy may have uncontrolled cancer growth, resulting in worse predictive performance. | Subgroup of patients who did not receive primary hormone therapy will have worse predictive performance due to the potential lack of hormonal control over the cancer growth. |
| 7 | Patients who underwent radical prostatectomy | The model might perform worse for these patients as the outcomes of radical prostatectomy might vary greatly among individuals, and this variability might not be well captured by the model. | Subgroup of patients who did not undergo radical radiotherapy will have worse predictive performance due to the potential lack of localized cancer control. | Radical radiotherapy is a treatment option that aims to destroy cancer cells in the prostate gland. Patients who do not undergo this treatment may have a higher likelihood of localized cancer progression, leading to worse predictive performance. |
| 8 | Patients with lower Gleason scores | The model might perform worse for these patients as lower Gleason scores indicate less aggressive cancer, which might progress slower and be harder to predict. | Radical prostatectomy involves the surgical removal of the prostate gland. Patients who do not undergo this procedure may have residual cancer cells in the prostate, which can contribute to worse predictive performance. | Subgroup of patients who did not undergo radical prostatectomy will have worse predictive performance due to the potential presence of residual cancer cells in the prostate gland. |
| 9 | Patients in stage 1 | The model might perform worse for these patients as stage 1 prostate cancer is early-stage and might be harder to predict due to lack of symptoms. | Subgroup of patients with higher prostate-specific antigen (PSA) levels (above 50) will have worse predictive performance due to the indication of more advanced cancer. | Higher PSA levels are often associated with more advanced prostate cancer. Patients with higher PSA levels may have a worse prognosis and higher mortality rates, leading to worse predictive performance. |
| 10 | Patients who underwent primary hormone therapy | The model might perform worse for these patients as the outcomes of primary hormone therapy might vary greatly among individuals, and this variability might not be well captured by the model. | The combination of older age, higher Gleason scores, and stage 4 cancer represents a subgroup with multiple negative prognostic factors. The cumulative effect of these factors is likely to result in the worst predictive performance. | Subgroup of patients with a combination of older age (above 70), higher Gleason scores (4 or 5), and stage 4 cancer will have the worst predictive performance due to the cumulative effect of advanced age, aggressive cancer, and metastasis. |

**Key takeaway**. Many of the hypotheses for both models are similar. The underlying difference is in the ordering and justification of the hypotheses.

### D.4.2 Effects of different tabular machine learning models

The primary task of the SMART method is to evaluate a given, trained machine learning model. Thus far, we have been using a logistic regression model as the basis for evaluation in the main experiments. However, the results are *not* sensitive to the type of tabular model. Therefore, we provide additional experiments in which we vary the tabular model for the task. Specifically, we consider the following models initialized with their default hyperparameters for evaluation: Logistic Regression, Support Vector Machines, Boosting (implemented with XGBoost) and a multi-layer perceptron with 2 hidden layers and RELU activation functions in the hidden layers.

**Goal**. The primary goal is to understand whether the framework generalizes to other models that operate under different mapping mechanisms (e.g. a logistic regression, which is a linear model, compared to a tree-based model).

**Setup**. Given that not all the slice discovery or model evaluation algorithms output multiple slices, we constrain the evaluation to only focus on a single slice, which might have discrepant performance. We use the UK prostate cancer dataset and evaluate the discrepancy of the top identified slice relative to the average across all identified model types. The discrepancy is calculated as the absolute differences in the average performance between the two groups, as well as the p-value associated with the difference. The results show the average performance +- standard deviation of 5 random splits. A higher absolute difference indicates that the model fails on one of the slices more than average.

Table D.9 The differences in accuracies between the top slice identified for each method on a testing dataset. The p-value computes the p-value associated with the difference in the accuracy. For the accuracy, higher values imply a greater ability to detect divergent slices (hence, higher is better). For the p-value, the lower is better. Averages +- standard deviations are shown across 5 runs with random seeds and data splits.

| | Logistic Regression | | SVM | | XGBoost | | Multi-layer Perceptron | |
|---|---|---|---|---|---|---|---|---|
| | $\lvert\Delta Acc\rvert$ | p-value | $\lvert\Delta Acc\rvert$ | p-value | $\lvert\Delta Acc\rvert$ | p-value | $\lvert\Delta Acc\rvert$ | p-value |
| Autostrat | $0.24 \pm 0.02$ | $0.00 \pm 0.00$ | $0.24 \pm 0.02$ | $0.00 \pm 0.00$ | $0.09 \pm 0.09$ | $0.33 \pm 0.46$ | $0.24 \pm 0.02$ | $0.00 \pm 0.00$ |
| $PSG_B$ | $0.23 \pm 0.01$ | $0.00 \pm 0.00$ | $0.23 \pm 0.01$ | $0.00 \pm 0.00$ | $0.11 \pm 0.07$ | $0.18 \pm 0.40$ | $0.23 \pm 0.01$ | $0.00 \pm 0.00$ |
| $PSG_A$ | $0.23 \pm 0.01$ | $0.00 \pm 0.00$ | $0.23 \pm 0.01$ | $0.00 \pm 0.00$ | $0.11 \pm 0.07$ | $0.19 \pm 0.42$ | $0.23 \pm 0.01$ | $0.00 \pm 0.00$ |
| Divexplorer | $0.05 \pm 0.11$ | $0.81 \pm 0.43$ | $0.09 \pm 0.13$ | $0.61 \pm 0.53$ | $0.14 \pm 0.15$ | $0.47 \pm 0.49$ | $0.02 \pm 0.05$ | $0.86 \pm 0.32$ |
| Slicefinder | $0.01 \pm 0.00$ | $0.00 \pm 0.00$ | $0.01 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.01 \pm 0.01$ | $0.00 \pm 0.00$ |
| Sliceline | $0.26 \pm 0.06$ | $0.00 \pm 0.00$ | $0.26 \pm 0.06$ | $0.00 \pm 0.00$ | $0.18 \pm 0.09$ | $0.00 \pm 0.01$ | $0.26 \pm 0.06$ | $0.00 \pm 0.00$ |
| *SMART_NSF* | $0.17 \pm 0.01$ | $0.00 \pm 0.00$ | $0.17 \pm 0.01$ | $0.00 \pm 0.00$ | $0.09 \pm 0.05$ | $0.07 \pm 0.16$ | $0.17 \pm 0.01$ | $0.00 \pm 0.00$ |
| SMART | $0.37 \pm 0.03$ | $0.00 \pm 0.00$ | $0.37 \pm 0.03$ | $0.00 \pm 0.00$ | $0.26 \pm 0.06$ | $0.00 \pm 0.00$ | $0.37 \pm 0.03$ | $0.00 \pm 0.00$ |

**Discretizing the inputs**. Many of the discovery methods, however, operate only on categorical data. The previously used dataset, however, has three continuous variables: age, prostate-specific antigen, and comorbidities. Therefore, we also assess the quality of these methods to discover slices on the testing dataset when these three variables are discretized

into 10 bins each. The following table reports the performance of all the methods (note: we did not discretize the dataset for SMART because it can work natively on continuous data).

Table D.10 The differences in accuracies between the top slice identified for each method on a testing dataset when the dataset's continuous features are discretized.

| | Logistic Regression | | SVM | | XGBoost | | Multi-layer Perceptron | |
|---|---|---|---|---|---|---|---|---|
| | $\lvert\Delta Acc\rvert$ | p-value | $\lvert\Delta Acc\rvert$ | p-value | $\lvert\Delta Acc\rvert$ | p-value | $\lvert\Delta Acc\rvert$ | p-value |
| Autostrat | $0.02 \pm 0.02$ | $0.38 \pm 0.27$ | $0.02 \pm 0.02$ | $0.45 \pm 0.37$ | $0.02 \pm 0.02$ | $0.52 \pm 0.32$ | $0.02 \pm 0.02$ | $0.43 \pm 0.33$ |
| *PSG_B* | $0.01 \pm 0.01$ | $0.56 \pm 0.29$ | $0.01 \pm 0.01$ | $0.62 \pm 0.38$ | $0.01 \pm 0.01$ | $0.55 \pm 0.25$ | $0.01 \pm 0.01$ | $0.57 \pm 0.23$ |
| *PSG_A* | $0.01 \pm 0.01$ | $0.55 \pm 0.29$ | $0.01 \pm 0.01$ | $0.61 \pm 0.38$ | $0.01 \pm 0.01$ | $0.55 \pm 0.26$ | $0.01 \pm 0.01$ | $0.59 \pm 0.26$ |
| Divexplorer | $0.10 \pm 0.08$ | $0.47 \pm 0.28$ | $0.11 \pm 0.10$ | $0.41 \pm 0.36$ | $0.13 \pm 0.11$ | $0.38 \pm 0.35$ | $0.13 \pm 0.10$ | $0.38 \pm 0.37$ |
| Slicefinder | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ | $0.00 \pm 0.00$ |
| Sliceline | $0.09 \pm 0.01$ | $0.05 \pm 0.06$ | $0.08 \pm 0.02$ | $0.14 \pm 0.09$ | $0.11 \pm 0.07$ | $0.25 \pm 0.24$ | $0.09 \pm 0.01$ | $0.06 \pm 0.08$ |
| *SMART_NSF* | $0.17 \pm 0.01$ | $0.00 \pm 0.00$ | $0.17 \pm 0.01$ | $0.00 \pm 0.00$ | $0.09 \pm 0.05$ | $0.07 \pm 0.16$ | $0.17 \pm 0.01$ | $0.00 \pm 0.00$ |
| SMART | $0.37 \pm 0.03$ | $0.00 \pm 0.00$ | $0.37 \pm 0.03$ | $0.00 \pm 0.00$ | $0.26 \pm 0.06$ | $0.00 \pm 0.00$ | $0.37 \pm 0.03$ | $0.00 \pm 0.00$ |

**Takeaway.** SMART is able to consistently identify the greatest-performing slices across a number of different tabular models.

### D.4.3 On the inductive biases of ML model testing

**Goal.** We further observe that data-only testing methods implicitly assume the existence of slices with discrepancies in performance. While indeed ML models do fail, it is equally problematic to highlight failures where there are none.

**Setup.** To evaluate this, we propose a fully synthetic setup. Here, both the dependent variable ($Y$) and the independent variables ($\mathbf{X}$) are sampled from a predefined random distribution. Specifically, we predict loan default ($Y \in \{0,1\}$) based on a set of independent variables $\mathbf{X} = \{N_{\text{runs}}, M_{\text{pref}}, A_{\text{rainfall}}, F_{\text{color}}, P_{\text{season}}\}$, which are conceptually and empirically independent of the outcome of interest. Ideally, if we account for context, we should be able to identify these disparate features that should not influence the account, and hence, without prior relationships, we should not flag spurious slices.

We compare data-only methods to SMART under three different data-generating processes (scenarios) that capture diverse underlying dynamics denoted as $\mathscr{S}_{\text{uniform}}$, $\mathscr{S}_{\text{skewed}}$, and $\mathscr{S}_{\text{interactions}}$, where each DGP has a focus on uniform, skewed, and interactive effects, respectively.

The first scenario is given by the variables sampled from the following DGPs:

$$
\begin{aligned}
N_{\text{runs}} &\sim \text{Uniform}\{1,499\},\\
M_{\text{pref}} &\sim \text{Uniform}\{1,5\},\\
A_{\text{rainfall}} &\sim \text{Uniform}\{20000,99999\},\\
F_{\text{color}} &\sim \text{Uniform}\{1,6\},\\
P_{\text{season}} &\sim \text{Uniform}\{0,3\},\\
Y &\sim \text{Uniform}\{0,1\}.
\end{aligned}
$$

The second scenario is given by the variables sampled from the following DGPs:

$$
\begin{aligned}
N_{\text{runs}} &\sim \text{Uniform}\{1,499\},\\
M_{\text{pref}} &\sim \text{Binomial}(1,0.5),\\
A_{\text{rainfall}} &\sim \text{Categorical}(0.1,0.3,0.4,0.2),\\
F_{\text{color}} &\sim \text{Binomial}(1,0.1),\\
P_{\text{season}} &\sim \text{Binomial}(1,0.05),\\
Y &\sim \text{Uniform}\{0,1\}.
\end{aligned}
$$

The third scenario is given by the variables sampled from the following DGPs:

$$
\begin{aligned}
N_{\text{runs}} &\sim \text{Uniform}\{1,499\},\\
M_{\text{pref}} &\sim \text{Binomial}(1,0.5),\\
A_{\text{rainfall}} &\sim \text{Categorical}(0.1,0.3,0.4,0.2),\\
A_{\text{music_hap}} &= M_{\text{pref}} \times A_{\text{rainfall}},\\
A_{\text{run_hap}} &= N_{\text{runs}} \times M_{\text{pref}},\\
Y &\sim \text{Uniform}\{0,1\}.
\end{aligned}
$$

**Analysis.** Table D.11 shows the number of slices spuriously discovered, while Table D.12 outlines the number of conditions within the slices. We can clearly see the pitfalls of data-only approaches that detect slices that in reality have no relation to one another — often surfacing with few conditions per group, which suggests they arise by chance. The rationale for this failure is simply because data-only approaches do not and cannot reason about the features and/or understand context and simply aim to find slices with discrepancies in performance,

which, of course, could arise by chance. In contrast, we see that SMART, by virtue of context-awareness, can avoid surfacing groups, which in reality have no relationships.

Table D.11 Number of discovered slices on a synthetic dataset with no prior relationships in three data generating process scenarios. slices, capped at most 20. Results are averaged over 50 runs and ± standard deviations are shown.

| Method | $\mathscr{S}_{\text{uniform}}$ | $\mathscr{S}_{\text{skewed}}$ | $\mathscr{S}_{\text{interactions}}$ |
|---|---|---|---|
| Autostrat | 1.00 ± .0 | 1.00 ±.0 | 1.00 ± .0 |
| *PSG_B* | 20.00 ± .0 | 20.00 ± .0 | 20.00 ± .0 |
| *PSG_A* | 20.00 ± .0 | 20.00 ± .0 | 20.00 ± .0 |
| divexplorer | 20.00 ± .0 | 20.00 ± .0 | 20.00 ± .0 |
| slicefinder | 20.00 ± .0 | 20.00 ± .0 | 20.00 ± .0 |
| SMART | **0.00 ± .0** | **0.00 ± .0** | **0.00 ± .0** |

Table D.12 Number of conditions per discovered slice (false positives) in three data generating process scenarios. Results are averaged over 50 runs and ± standard deviations are shown. Lower is better.

| Method | $\mathscr{S}_{\text{uniform}}$ | $\mathscr{S}_{\text{skewed}}$ | $\mathscr{S}_{\text{interactions}}$ |
|---|---|---|---|
| Autostrat | 2.17 ± 0.46 | 1.73 ± 1.01 | 1.13 ± 0.35 |
| *PSG_B* | 1.03 ± 0.18 | 1.90 ± 0.71 | 1.27 ± 0.45 |
| *PSG_A* | 1.03 ± 0.18 | 1.90 ± 0.71 | 1.27 ± 0.45 |
| divexplorer | 2.10 ± 0.31 | 2.87 ± 0.68 | 2.40 ± 0.50 |
| slicefinder | 1.40 ± 0.50 | 1.50 ± 0.57 | 1.00 ± 0.00 |
| SMART | **0.00 ± 0.00** | **0.00 ± 0.00** | **0.00 ± 0.00** |

### D.4.4 Cost of SMART

We assess the cost of LLM hypothesis generation and scalability to larger datasets. Specifically, we demonstrate not only that SMART is cheap but also that it is easily scalable to large datasets.

- Scalability: SMART's scalability depends on the number of hypotheses generated, not dataset size (unlike data-only methods). This allows SMART to scale easily to arbitrarily large datasets.
- Cost Analysis: In practical terms, cost then also scales primarily with the number of hypotheses generated, not dataset size. We provide a rough estimate based on token counts of input and outputs for 2 datasets (SEER and OULAD) in Table D.13. This would be less than 0.1 USD for 5 hypotheses and less than 0.5 USD for 100 hypotheses for state-of-the-art models.

Table D.13 Cost of SMART (USD) for different GPT LLMs and different numbers of hypotheses generated. The cost is estimated based on token counts. Note: GPT-4o models are even cheaper.

| Model | Cost SEER 5 Hypothesis (USD) | Cost SEER 100 Hypothesis (USD) | Cost OULAD 5 Hypothesis (USD) | Cost OULAD 100 Hypothesis (USD) |
|---|---|---|---|---|
| GPT-4 | 0.017 | 0.249 | 0.022 | 0.316 |
| GPT-3.5 | 0.004 | 0.050 | 0.005 | 0.064 |
| GPT-4o Mini | 0.0003 | 0.005 | 0.0005 | 0.006 |
| GPT-4o | 0.008 | 0.125 | 0.011 | 0.158 |

### D.4.5 SMART with open-weight models

SMART should ideally be used with the most capable LLM possible. That said, we assess the differences in hypotheses between open-weight models and GPT-4.

We assess Mistral-7b, Qwen-1.5-7b, Llama-3-8b, Llama-70b, whereas for the OULAD and SEER datasets, we generate 5 hypotheses and assess the overlap with the hypotheses generated by GPT-4. This is presented in Table D.14.

To summarize, the overlap between open-source models and GPT-4 is between 60-80%. We find that open-source models propose similar hypotheses, but they are not replacements for more capable models. This highlights that less capable models might propose similar hypotheses, yet they still catch fewer model failures.

Table D.14 Comparison hypotheses by GPT-4 and overlap w/ open-weight models

| Dataset | Factors (GPT Hypotheses) | Mistral-7b | Llama 3-8b | Qwen 1.5-7b | Llama 70b |
|---|---|---|---|---|---|
| **Oulad Dataset** | Disability | ✓ | ✓ | ✓ | ✓ |
| | IMD band | ✓ | ✓ | | ✓ |
| | Age | | ✓ | ✓ | ✓ |
| | Number of previous attempts | ✓ | ✓ | ✓ | |
| | Test (boolean) | | | | |
| | **Oulad Overlap Percentage** | 60% | 70% | 60% | 60% |
| **SEER Dataset** | Age | | ✓ | ✓ | ✓ |
| | Prostate-specific antigen (PSA) | | | | |
| | Comorbidities | ✓ | ✓ | ✓ | ✓ |
| | Treatment (conservative management) | ✓ | ✓ | ✓ | ✓ |
| | Cancer stage | | | ✓ | ✓ |
| | **SEER Overlap Percentage** | 40% | 60% | 80% | 80% |

### D.4.6 Understanding the importance of feature names

SMART uses the implicit context encoded in the interpretable feature names as a source of contextual information to guide hypothesis generation. For instance, in a medical dataset, features with names like age, sex, or patient covariate features provide context to guide LLM hypothesis generation. This contrasts with data-only approaches, which use the numerical data values alone and ignore the context surrounding the feature names.

We aim to assess the sensitivity to interpretable feature names to provide guidance on the use of SMART. First, we perform a qualitative study where we limit the data schema by hiding the feature names (such that they become uninformative) and inspect the hypotheses and justifications generated. We find that in the limited-schema case, SMART generates hypotheses based on inferences about the feature information (e.g. "the model might fail on feature_4 if feature_4 represents gender"). In contrast, informative names guide meaningful hypothesis generation. Such hypotheses and justifications are illustrated in Table D.15.

Second, we evaluate whether limiting the data schema by hiding some feature names and leaving minimal external context affects the detection rates of model failures. We compare two versions of SMART, original and with corrupted feature labels, in identifying data slices with high performance discrepancies from the average (Figure D.4. We find that across two real-world private datasets, hiding the feature names hinders model evaluation. This highlights that feature names play an important role in finding model failures.

These results highlight that while SMART does not rely on any additional feature descriptions, feature names play an important role in finding model failures, just as any human requires interpretable feature names to understand the data. That said, feature names (e.g. column labels such as sex, age, race etc) are present in almost all tabular datasets in the research field and in the real-world where data is stored in SQL tables with column names.

Table D.15 Example hypotheses and justifications when dataset column names are hidden. The hypotheses and justifications for features that do not encode meaningful information lose practical meaning. Context-aware testing benefits from rich feature names that can be used to guide meaningful hypothesis generation.

| Model type | Hypothesis | Justification |
|---|---|---|
| Base | The model will perform worse on predicting the risk of disease for older individuals. | Age is a significant factor in many diseases, and older individuals often have more complex health situations with multiple comorbidities. This complexity could make it more difficult for the model to accurately predict disease risk. |
| Base | The model will perform worse on predicting the risk of disease for individuals with lower Gleason scores. | The Gleason score is a grading system used to determine the aggressiveness of prostate cancer. Lower scores indicate less aggressive cancer, which may be more difficult to predict due to its slower progression and less obvious symptoms. |
| Corrupted | {'1': ['feature_0']} | If feature_0 represents a characteristic such as race or ethnicity, the model might perform worse for certain subgroups due to societal biases. For instance, if the dataset is biased towards a particular race or ethnicity, the model's predictions for other races or ethnicities might be less accurate. |
| Corrupted | {'2': ['feature_4']} | If feature_4 represents a characteristic such as gender, the model might perform worse for certain subgroups due to societal biases. For example, if the dataset is biased towards one gender, the model's predictions for the other gender might be less accurate. |

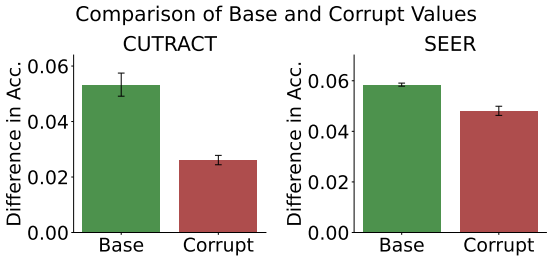



Fig. D.4 Identifying the importance of feature names as a source of information for context-aware testing. We compare two versions of SMART on CUTRACT and SEER datasets. The first version, "base", is the normal SMART method. The second version, "corrupt", has feature names changed to uninformative ones. The goal is to identify slices with the highest performance discrepancies between the average & subgroup prediction, calculated as the difference in their respective accuracies (y-axis). We see that showing feature names enables us to find subgroups with a higher performance discrepancy from baseline performance than without feature names. ↑ is better.

### D.4.7 Example of a model report

We show an example of a model report, which is automatically generated by SMART. A model report includes a summary of key hypotheses, justifications, sources, and summary findings as a written report. A model report fully satisfies the requirements of the user.

In this case, the model is generated to directly match latex output requirements. It is provided in the shaded box below as a part of the US prostate cancer (SEER) [47] evaluation.

A report on the performance of the model has been concluded. The following are the hypotheses, their justifications tested on the model with their conclusions on whether the hypothesis was supported.

| Hypothesis | Justification | Operationalization | Hypothesis Supported |
|---|---|---|---|
| The model may perform worse for older patients | Older patients may have more comorbidities and complex health situations that are not fully captured by the dataset. Additionally, societal biases may lead to less aggressive treatment options being pursued for older patients, which could affect the model's predictions. | $age > 75$ | Yes |
| The model may perform worse for patients with lower prostate-specific antigen levels | Lower levels of prostate-specific antigen may be associated with earlier stages of prostate cancer, which may be harder to predict due to less data and less obvious symptoms. | $prostate_specific_antigen < 10$ | Yes |
| The model may perform worse for patients who have undergone conservative management | Conservative management is a less aggressive form of treatment, which may be chosen due to a variety of factors not captured in the dataset, such as patient preference or other health considerations. This could introduce additional complexity into the model's predictions. | $treatment_conservative_management == 1$ | Yes |
| The model may perform worse for patients with a higher number of comorbidities | Patients with more comorbidities may have more complex health situations that are not fully captured by the dataset. Additionally, these patients may be more likely to die from causes other than prostate cancer, which could confuse the model's predictions. | $comorbidities > 2$ | No |
| The model may perform worse for patients with a higher Gleason score | A higher Gleason score indicates more aggressive cancer, which may be harder to predict due to its rapid progression and the potential for other health factors to influence outcomes. | $gleason_score == 4$ | No |

A further analysis was run on the training and testing datasets to understand the effects of each identified slice. The following are two tables that summarize a part of this information:

| | group_size | support | p_value_bootstrap | num_criteria | outcome_diff | accuracy_diff |
|---|---|---|---|---|---|---|
| H0 | 4915.00 | 0.31 | 0.00 | 1.00 | 0.23 | 0.05 |
| H1 | 7749.00 | 0.48 | 0.00 | 1.00 | 0.26 | 0.01 |
| H2 | 1456.00 | 0.09 | 0.00 | 1.00 | 0.06 | 0.09 |
| H3 | 574.00 | 0.04 | 0.94 | 1.00 | 0.08 | 0.00 |
| H4 | 94.00 | 0.01 | 0.14 | 1.00 | 0.36 | 0.09 |

| | odds_ratio_outcome | odds_ratio_acc | lift_outcome | lift_acc | weighted_relative_y | weighted_relative_acc |
|---|---|---|---|---|---|---|
| H0 | 0.82 | 1.39 | 1.46 | 0.94 | 0.07 | -0.02 |
| H1 | 0.96 | 1.12 | 0.48 | 0.99 | -0.13 | -0.01 |
| H2 | 0.99 | 1.41 | 1.11 | 0.89 | 0.01 | -0.01 |
| H3 | 0.98 | 1.01 | 0.85 | 1.00 | -0.00 | -0.00 |
| H4 | 0.48 | 0.55 | 1.72 | 1.10 | 0.00 | 0.00 |

Recommendations:

1. The model appears to be less reliable for older patients ($age > 75$), patients with lower levels of prostate-specific antigen ($< 10$), and those who have undergone conservative management treatment. It also shows decreased performance for patients with more than two comorbidities and those with a Gleason score of 4. However, the model is more reliable when these conditions are not met.

2. Before deploying the model, the end user should be aware of the following: - The model's performance may be compromised for older patients and those with multiple comorbidities. Consider additional validation or alternative models for these groups. - Patients with lower prostate-specific antigen levels and those who have undergone conservative management treatment may also experience less accurate predictions. Additional clinical insights may be needed for these cases. - Although the model shows decreased performance for patients with a Gleason score of 4, this group is relatively small, so the impact on overall model performance may be limited. However, caution should be exercised when interpreting results for these patients.

Remember, these recommendations are based on the training and test datasets from the UK. If deploying in a different geographical context, consider revalidating the model with local data.

—

Definitions of the metrics:

- **Group Size**:

$$\text{group_size} = |\text{slice}|$$

- **Support**:

$$\text{support} = \frac{|\text{slice}|}{|\text{dataset}|}$$

- **Number of Criteria**:

$$\text{num_criteria} = \text{Count}(\text{"and"}) + 1$$

- **Outcome Difference**:

$$\text{outcome_diff} = |\text{avg_outcome_dataset} - \text{avg_outcome_slice}|$$

- **Accuracy Difference**:

$$\text{accuracy_diff} = |\text{accuracy_dataset} - \text{accuracy_slice}|$$

- **Odds Ratio (Outcome)**:

$$\text{odds_ratio_outcome} = \frac{p_1(1-p_1)}{p_0(1-p_0)}$$

- **Odds Ratio (Accuracy)**:

$$\text{odds_ratio_acc} = \frac{p_1(1-p_1)}{p_0(1-p_0)}$$

*(where $p_1$ and $p_0$ are accuracies in the slice and the rest of the dataset, respectively)*

- **Lift (Outcome)**:

$$\text{lift_outcome} = \frac{p_1}{p}$$

- **Lift (Accuracy)**:

$$\text{lift_acc} = \frac{p_1}{p}$$

*(where $p_1$ is accuracy in the slice and $p$ is accuracy in the entire dataset)*

- **Weighted Relative Outcome**:

$$\text{weighted_relative_outcome} = \text{support} \times \text{diff_outcomes}$$

- **Weighted Relative Accuracy**:

$$\text{weighted_relative_accuracy} = \text{support} \times \text{diff_accuracy}$$

# Appendix E

# Appendix for Chapter 6

## E.1 Comment on SSCP Coverage

Conformal prediction was originally introduced in [250]. However, this so-called *transductive* method is impractical for naive use with most modern algorithms due to multiple refits of the model being required per test point. *Inductive* (or *split*) conformal prediction [165] provides a satisfying solution to this limitation by separating the fitting of the model and the calibration of the conformal predictor across two disjoint datasets. As noted in [240, 267], by considering the trained model $f$ fixed, the non-conformity scores on the calibration and test data are clearly exchangeable, provided the data itself is exchangeable. Of course, this method results in constant interval width across all test points, which is determined by the critical non-conformity score $\varepsilon$. In order to provide adaptive width intervals that reflect the difficulty of a particular example, [167] introduced a residual estimator $\sigma$, resulting in CRF. By adjusting the denominator of the non-conformity score function to $\gamma(\mathbf{x}, y) = \frac{|y - f(\mathbf{x})|}{\sigma(\mathbf{x})}$, the prediction intervals become $[f(\mathbf{x}) - \varepsilon \cdot \sigma(\mathbf{x}), f(\mathbf{x}) + \varepsilon \cdot \sigma(\mathbf{x})]$ and still enjoy marginal coverage in the inductive setting. We emphasize that it is only required that the calibration data and test data are exchangeable; both $\mu$ and $\sigma$ can be trained on any disjoint data provided they are subsequently held fixed.

It is straightforward to check that the method proposed in this work (SSCP) can be viewed as a special case of CRF as described previously and, therefore, maintains the same coverage guarantees. The additional self-supervised function $f_{\text{ss}}$ can simply be viewed as a sub-component of an alternative residual model, defined as $\hat{\sigma}(\mathbf{x}) = \sigma([\mathbf{x}, \mathscr{L}_{\text{ss}}(\mathbf{x})])$. Just as before, and provided this model remains fixed during calibration and testing, the same conformal prediction coverage guarantees hold.

## E.2 Experimental details

We now outline further details of the experimental setup used in Chapter 6.5.

We assess SSCP on multiple open-source regression datasets from the UCI repository [12]. These datasets vary both in the number of features and in the number of samples (documented in Table 6.1). Similar to [196], we standardize the features to have zero mean and unit variance. The output $y$ is rescaled by the mean absolute value. We compute the performance metrics averaged over 5 random runs. In each run, we change the train-test split sampling, where the split proportion is 80-20. $\mathscr{D}_{\text{res}}$ is then split 80-20 with $\mathscr{D}_{\text{train}}$ and then subsequently $\mathscr{D}_{\text{cal}}$ split 80-20 with the remaining $\mathscr{D}_{\text{train}}$. We fix $\alpha$=0.1.

### E.2.1 Predictive and conformal normalization models.

Our baseline models, both the predictive model and conformal normalization model, are implemented using neural networks. Our architectures are similar to those used in [196].

The neural network is an MLP with three fully connected layers, each with ReLU activations. The input is a $d$-dimensional feature vector $X$ passed to an input layer with 64 hidden units. The hidden layer also contains 64 units. Finally, the output layer is a single neuron corresponding to the regression output $y$.

Mean squared error (MSE) loss is used throughout our experiments, optimized using Adam with a learning rate of 5e-4. The batch size is 128, and we apply dropout regularization with a dropout probability of 0.1. Finally, we train using early stopping with patience of 20.

We extract up to the second fully connected layer as the encoder $e$.

### E.2.2 Self-supervised learning

**VIME.** We specifically leverage the self-supervised component of VIME. This consists of an encoder together with an output mask estimator and feature estimator, which are jointly optimized. We keep the default architecture as per [263] with a dense layer for each.

We train with a batch size of 128 for 500 epochs with early stopping enabled. The probability of corruption used is 0.3 and the weight of the feature to mask loss $\alpha$ is set to 2. Future work could assess how to optimally set these parameters specific to the conformal prediction setting.

A minor difference from standard VIME in our application is that instead of the input being raw features, we provide the output from encoder $e$ as the task input and train VIME as the self-supervised predictor $h$.

**Autoencoder.** Similar to VIME, instead of the input being raw features, we provide the output from the encoder $e$ as the input and simply train the autoencoder decoder portion as the self-supervised predictor $h$.

The decoder has the same layers and dimensionality as the encoder $e$, simply being reversed such that the reconstructed output matches the dimensionality of the raw features.

We train with a batch size of 128 for 500 epochs with early stopping enabled. MSE is the loss function. Similar to VIME, the nature and depth of the autoencoder could be assessed as future work concerning how to optimally set these parameters specific to the conformal prediction setting.

### E.2.3 Code details

Code for this work can be found at: https://github.com/seedatnabeel/SSCP

## E.3 Additional Experiments

### E.3.1 Ablation: unlabeled data experiment

**Goal.** This experiment seeks to understand the benefit of the unlabeled data to SSCP. To do so, we perform an ablation of the unlabeled data when used with SSCP for varying proportions $p$ of labeled data. In the ablation variant, i.e., SSCP (Labeled), we only re-use the fully labeled data for the self-supervised task and compare it to SSCP (ALL), which augments the self-supervised dataset with the unlabeled data.

**Analysis.** The results outlined in Table E.1 highlight the benefit of the unlabeled data, especially true in the low data regime when the labeled dataset is small (i.e. low p), where the ablation SSCP (Labeled) showcases an increased average interval width, i.e. when not including the unlabeled data.

Table E.1 Avg. width of intervals with and without unlabeled data (↓ is better)

| p | SSCP (ALL) | SSCP (Labeled) | Difference |
|---|---|---|---|
| 0.1 | 3.603 | 3.893 | 0.290 |
| 0.2 | 2.878 | 3.000 | 0.122 |
| 0.3 | 1.796 | 1.752 | -0.044 |
| 0.4 | 1.505 | 1.459 | -0.045 |
| 0.5 | 1.465 | 1.490 | 0.025 |

### E.3.2 Sanity check: alternative sources of signal

**Goal.** We aim to assess alternative sources of signal with respect to the residual prediction. The goal is to demonstrate the value imbued by self-supervision compared to other alternatives. We assess (i) using an isolation forest instead of SSL and (ii) using the self-supervised signal directly, rather than as input to a residual/normalization model $\sigma$.

**Analysis.** We analyze adding an isolation forest in place of VIME i.e. SSCP (IF) and simply using the SSL signal for normalization, instead of training $\sigma$ (SSL-normalizer). The results are shown in Table E.2 and indicate that indeed overall the usage of VIME provides the best intervals in the most stable manner.

Table E.2 Sanity check comparing alternative self-supervised signals to SSCP (VIME). We report mean average interval width over 5 runs. (↓ better)

| Dataset | SSCP (VIME) | SSCP (IF) | SSCP (SSL-normalizer) |
|---|---|---|---|
| Overall | **1.587** | 1.652 | 4.139 |
| concrete(n=1030) | **0.743** | 0.904 | 0.761 |
| community(n=1994) | **2.462** | 2.581 | 2.468 |
| star(n=2161) | 0.263 | 0.230 | **0.207** |
| bike(n=10886) | **0.690** | 0.711 | 1.016 |
| Blog data (n=52397) | **3.360** | **3.360** | 11.823 |
| Facebook (n=81311) | **1.860** | 2.127 | 8.562 |

### E.3.3 Conformalized Quantile Regression

Conformalized Quantile Regression (CQR) [196] provides a powerful alternative to CRF for achieving locally adaptive intervals for conformal prediction. The method builds upon conditional quantile regression [115] in which the pinball loss objective is utilized such that a model predicts upper and lower quantiles around a target by enforcing non-symmetric and opposite loss functions for the two model outputs. This method was then conformalized in [196] using an inductive conformal prediction procedure.

Although CQR removes the flexibility of post hoc conformalization of an arbitrary model, it often benefits from more efficient intervals in practice. Therefore, one might wish to consider it as an alternative to CRF in our SSCP framework. The key difference between the two methods is that CQR no longer requires a separate conformal normalizer model, as the prediction intervals $[l(\mathbf{x}), r(\mathbf{x})]$ are derived from the main model's quantile predictions. Instead, we pass the self-supervised errors as an additional input into the encoder of the main model. Sharing the encoder between the self-supervised model $f_{ss}$ and the main model $f$ (i.e. $e = e_{ss}$) can still be achieved by passing a placeholder (0) for the additional feature during the self-supervised phase. In our experiments, we consider both a shared encoder between the two models (Shared) and an independent encoder per model (Indep). This procedure is summarized in Algorithm **??**.

We include the results of applying SSCP with CQR to the same experiments described in Chapters 6.5.2 & 6.5.2 in Table E.3 & Figure E.1 respectively.

**Algorithm 7:** Self-supervised Conformal Prediction with CQR

**Input:** $\mathscr{D}_{\text{train}}, \mathscr{D}_{\text{cal}}, e, e_{ss}, g, h$
Train $f_{ss} = h \circ e_{ss}$ on $\mathscr{D}_{\text{train}}$;
Calculate self-supervised errors $\mathscr{L}_{ss}(x_i)$ for all $x_i$ in $\mathscr{D}_{\text{train}}$ and $\mathscr{D}_{\text{cal}}$;
Construct augmented datasets $\mathscr{D}^{\text{aug}}_{\text{train}} = \{([x_i, \mathscr{L}_{ss}(x_i)], y_i) \mid (x_i, y_i) \in \mathscr{D}_{\text{train}}\}$ and $\mathscr{D}^{\text{aug}}_{\text{cal}} = \{([x_i, \mathscr{L}_{ss}(x_i)], y_i) \mid (x_i, y_i) \in \mathscr{D}_{\text{cal}}\}$;
Train $f = g \circ e$ on $\mathscr{D}^{\text{aug}}_{\text{train}}$;
Apply standard CQR calibration on $\mathscr{D}^{\text{aug}}_{\text{cal}}$;
**Output:** $e, e_{ss}, f, g, h$ for test evaluation

In the fully labeled setting, we find that (1) independent encoders outperform shared encoders using the method we describe, and (2) CQR augmented with a self-supervised signal is competitive with and often improves vanilla CQR in terms of average interval width.

In the partially labeled setting, we observe a similar trend, that for lower quantities of labeled data, SSCP provides the largest gains over standard CQR. The trend between the four methods indicates, consistent with the labeled setting, that independent encoders outperform a shared representation. However, evidence for improved performance with additional unlabeled data is inconclusive in these experiments.

Hence, further research on more effective ways to (1) share a representation in this context and (2) integrate unlabeled data provides exciting directions for future work.

Table E.3 Repeating the fully labeled experiments from Chapter 6.5.2 with CQR. We report the mean average interval width (↓ better) over 5 runs. These results indicate that using CQR augmented with a self-supervised loss (SSCP-CQR) often improves vanilla CQR.

| Dataset | CQR | SSCP-CQR (Indep) | SSCP-CQR (Shared) |
|---|---|---|---|
| Overall | 0.904 | **0.866** | 0.894 |
| concrete(n=1030) | 0.532 | **0.505** | 0.512 |
| community(n=1994) | 1.733 | **1.603** | 1.685 |
| star(n=2161) | **0.196** | 0.198 | 0.208 |
| bike(n=10886) | **0.620** | 0.623 | 0.650 |
| Blog data (n=52397) | 1.441 | **1.402** | 1.405 |
| Facebook (n=81311) | 1.274 | **1.220** | 1.292 |

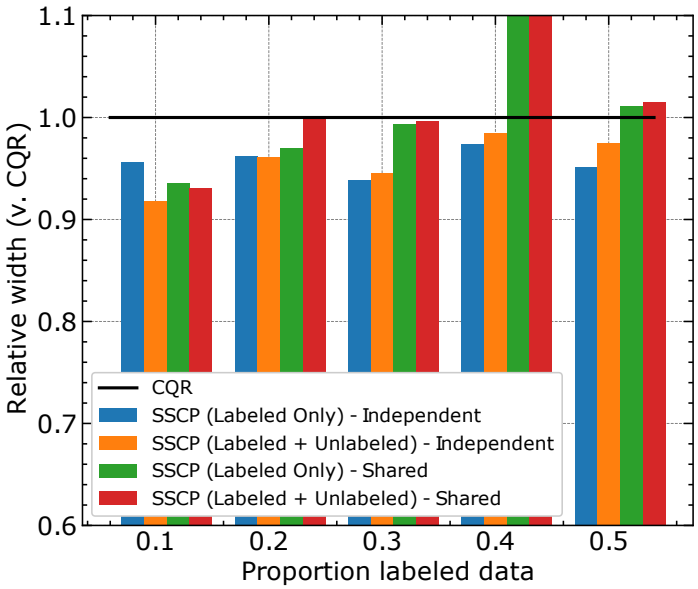



Fig. E.1 Repeating the partially labeled experiments from Chapter 6.5.2 with CQR. Evaluating independent vs. shared representations and the inclusion of additional unlabeled data for various proportions of labeled data. Lower proportions of labeled data achieve the largest gains over standard CQR.

### E.3.4 Evaluating alternative self-supervised tasks & the source of gain

**High-level.**

**Goal.** In the main thesis, we have examined VIME as an example self-supervised task coupled with SSCP. We extend this analysis and explore a reconstructing autoencoder (AE) as an alternative self-supervised task. We refer to these as SSCP (AE) and SSCP (VIME).

**Analysis.** We compare the performance of SSCP (VIME) and SSCP (AE) to CRF in Table E.4. We find that, in general, the addition of the self-supervised loss is a useful feature, resulting in more adaptive intervals.

That being said, VIME seems to provide greater performance improvements across a variety of datasets, compared to the AE. This is reasonable, as VIME has been shown to be a more performant self-supervised task in general for tabular data. Next, we take a deep dive to understand the impact of the self-supervised task and where the improvements lie.

Table E.4 Assessing the impact of the SSL task on improving CRF for two different self-supervised tasks (i.e., VIME and AE). We find in general that the self-supervision helps with VIME being more performant than AE as the self-supervised task. (↓ better)

| Dataset | Method | CRF | SSCP(AE) | SSCP(VIME) |
|---|---|---|---|---|
| concrete(n=1030) | Avg.Width | 0.768 | 0.794 | **0.743** ↓ |
| | Avg.Deficit | 0.099 | 0.117 | **0.098** ↓ |
| | Avg.Excess | 0.263 | 0.276 | **0.253** ↓ |
| community(n=1994) | Avg.Width | 2.602 | 2.667 | **2.462** ↓ |
| | Avg.Deficit | 0.355 | 0.382 | **0.361** ↓ |
| | Avg.Excess | 1.030 | 1.068 | **0.967** ↓ |
| star(n=2161) | Avg.Width | 0.293 | **0.261** ↓ | 0.263 ↓ |
| | Avg.Deficit | 0.036 | 0.040 | **0.031** ↓ |
| | Avg.Excess | 0.120 | 0.110 ↓ | **0.100** ↓ |
| bike(n=10886) | Avg.Width | 0.720 | 0.706 ↓ | **0.690** ↓ |
| | Avg.Deficit | 0.164 | 0.161 ↓ | **0.161** ↓ |
| | Avg.Excess | 0.244 | 0.241 ↓ | **0.232** ↓ |
| Blog data (n=52397) | Avg.Width | 3.474 | 3.400 ↓ | **3.360** ↓ |
| | Avg.Deficit | **3.084** | 3.095 | 3.155 |
| | Avg.Excess | 1.292 | 1.246 ↓ | **1.227** ↓ |
| Facebook (n=81311) | Avg.Width | 1.917 | 1.872 ↓ | **1.860** ↓ |
| | Avg.Deficit | **1.956** | 2.010 | 1.998 |
| | Avg.Excess | 0.584 | 0.563 ↓ | **0.554** ↓ |

**Deep-dive.**

**Goal.** We now take a deep dive to better understand the differences in SSCP with different self-supervised tasks, SSCP (VIME) and SSCP (AE). In addition, we seek to understand the gains from SSCP better. The analysis is performed on the Star dataset.

**Analysis.** We first compare the prediction intervals in low-dimensional space. Similar to the main thesis, we reduce the dimensionality of the input features using PCA to facilitate visualization. We plot PC1 vs. output $y$ and compare the prediction intervals for SSCP (VIME) vs. SSCP (AE) vs. CRF. The results can be seen in Figure E.2.

We see that, in general, CRF results in wider intervals compared to the SSCP variants. Additionally, SSCP (VIME) generally outperforms SSCP (AE) with narrower intervals. In the analysis that follows, we show that the majority of this gain in performance is achieved on the most uncertain examples in the more sparsely populated regions of input space. The

implication is that the self-supervised loss is useful in informing the interval widths on the most uncertain examples where the highest inefficiency occurs.

We also seek to better understand why SSCP (VIME) outperforms SSCP (AE). Beyond simply being a better self-supervised task in general [263], we find that VIME loss also appears to have a stronger relationship with the predictive model's errors compared to the autoencoder, with correlations of **0.33** and **0.06**, respectively.

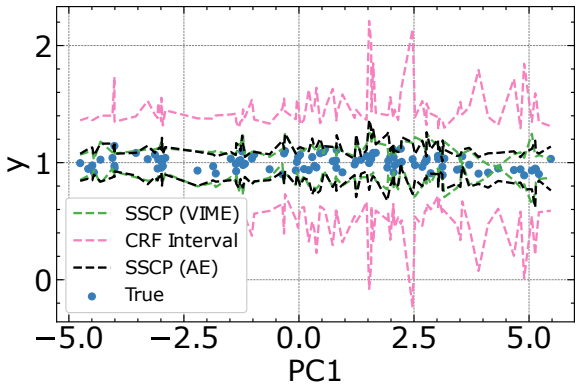



Fig. E.2 Comparing the prediction intervals in different regions. All variants of SSCP provide benefits in the sparser regions. The intervals are similar for SSCP (VIME) and SSCP (AE) in the denser regions, with SSCP (VIME) generally performing better. This is especially the case in the sparser regions with narrower intervals.

Looking beyond the mean correlation of SS loss with the main predictive model error, we find that the largest improvements in reducing interval width occur for samples that have the largest correlation of SS loss with the predictive model error ($\pm$0.35-45). We find limited improvement for samples with the lowest correlation ($\pm$0.11-0.20), highlighting the importance of the relationship between SS loss and the predictive error itself.

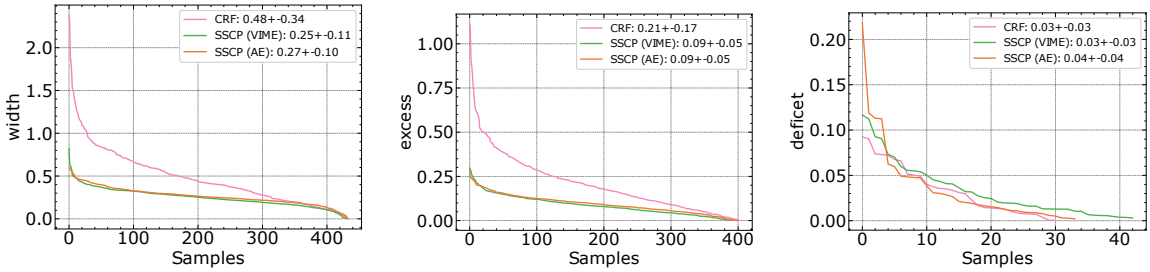
As described in the main thesis, a stronger relationship between the self-supervised loss and the predictive model errors should typically result in a better residual model and, in turn, more efficient prediction intervals. This result confirms that hypothesis. The broader implication is that such considerations should be kept in mind when selecting the self-supervised task. Furthermore, these findings could be leveraged in future works for designing specific self-supervised tasks in the SSCP setting such that the task should seek to explicitly increase this correspondence.

We then continue the deep dive to better understand the performance improvements. We specifically analyze the types of samples for which the self-supervised loss provides the greatest gains. Figure E.3 orders the metrics sample-wise from largest to smallest. We can clearly see that the SSCP improvements are, in general, for those samples for which the metric is the highest in CRF. In other words, the self-supervision helps to improve the

prediction intervals where the greatest inefficiency occurs on the most challenging samples with the greatest uncertainty. This is, of course, also the most natural target, as it is more challenging to improve prediction intervals that are already close to optimal.

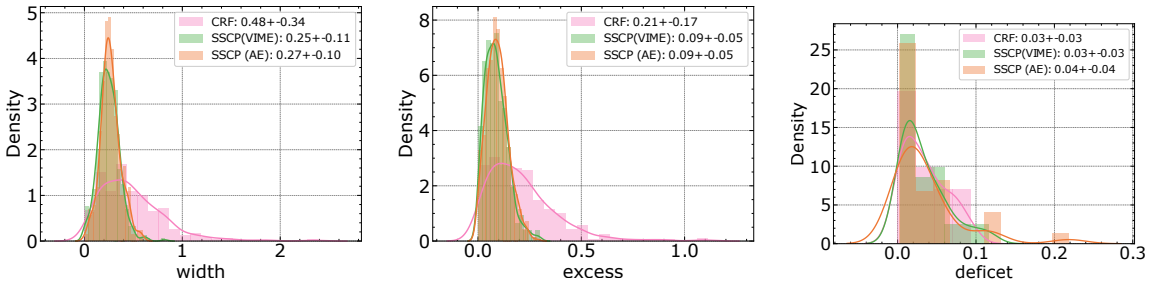
Figure E.4, which illustrates the distribution of said metrics, also confirms this insight. We note the reduction in the long tails of the distributions when augmenting with the self-supervised loss.

To summarize, in general, SSCP largely improves the intervals on samples with higher uncertainty (i.e. larger intervals), which often correspond to regions of sparsity in the input space. These fine-grained improvements at the sample level then translate into dataset-level improvements to the prediction intervals as per our experiments (Chapter 6.5).




(a) Width (b) Excess (c) Deficet

Fig. E.3 Comparing the metrics - ordered from the largest to smallest. We note that the benefit of SSCP for both VIME and AE is largely in reducing the excess width of the intervals on those samples for which it is greatest (i.e. the most inefficient intervals).




(a) Width (b) Excess (c) Deficet

Fig. E.4 Comparison of the distributions of the metrics. We can clearly see that the addition of the self-supervision helps to reduce the long tails of the distributions corresponding to the most inefficient intervals.